

CompTIA Server+

Server Operating Systems Troubleshooting Techniques

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Introduction

Server+

Services

Windows Defender Firewall

Schedule Reboots

Reload OS

Sudo

Snapshots

Safe Mode and Single User Mode

Welcome to the **Server Operating Troubleshooting Techniques** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Common Problems & Causes
- Exercise 2 - Services and Processes
- Exercise 3 - Scheduled Reboots
- Exercise 4 - Services

- Exercise 5 - Privilege Escalation
- Exercise 6 - Recovery

After completing this module, you should be able to:

- Create a Rule Blocking a Port
- Allow Secure Connection through the Windows Defender Firewall
- Allow a Program through the Windows Defender Firewall
- Block a Port with Group Policy Management
- Use Task Scheduler to Schedule Reboots
- Start and Stop a Service
- Use Run as Administrator
- Use the Sudo/Su Command
- Explore Safe Mode
- Work with Snapshots
- Reload OS

After completing this module, you should have further knowledge of:

- Defining Common Problems
- Causes of Common Problems

Exam Objectives

The following exam objectives are covered in this module:

4.4 Given a scenario, troubleshoot common OS and software problems

- Common problems
- Causes of common problems
- OS and software tools and techniques

Lab Duration

It will take approximately **1 hour and 30 minutes** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this pane.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABDM01** - (Windows Server 2022 - Domain Member Server)
- **PLABWIN10** - (Windows 10 - Domain Member Workstation)
- **PLABALMA** - (Alma Linux 8.6 - Stand-alone Linux Workstation)
- **PLABUBUNTU** - (Linux Ubuntu - Stand-alone Linux Server)

Click **Next** to proceed to the first exercise.

Exercise 1 - Common Problems & Causes

Computers are known to have various recurring problems, each of which has a corresponding cause. In this exercise, a variety of common problems and the underlying causes of those problems will be discussed.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Defining Common Problems
- Causes of Common Problems

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Define Common Problems

Unable to Log On

This occurs when a user can't log on to the computer. It could be a local issue with the device itself, or it may be an issue with the network. Some common steps for troubleshooting this issue would be:

- Restart the device - You should try restarting the computer as a first resort. If you can enter the user interface where you would sign in, select "**Restart**" from the power icon in the lower right corner. This will force a restart of your computer, allowing you to see if the problem persists after the reboot.
- Run the Start-up repair program.
- Boot the device in Safe mode - This only loads necessary drivers and can help if something is stopping the ability to log on.
- Create a new user account.
- Disable fast start-up - Fast start-up can be a feature that stops the computer from being able to log into an account. Fortunately, it's simple to turn off this feature in the System and Security settings if a user can log in with a different account or can get in with safe mode.
- Ensure Windows is updated - Updating windows fixes issues and bugs in the system.

Unable to Access Domain Resources

If you cannot access domain resources, it is likely due to outdated network drivers, incorrect configuration, or a problem communicating with the DNS provider. Proxy servers can cause this problem. First, try disabling the proxy server to see if the computer can reconnect to the Internet. It's advised not to modify the settings if a restricted Internet connection is used, like at school or work. These connections can have their own unique settings and restrictions. Conflicting configurations might prevent you from connecting to the desired resources. Some common steps for troubleshooting this problem would be:

- Update the drivers - Maintaining updated drivers will help ensure the hardware and software function together.
- Clear the DNS cache - Data caching is helpful and can improve the speed of numerous services and programs. On the other hand, it may be the cause of the issue if it stores inaccurate or corrupted information. To clear the cache, access the Windows command line and type **ipconfig /flushdns**.

- Reset the connection - Perhaps just flushing the DNS cache is not enough. Other commands like **ipconfig /registerdns**, **ipconfig /release**, **ipconfig /renew** can be used in conjunction with the **ipconfig /flushdns** command.
- Changes in network IP settings - Changes to the Wi-Fi network settings might cause unexpected problems with the connection. Users might be unable to connect because of static IPv4 or IPv6 address conflicts.

Unable to Access Files

This occurs when users do not have access to their files. It could be due to a network misconfiguration, network access issues, or changed permissions. Some common steps for troubleshooting this problem would be:

- Ensure the proper network is selected - If users are attempting to join the wrong network, they will not have access to their usual files.
- Check the permissions of the file - If file permissions have changed, users may be removed from accessing them.

System File Corruption

Over time and use, the system files on a computer might become corrupted. A simple restart can often fix many issues with the device because it restarts those system files. Several built-in tools can help diagnose and repair these issues if restart does not work. Some common steps for troubleshooting this problem would be:

- Sfc - The System file checker will verify the system files are not corrupt and repair them.
- Chkdsk - Attempts to find errors on the hard disk and repair bad sectors.
- Dism - Conducts a scan of the computer image to see if any corrupted or damaged files may be repaired automatically.

End of Life/End of Support

Each vendor will support their products for a time span, providing patching and support for the software. Once that time has passed, if the person or organization would like to continue using the product, they will be responsible for those security updates. Some common steps for troubleshooting this problem would be:

- Microsoft - 5 years of mainstream support and 5 additional years of extended support for their operating systems.

- Apple - There is no EOL for their operating systems. If the hardware can support the new operating system, the customer can upgrade.
- Linux - There is no set rule for Linux. It will depend on the creators of the distribution on how they will support their operating system.

Slow performance

Many issues can cause slow performance in a device. Some common steps for troubleshooting this problem would be:

- Look for hardware, applications and programs that use a lot of resources on the device and shut them down if possible.
- Check connections like the Ethernet and Wi-Fi.
- The hard drive might be fragmented and in need of defragmentation.
- Upgrade to a Solid-State Drive (SSD).
- Add more RAM.
- Perform a hard reset if data is backed up to start with a clean device.

Cannot Write to System Logs

Event Log Service is the service program that comes built-in with Windows. The Event Log program writes all the information and error messages to a text file. The Windows Event Log Service is used for many of the tasks that Windows does on its own. Without this service, several Windows programs like Task Scheduler, Calendar, and Mail won't function correctly. Some common steps for troubleshooting this problem would be:

- Access the Service MMC and Start the service if not started or restart it if it is currently running.
- Edit the registry to fix ownership of the logs.
- Check the Permissions to ensure user access is allowed.
- Check log retention to ensure proper configuration.
- Clean up old logs that might be using up all the space.
- Backup and reinstall Windows.

Service Failures

Services run in the background of the computer and allow the use of many Windows services and third-party applications. When they do not work, not just that service but other dependent

services will not function either. Some common steps for troubleshooting this problem would be:

- Update the drivers to ensure the latest ones are used.
- Ensuring the service is started will be especially important if it is a dependent service.
- Check the start-up method for the service to ensure proper configuration.
- Restart the service if necessary.
- Restart the device if necessary.

System or Application Hanging

Sometimes closing an application does not work and is known as a runaway application. This can cause resource utilization and cause further issues for other programs that might need to use the same system file. Some common steps for troubleshooting this problem would be:

- Use Task Manager to find and force close the application.
- Use the command line to kill the process.
- Restart the device.

Freezing

When computer resources become overwhelmed, it can cause the device to freeze. Ensuring unnecessary applications are closed will help to keep the computer running smoother. Often a restart will help fix freezing issues.

Causes of Common Problems

Incompatible Drivers/Modules and Unstable Drivers or Software

Verify you are not using outdated or incorrect drivers with a piece of hardware or software. Ensure the drivers are updated to the most recent versions. Use manufacturer supplied driver over generic whenever possible.

Not Joined to Domain

Often this is a configuration issue made when setting up the server, active directory, or the account.

Clock Skew

Over time the system clock can start to lose time. This is due to the CMOS battery in the computer starting to lose charge. It will need to be replaced, and the time will need to be fixed in the BIOS. Common signs of this will be certificate errors on websites that can cause authentication issues.

Memory Leaks

Memory leaks occur when an application fails to release its allocated memory and other system resources after its use has ended. This prevents users from doing other things on the computer since there isn't enough RAM. Fixing a memory leak can be as simple as closing resource-intensive programs or turning off some starting programs.

Buffer Overrun/Overflow

Buffer overflow occurs when a software or a process tries to store more information in a buffer than the buffer can hold. All data that goes over the buffer's capacity will be lost, and the values stored at the memory addresses right next to the buffer's location will be overwritten.

Incompatibility

Insecure Dependencies

Since applications have expanded in size and complexity, the average number of external services they rely on has increased. It is the developer's responsibility to upgrade to the most recent version of any third-party dependencies once a security flaw is discovered and a patch is provided. It is not uncommon for malicious packages to be published on package management systems to take advantage of security holes introduced by errors in the source code of third-party dependencies or by the absence of names for deprecated packages.

Version Management

If the version that you are using is not up to date, this can cause it to become unstable and incompatible. Once tested, updates and patches should be applied to help close those security gaps.

Architecture

Everything in computing has a structure and a way that data and information are supposed to flow. If the operating system is not built correctly, the device won't work. If the network is not connected correctly, there will be no connectivity.

Update Failures

This often can be attributed to file corruption, insufficient storage, driver, and hardware incompatibilities.

Missing Updates

This can be caused by not updating the device. Through group policy, automatic updates will likely be disabled because organizations want to test them to ensure they will not cause issues with the production environment.

Missing Dependencies

A software dependency is a previously developed code library or package that is called upon by another program. To construct models, a machine learning project may use a JavaScript library. Software dependencies let developers speed up their work by reusing parts of it in new projects. Not having these files prevents the applications that rely on them from working.

Downstream Failures due to Updates

This term refers to one process needing another process to function and not having it due to improper updates.

Inappropriate Application-level Permissions

When users can't access information or applications that they should be able to, they may have improperly set up permissions.

Improper CPU Affinity and Priority

Applications can use processor affinity or CPU pinning to pin a process or thread to a single core or a set of cores. If a thread is pinned to a certain CPU core, the operating system will only

schedule it to execute on that core when it is scheduled. If it is improperly configured, it can disrupt the functions of the device.

Exercise 2 - Services and Processes

Windows Firewall is a host-based firewall that applies security on a computer by blocking unauthorized access to its services and decreasing its exposure to potentially destructive network probes when connected to the Internet or a local Intranet.

Windows Firewall is managed using two applications. Namely, Windows Firewall (known as Windows Defender Security Center in Windows 10) and Windows Defender Firewall with Advanced Security. Windows Defender Firewall with Advanced Security includes features for creating rules for granular control of Inbound and Outbound traffic. It has Connection Security rules for creating IPsec policies and network isolation. It has been available in versions of Windows dating back to Windows Vista.

Internet Protocol security (IPsec) is supported by Windows Defender Firewall, allowing you to require authentication from any device attempting to communicate with your device. If another device isn't able to verify its identity as a trusted device, it won't be able to exchange data with others. You can also use IPsec to encrypt network traffic to prevent it from being read by network packet analyzers that a malicious user could use to connect to the network.

The Windows Defender Firewall has the following advantages:

- Defend against network security attacks. The Windows Defender Firewall minimizes the device's attack surface, adding another layer to the defense-in-depth approach. Reducing a device's attack surface improves manageability and reduces the chances of a successful attack.
- Protects sensitive information and intellectual property.
- Increases the value of current investments. There is no need for additional hardware or software because Windows Defender Firewall is a host-based firewall that comes with the operating system.

In this exercise, you will activate and deactivate the Windows Defender Firewall. You will also create a rule to block a port and block/allow a program through the firewall.

Learning Outcomes

After completing this exercise, you should be able to:

- Create a Rule Blocking a Port
- Allow Secure Connection through the Windows Defender Firewall
- Allow a Program through the Windows Defender Firewall
- Block a Port with Group Policy Management

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)

Task 1 - Create a Rule Blocking a Port

In this task, the Windows Defender Firewall MMC will be accessed, and its additional capabilities will be discussed. You will create a rule for the firewall to block a port and view the various profiles that are available to set rules up for.

Step 1

Connect to **PLABDC01**.

The **Server Manager** window is displayed.

Select **Tools** and then select **Windows Defender Firewall with Advanced Security**.

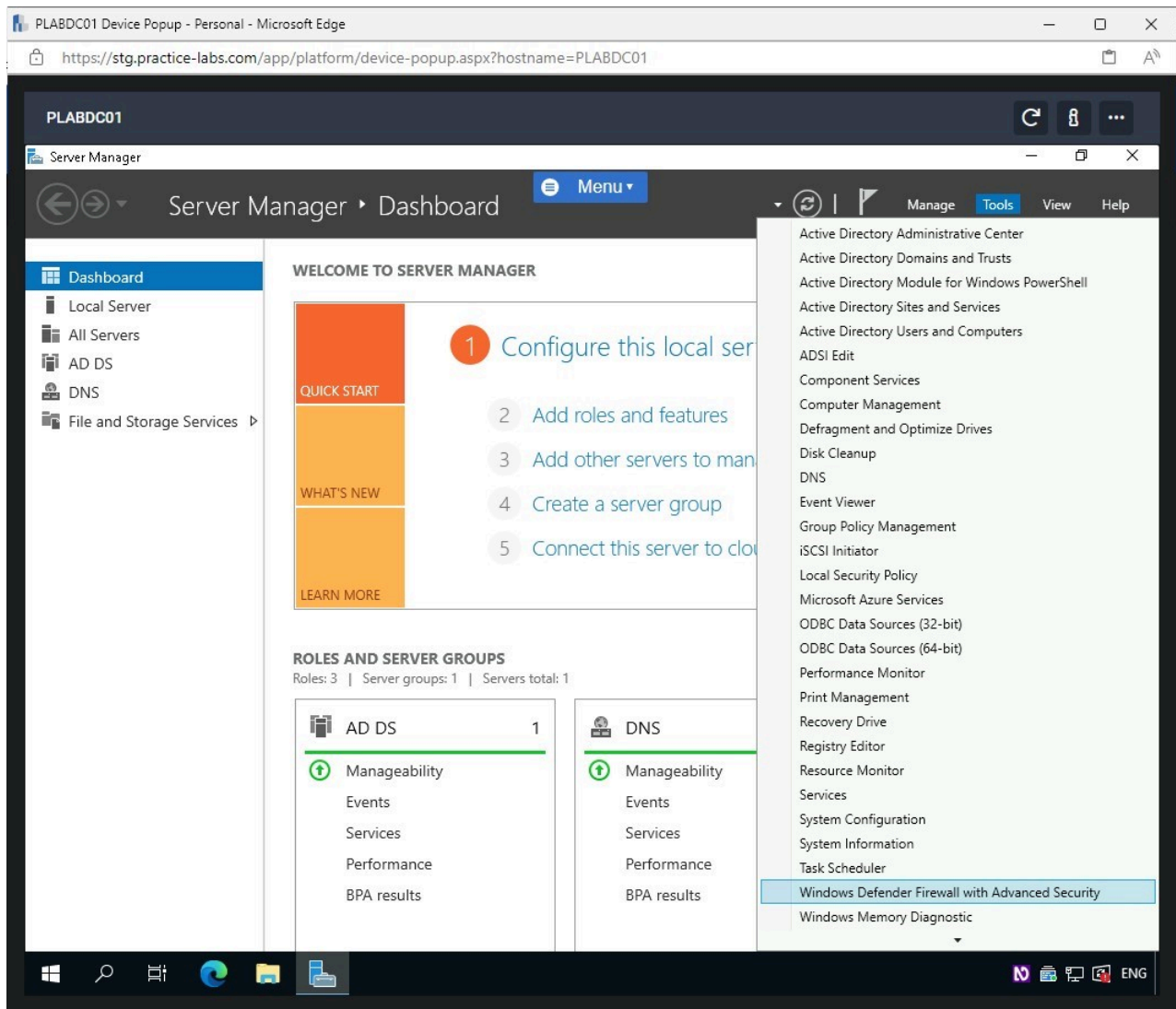


Figure 2.1 Screenshot of PLABDC01: Displaying the Server Manager window. The Tools menu is accessed, and Windows Defender Firewall with Advanced Security is selected.

Step 2

In the **Windows Defender Firewall with Advanced Security** window, the following can be configured:

- **Inbound Rules** - Used to configure rules for traffic coming into the system
- **Outbound Rules** - Used to configure rules for traffic leaving the system
- **Connection Security Rules** - Used to configure extra layers of authentication and security
- **Monitoring** - Shows each profile along with pertinent information to logging and monitoring

Select **Inbound Rules** on the left pane.

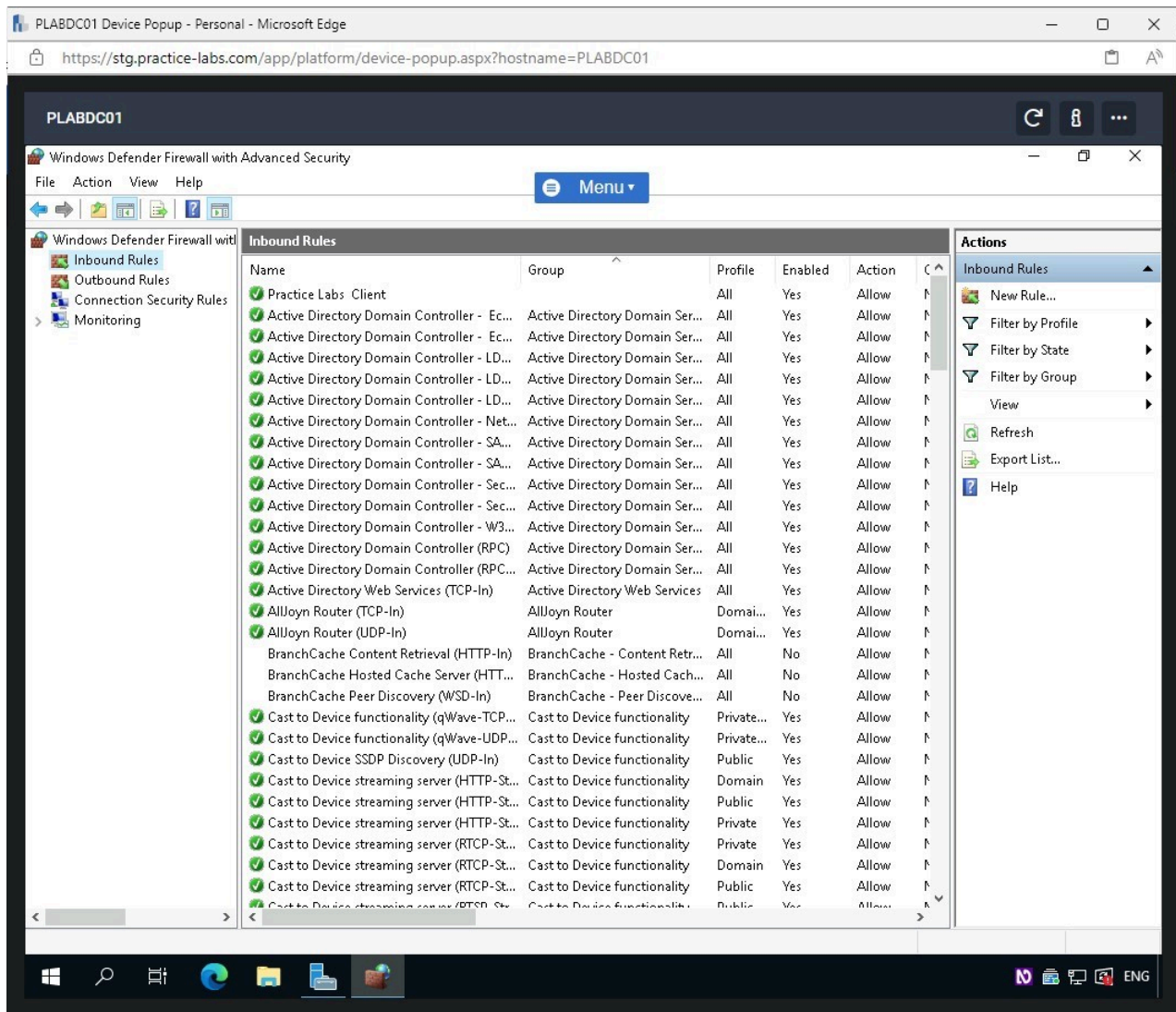


Figure 2.2 Screenshot of PLABDC01: Displaying the Windows Defender Firewall with Advanced Security window, showing Inbound Rules selected on the left pane.

Step 3

In the **Actions** pane on the right, select **New Rule**.

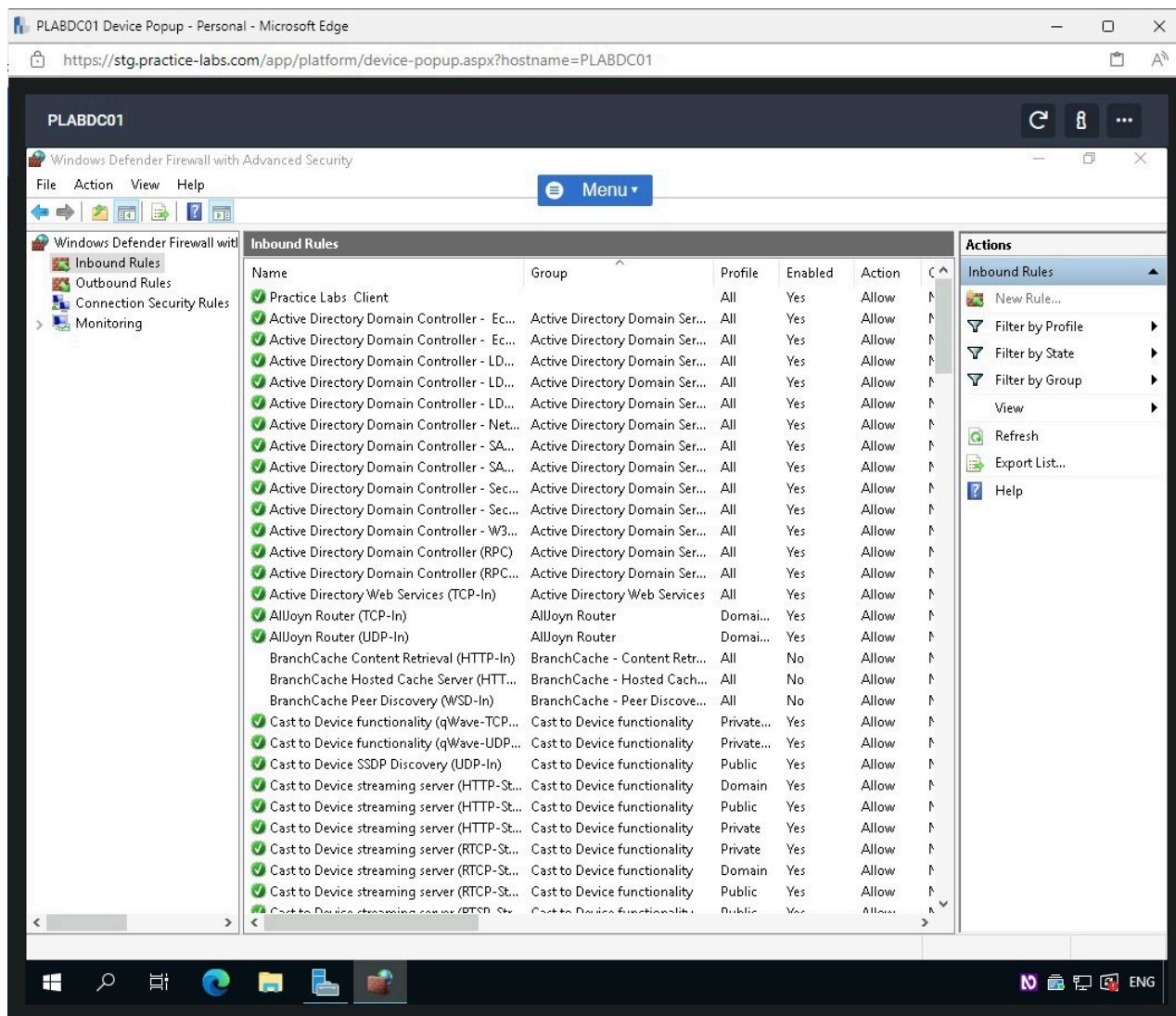


Figure 2.3 Screenshot of PLABDC01: Displaying the Windows Defender Firewall with Advanced Security window. New Rule is selected on the Actions pane.

Step 4

On the **New Inbound Rule Wizard - Rule Type** window, rules can be set up to control program and port access. Predefined rules can be configured as well as custom rules.

You will now configure the system to block inbound requests from Telnet.

Select the **Port** radio button and click **Next**.

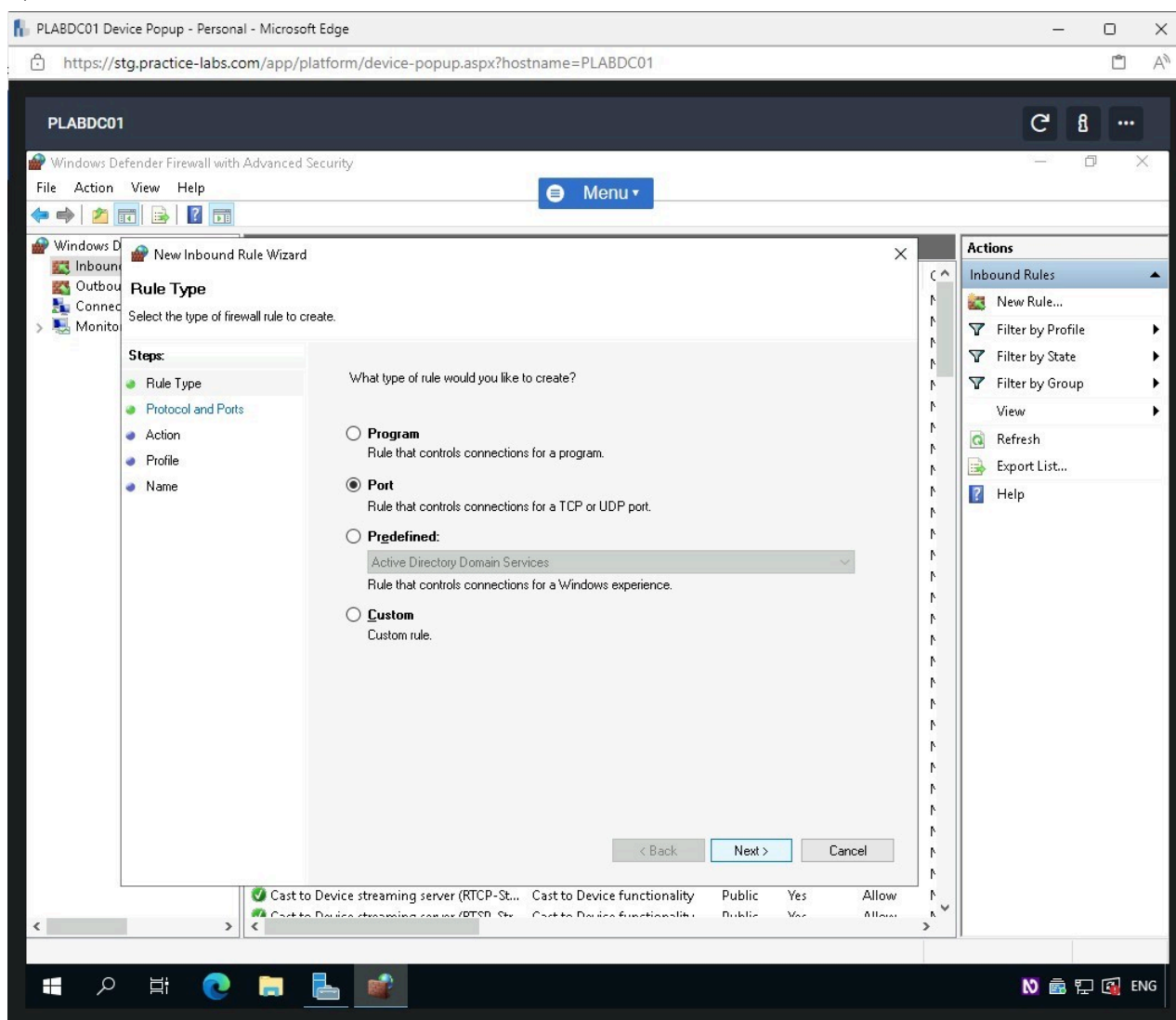


Figure 2.4 Screenshot of PLABDC01: Displaying the New Inbound Rule Wizard - Rule Type window, showing the Port option selected and the Next button highlighted.

Step 5

On the **Ports and Protocols** page, the method of data movement is specified.

Select the **TCP** option.

Under the **Does this rule apply to all local ports or specific local ports?** section, select **Specific local ports** and type the following:

20, 21

Click **Next**.

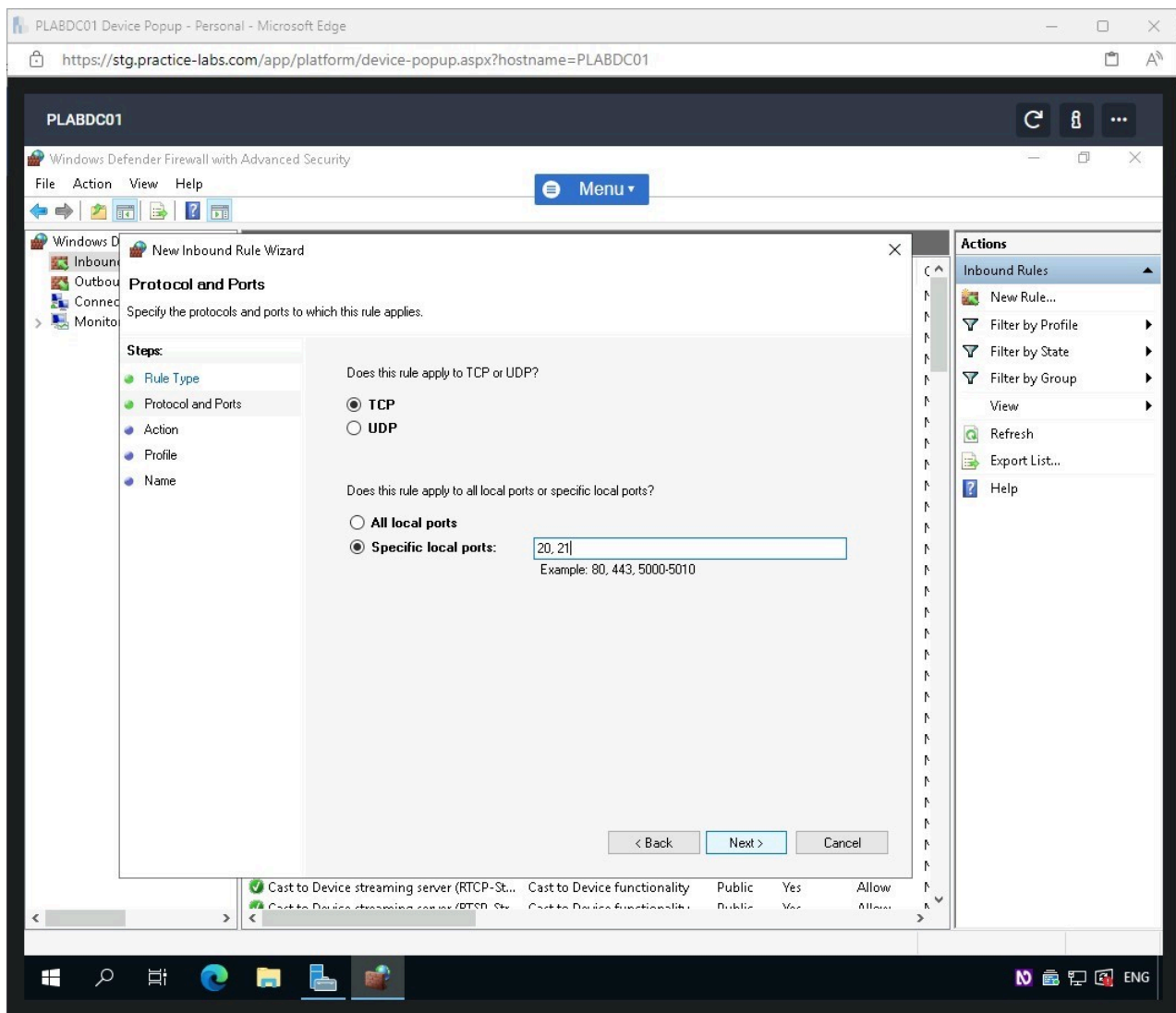


Figure 2.5 Screenshot of PLABDC01: Displaying the Ports and Protocols page, showing the required settings performed and the Next button selected.

Step 6

In the **Action** page, select **Block the connection**.

Click **Next**.

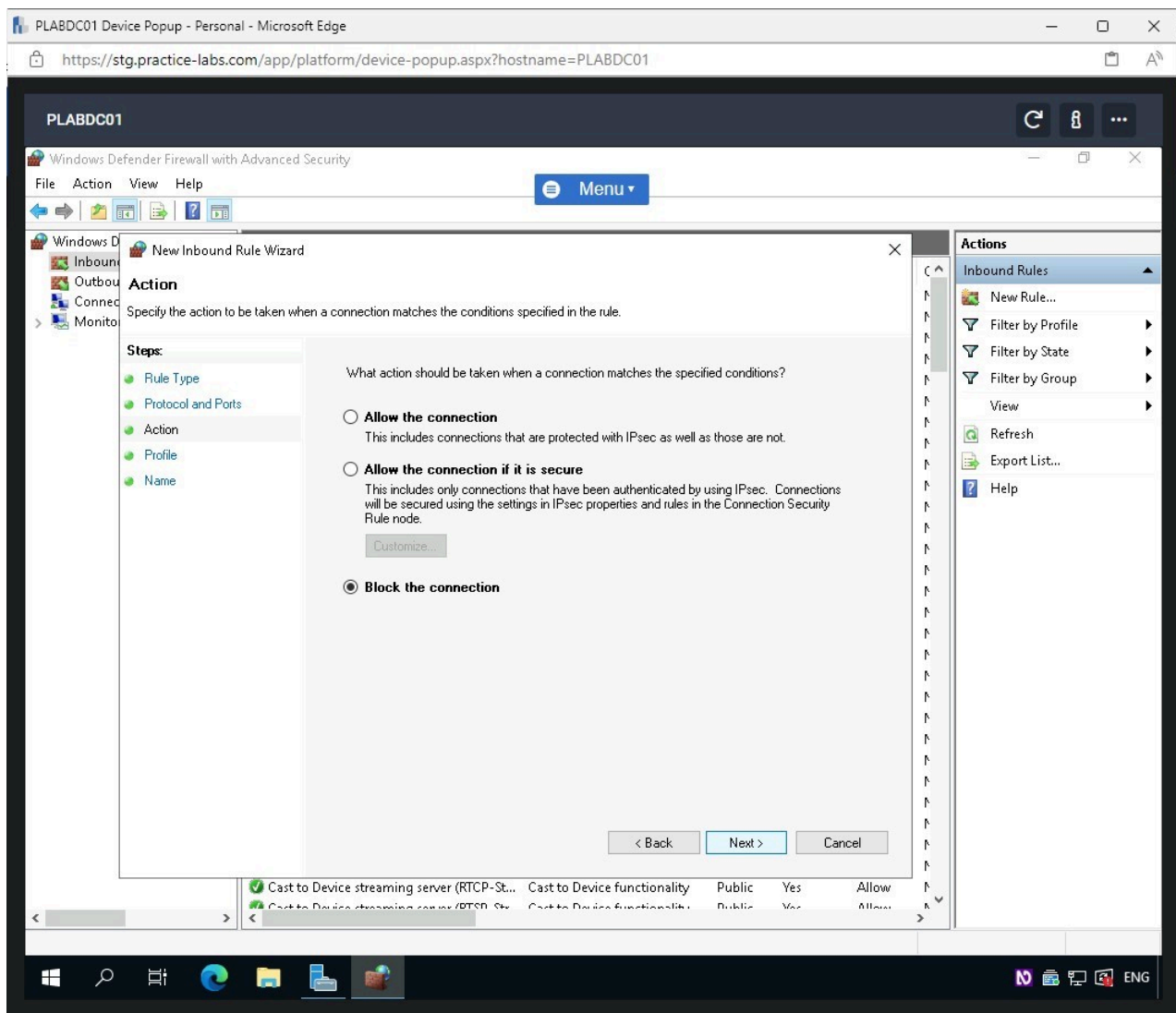


Figure 2.6 Screenshot of PLABDC01: Displaying the Action page, showing the required settings performed and the Next button selected.

Step 7

From the **Profile** page, leave the default selections and click **Next**.

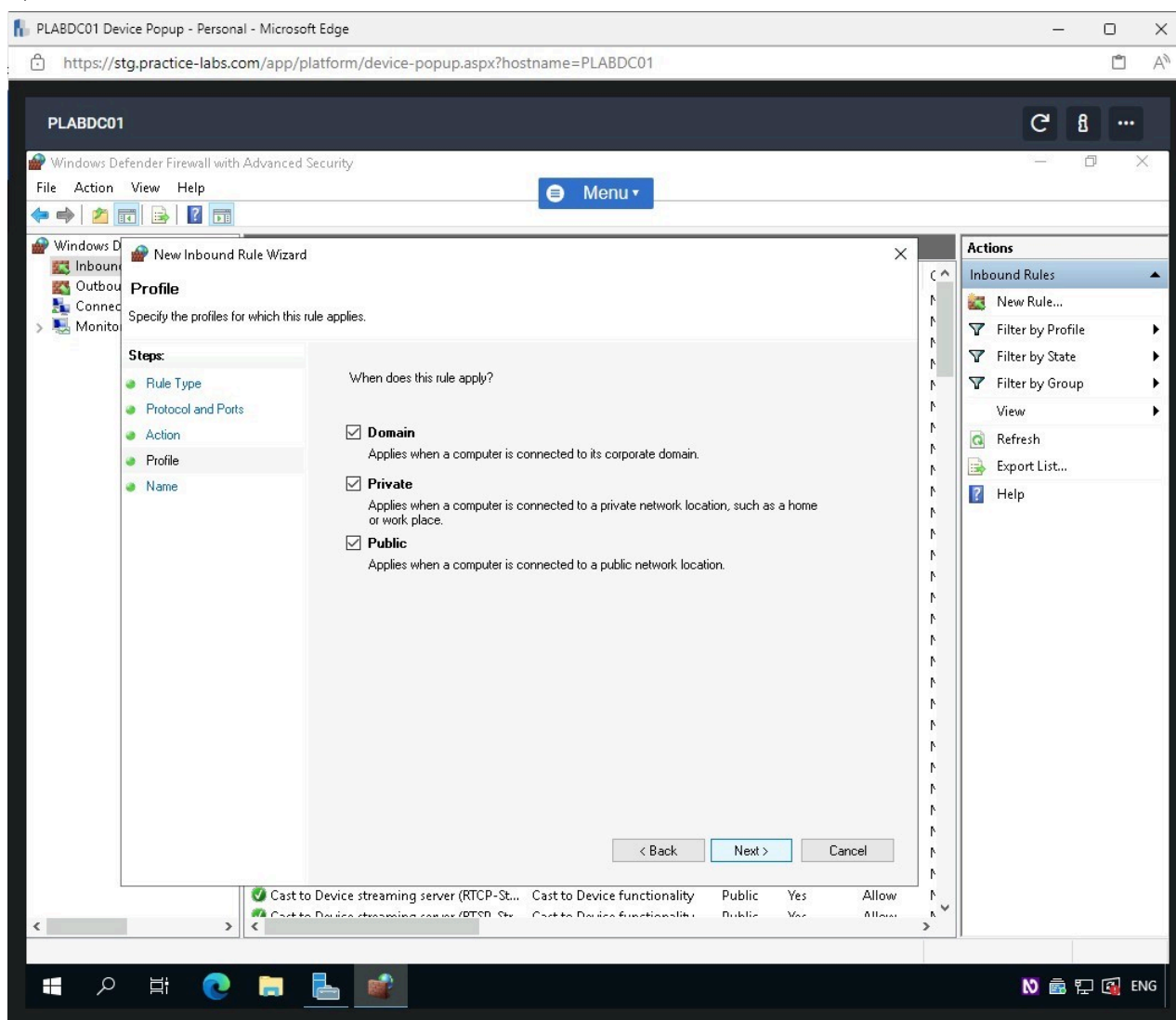


Figure 2.7 Screenshot of PLABDC01: Displaying the Profile page, showing the Next button selected.

Step 8

In the **Name** page, type the following for the **Name** field:

Block Inbound FTP

Type the following for the **Description** field:

Block inbound FTP port number 20,21

Click **Finish**.

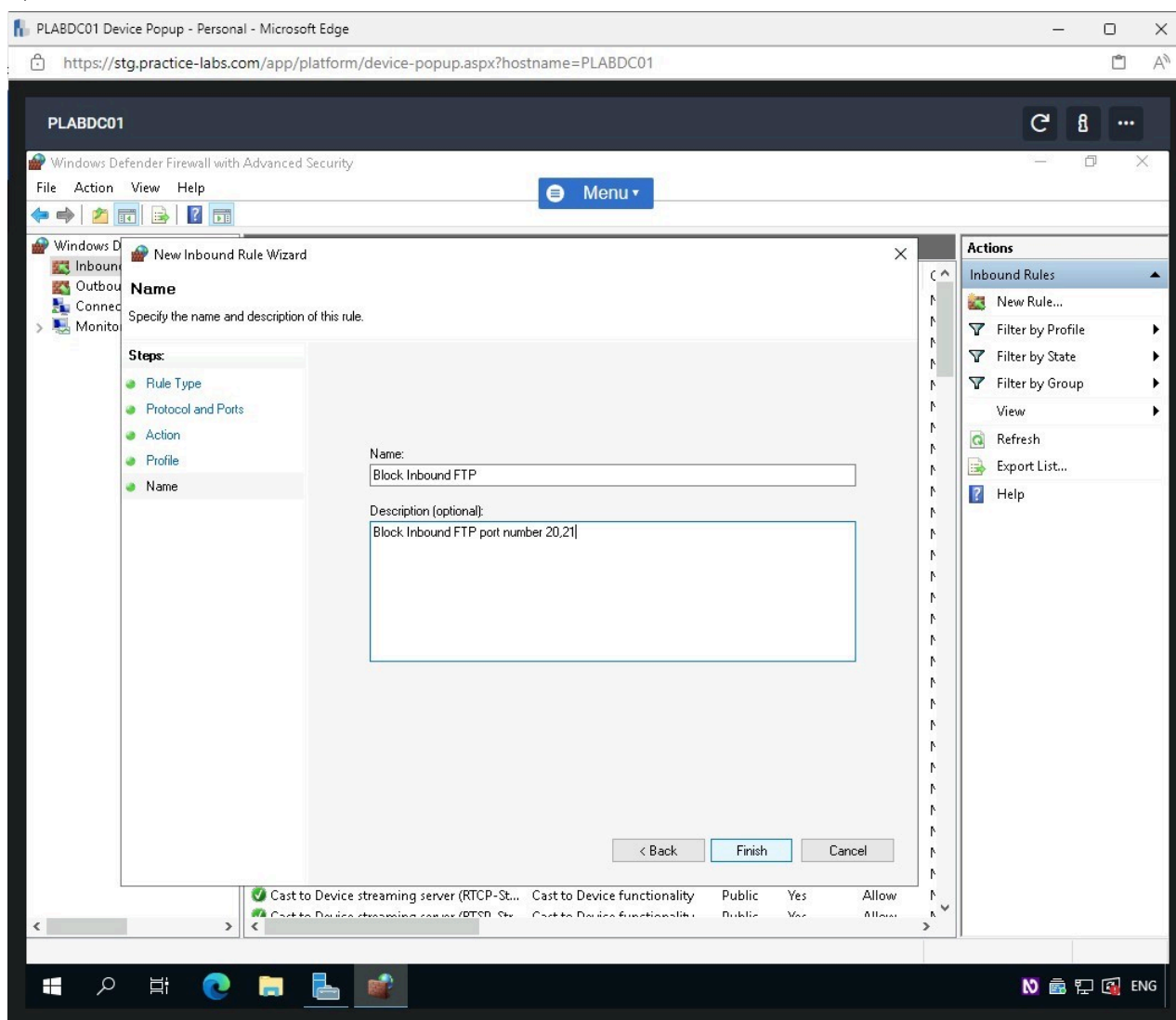


Figure 2.8 Screenshot of PLABDC01: Displaying the Name page, showing the required settings performed and the Finish button selected.

Step 9

Back on the **Windows Defender Firewall with Advanced Security** window, notice the top rule that is now listed on the **Inbound Rules** pane is the **Block Inbound FTP** rule that was just created.

Note: You can cut, copy, disable or view the properties of a rule by right-selecting on the rule and selecting the required option. Selecting **Properties** allows you to further configure and adjust the rule.

Right-click on the **Block Inbound Telnet** rule and select **Delete** to remove the rule.

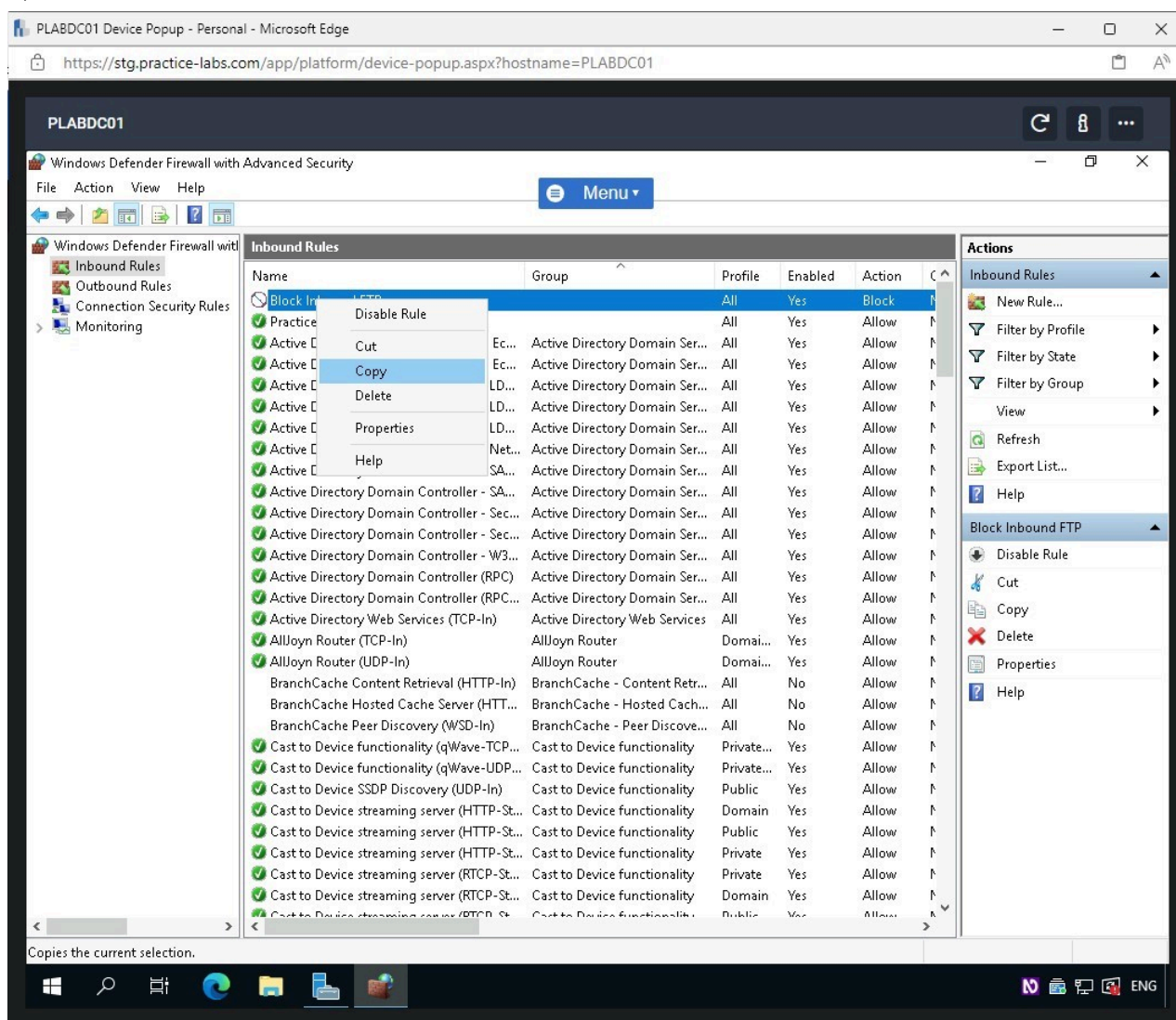


Figure 2.9 Screenshot of PLABDC01: Displaying the Windows Defender Firewall with Advanced Security window. Block Inbound Telnet rule is right-clicked and Delete selected from the context menu.

Keep the **Windows Defender Firewall with Advanced Security** window open.

Task 2 - Allow Secure Connection through the Windows Defender Firewall

In this task, the Windows Defender Firewall will be accessed. You will allow a program through the firewall if the connection is secure.

Step 1

In **PLABDC01**, the **Windows Defender Firewall with Advanced Security** window is open.

Select **Inbound Rules** on the left pane, and then select **New Rule** in the **Actions** pane on the right.

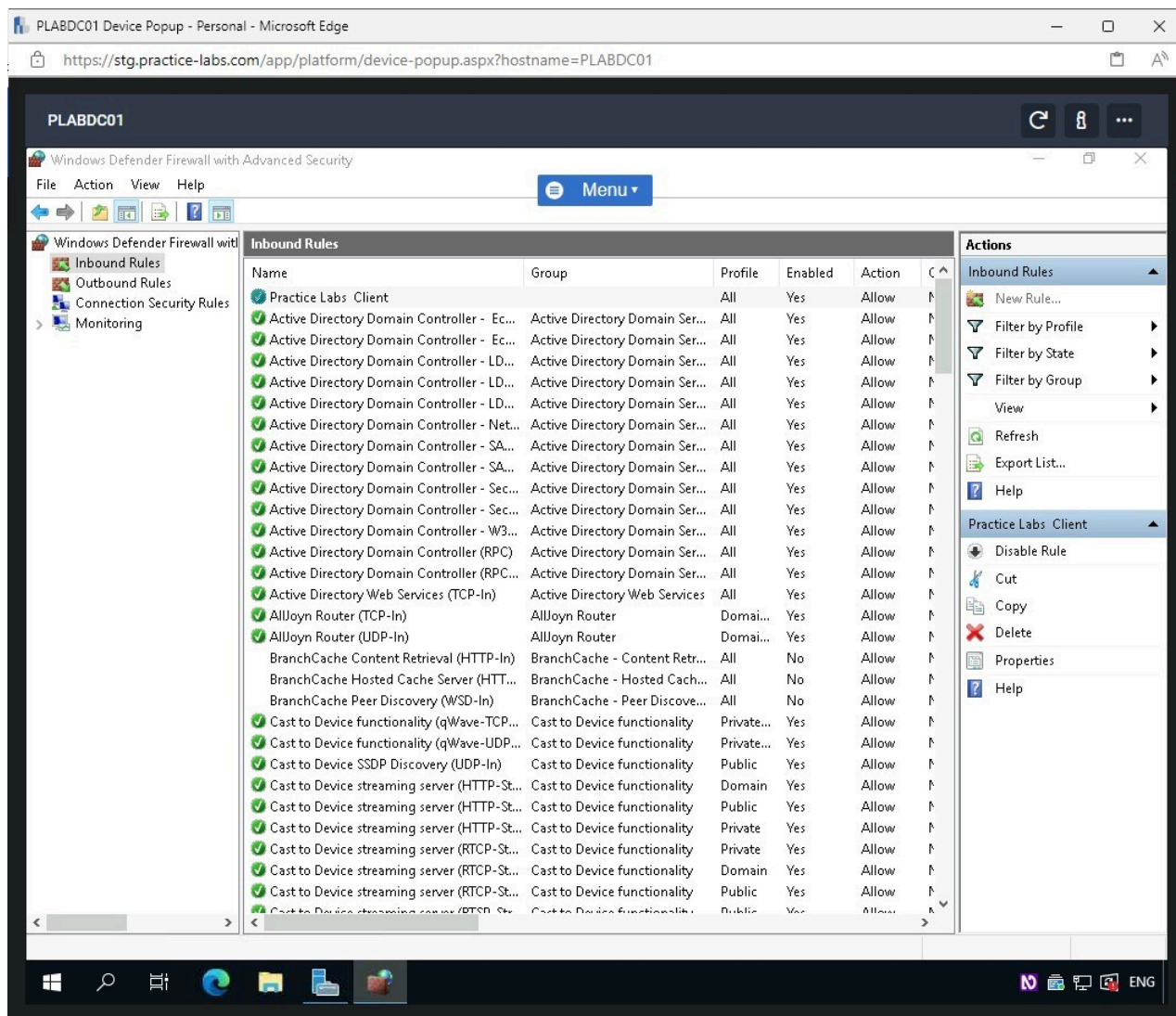


Figure 2.11 Screenshot of PLABDC01: Displaying the Windows Defender Firewall with Advanced Security window. New Rule is selected on the Actions pane.

Step 2

In the **New Inbound Rule Wizard - Rule Type** window, ensure that the **Program** option is selected.

Click **Next**.

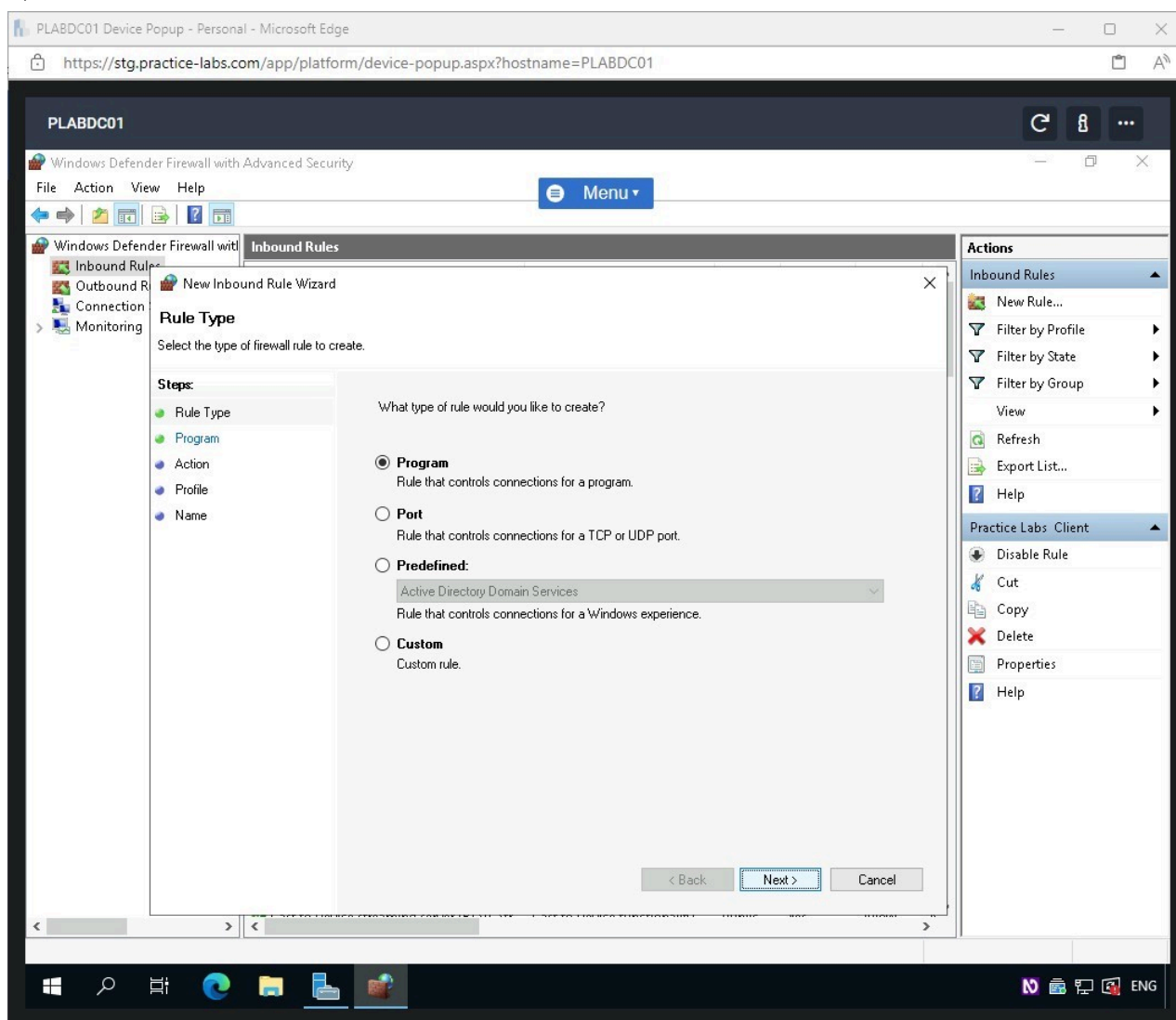


Figure 2.12 Screenshot of PLABDC01: Displaying the New Inbound Rule Wizard - Rule Type page, showing the Program option selected and the Next button highlighted.

Step 3

On the **Program** page, you can either choose to select all programs and apply rules to all connections to the system that match other rule properties, or you can choose an individual application.

Select the **This program path** option and click **Browse**.

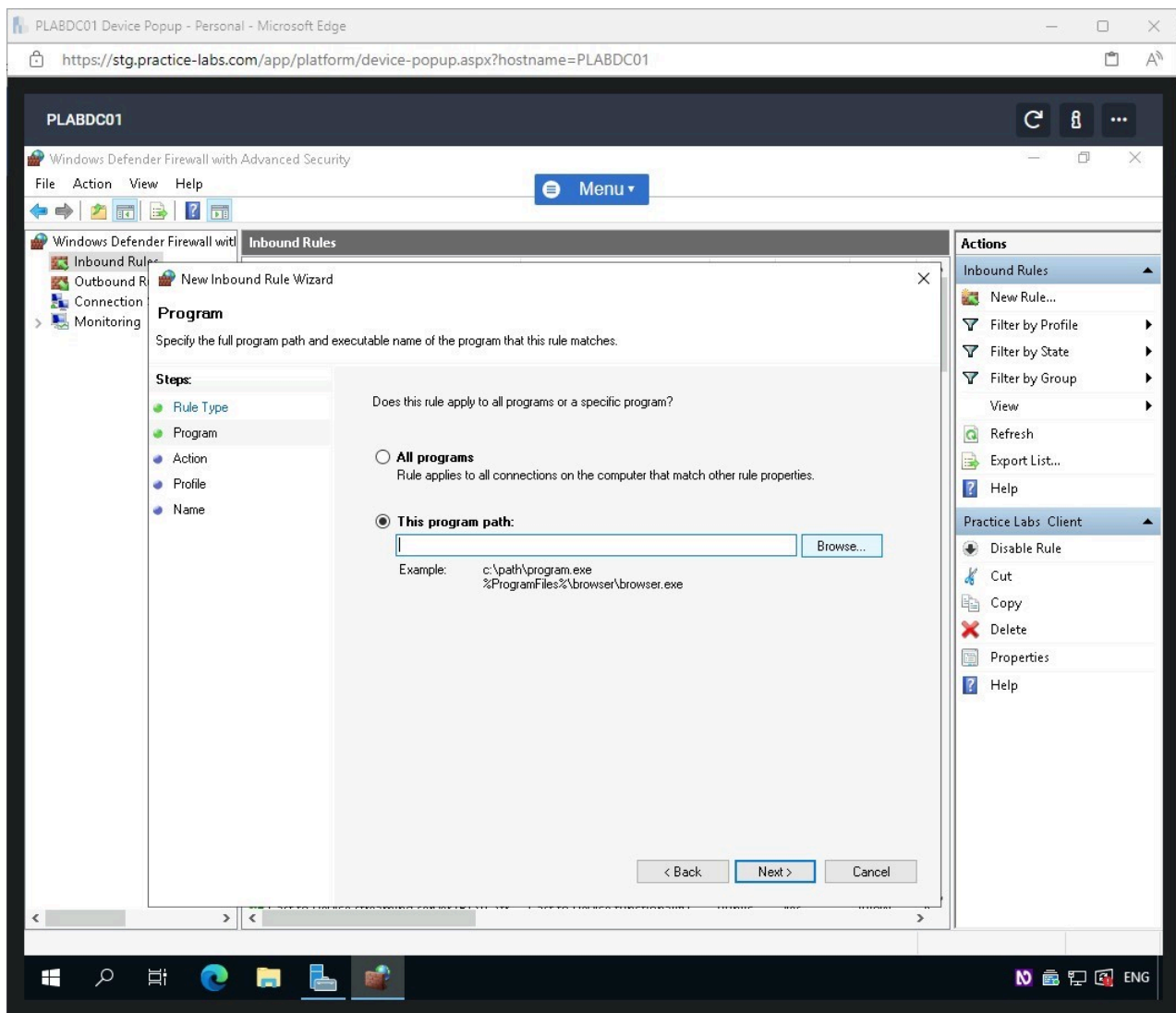


Figure 2.13 Screenshot of PLABDC01: Displaying the Program page, showing the Browse button selected.

Step 4

In the **Open** dialog box, ensure **This PC** is selected on the right pane.

Scroll down and select **ipconfig** and click **Open**.

Note: *ipconfig* was chosen to demonstrate here because it is a non-essential application that will not interfere with the system's operation.

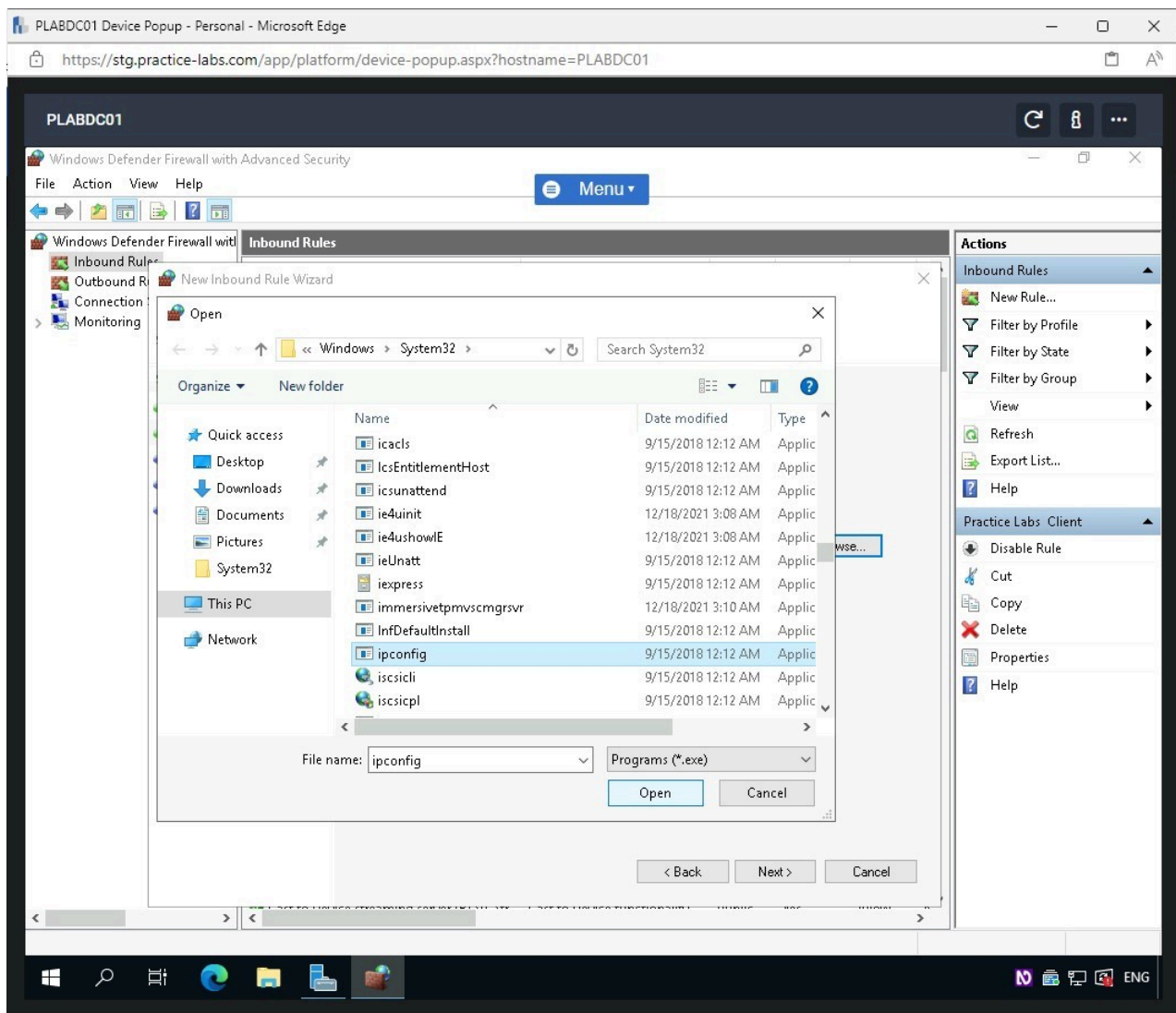


Figure 2.14 Screenshot of PLABDC01: Open dialog box is displayed, showing the required Program highlighted and the Open button selected.

Step 5

Back on the **Program** page, select **Next**.

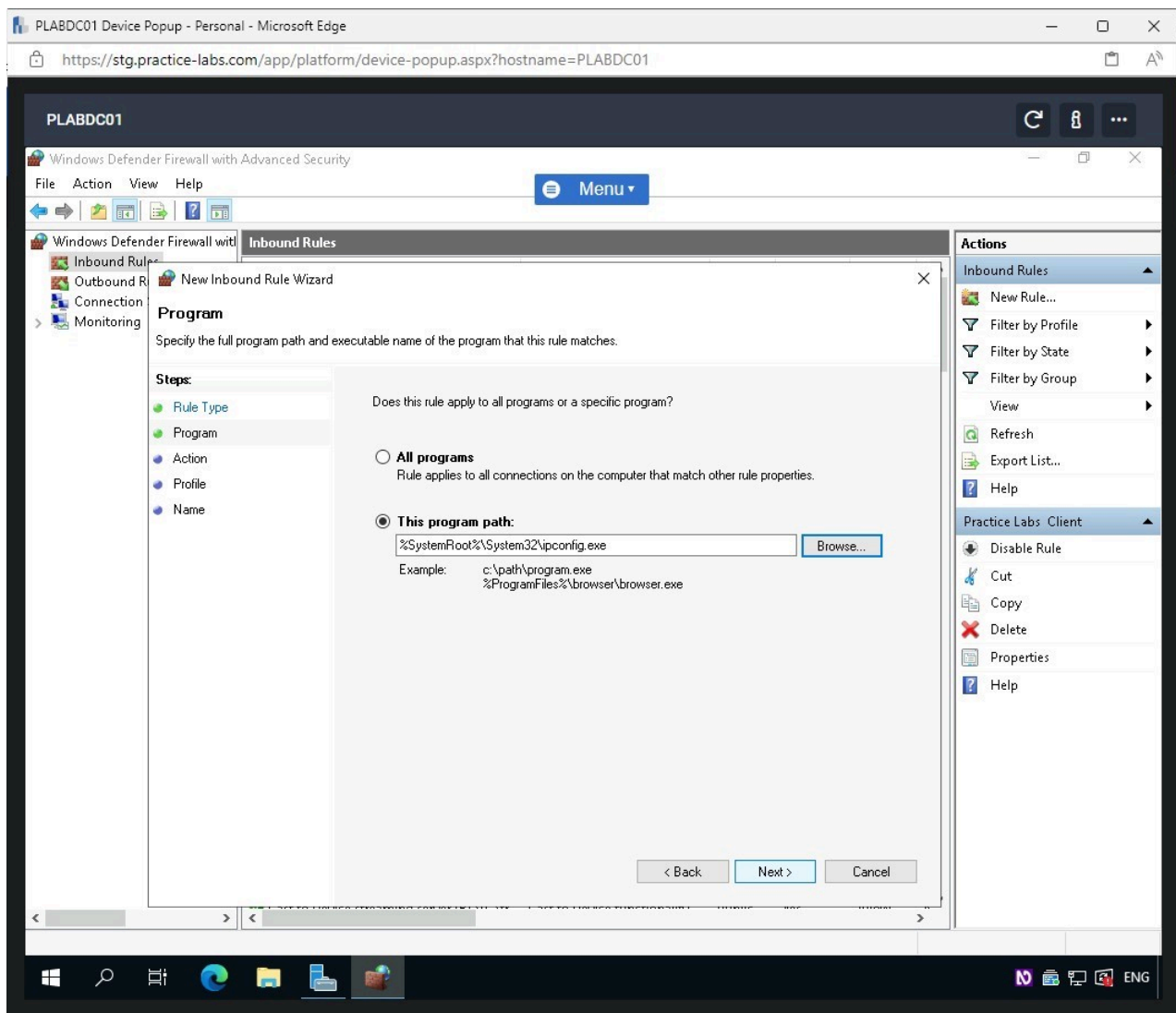


Figure 2.15 Screenshot of PLABDC01: Displaying selecting Next on the Program page.

Step 6

In the **Action** page, select the **Allow the connection if it is secure** option.

Click **Next**.

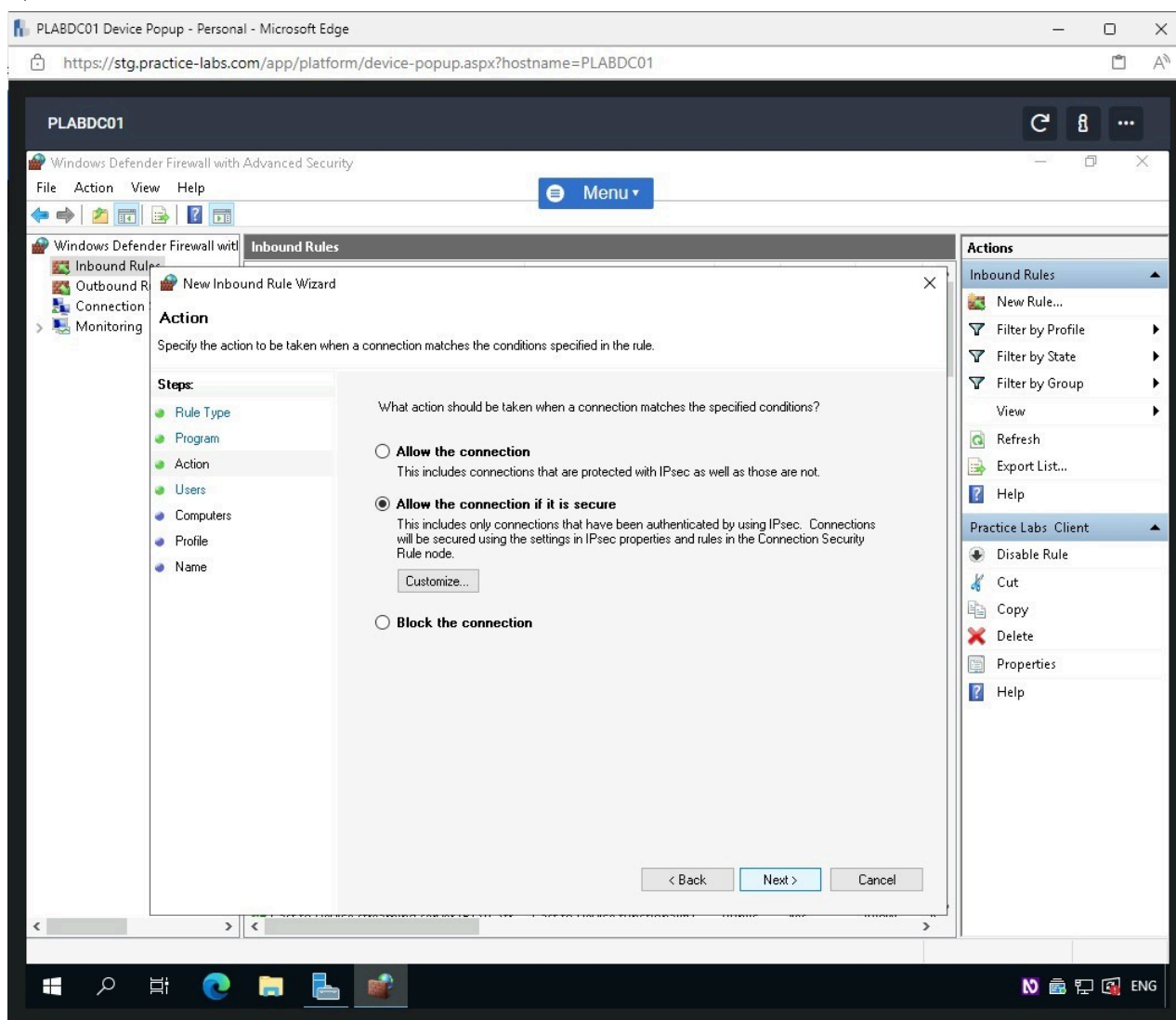


Figure 2.16 Screenshot of PLABDC01: Displaying the Action page, showing the required option selected and the Next button highlighted.

Step 7

From the **Users** page, enable the **Only allow connections from these users** checkbox under the **Authorized users** section.

Click **Add**.

Note: Users can be authorized or excluded. This field is not a mandatory field to be filled.

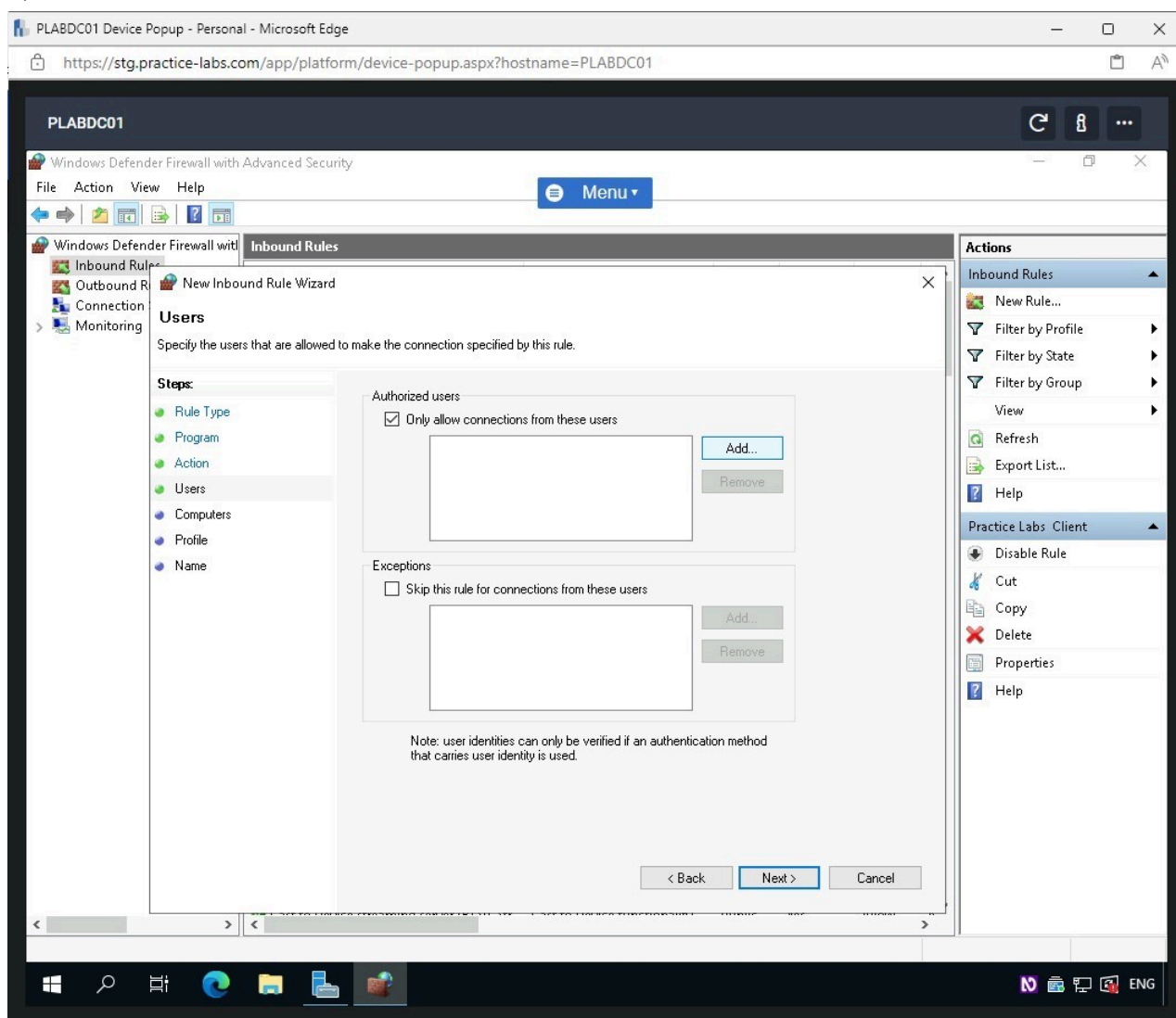


Figure 2.17 Screenshot of PLABDC01: Displaying the Users page, showing the required checkbox enabled and the Add button selected.

Step 8

In the **Select Users, Computers, or Groups** dialog box, type the following in the **Enter the object names to select** field:

Admin

Click **Check Names**.

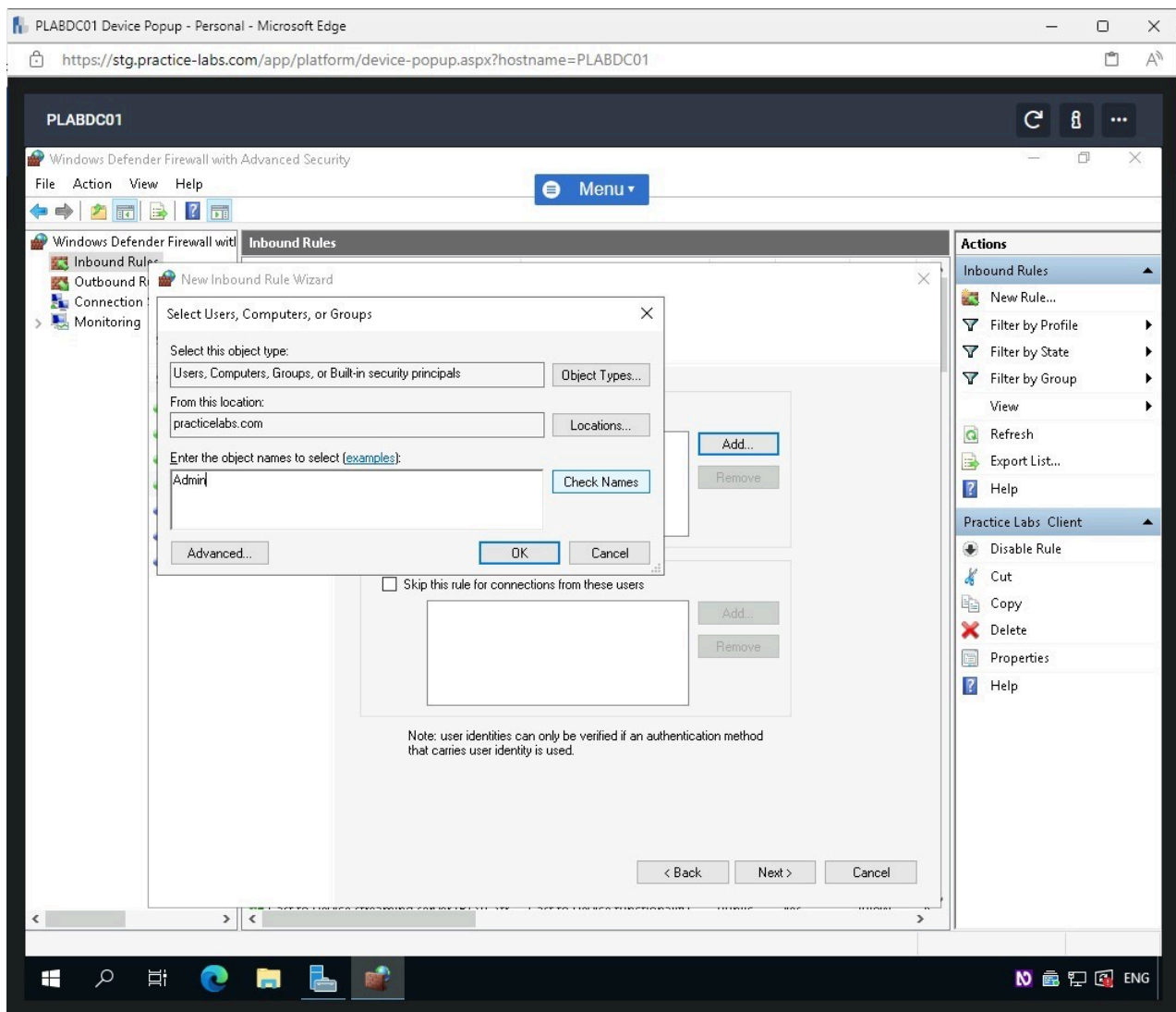


Figure 2.18 Screenshot of PLABDC01: Displaying the Select Users, Computers, or Groups dialog box, showing the required object name entered and the Check Names button selected.

Step 9

From the **Multiple Names Found** window, select the first **Administrator** option.

Click **OK**.

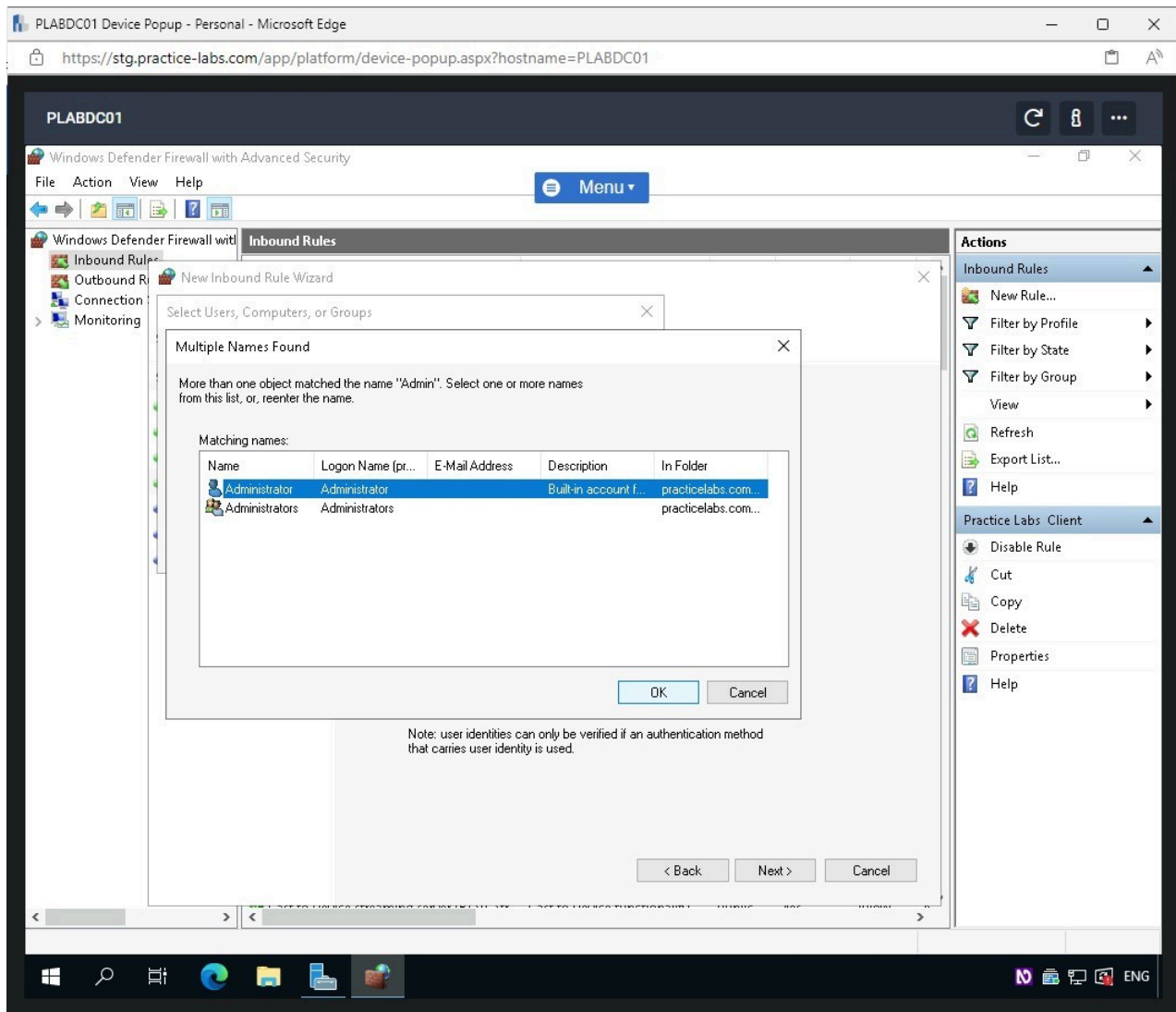


Figure 2.19 Screenshot of PLABDC01: Displaying the Multiple Names Found window, showing Administrator selected and the OK button highlighted.

Step 10

Back in the **Select Users, Computers, or Groups** dialog box, click **OK**.

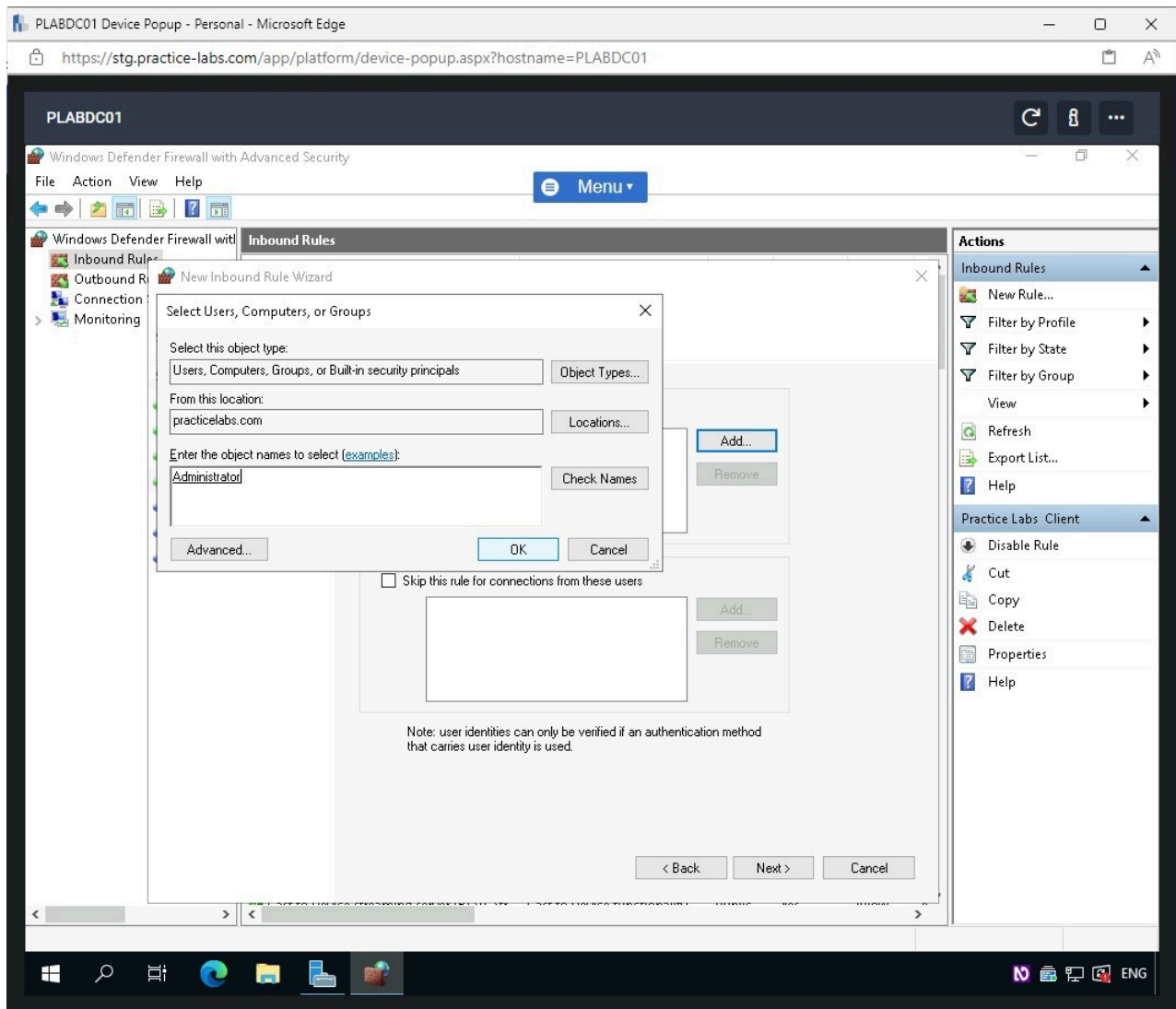


Figure 2.20 Screenshot of PLABDC01: Displaying selecting OK in the Select Users, Computers, or Groups dialog box.

Step 11

On the **Users** page, click **Next**.

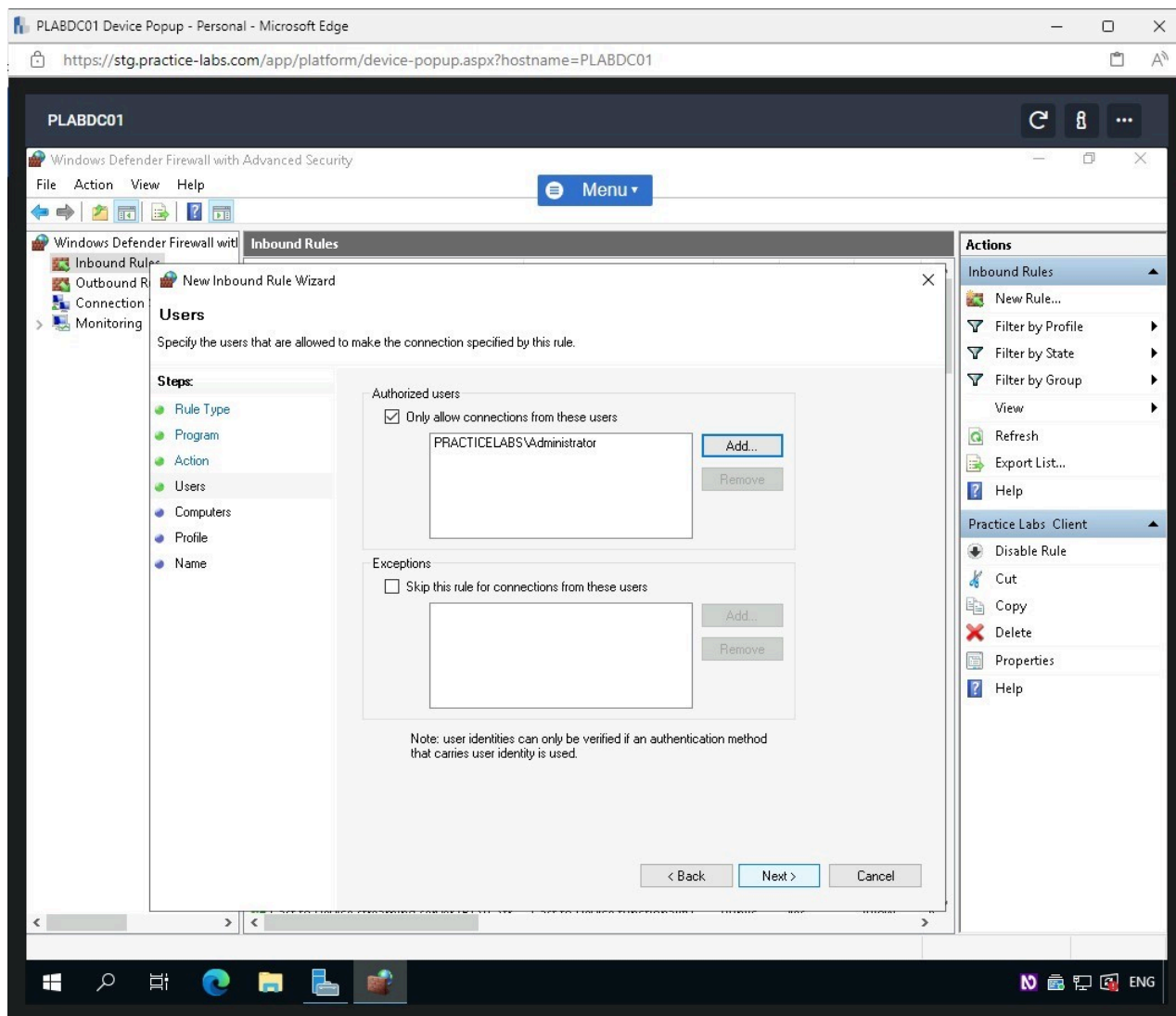


Figure 2.21 Screenshot of PLABDC01: Displaying selecting Next on the Users page.

Step 12

On the **Computers** page, users can select specific computers to apply the rule.

Leave the default settings and click **Next**.

Note: Computers can be authorized or excluded. This field is not a mandatory field to be filled.

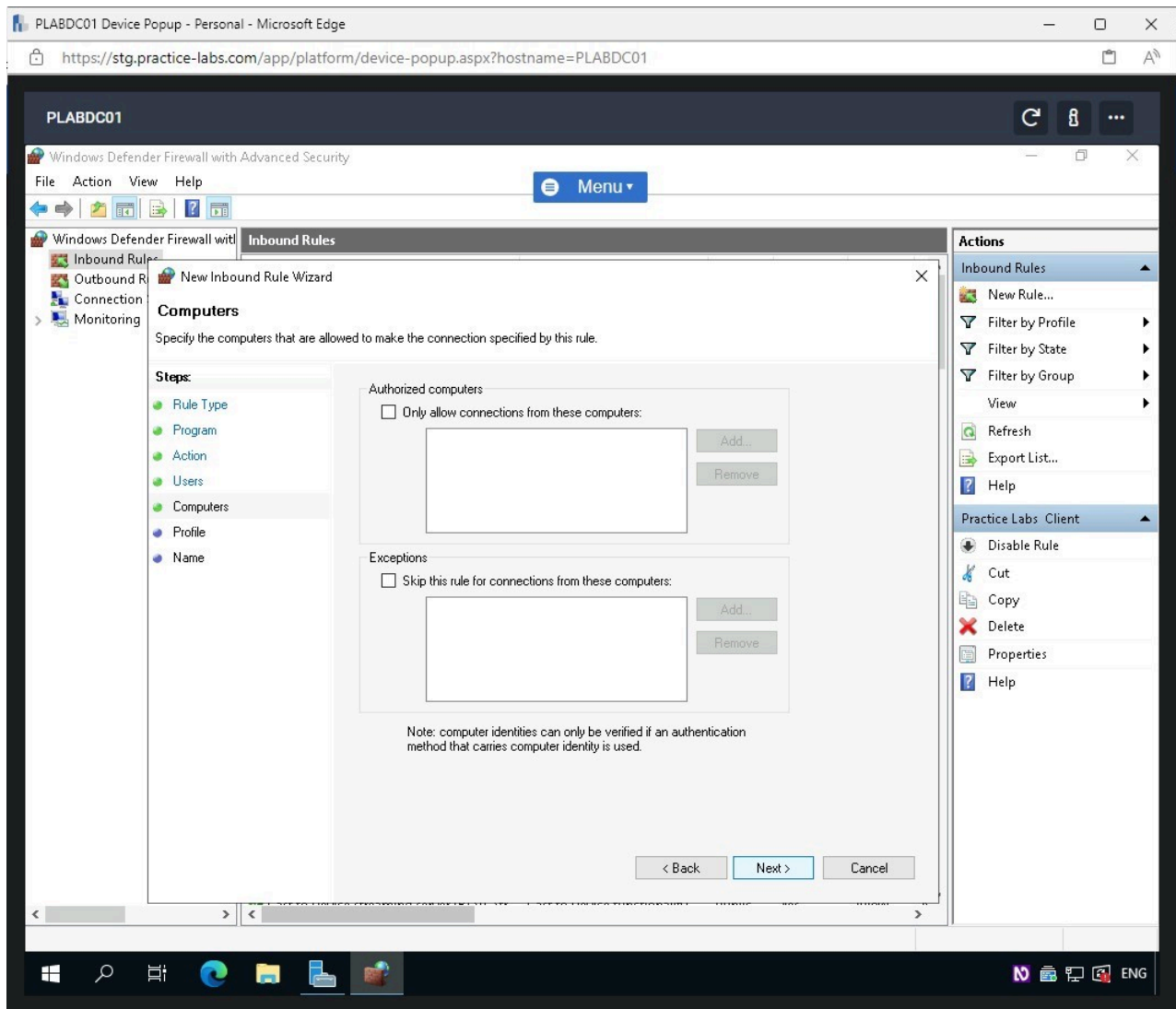


Figure 2.22 Screenshot of PLABDC01: Displaying selecting Next on the Computers page.

Step 13

Click **Next** on the **Profile** page.

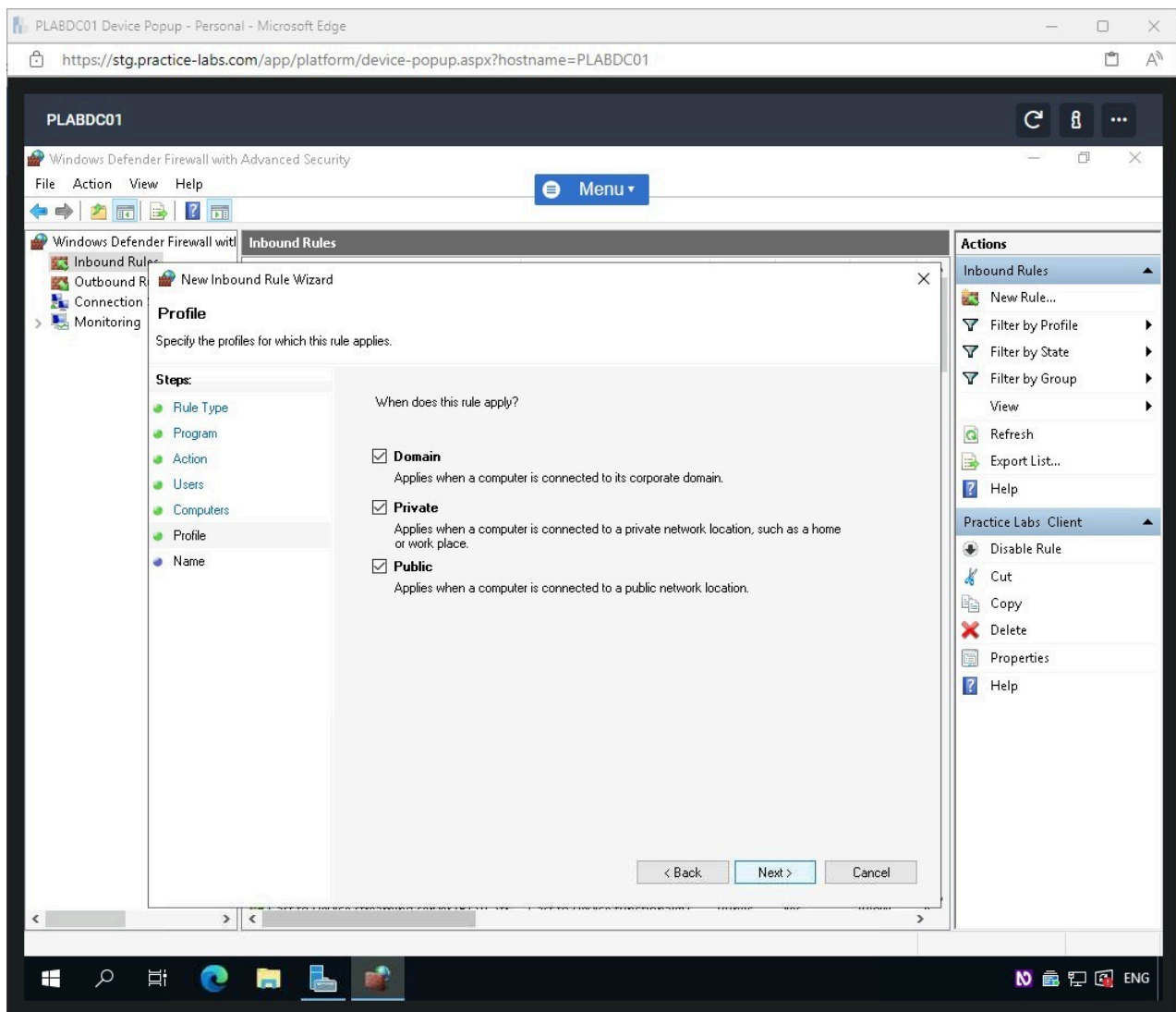


Figure 2.23 Screenshot of PLABDC01: Displaying selecting Next on the Profile page.

Step 14

In the **Name** page, type the following for the **Name** field:

Allow ipconfig Admin Secure

Type the following for the **Description** field:

Allow ipconfig for administrator if the connection is secure

Click **Finish**.

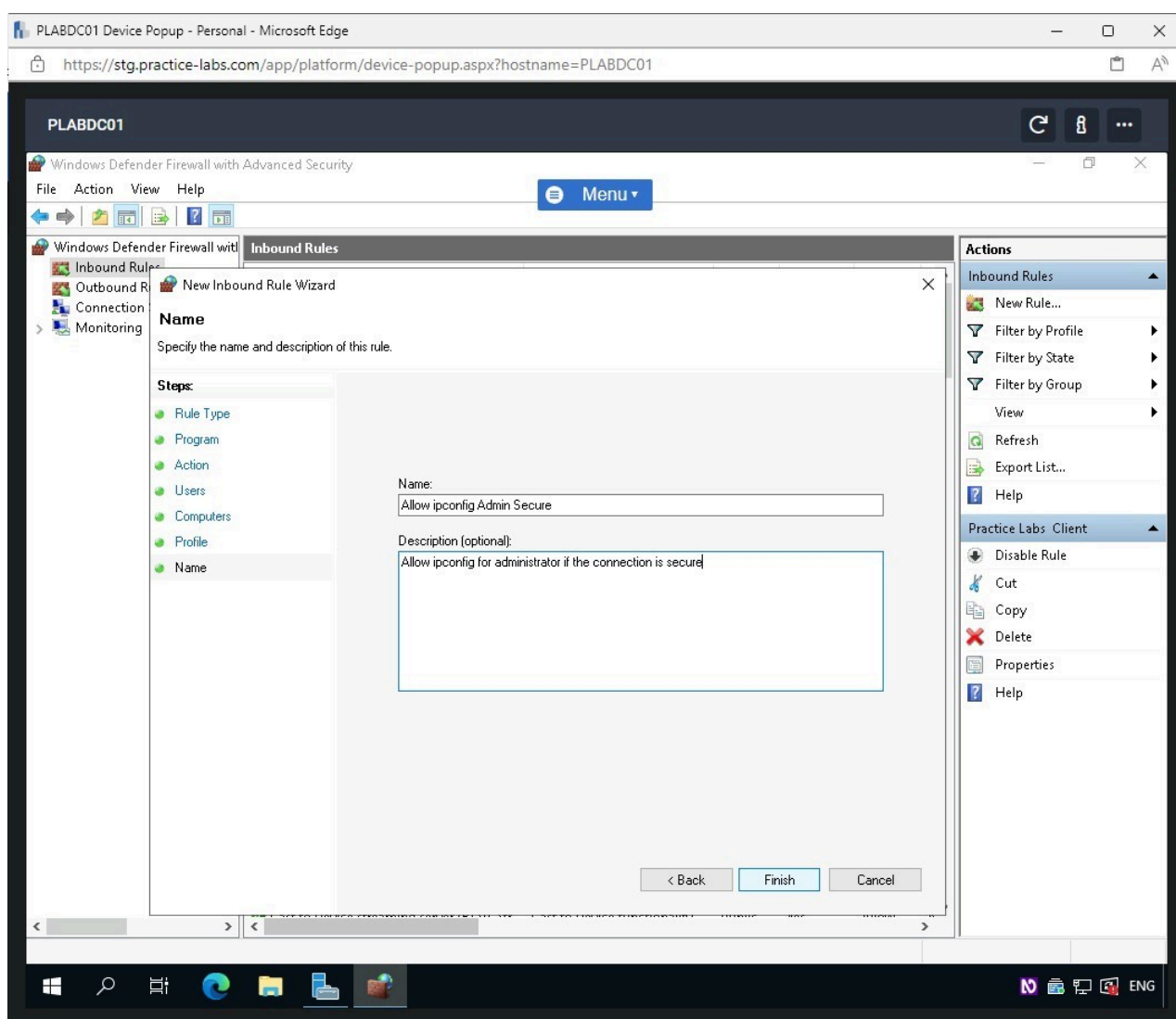


Figure 2.24 Screenshot of PLABDC01: Displaying the Name page, showing the required settings performed, and the Finish button highlighted.

Step 15

Notice the top rule that is now listed on the **Inbound rules** pane is the **Allow ipconfig Admin Secure** rule that was just created.

Right-click on the rule and select **Delete** to remove the rule.

Close the **Windows Defender Firewall with Advanced Security** window.

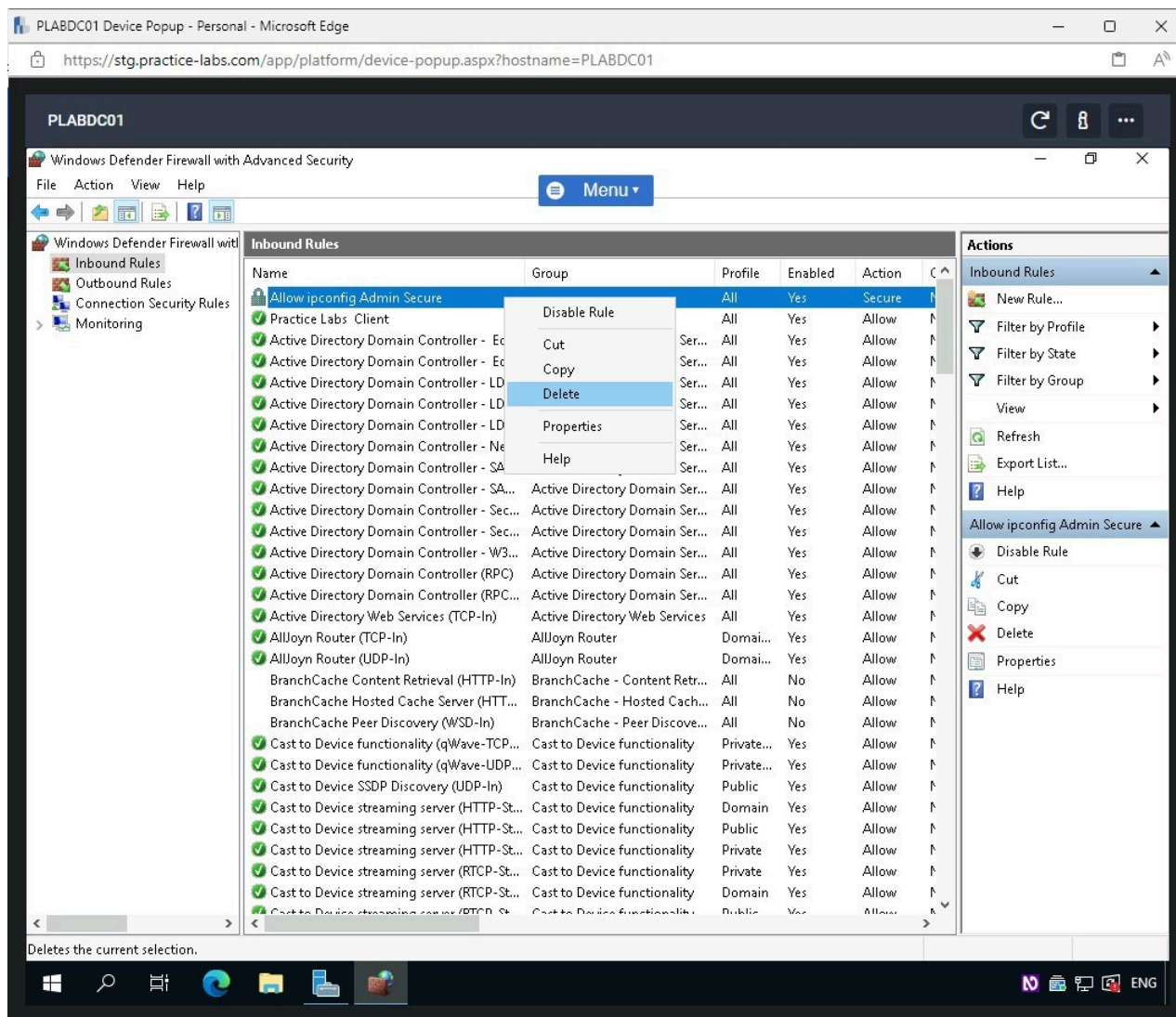


Figure 2.25 Screenshot of PLABDC01: Displaying the Windows Defender Firewall with Advanced Security window. Allow ipconfig Admin Secure rule is right-clicked and Delete selected from the context menu.

Task 3 - Allow a Program through the Windows Defender Firewall

In this task, you will learn how the graphical user interface can be used to allow a program through the Windows Defender Firewall.

Step 1

Ensure you are connected to **PLABDC01**.

Click the **Start** charm and type the following:

Windows Defender Firewall

Select **Windows Defender Firewall** from the **Best match** pop-up menu.

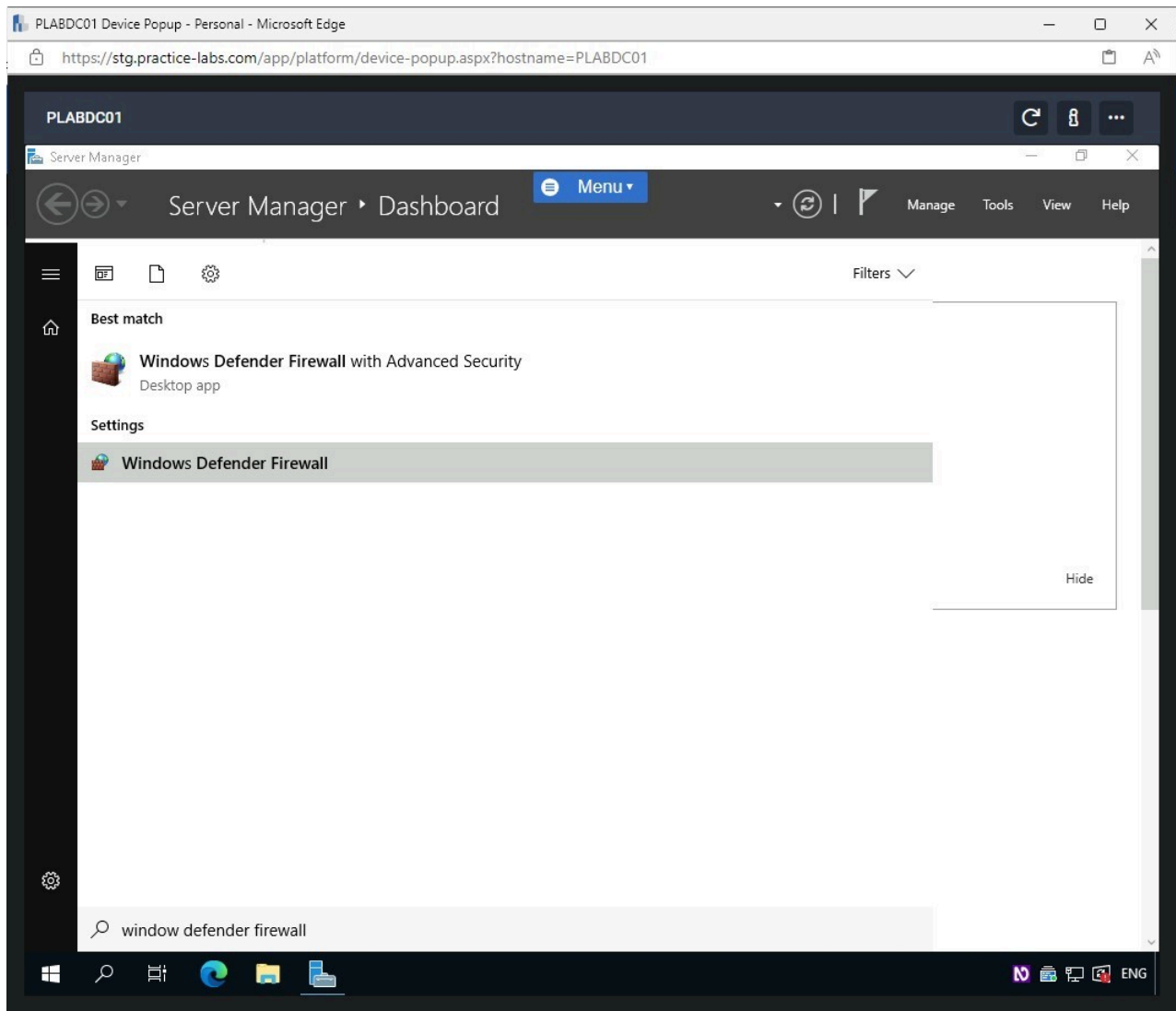


Figure 2.26 Screenshot of PLABDC01: Displaying selecting Windows Defender Firewall from the Best match pop-up menu.

Step 2

In the **Windows Defender Firewall** window, select the **Allow an app or feature through Windows Defender Firewall** link on the left pane.

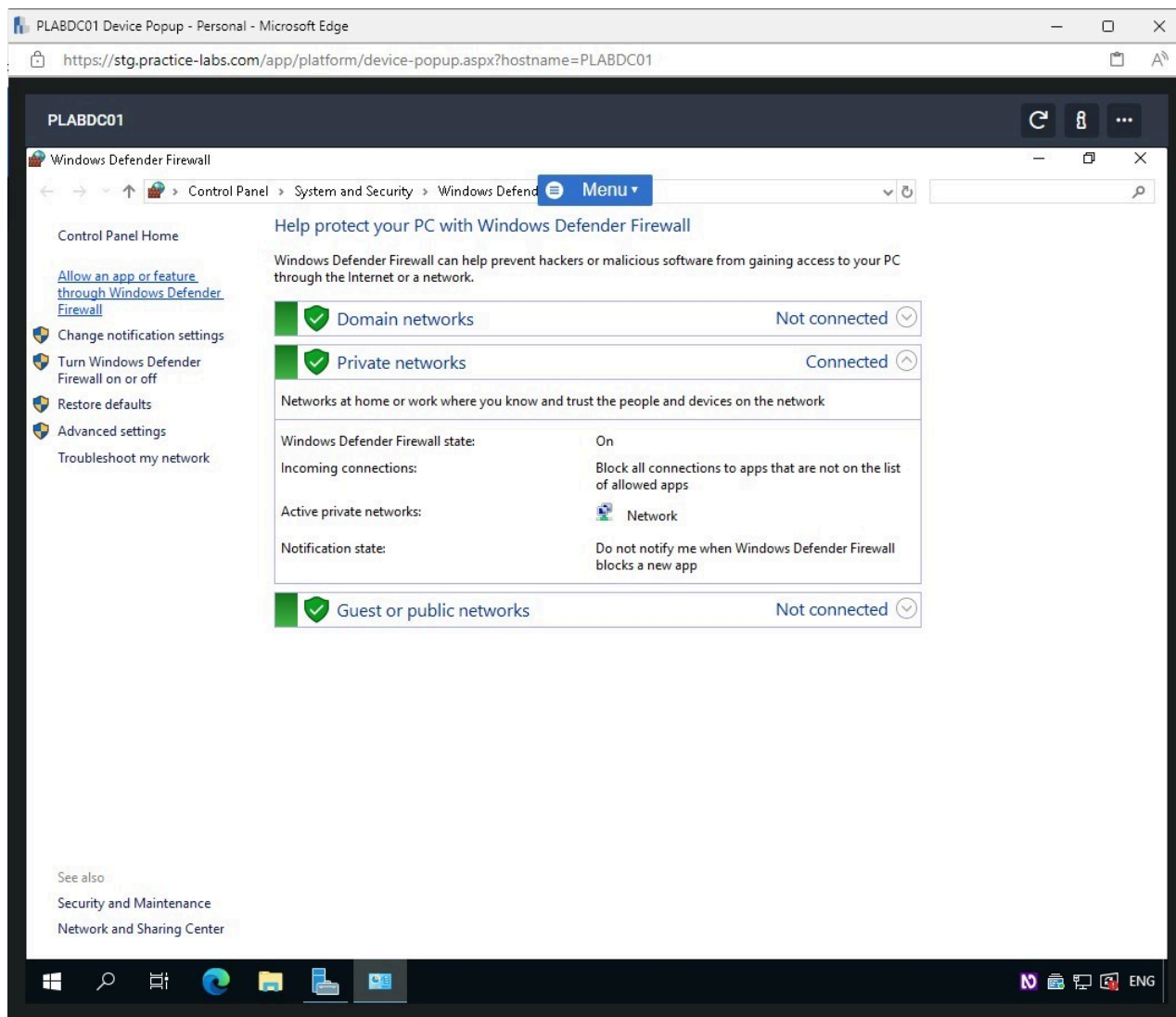


Figure 2.27 Screenshot of PLABDC01: Displaying selecting the Allow an app or feature through Windows Defender Firewall link in the Windows Defender Firewall window.

Step 3

In the **Allowed apps** window, review the list of applications you can choose to allow to communicate through the firewall. You can choose each profile type and allow through only when those conditions are met.

Once you have reviewed the applications, click **Cancel**.

Close the **Windows Defender Firewall** window.

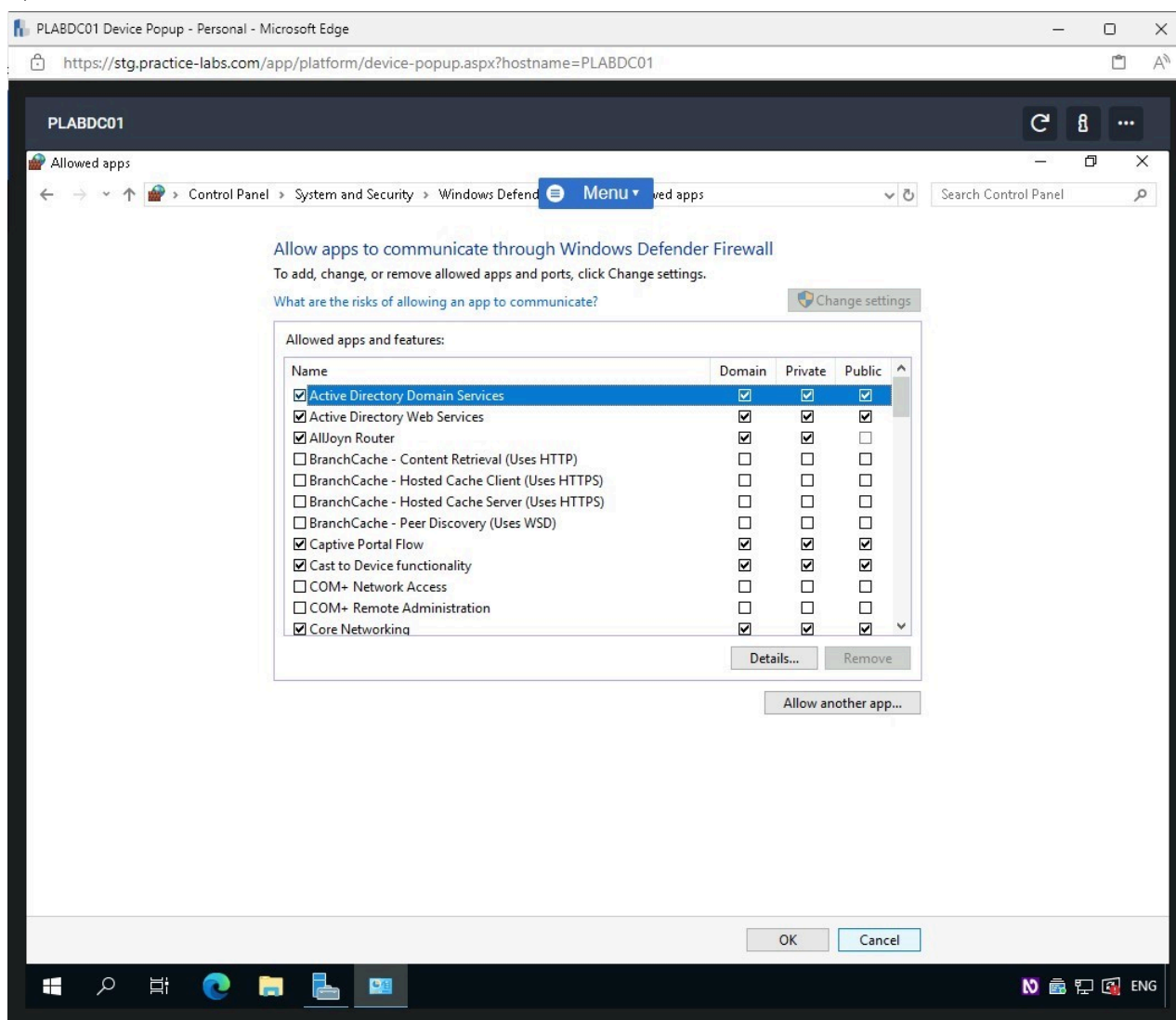


Figure 2.28 Screenshot of PLABDC01: Displaying selecting Cancel in the Allowed apps window.

Task 4 - Block a Port with Group Policy Management

Group Policy Management can be used to set up both users and their machines. It has settings for Software, Windows, and Administrative Templates that are unique to each user and computer. Administrative Templates shows the Group Policy client subsystems for which registry settings can be set, and Software Settings lets the Group Policy administrator choose which software programs will be installed on Group Policy client workstations. Utilizing group policy can help with maintaining the rules that are intended, including the management of the firewall.

In this task, the Group Policy Management MMC will be accessed. You will block a program in the firewall and view the various profiles that are available to set rules up for.

Step 1

Ensure you are connected to **PLABDC01**.

Click the **Server Manager** icon on the **Taskbar**.

In the **Server Manager** window, select **Tools** and then select **Group Policy Management**.

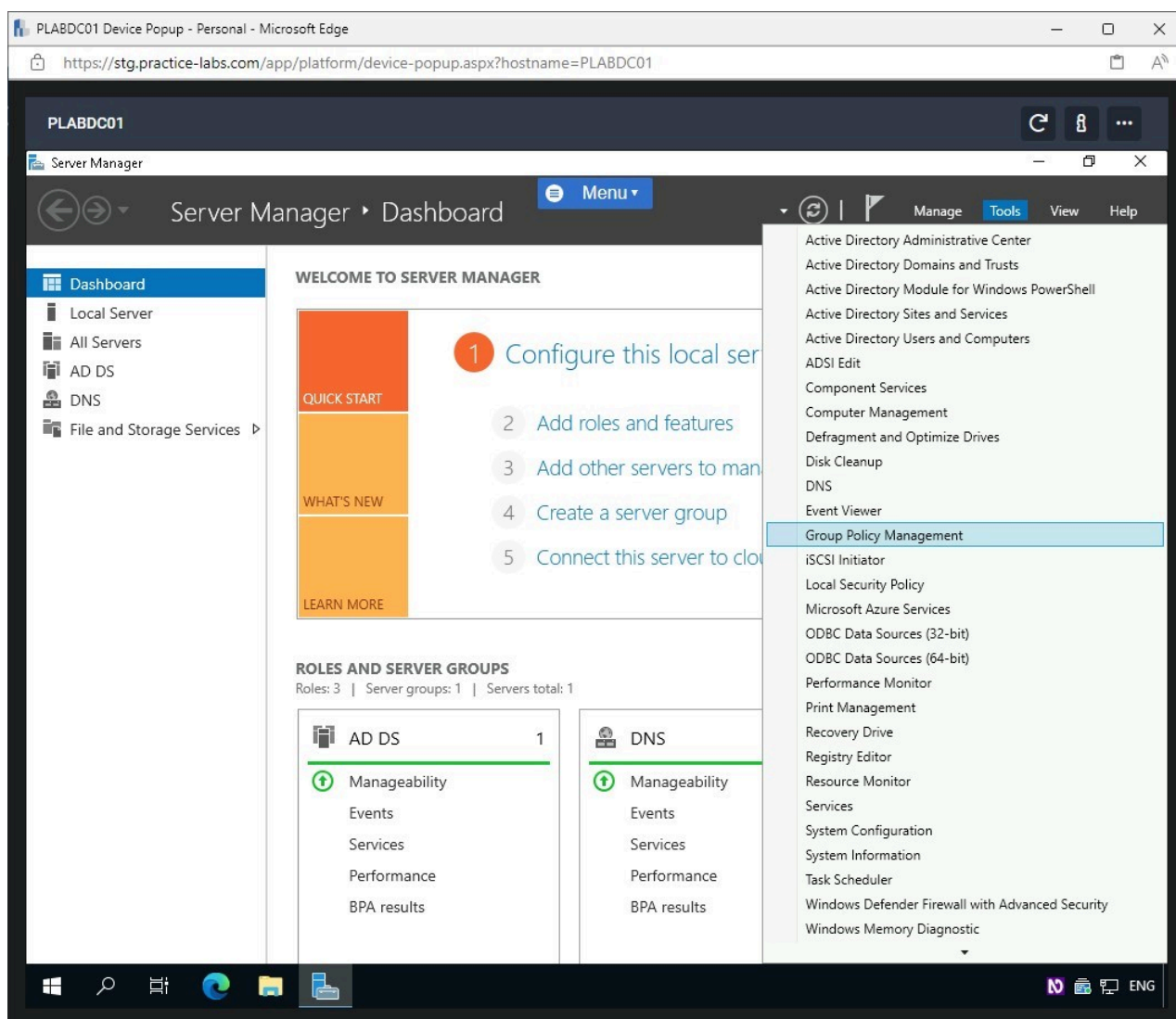


Figure 2.29 Screenshot of PLABDC01: Displaying the Server Manager window. The Tools menu is accessed, and Group Policy Management is selected.

Step 2

In the **Group Policy Management** window, expand **Forest: practicelabs.com > Domains**.

Right-click on the **practicelabs.com** domain and select **Create a GPO in this domain, and Link it here**.

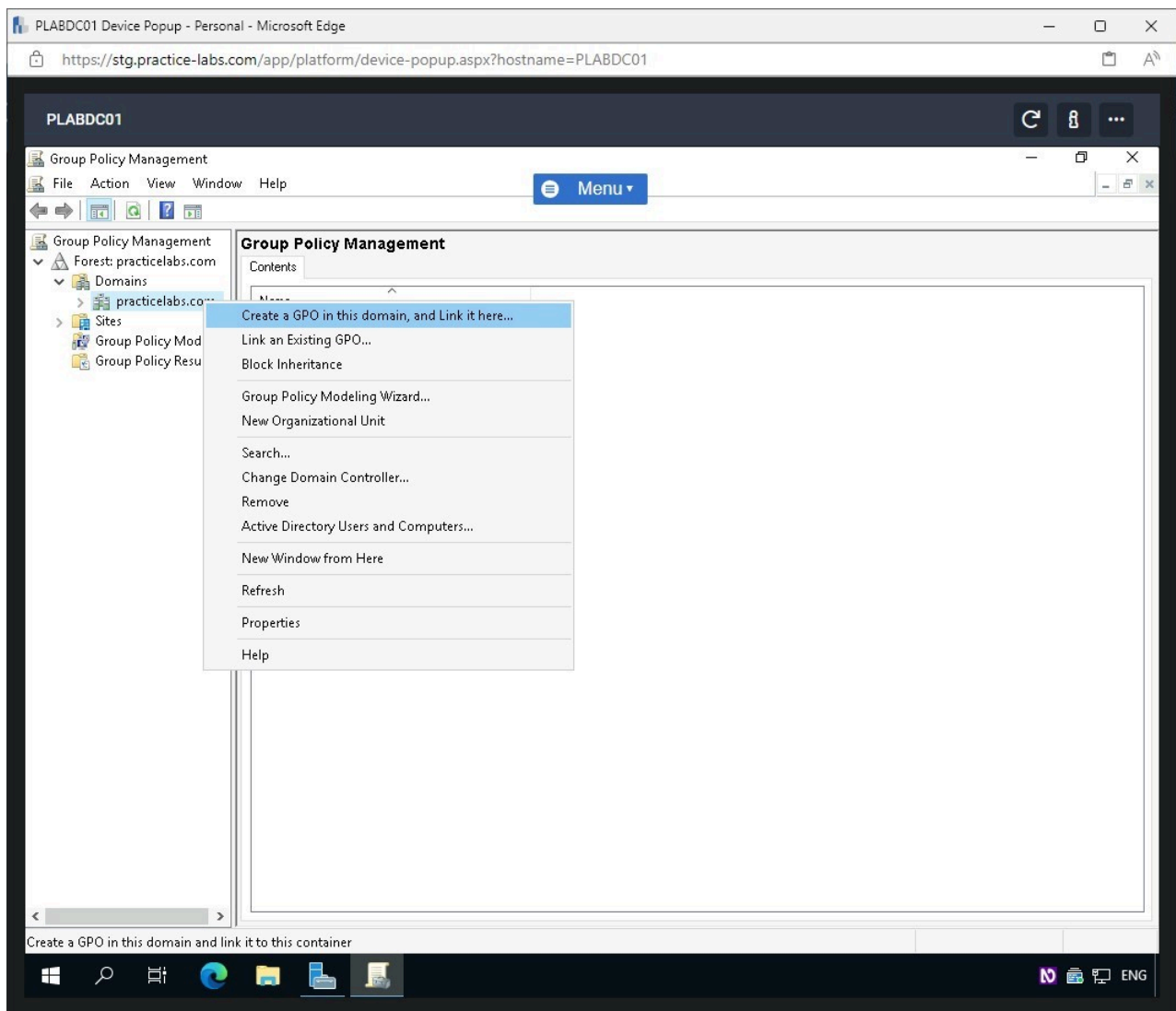


Figure 2.30 Screenshot of PLABDC01: Displaying the Group Policy Management window. practicelabs.com domain is right-clicked, and Create a GPO in this domain, and Link it here is selected.

Step 3

In the **New GPO** pop-up window, type the following for the **Name** field:

Firewall Management

Click **OK**

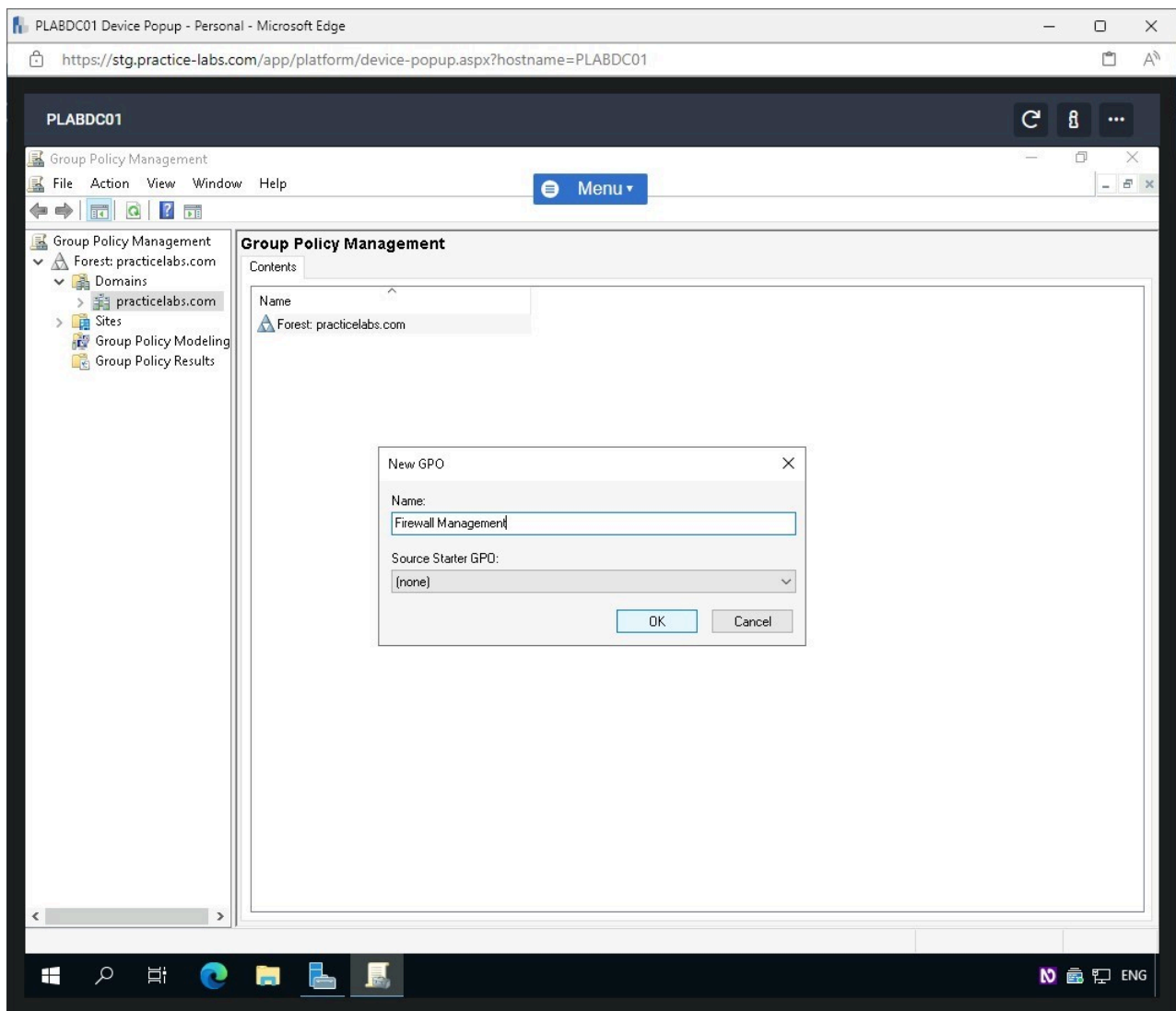


Figure 2.31 Screenshot of PLABDC01: Displaying the New GPO pop-up window, showing the required Name entered and the OK button selected.

Step 4

Back on the **Group Policy Management** window, expand **practicelabs.com**.

Right-click on the **Firewall Management** GPO and select **Edit**.

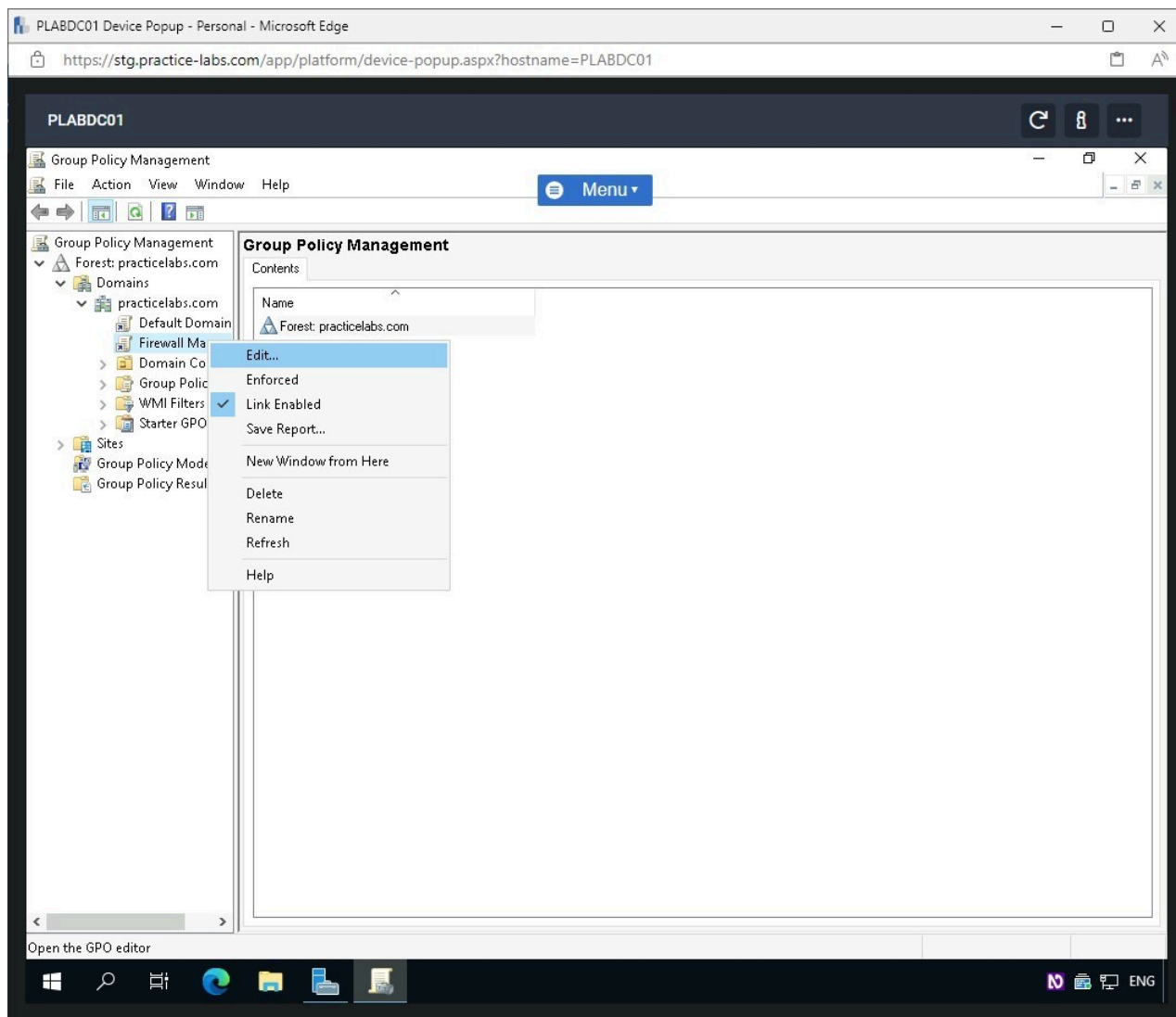


Figure 2.32 Screenshot of PLABDC01: Displaying the Group Policy Management window. Firewall Management is right-clicked, and Edit selected.

Step 5

In the **Group Policy Management Editor** window, expand **Computer Configuration > Policies > Windows Settings > Security Settings > Windows Defender Firewall with Advanced Security > Windows Defender Firewall with Advanced Security**.

Select the **Inbound Rules** link under the **View and create firewall rules** section on the right pane.

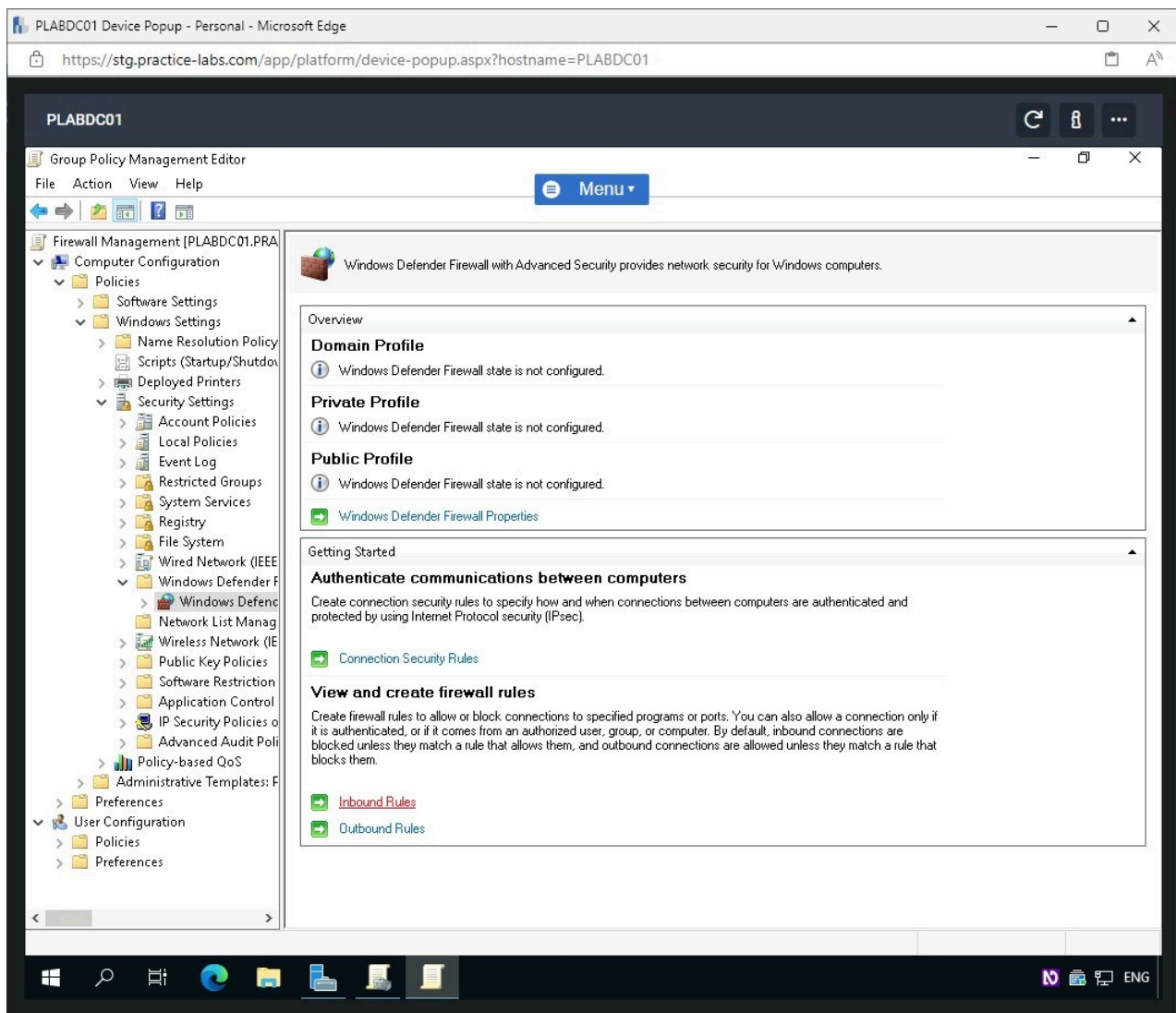


Figure 2.33 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window. Inbound Rules link is selected on the right pane.

Step 6

Notice there are no inbound rules configured yet.

Right-click anywhere on the details pane and select **New Rule**.

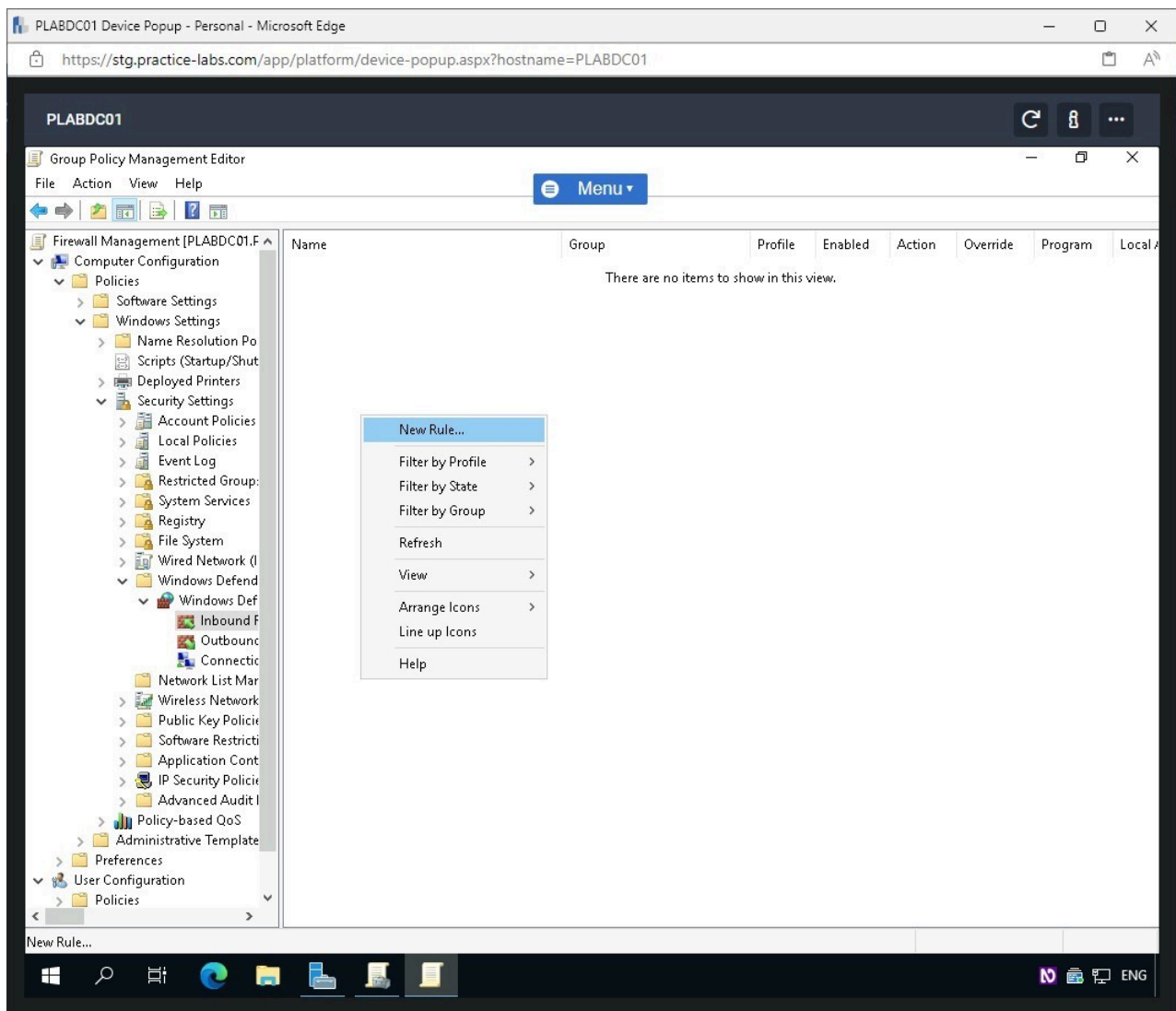


Figure 2.34 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window. New Rule is selected from the right-click context menu

Step 7

In the **New Inbound Rule Wizard - Rule Type** window, select **Port**.

Click **Next**.

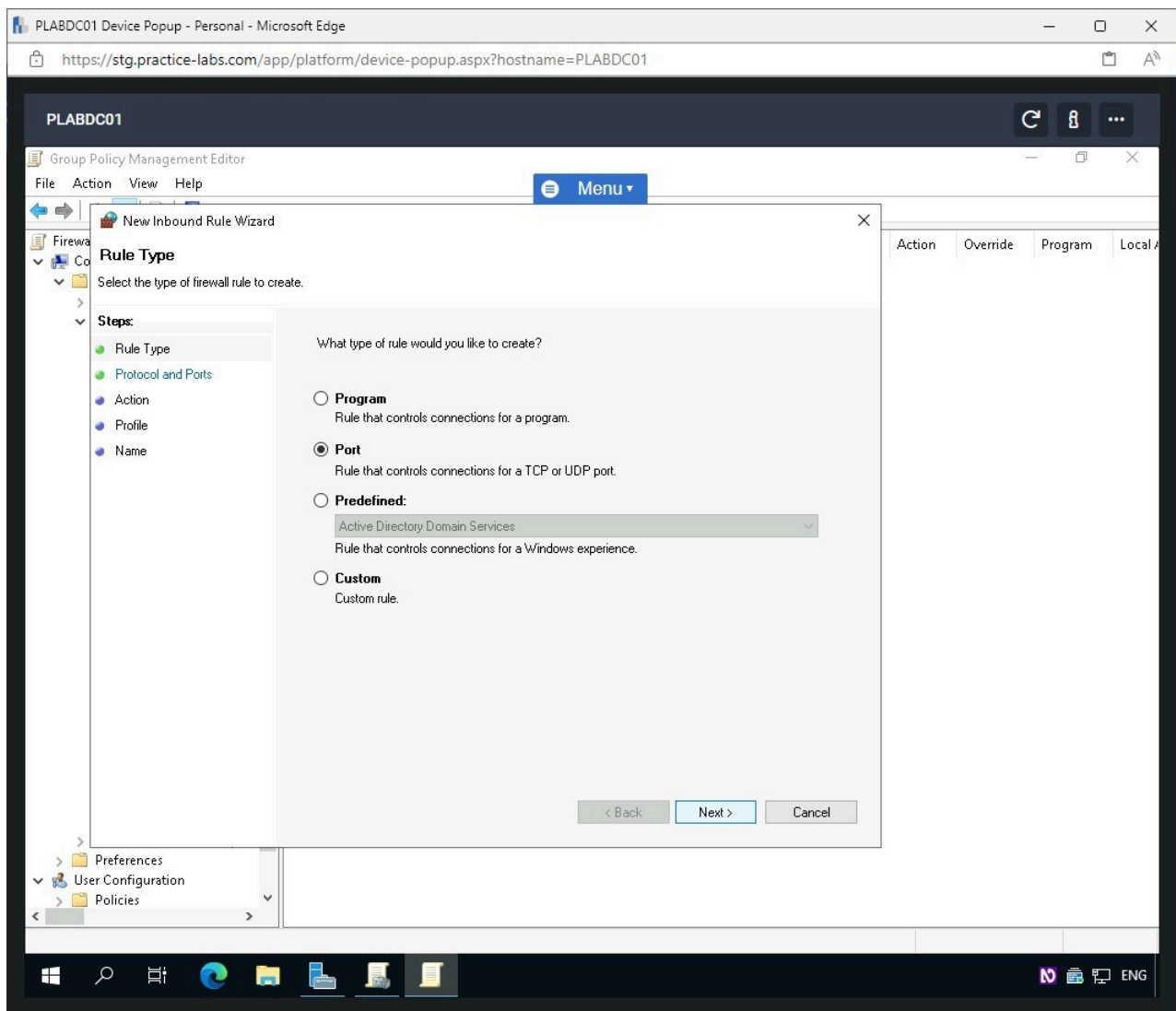


Figure 2.35 Screenshot of PLABDC01: Displaying the New Inbound Rule Wizard - Rule Type window, showing Port selected and the Next button highlighted.

Step 9

From the **Protocol and Ports** page, select the **Specific local ports** option and type the following:

23

Click **Next**.

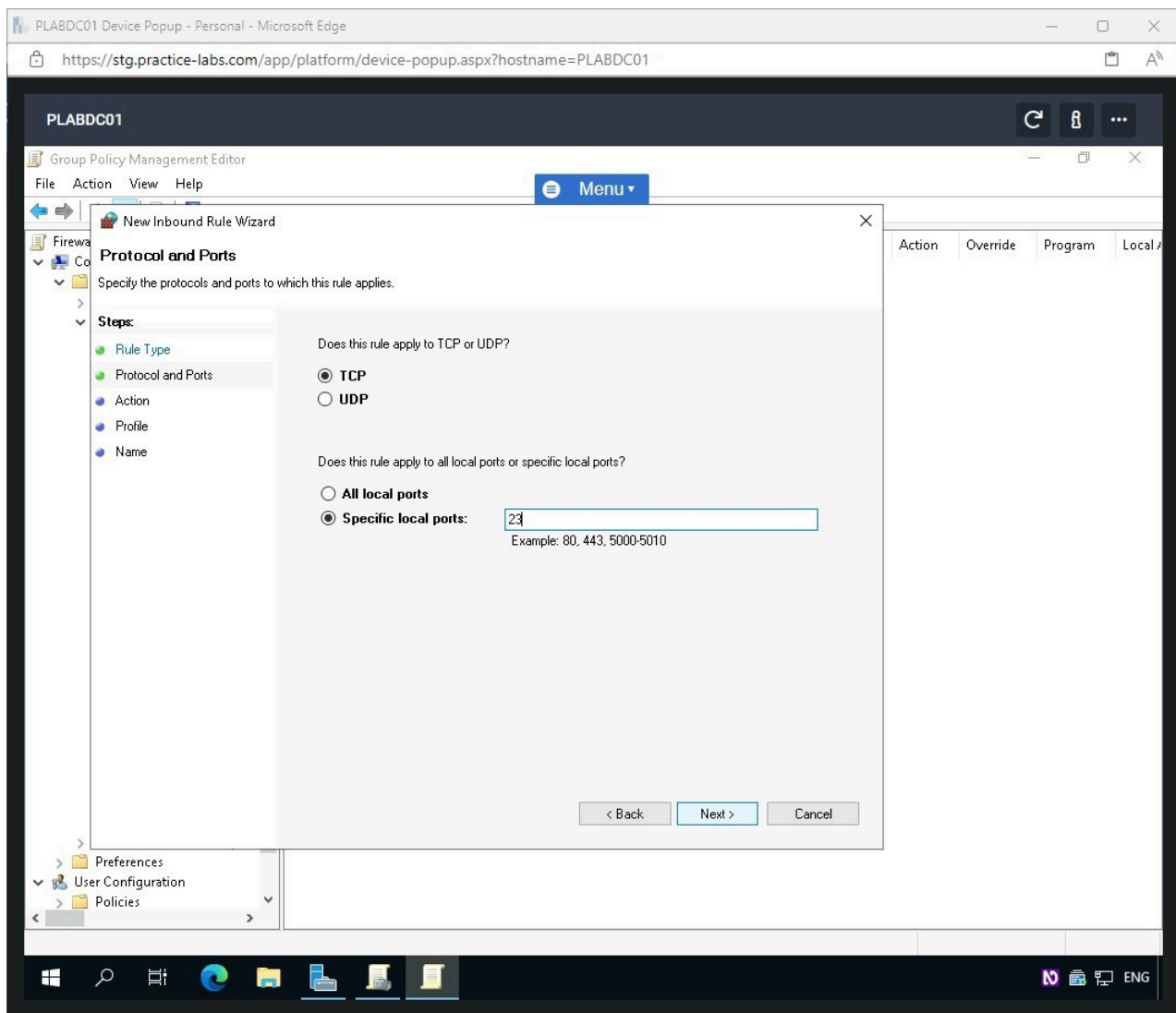


Figure 2.36 Screenshot of PLABDC01: Displaying the Protocol and Ports page, showing the required settings performed and Next selected.

Step 10

In the **Action** page, select **Block the connection** and click **Next**.

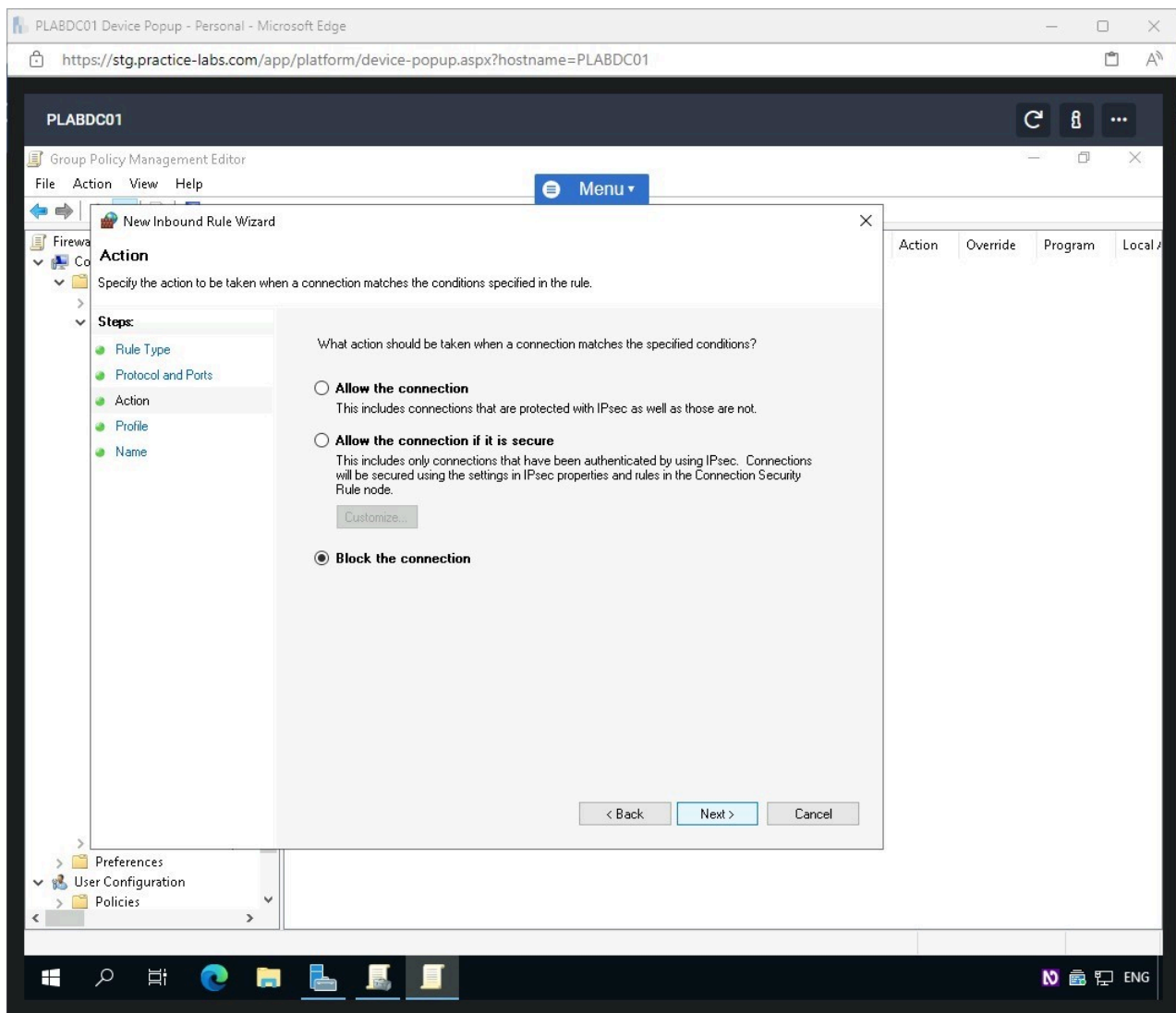


Figure 2.37 Screenshot of PLABDC01: Displaying the Action page, showing the required option selected and Next highlighted.

Step 11

On the **Profile** page, click **Next**.

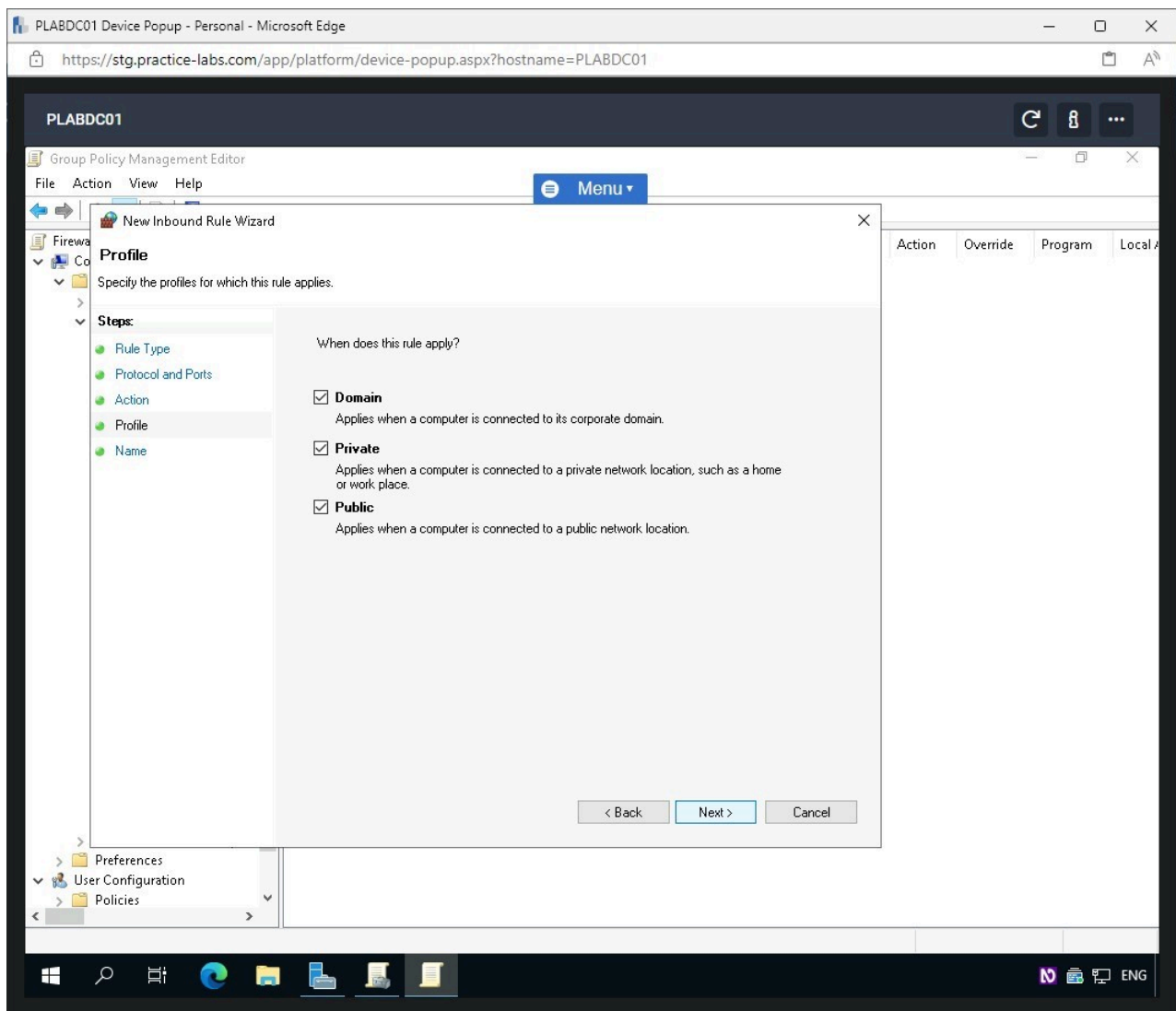


Figure 2.38 Screenshot of PLABDC01: Displaying selecting Next on the Profile page.

Step 12

On the **Name** page, type the following for the **Name** field:

Inbound Telnet Block

Click **Finish**.

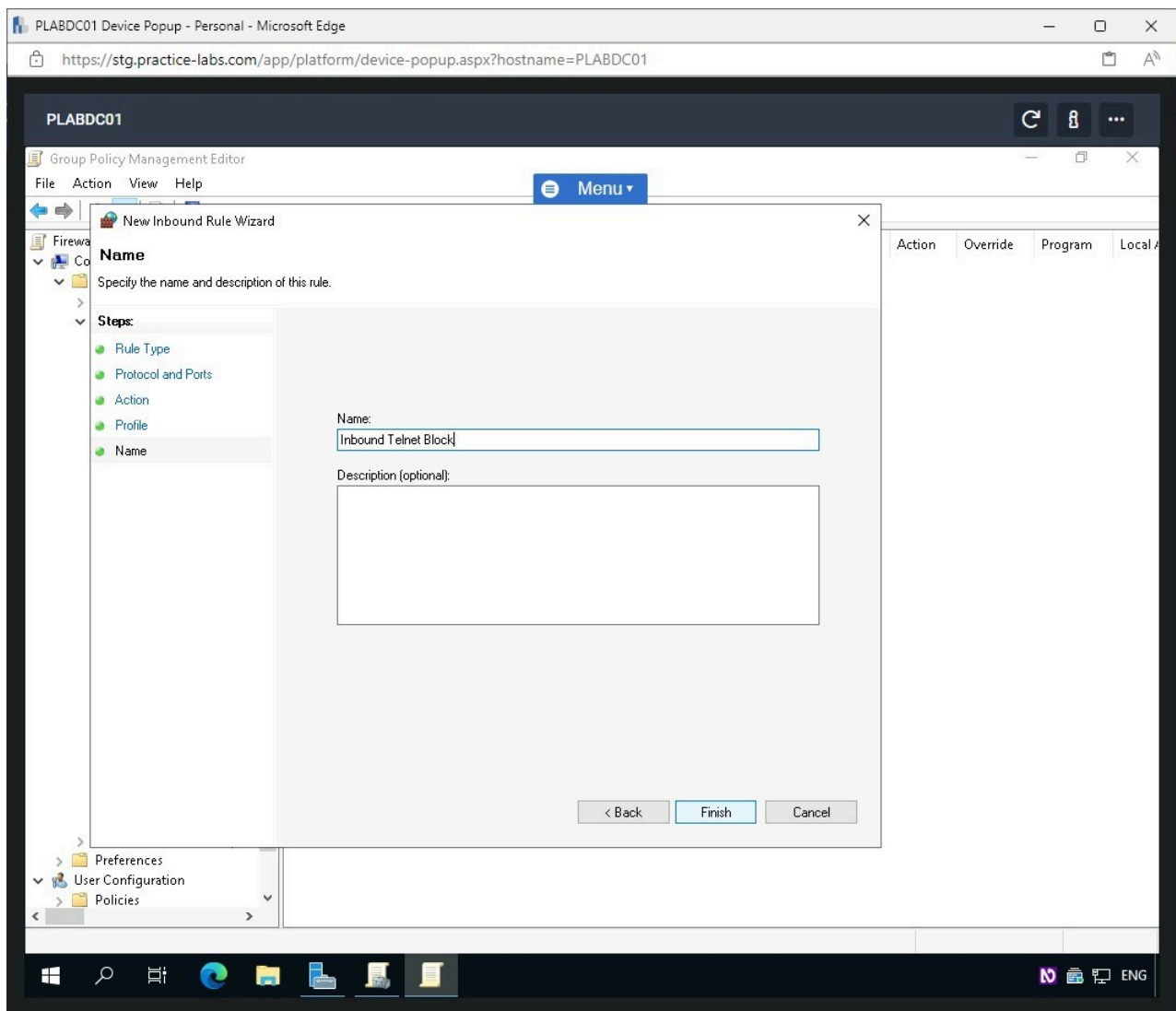


Figure 2.39 Screenshot of PLABDC01: Displaying the Name page, showing the required Name typed in and Finish selected.

Step 13

Back on the **Group Policy Management Editor** window, notice the rule is now created.

Note: Any port or program can be configured here and applied to different groups and individuals.

Close the **Group Policy Management Editor** window.

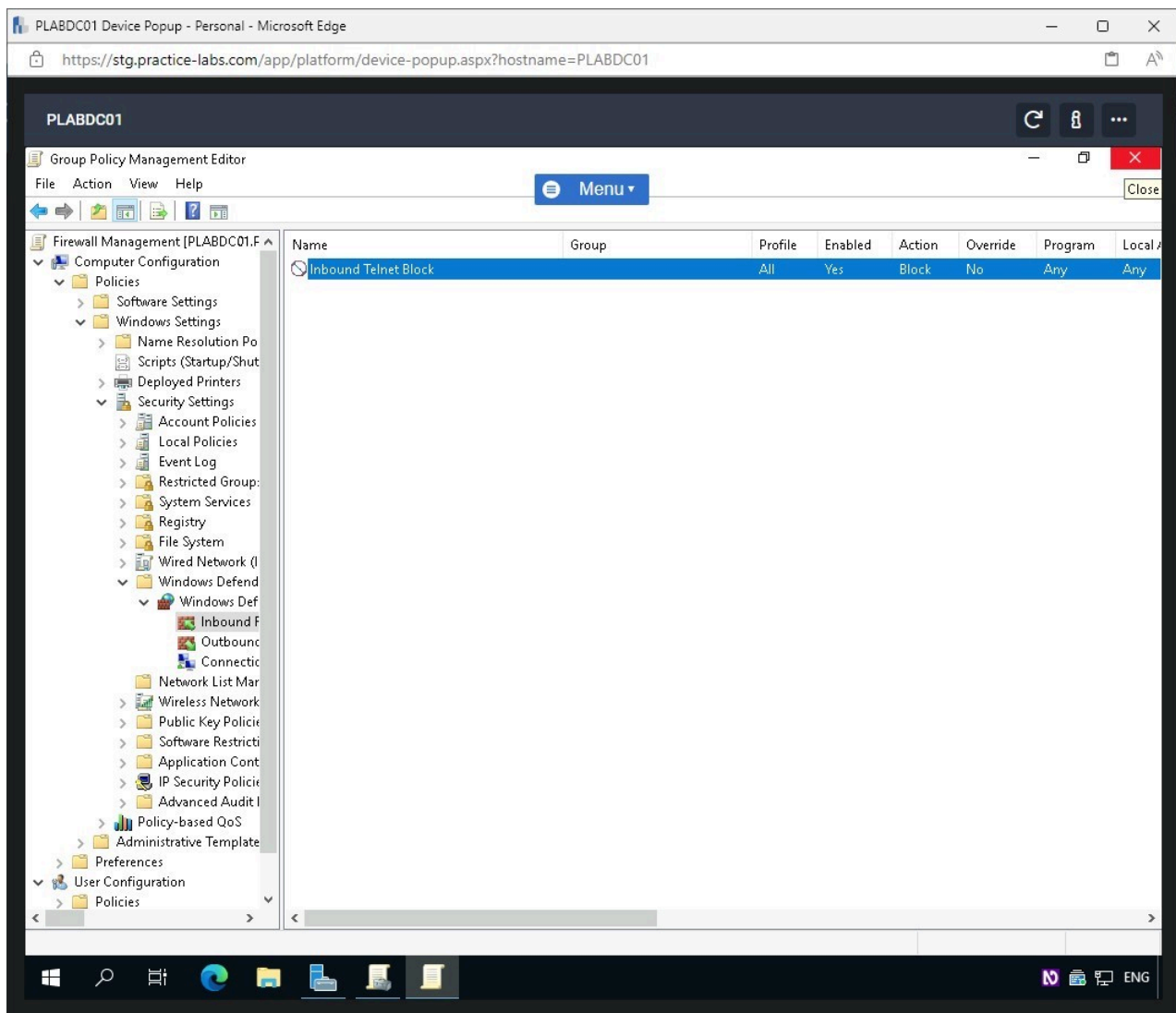


Figure 2.40 Screenshot of PLABDC01: Displaying closing the Group Policy Management Editor window.

Step 14

Close the **Group Policy Management** window.

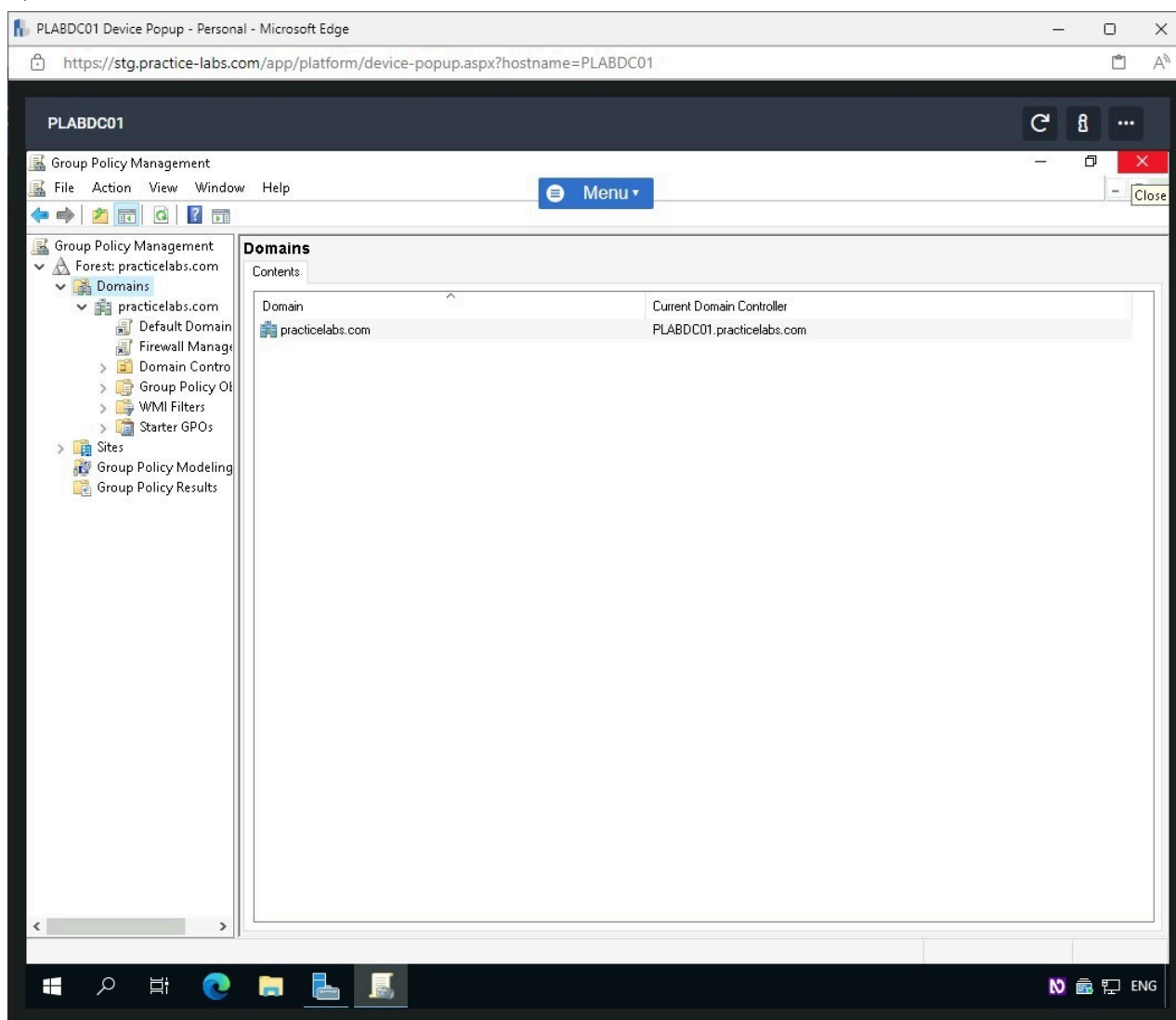


Figure 2.41 Screenshot of PLABDC01: Displaying closing the Group Policy Management Editor window.

Minimize the **Server Manager** window.

Exercise 3 - Scheduled Reboots

If you don't restart the server on a regular basis, it may slow down over time. A delay of even a few extra seconds in an application's response time may go unnoticed, but it can have a significant impact on productivity over the course of a workday. A poor Internet connection, a browser that takes a little extra time to reply, or a text search in an application that takes an extra five seconds can all negatively influence productivity. If you want to maintain the Server running at peak speed and prevent slowdowns and productivity losses, rebooting them at least once a week is a must. Restarting the device does several useful tasks that keep it working smoothly, securely, and with fewer issues that hold you down during the day.

Memory is being consumed by each of the computer's processes as it is utilized for various purposes. Everything that is done in a document, in a browser, in an app, etc., is recorded. Using more RAM might cause the device to run more slowly than normal, perhaps leading to frozen programs and longer load times for online pages. The computer's RAM is not released when it is put in sleep mode. This can only be achieved by a full shutdown and restart. Rebooting can clear that memory, making it quicker and more responsive than before.

Memory leaks are another common cause of server slowdowns. This occurs when an application is supposed to be closed, but an error prevents it from doing so, and the application continues to use memory. Leaks can be fixed by restarting the system, which clears the whole memory cache.

One thing to keep in mind is that when restarting a server, always refer to procedures and protocols because servers provide services, and those services always need to be running.

In this exercise, you will schedule a restart using the Task Scheduler.

Learning Outcomes

After completing this exercise, you should be able to:

- Use Task Scheduler to Schedule Reboots

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)

Task 1 - Use Task Scheduler to Schedule Reboots

In this task, you will use the Task Scheduler to schedule a weekly reboot.

Step 1

Connect to **PLABDC01**.

Click the **Start** charm and type the following:

Task Scheduler

Select **Task Scheduler** from the **Best match** pop-up menu.

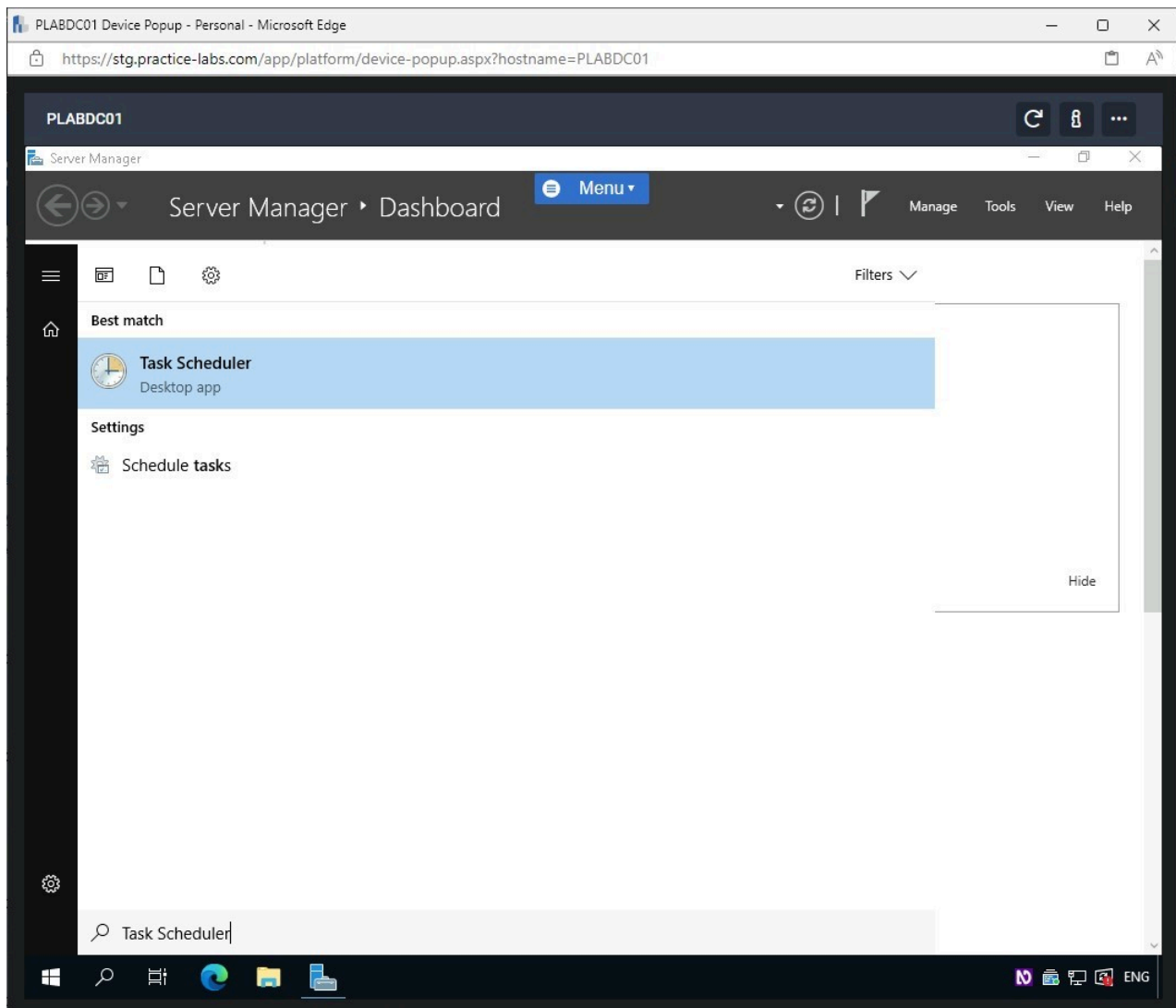


Figure 3.1 Screenshot of PLABDC01: Displaying selecting Task Scheduler from the Best match pop-up menu.

Step 2

In the **Task Scheduler** window, select and right-click on **Task Scheduler Library** and select **New Folder**.

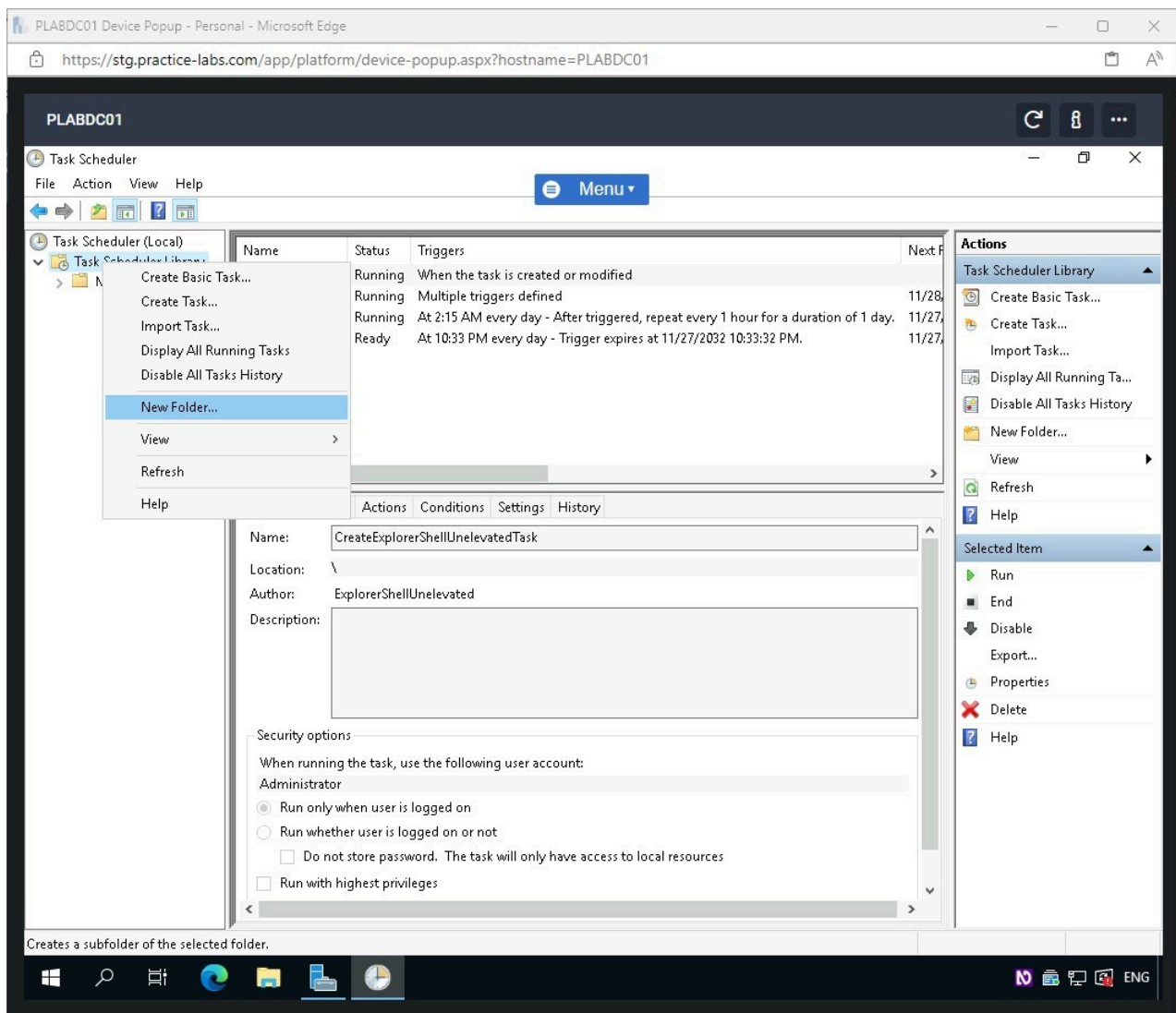


Figure 3.2 Screenshot of PLABDC01: Displaying the Task Scheduler window. Task Scheduler Library is right-clicked, and New Folder is selected.

Step 3

In the **Enter name of the new folder** pop-up window, type the following for the **Name** field:

Weekly Restart

Click **OK**.

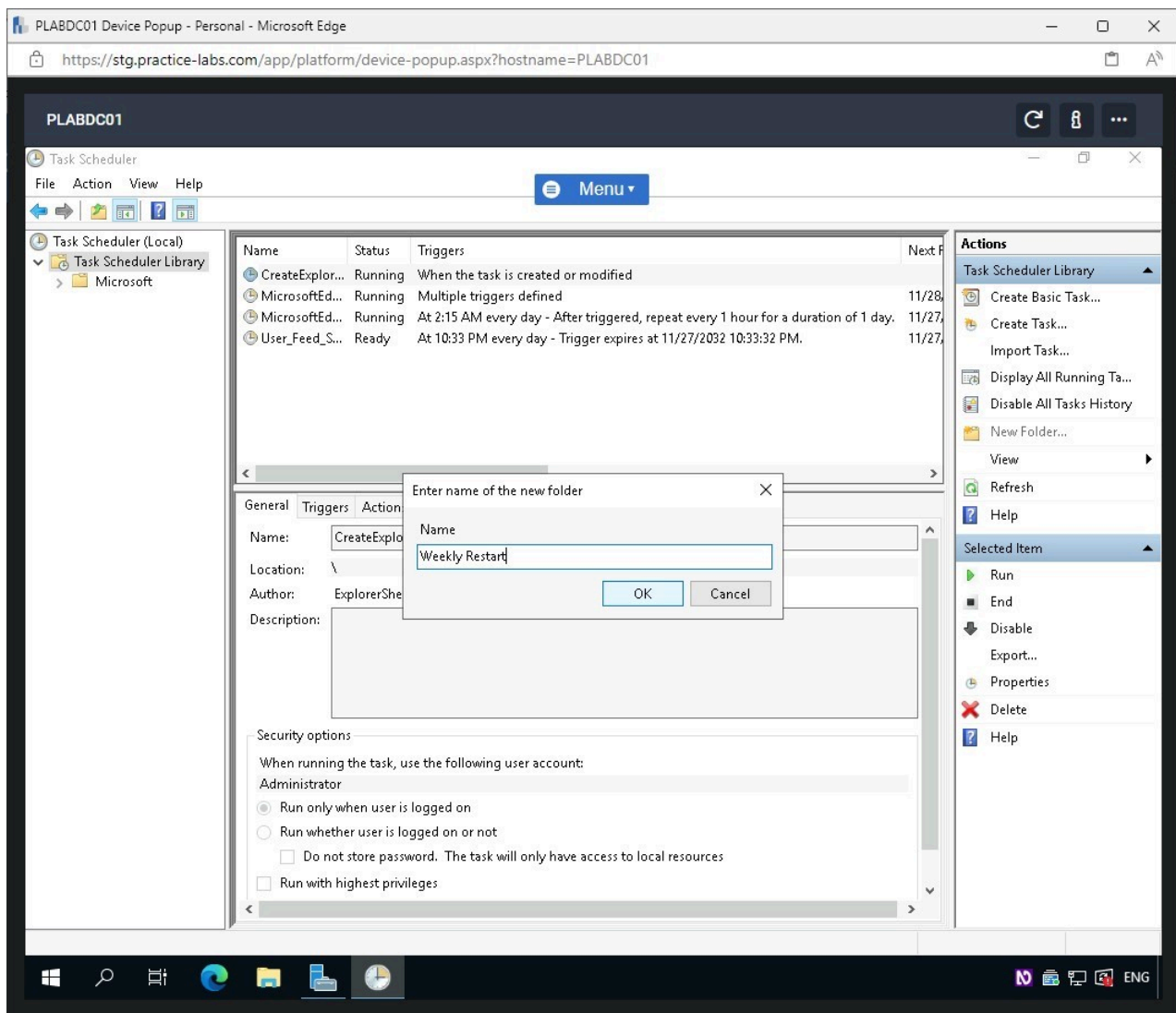


Figure 3.3 Screenshot of PLABDC01: Displaying the Enter name of the new folder pop-up window, showing the required Name typed in and OK selected.

Step 4

In the **Create Basic Task Wizard** window, select **Create a Basic Task** on the left pane.

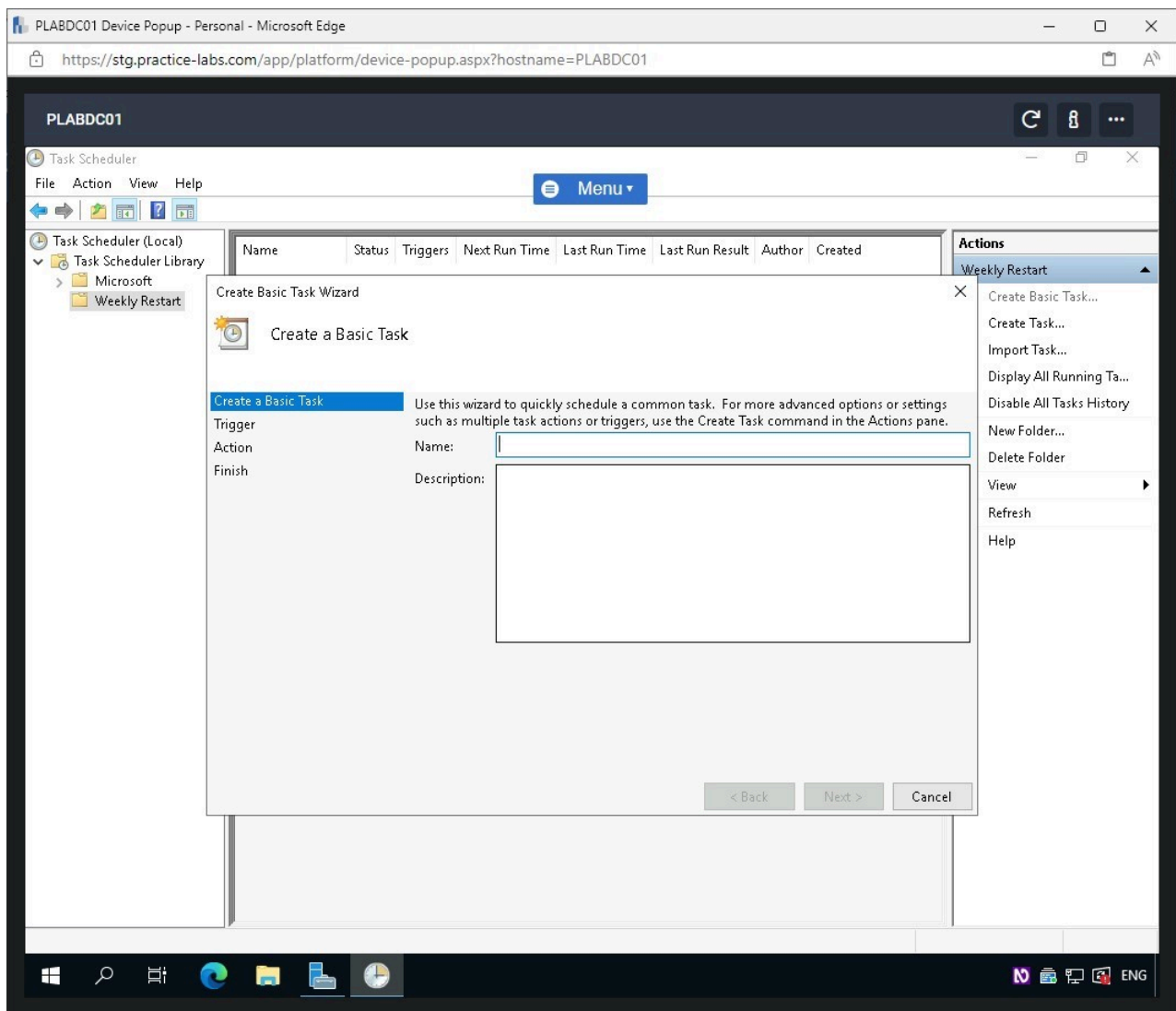


Figure 3.4 Screenshot of PLABDC01: Displaying the Create Basic Task Wizard window, showing Create a Basic Task selected on the left pane.

Step 5

In the **Create a Basic Task** page, type the following into the **Name** field:

Weekly Restart

Click **Next**.

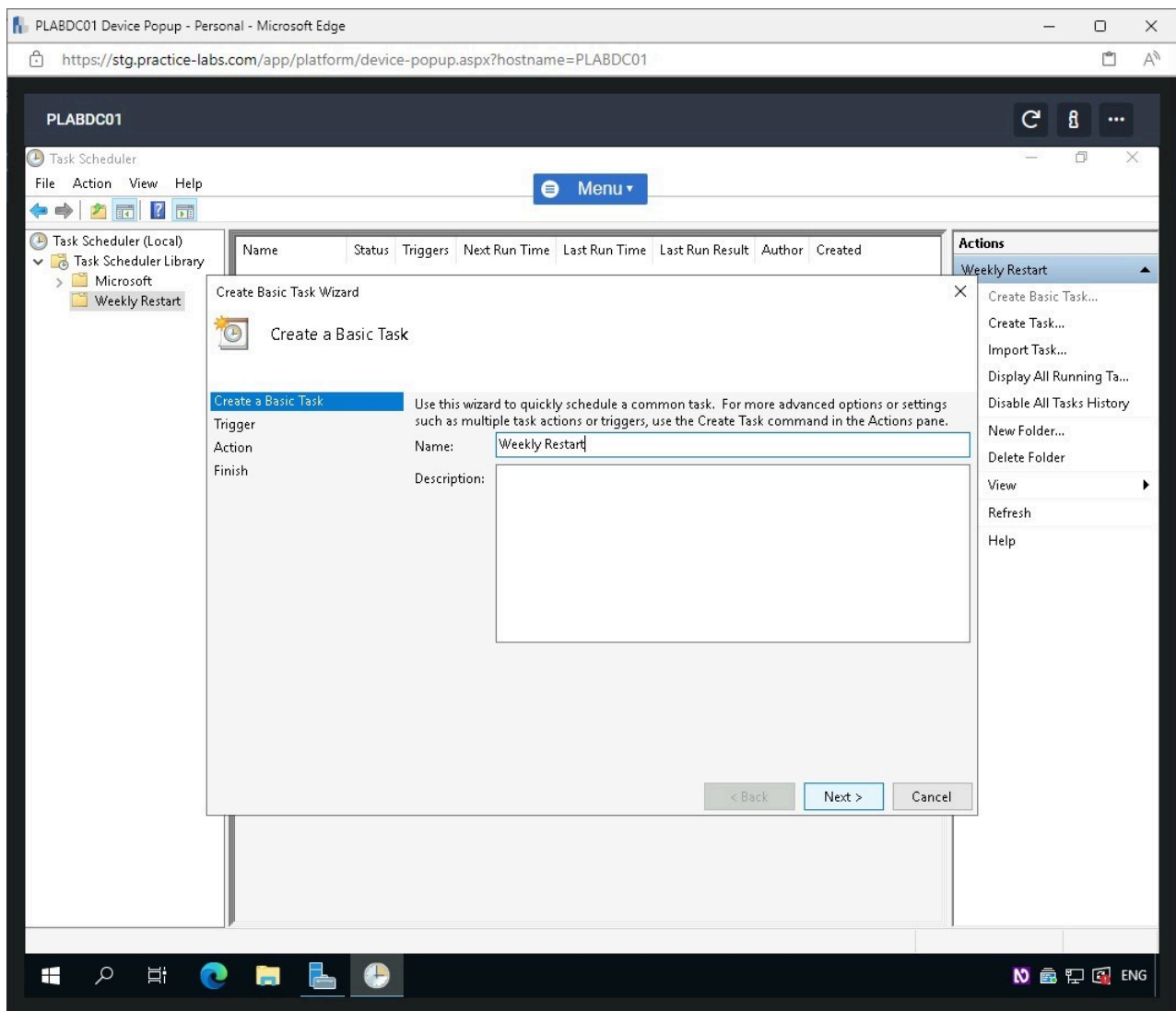


Figure 3.5 Screenshot of PLABDC01: Displaying the Create a Basic Task page, showing the required Name typed in and Next selected.

Step 6

In the **Task Trigger** page, select the **Weekly** option and click **Next**.

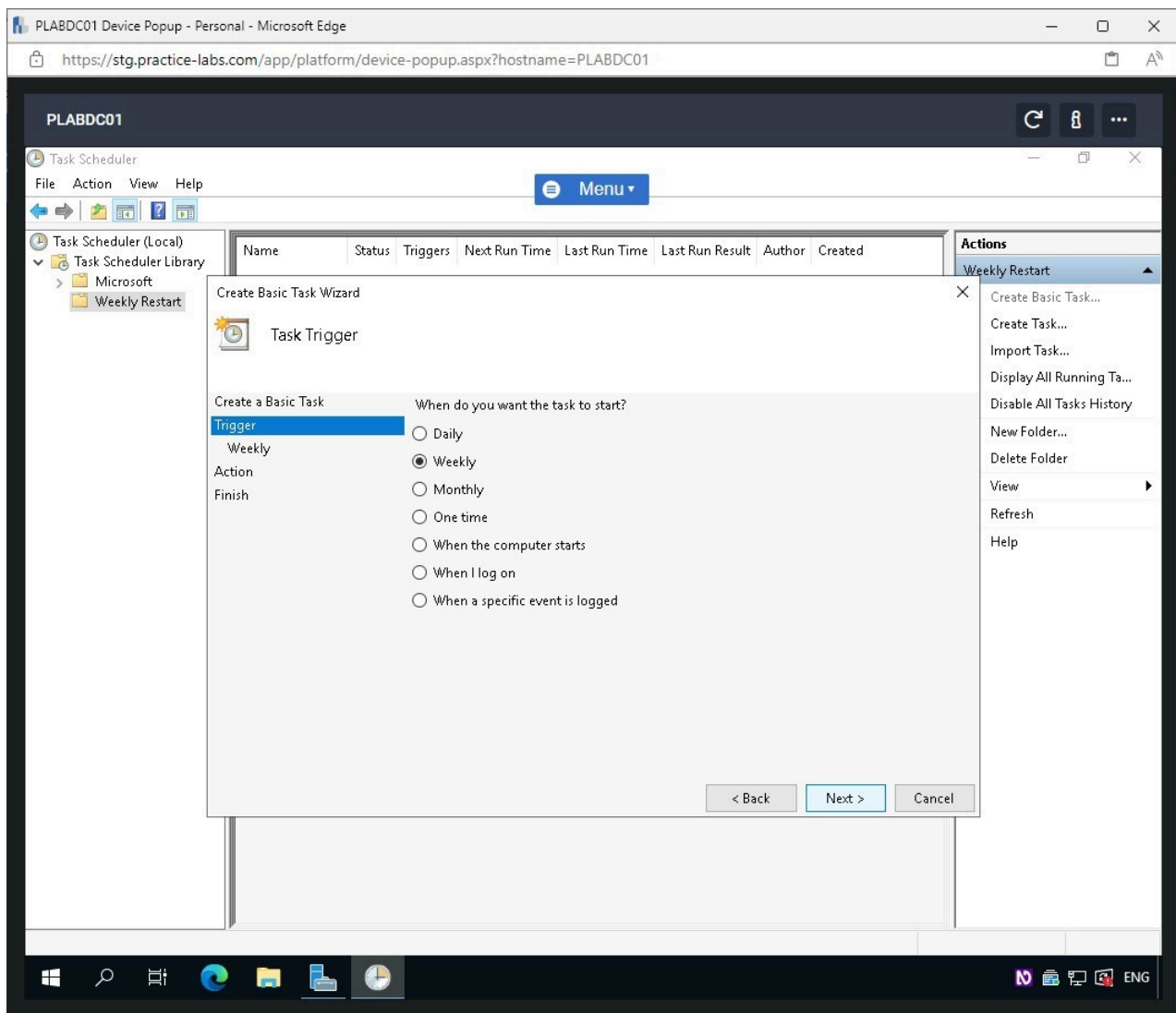


Figure 3.6 Screenshot of PLABDC01: Displaying the Task Trigger page, showing the Weekly option selected and Next highlighted.

Step 7

From the **Weekly** page, configure the following:

- For the **Start** date, select the date for the upcoming Sunday
- For the **Start** time, select **11:00 PM**
- Enable the **Synchronize across time zones** checkbox
- For the **Recur every** field, type **1**
- Enable the **Sunday** checkbox

Click **Next**.

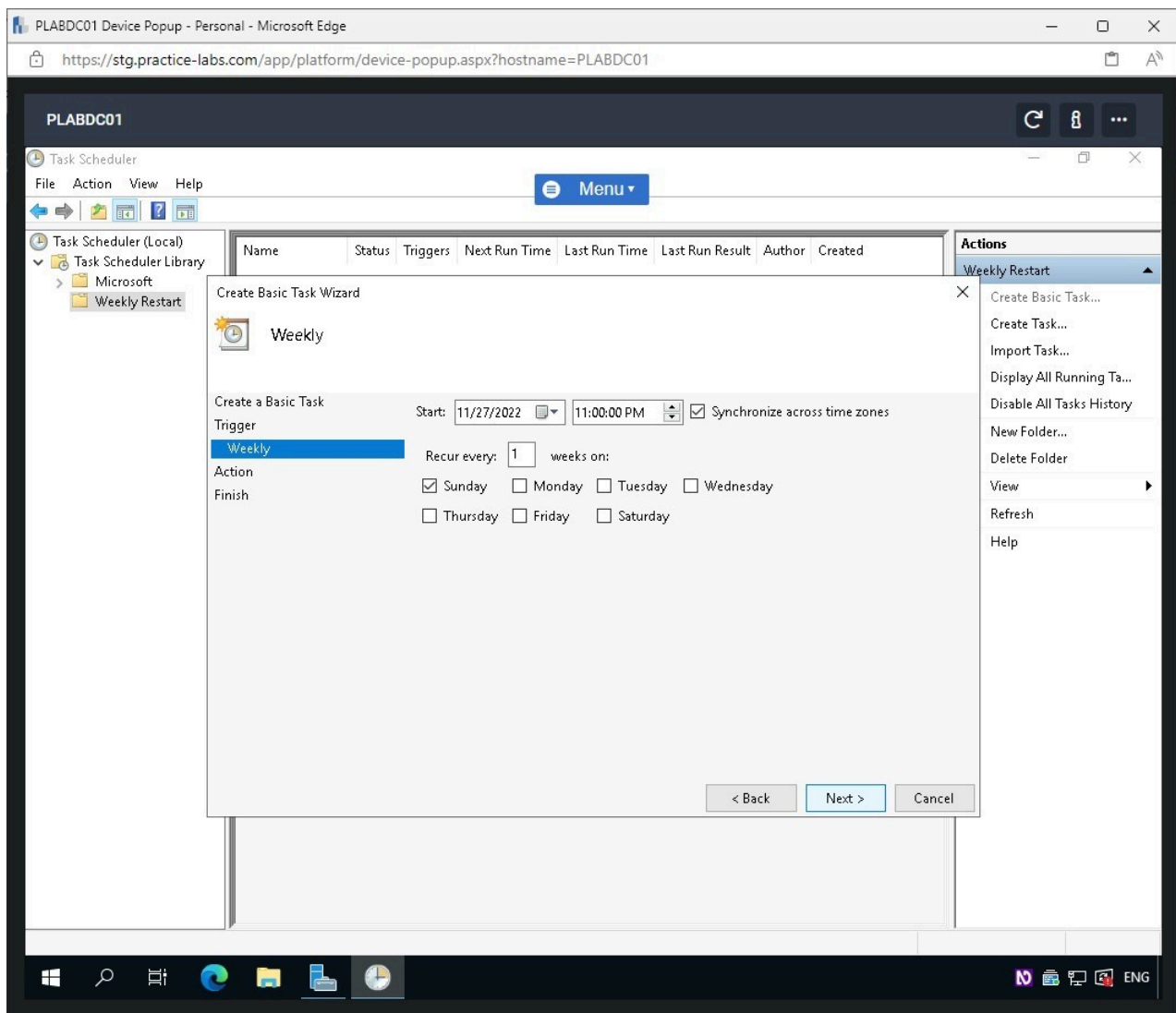


Figure 3.7 Screenshot of PLABDC01: Displaying the Weekly page, showing the required settings performed and Next selected.

Step 8

In the **Action** page, leave the **Start a program** option selected.

Click **Next**.

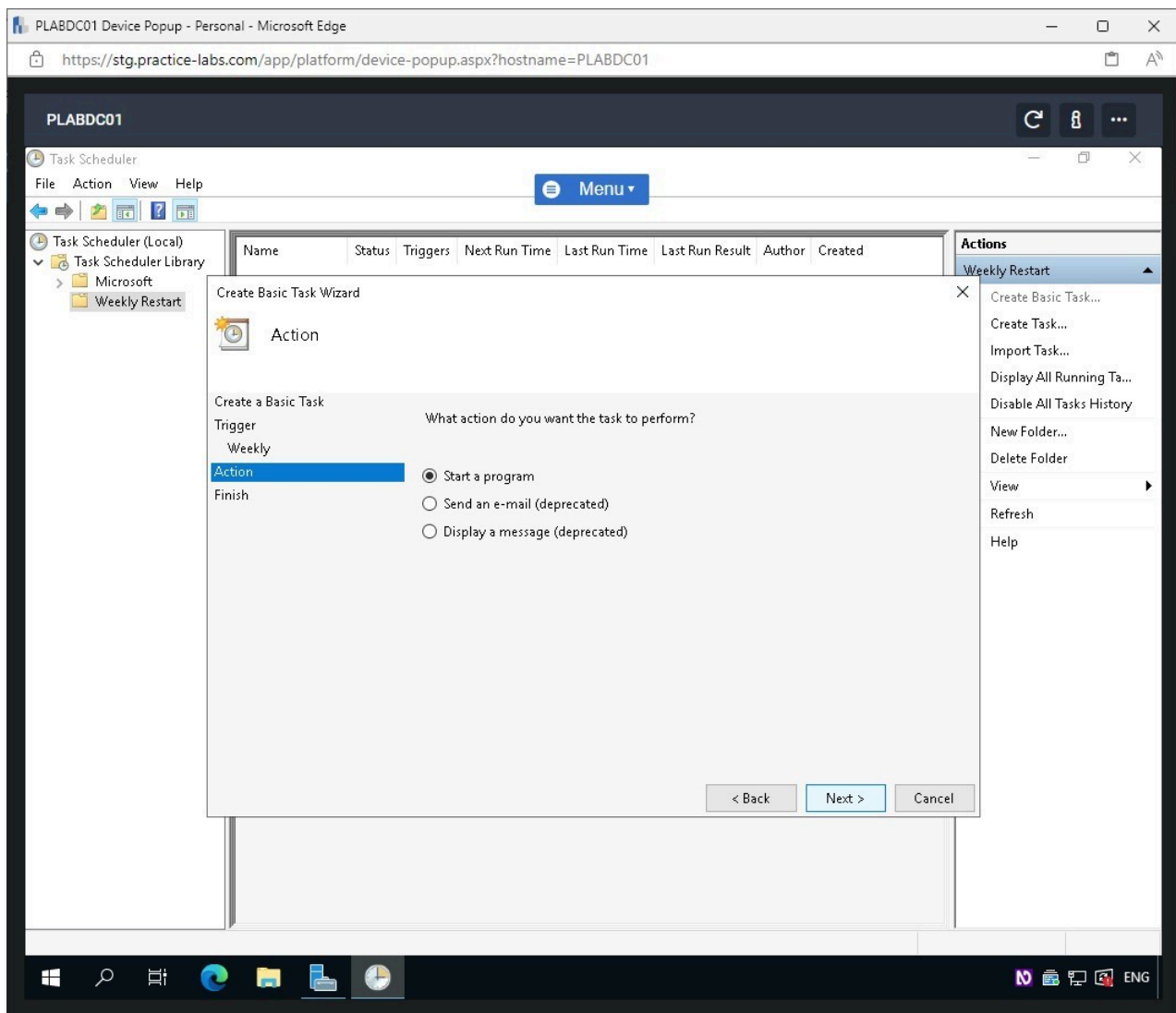


Figure 3.8 Screenshot of PLABDC01: Displaying selecting Next in the Action page.

Step 9

In the **Start a Program** page, type the following under the **Program/script** section:

```
shutdown /r
```

Click **Next**.

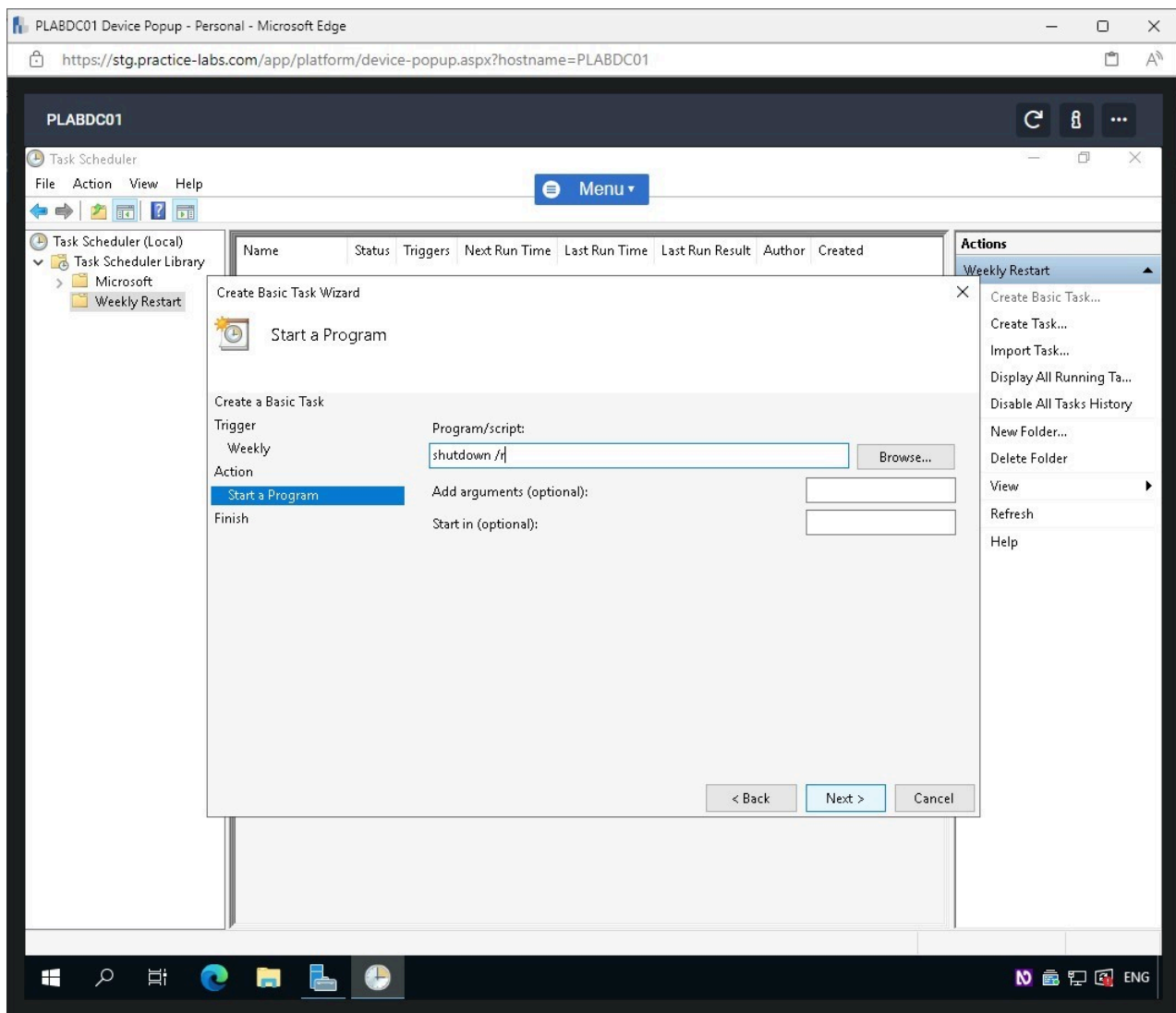


Figure 3.9 Screenshot of PLABDC01: Displaying the Start a Program page, showing the required settings performed and Next selected.

Step 10

Click **Yes** on the **Task Scheduler** pop-up window.

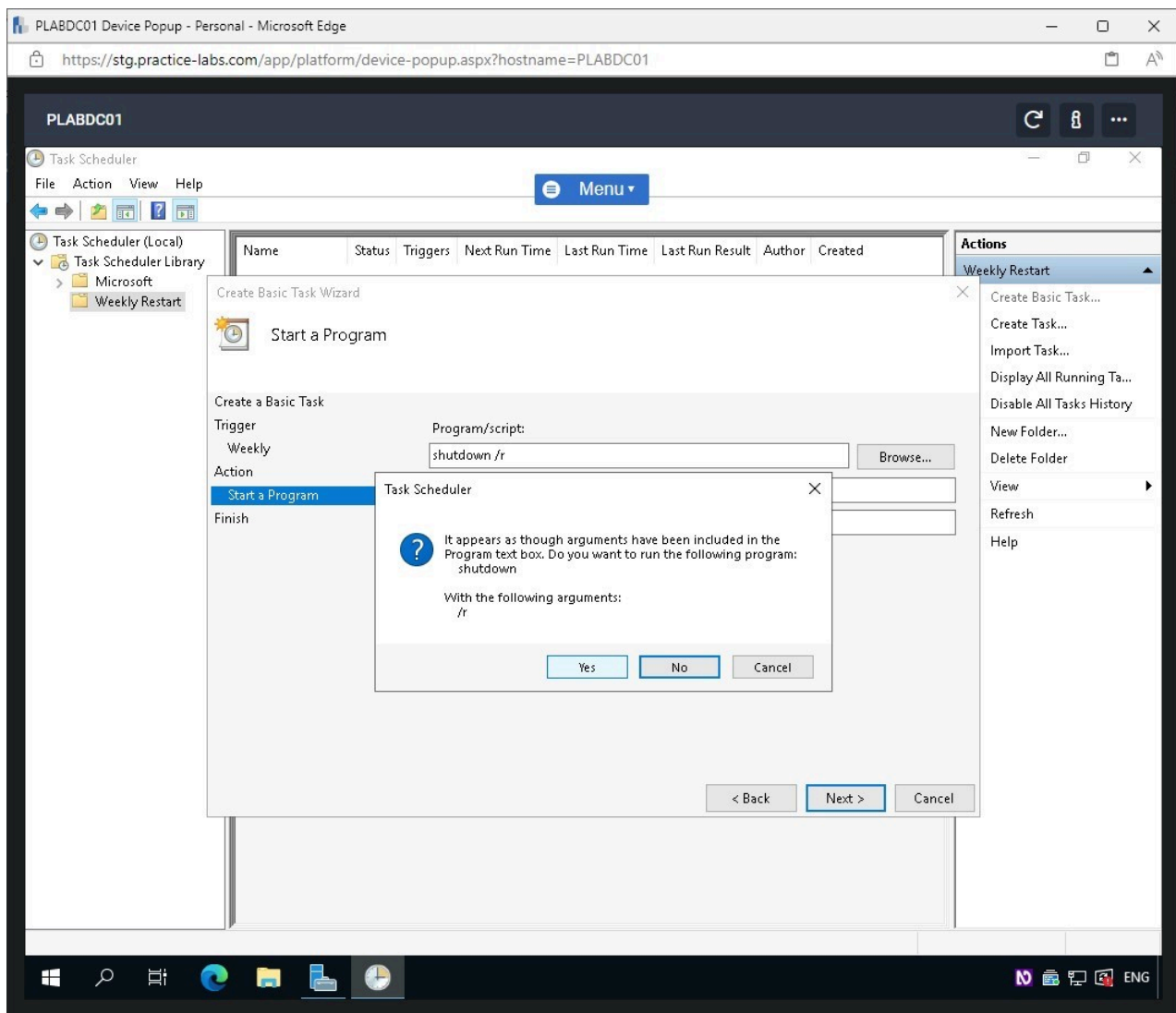


Figure 3.10 Screenshot of PLABDC01: Displaying selecting Yes on the Task Scheduler pop-up window.

Step 11

On the **Summary** page, click **Finish**.

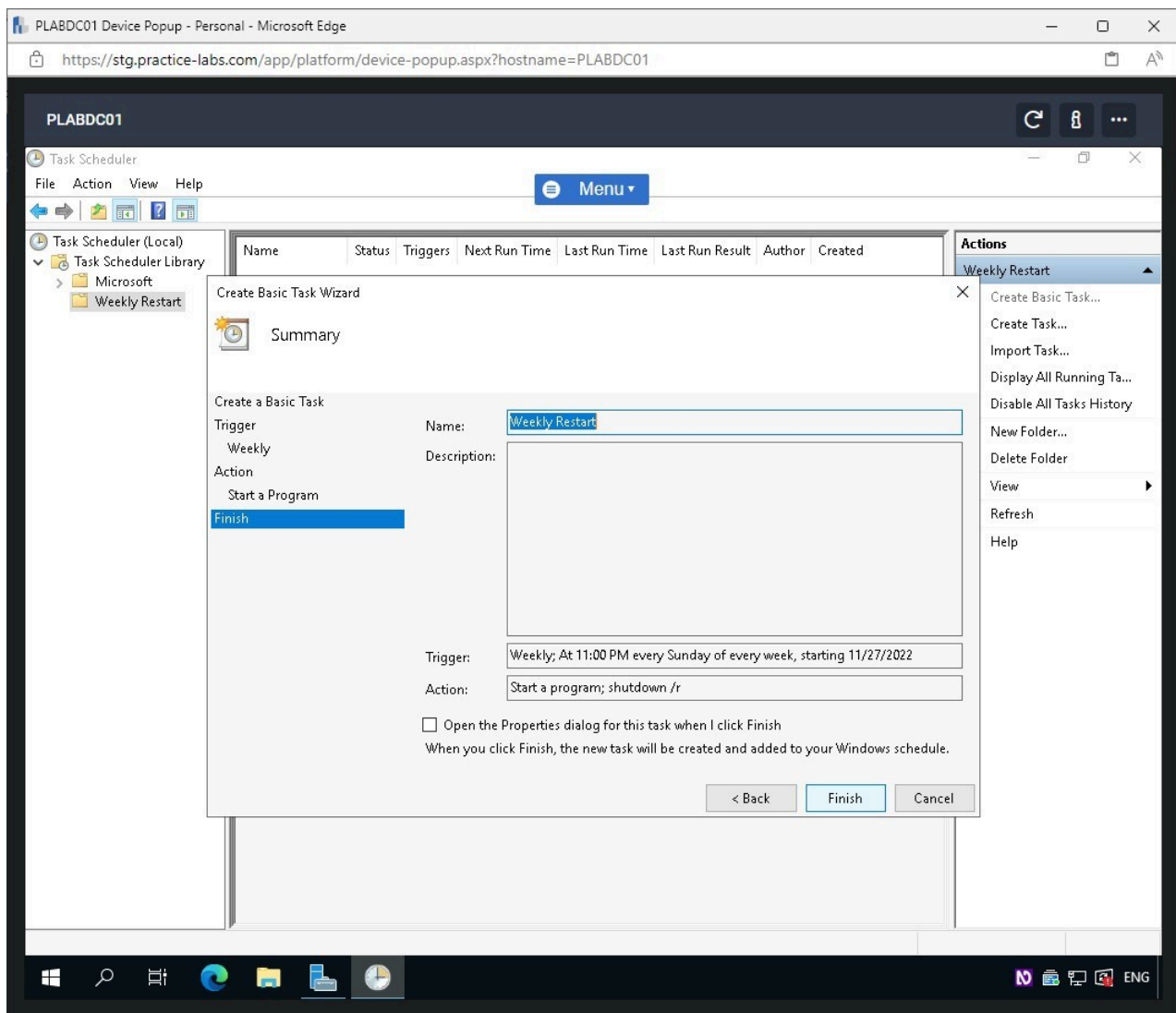


Figure 3.11 Screenshot of PLABDC01: Displaying selecting Finish on the Summary page.

Step 12

You have now configured a task to restart the Server each week at 11 PM.

Close the **Task Scheduler** window.

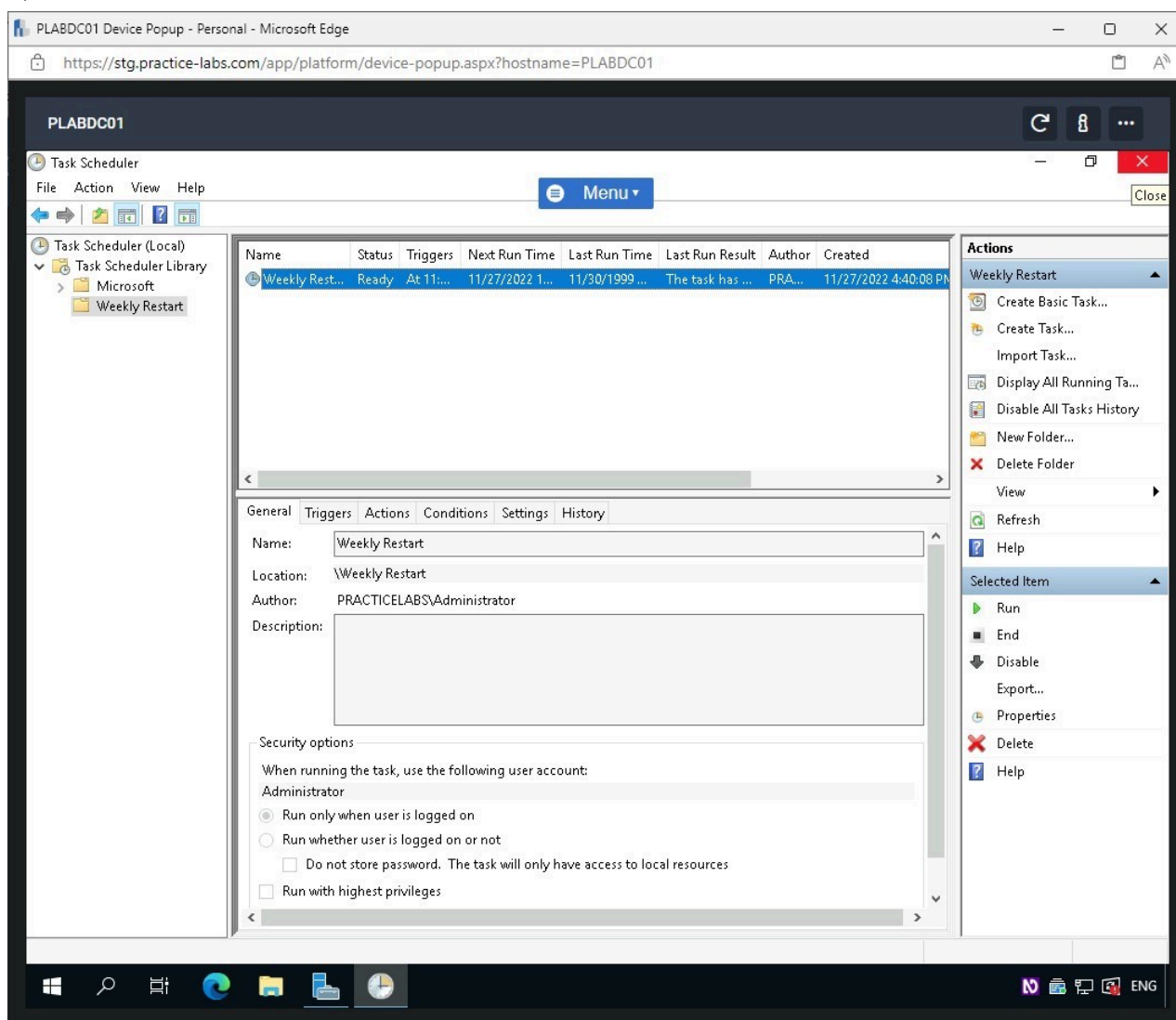


Figure 3.12 Screenshot of PLABDC01: Displaying closing the Task Scheduler window.

Exercise 4 - Services

In computing, services are applications that track and react to data transmissions across a network. Access to the computer and its contents can also be shared through other services, including web servers, file servers, FTP servers, email, proxies, open ports, and Remote Desktops. These services can occasionally create problems.

In this exercise, you will learn how to Start and Stop a Service.

Learning Outcomes

After completing this exercise, you should be able to:

- Start and Stop a Service

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)

Task 1 - Start and Stop a Service

In this task, you will access the Services MMC and see how to start and restart a service.

Step 1

Connect to **PLABDC01**.

Restore the **Server Manager** window from the **Taskbar**.

Click the **Tools** menu and select **Services**.

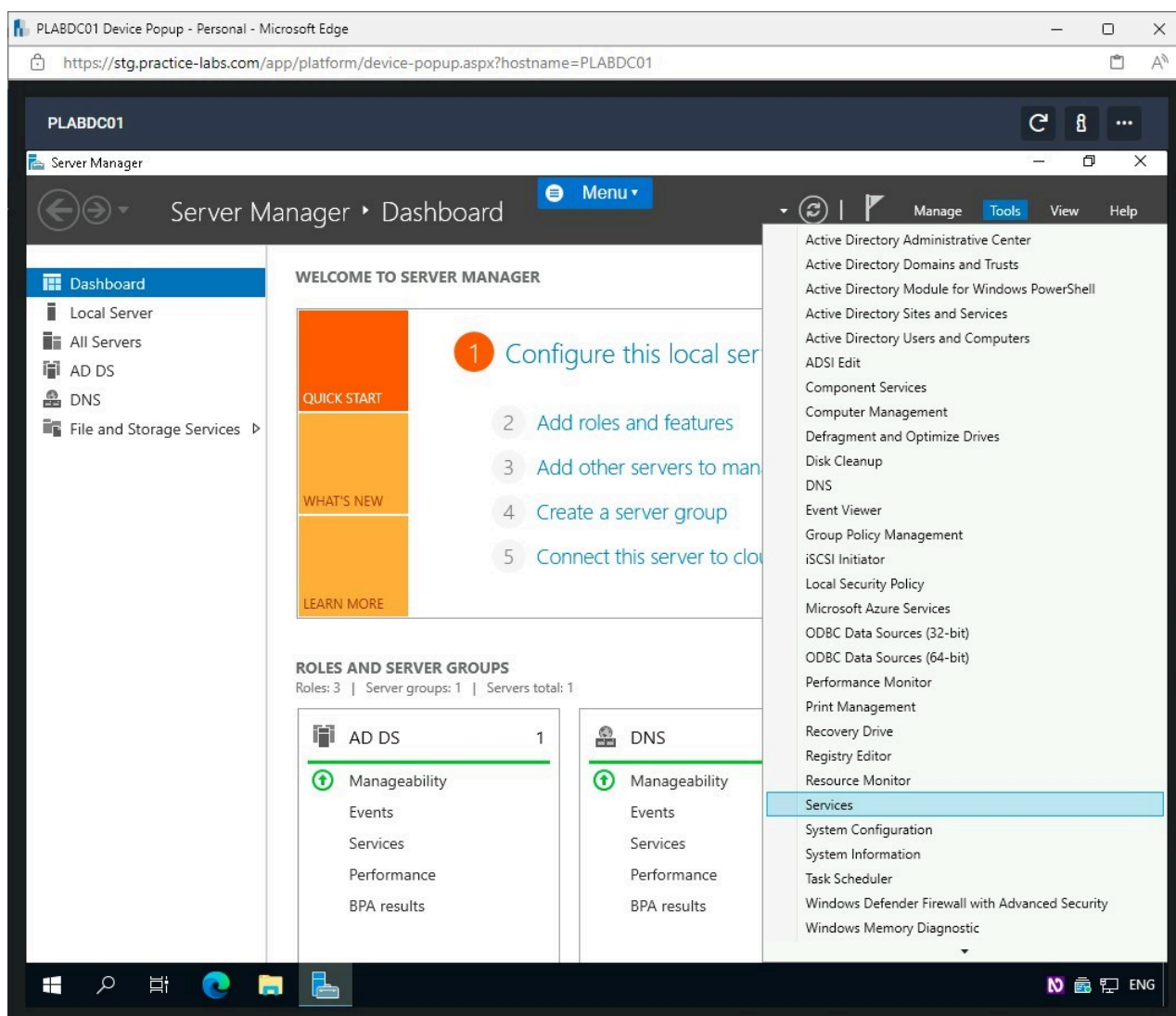


Figure 4.1 Screenshot of PLABDC01: Displaying the Server Manager window. The Tools menu is accessed, and Services is selected.

Step 2

In the **Services** window, configuring services that are specific to this machine is indicated by **Services (Local)**.

In the right pane, scroll down and double-click on **Print Spooler**.

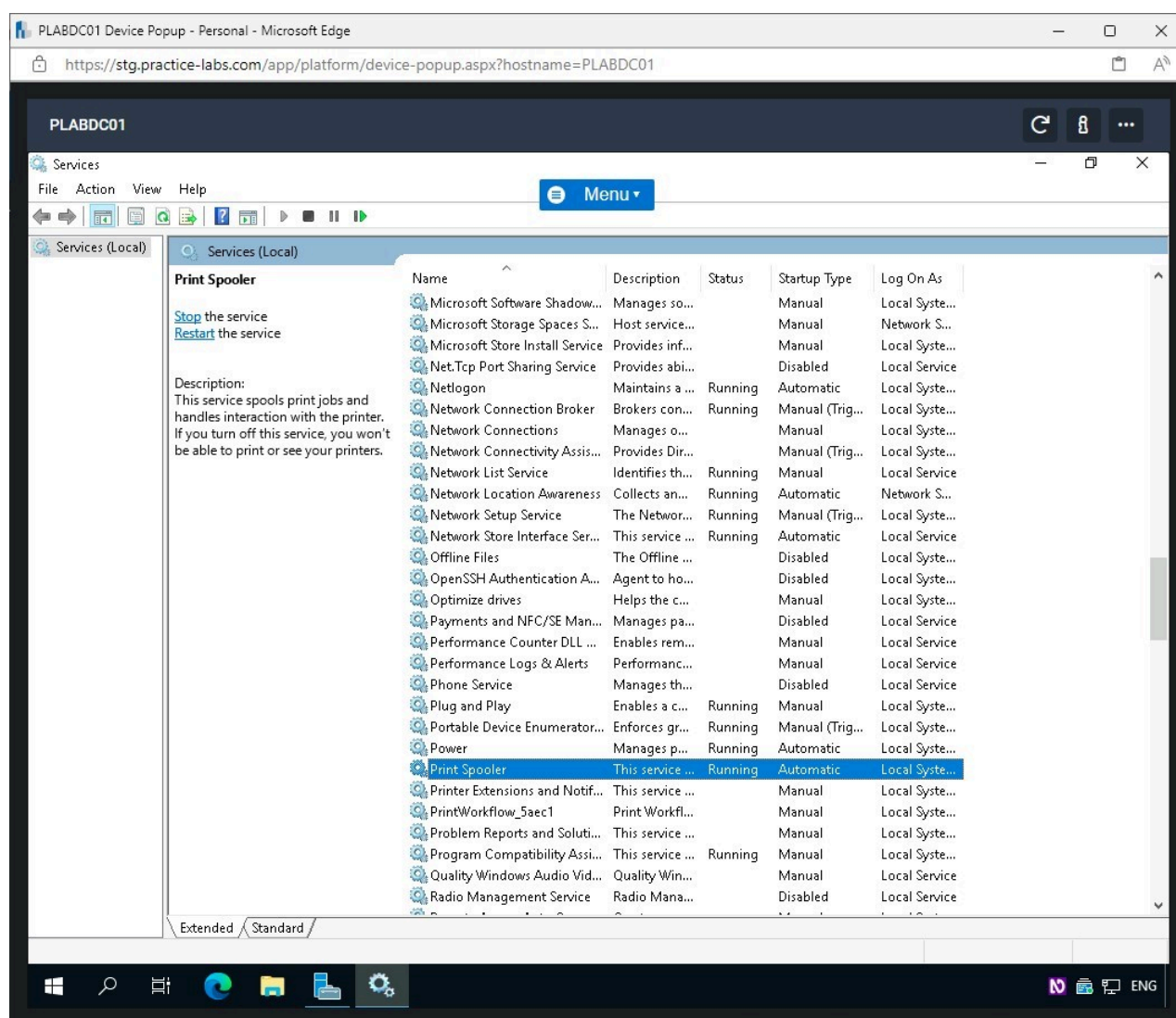


Figure 4.2 Screenshot of PLABDC01: Displaying the Services window, showing Print Spooler selected.

Step 3

In the **Print Spooler Properties (Local Computer) - General** tab, you can view the **Services name**, **Display name**, **Description**, **Path to executable**, **Startup type**, and **Service status**.

The **Path to executable** is where the file needed to start the service resides.

The **Startup type** tells the service how it is to be initiated. The options are:

- **Automatic:** Starts automatically with the operating system
- **Automatic Delayed:** Starts automatically once the Automatic Services are all started. Helps reduce bottlenecks and speeds up the boot process
- **Manual:** Needs to be executed to start and is not started with the operating system
- **Disabled:** The service is shutdown and will not start

Click the **Stop** button.

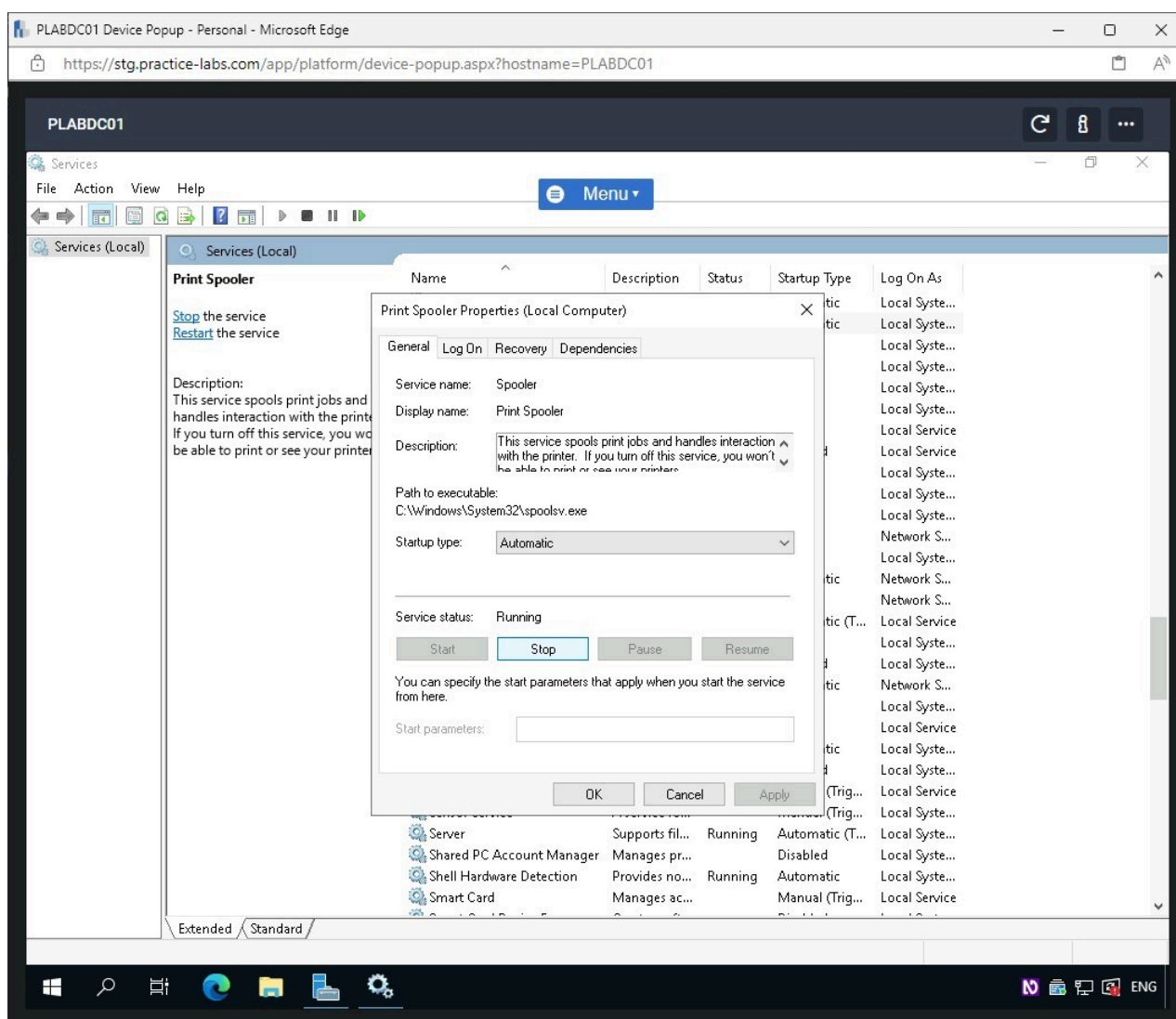


Figure 4.3 Screenshot of PLABDC01: Displaying the Print Spooler Properties window, showing the Stop button selected.

Step 4

Notice the **Print Spooler** service is now stopped.

Click **Start**.

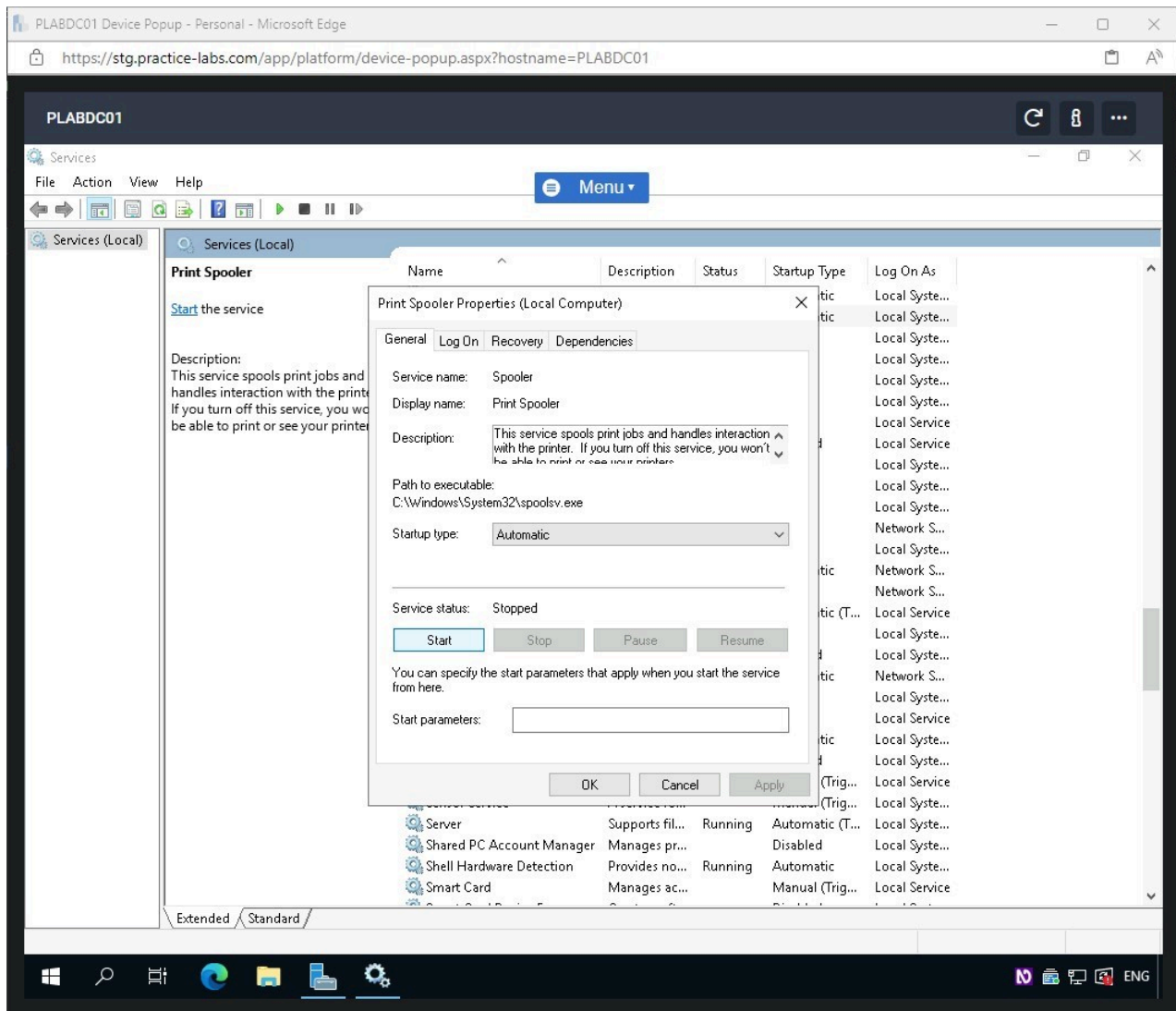


Figure 4.4 Screenshot of PLABDC01: Displaying selecting Start in the Print Spooler Properties (Local Computer) window.

Step 5

The **Print Spooler** has now been restarted, and the service can resume. This measure is often taken when there are issues with the printer's processing jobs.

Select the **Log On** tab.

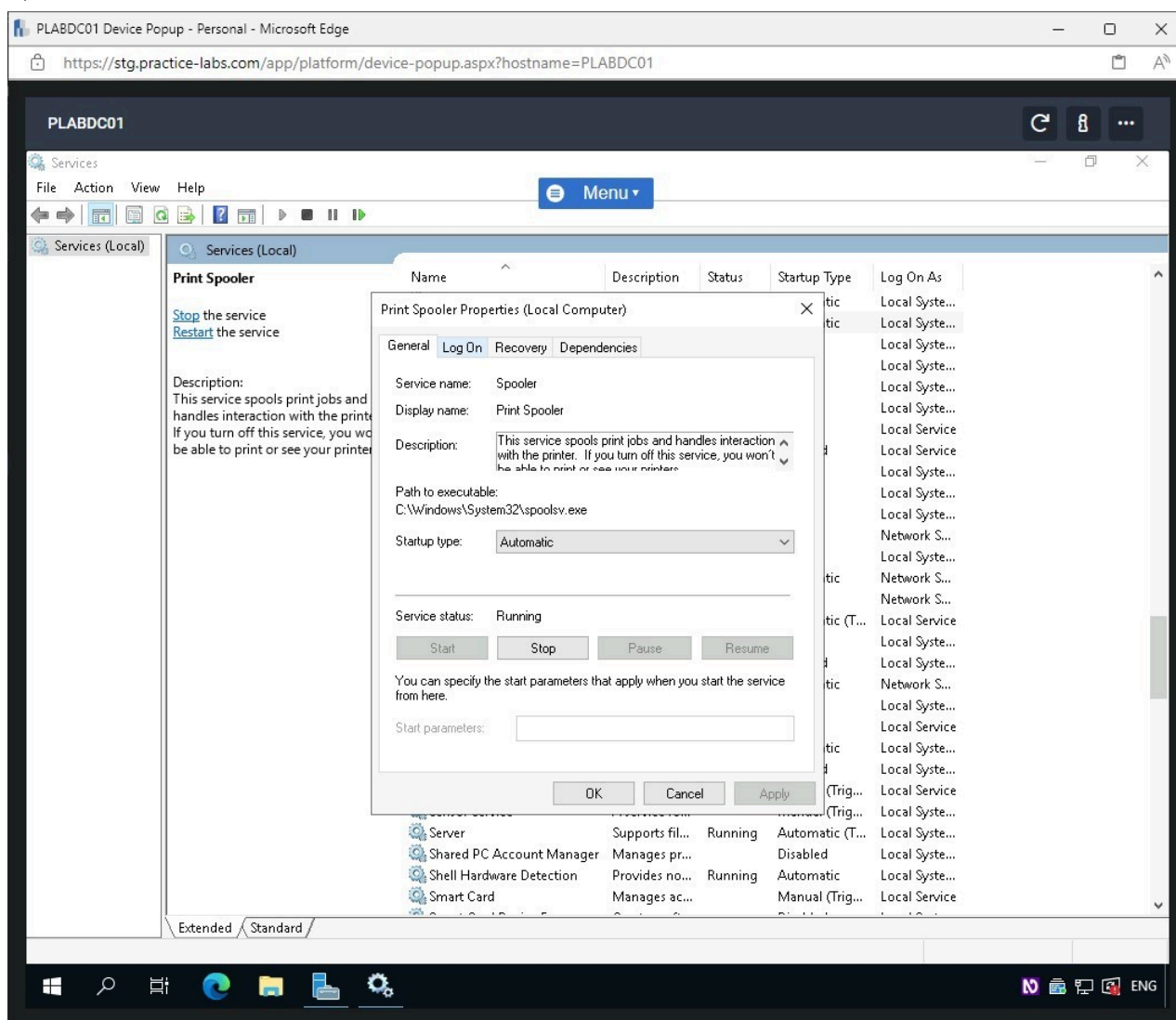


Figure 4.5 Screenshot of PLABDC01: Displaying the Print Spooler Properties (Local Computer) window, showing the Log On tab selected.

Step 6

In the **Print Spooler Properties (Local Computer) - Log On** tab, you can provide the services permission to log in as a certain user on the system to access a protected resource only the user can access. Because **Windows Services** do not end when a user logs out, access to these resources remains a capability. If you want the service to be able to access a certain user's protected resources even when they aren't signed in, you must supply the user credentials under the Service's **Log On** tab.

Select the **Recovery** tab.

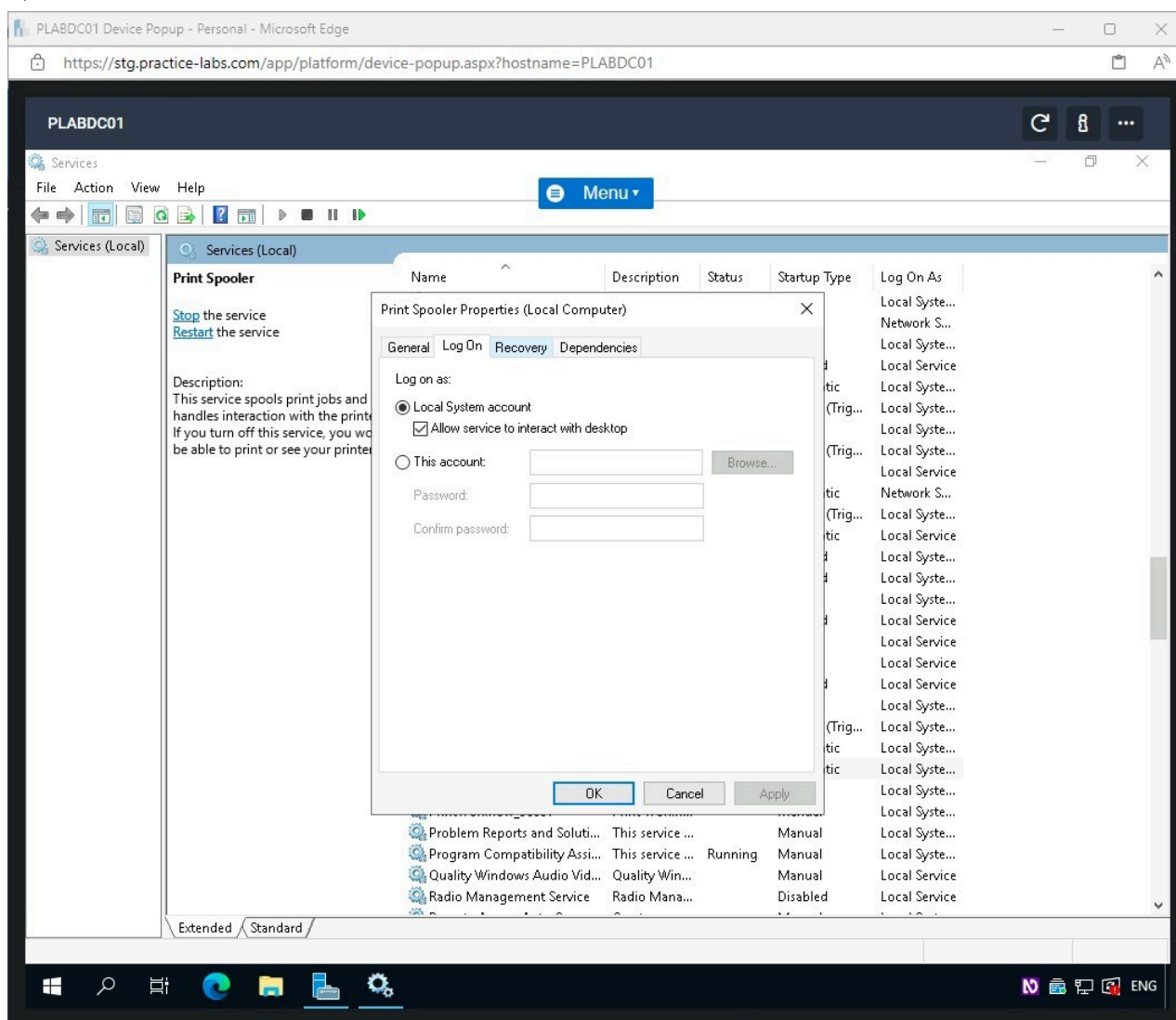


Figure 4.6 Screenshot of PLABDC01: Displaying the Print Spooler Properties (Local Computer) window, showing the Recovery tab selected.

Step 7

In the **Print Spooler Properties (Local Computer) - Recovery** tab, you can configure the actions the computer will take if the service start-up fails.

There are options for first failure, second failure, subsequent failures, and reset/restart counters. The options for each of them from the drop-down menu are **Take No Action**, **Restart the Service**, **Run a Program**, and **Restart the Computer**.

Select the **Dependencies** tab.

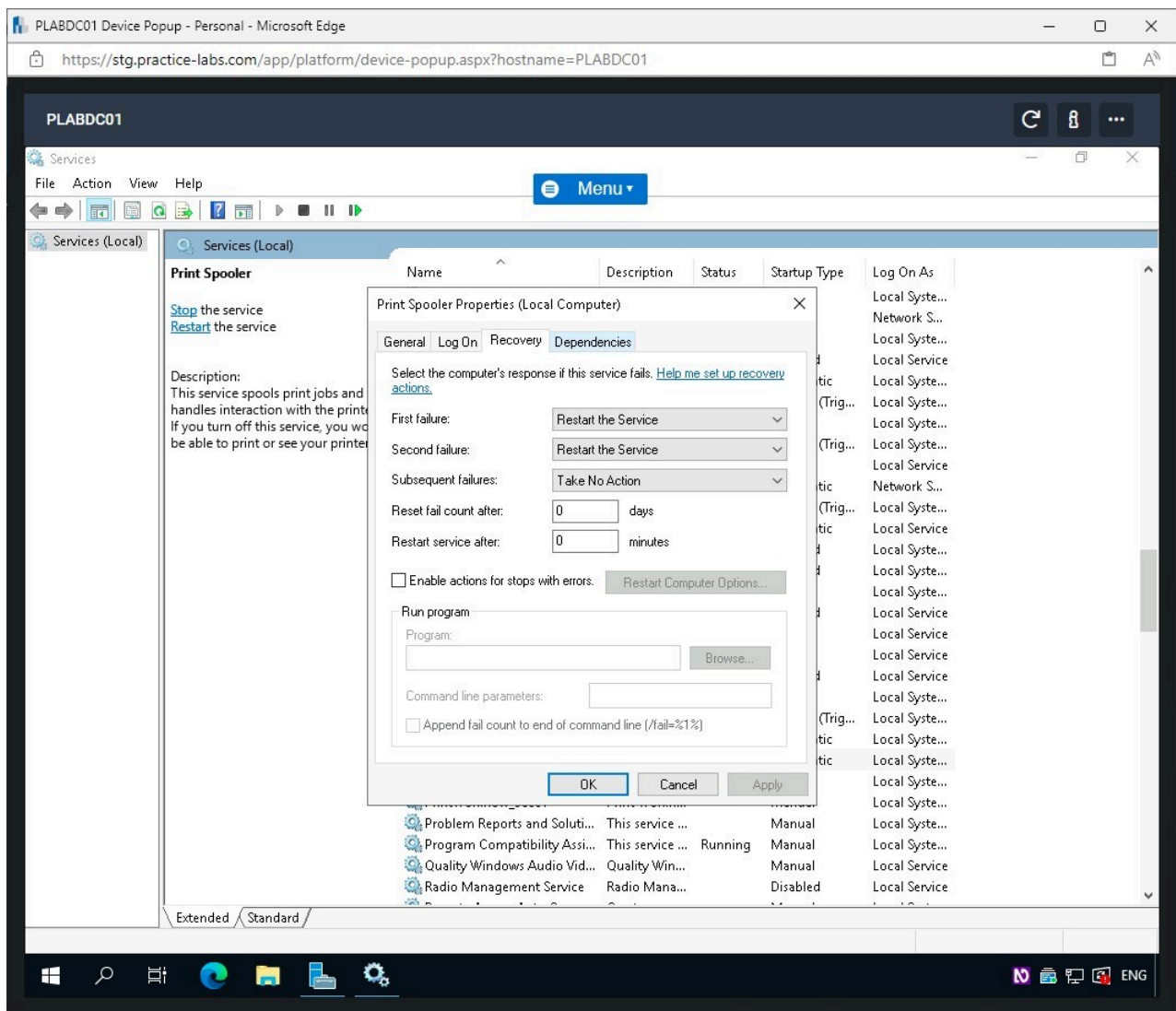


Figure 4.7 Screenshot of PLABDC01: Displaying selecting the Dependencies tab in the Print Spooler Properties (Local Computer) window.

Step 8

In the **Print Spooler Properties (Local Computer) - Dependencies** tab, other services that the Print Spooler needs running to function are displayed.

Click **OK**.

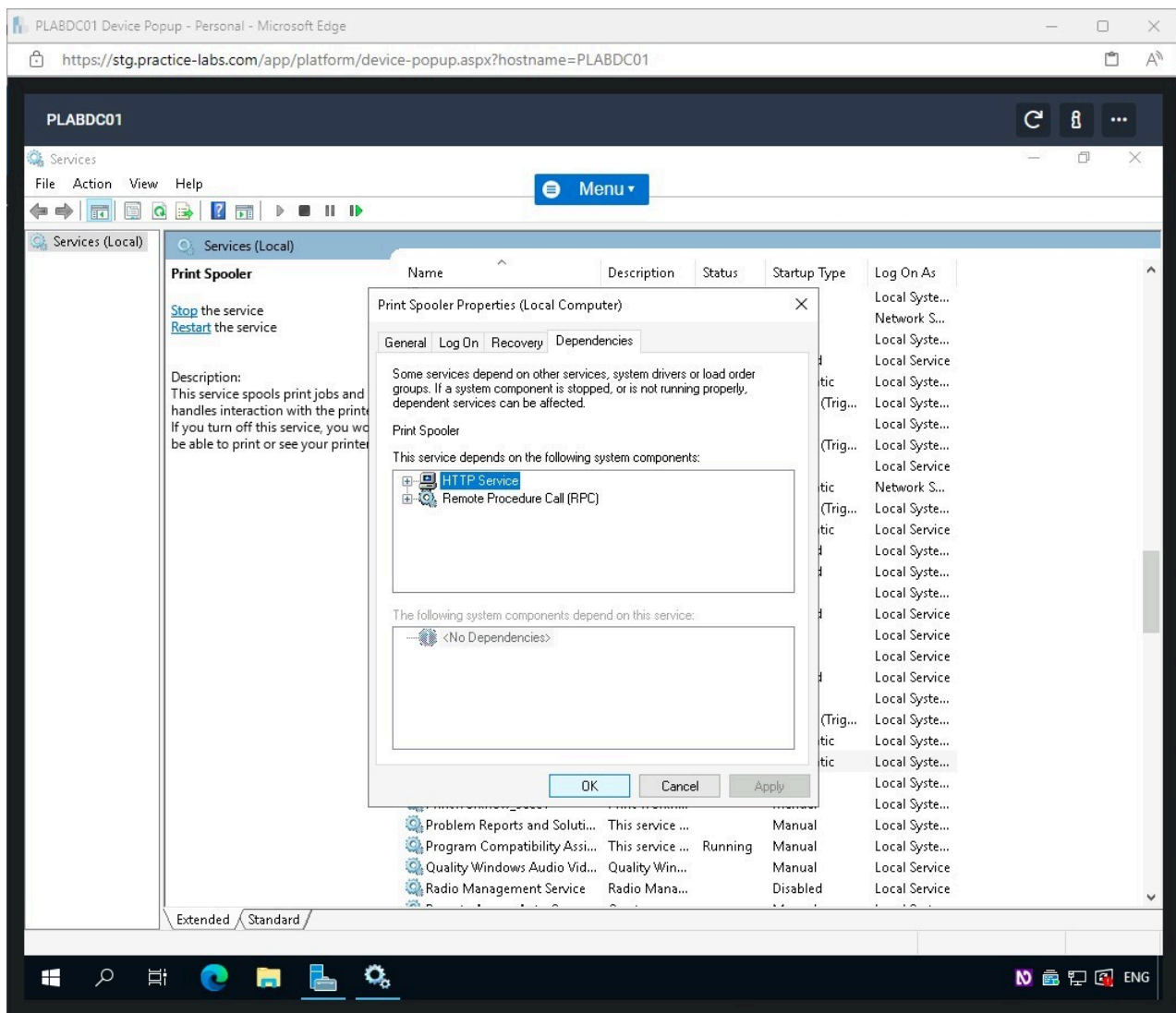


Figure 4.8 Screenshot of PLABDC01: Displaying selecting OK in the Print Spooler Properties (Local Computer) - Dependencies tab.

Step 9

Close the **Services** window.

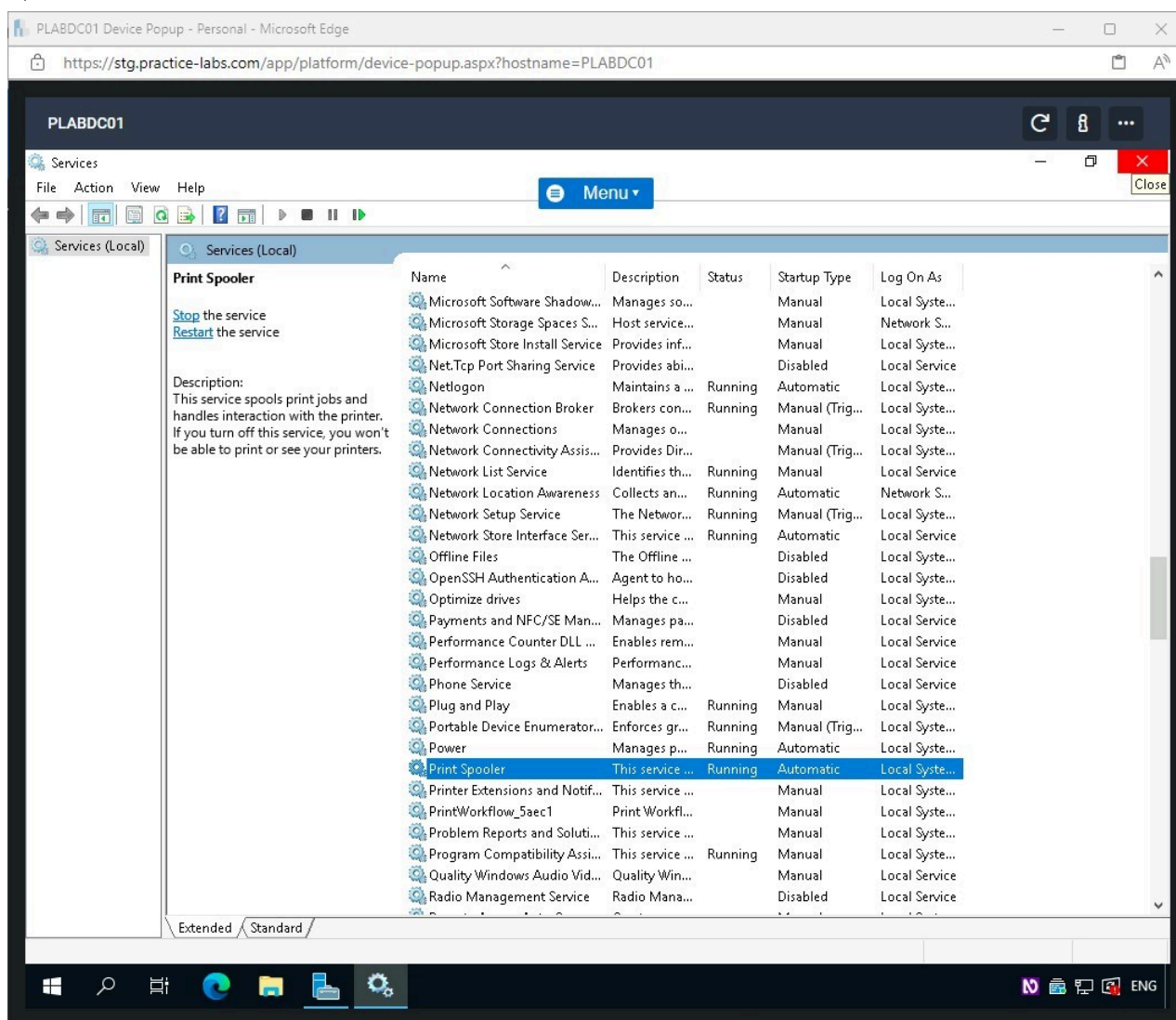


Figure 4.9 Screenshot of PLABDC01: Displaying closing the Services window.

Exercise 5 - Privilege Escalation

Standard Windows user accounts and administrator accounts are the two most common forms of user profiles. When using computers, users' permissions are based on their account type. A "regular" user account is recommended for typical, everyday computer use. Standard users cannot do complex system adjustments with a regular account but can still accomplish routine activities.

If a user has a regular account, everything they do, including downloading programs, making changes to the registry, and so on, will be met with a prompt from User Account Control (UAC). When prompted by UAC, regular user accounts must provide the credentials of an administrator account to continue. An administrator account functions similarly to a regular account but with elevated permissions. Once you have an administrator account, you can access all files on the computer, regardless of who owns them. Using a normal account rather

than an administrator account is strongly recommended for several reasons, including security and privacy.

In this exercise, administrator privileges in Windows and Linux will be discussed.

Learning Outcomes

After completing this exercise, you should be able to:

- Use Run as Administrator
- Use the Sudo/Su Command

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABWIN10** - (Windows 10 - Domain Member Workstation)
- **PLABUBUNTU** - (Linux Ubuntu - Stand-alone Linux Server)

Task 1 - Run as Administrator

While using a regular user account for Windows, users can temporarily elevate their status by selecting an application they want to run and then selecting the **Run as administrator** option.

In this task, you will learn to use the run as administrator option.

Step 1

Connect to **PLABWIN10**.

Click the Windows **Start** charm and type the following:

```
cmd
```

Select **Run as administrator** from the pane on the right.

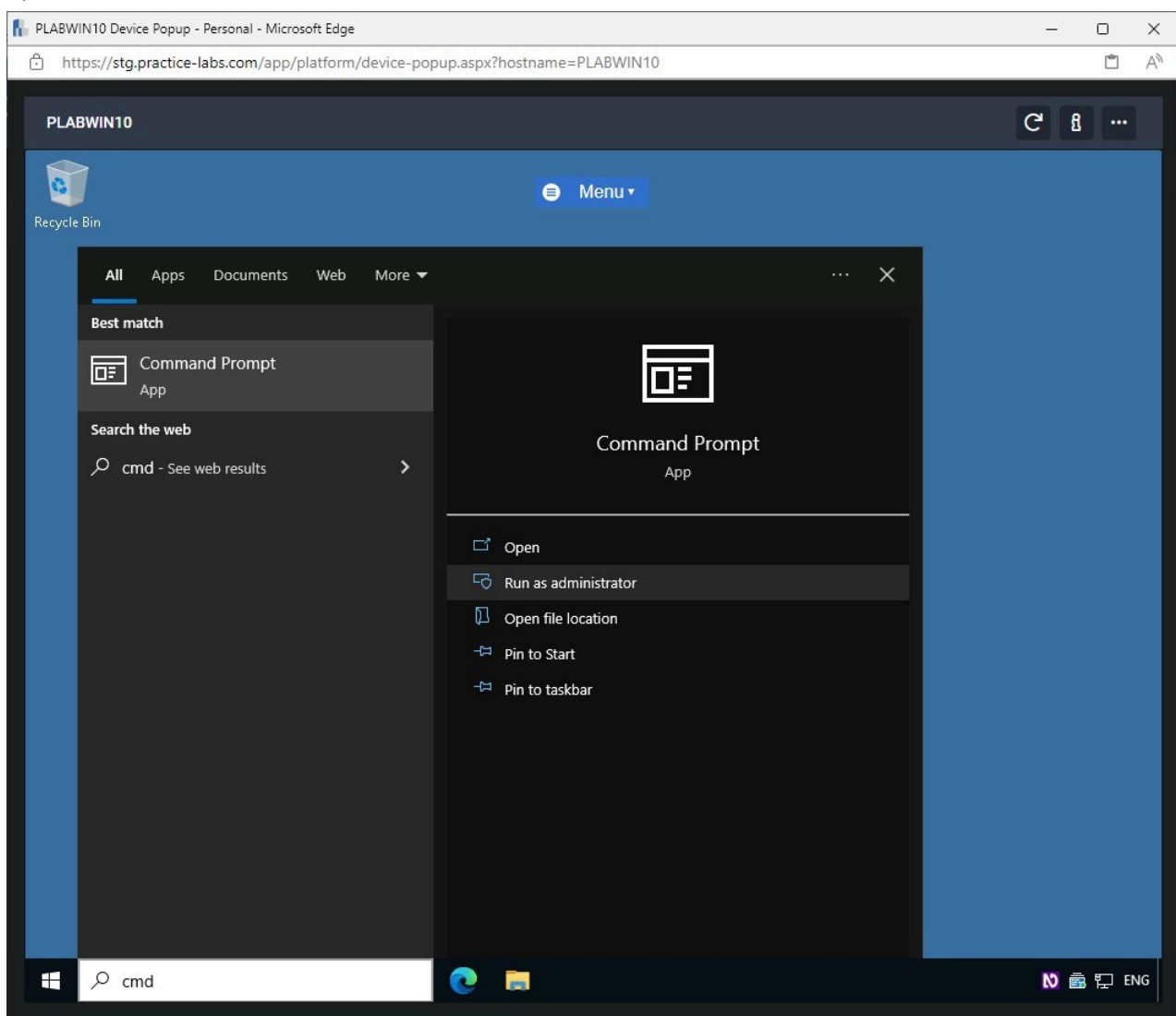


Figure 5.1 Screenshot of PLABWIN10: Displaying selecting Run as administrator option for the Command Prompt application.

Step 2

In the **Administrator Command Prompt** window, type the following command and press **Enter**:

```
chkdsk
```

Note: Since you are logged in as the administrator in the **PLABWIN10** device, you will not have to re-authenticate.

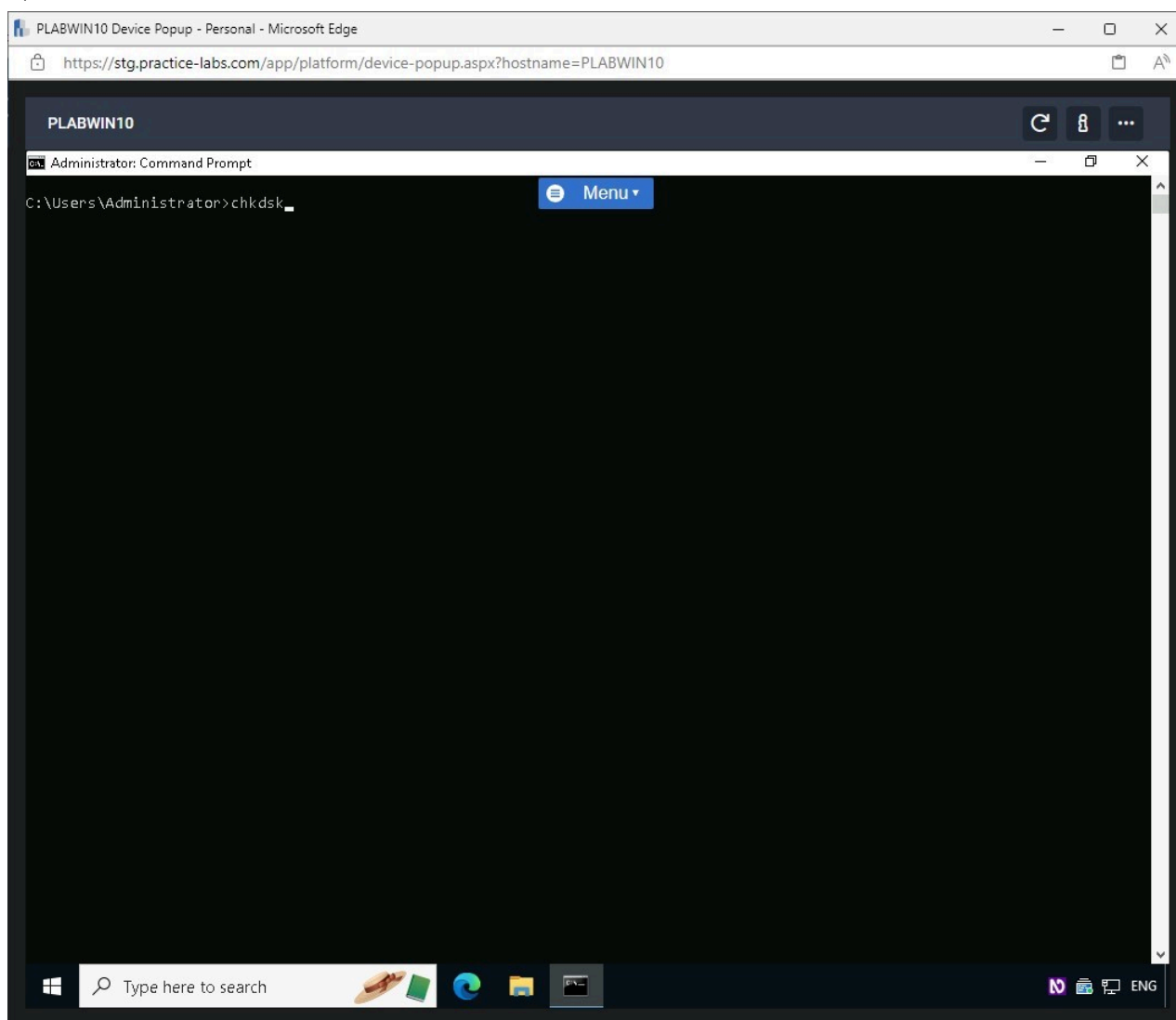


Figure 5.2 Screenshot of PLABWIN10: Displaying entering the required command in the Administrator Command Prompt window.

Step 3

The **chkdsk** command runs and checks the state of the disk.

Note: Due to being at a command prompt with administrative privileges, this command was executed. Without the proper authorization, the **chkdsk** command would have failed to complete. **Diskpart**, **net use**, and **sfc** are some more common commands that require administrative access.

Close the **Administrator Command Prompt** window.

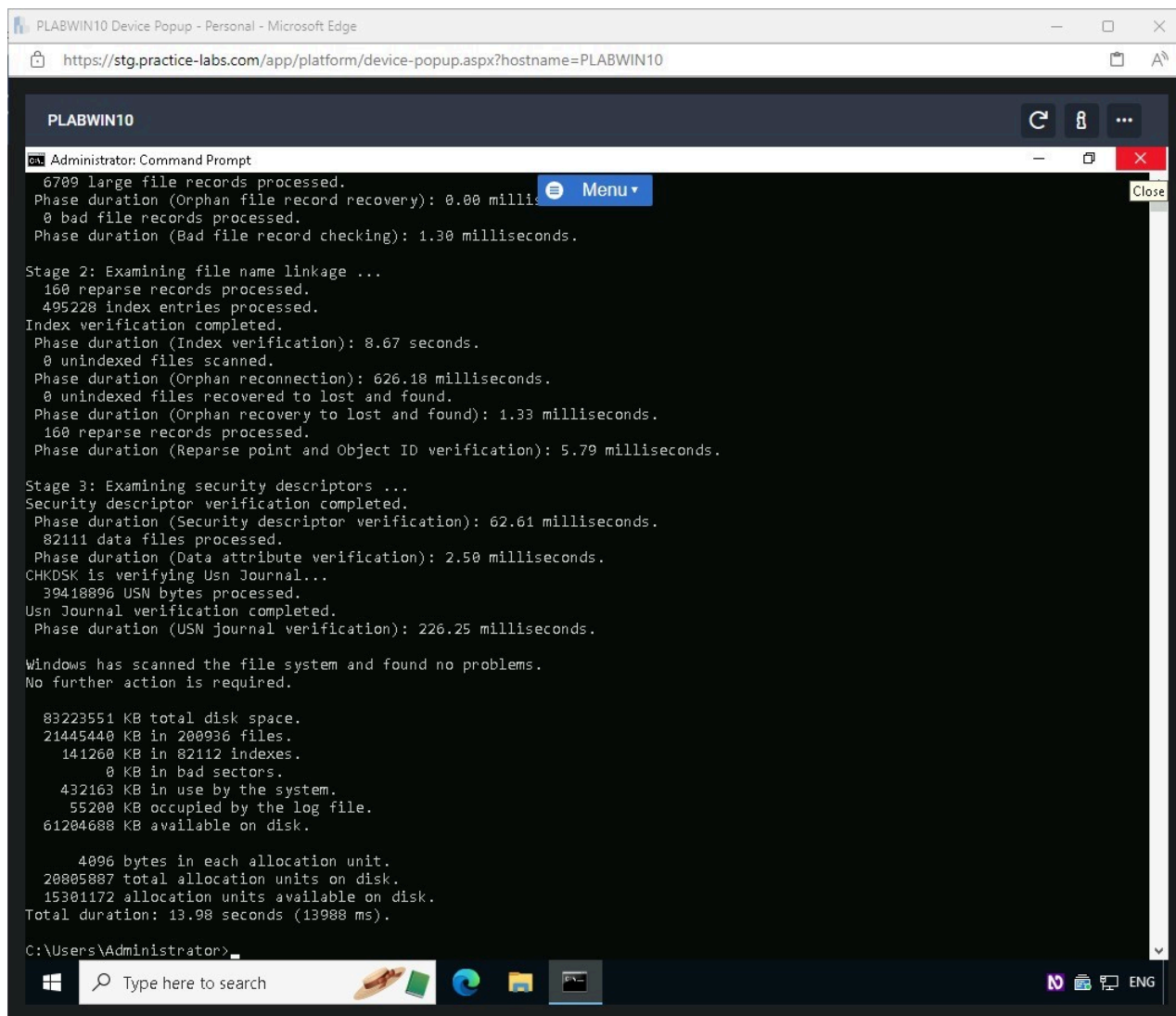


Figure 5.3 Screenshot of PLABWIN10: Displaying the Administrator Command Prompt window. Chkdsk command is executed, and the close icon on the top right corner of the window is selected.

Task 2 - Sudo/Su Command

Sudo is the Linux terminal command that temporarily elevates the user to root, which is the equivalent of an administrator in Windows. By using the **sudo** command in Linux, we can switch users.

In this task, you will learn about the sudo command.

Step 1

Connect to **PLABUBUNTU**.

Select the **Terminal** icon on the left side of the screen.

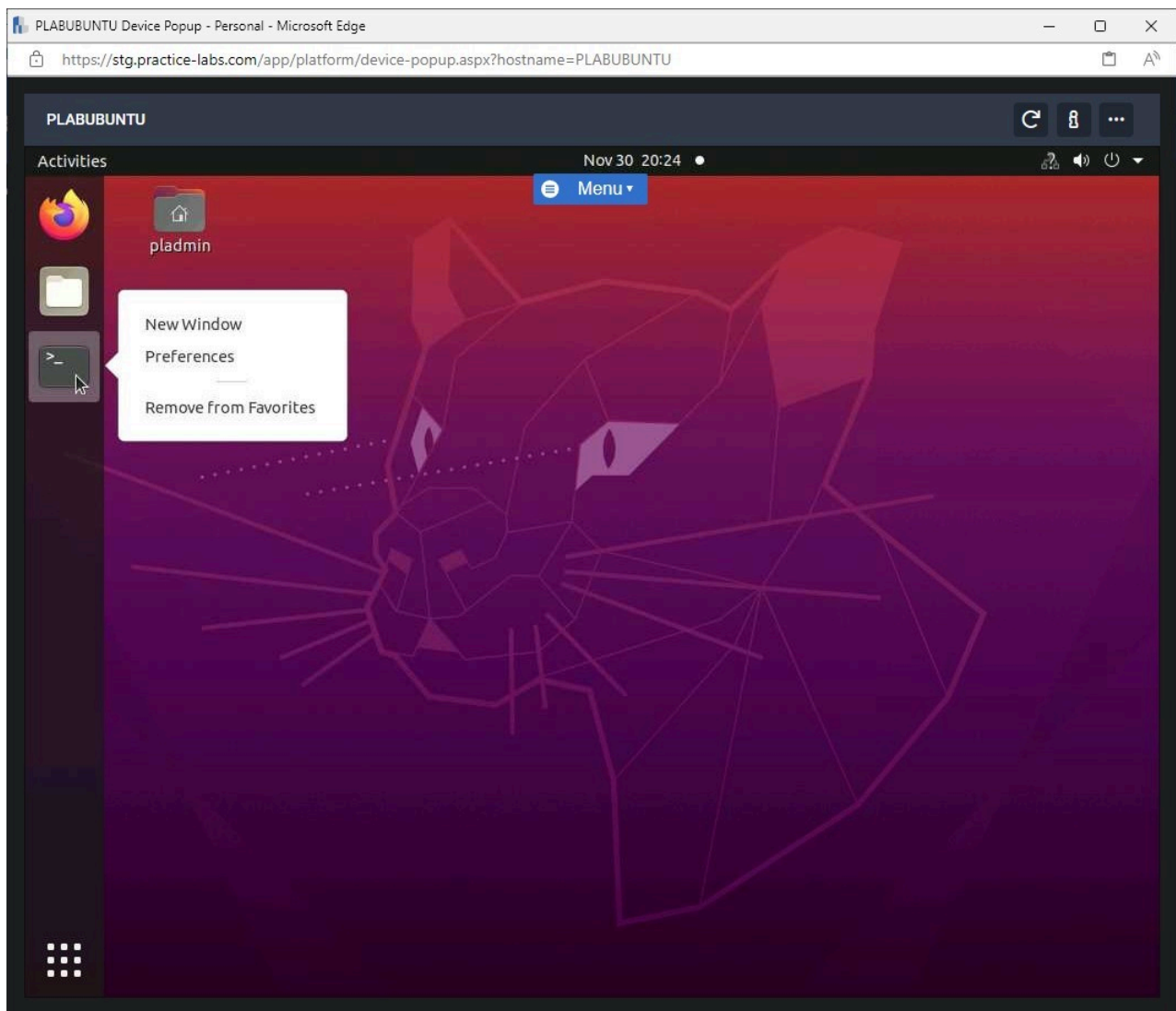


Figure 5.5 Screenshot of PLABUBUNTU: Displaying selecting the Terminal icon.

Step 2

In the **Terminal** window, type the following command and press **Enter**:

```
apt install handbrake
```

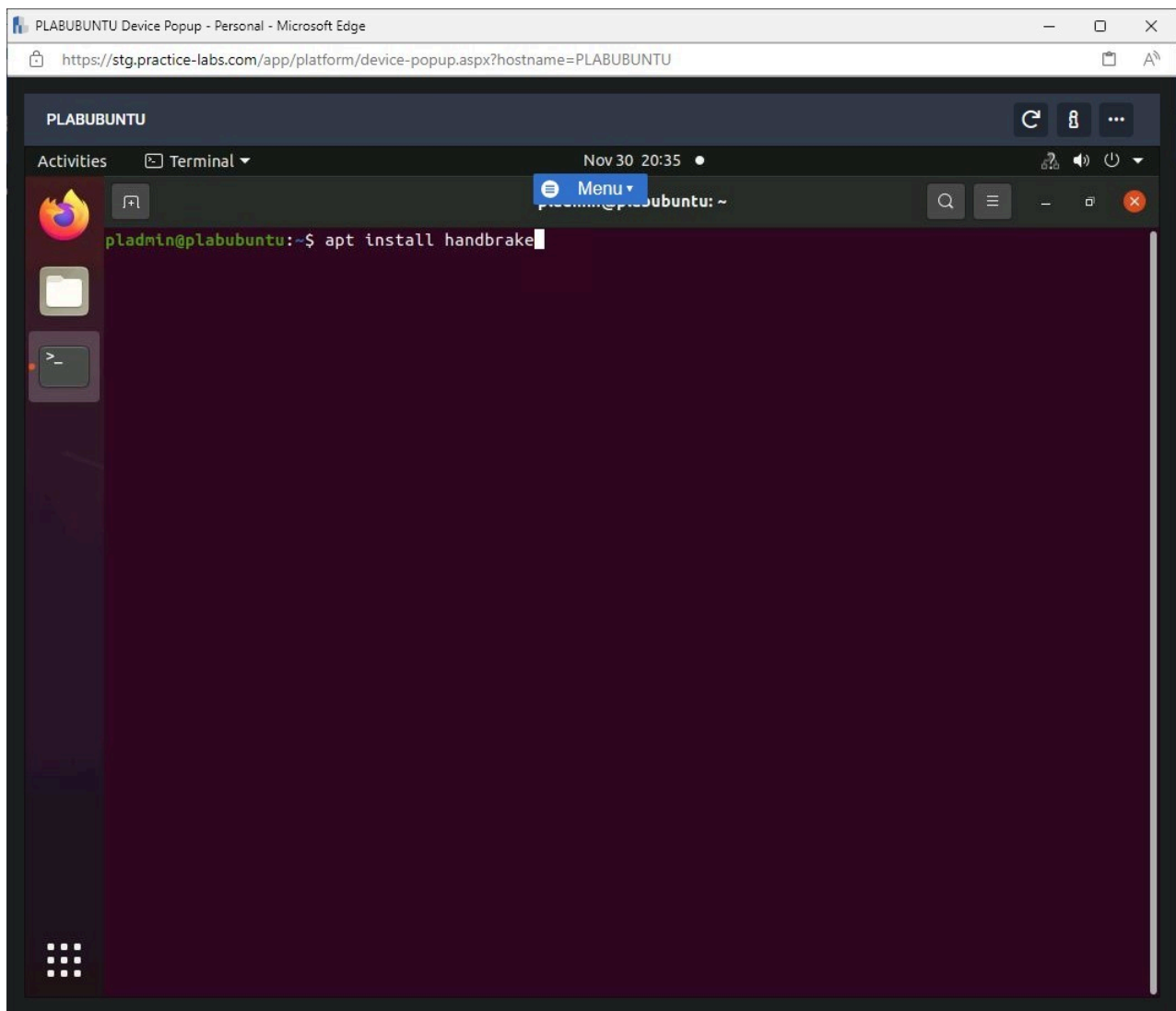


Figure 5.6 Screenshot of PLABUBUNTU: Displaying entering and executing the required command in the Terminal window.

Note: The command has failed because proper privileges are needed to install the application.

Step 3

Type the following command and press **Enter**:

```
sudo apt install handbrake
```

When prompted for the password, type the following and press **Enter**:

Passw0rd

Note: When entering a password in Linux, you will not see any characters as you type.

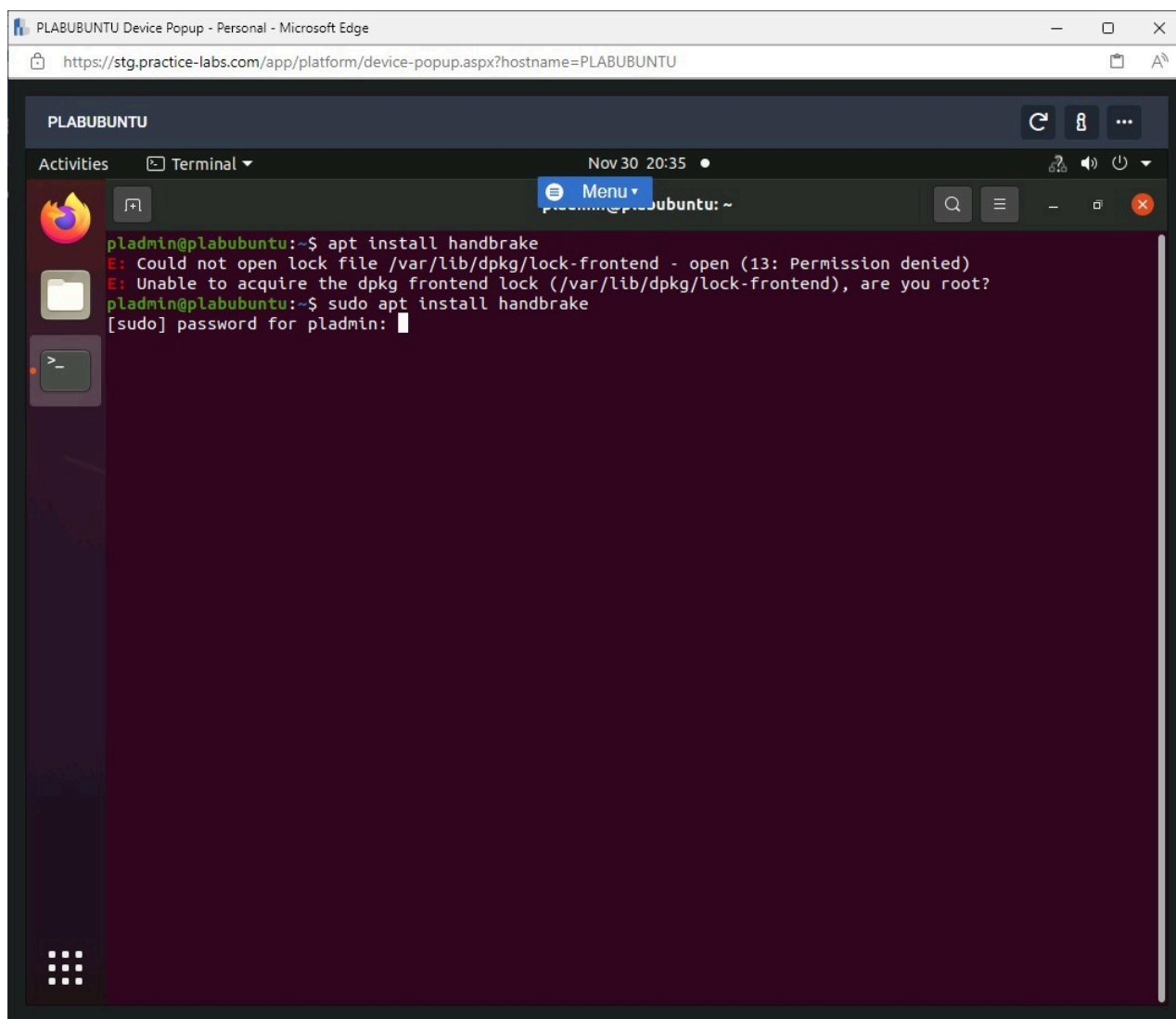


Figure 5.7 Screenshot of PLABUBUNTU: Displaying executing the required command in the Terminal window.

Note: If the command fails, please enter the following command and redo Steps 1-3:
`sudo apt-get update`
When prompted for a password, enter the following and press Enter:
Passw0rd

Step 4

The **apt package manager** looks for the required information and prompts you to ensure that you want to install these files.

Type the following and press **Enter**:

y

```

PLABUBUNTU Device Popup - Personal - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABUBUNTU

PLABUBUNTU
Activities Terminal Nov 30 20:35
pladmin@plabubuntu:~$ apt install handbrake
E: Could not open lock file /var/lib/dpkg/lock-frontent - open (13: Permission denied)
E: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontent), are you root?
pladmin@plabubuntu:~$ sudo apt install handbrake
[sudo] password for pladmin:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libllvm10 libllvm8 libxmlb1 shim
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  gstreamer1.0-libav i965-vd-driver intel-media-vd-driver libaacs0 libaom0 libass9 libavcodec58
  libavfilter7 libavformat58 libavutil56 libbdplus0 libbluray2 libbs2b0 libchromaprint1 libcodec2-0.9
  libdvdnav4 libdvdread7 libflite1 libgme0 libgsm1 libigdgmm1 liblilv-0-0 libmysofa1 libnorm1
  libopenmpt0 libpgm-5.2-0 libpostproc55 librubberband2 libserd-0-0 libshine3 libsnappy1v5 libsord-0-0
  libsratom-0-0 libssh-gcrypt-4 libswresample3 libswscale5 libva-drm2 libva-x11-2 libva2 libvdpau1
  libvidstab1.1 libx264-155 libx265-179 libxvidcore4 libzmq5 libzvi-common libzvi0 mesa-vd-drivers
  mesa-vdpau-drivers ocl-icd-libopencl1 va-driver-all vdpau-driver-all
Suggested packages:
  i965-vd-driver-shaders libbluray-bdj libdvdcss2 serdi sordi openc1-icd libvdpau-va-g1
  nvidia-vdpau-driver nvidia-legacy-340xx-vdpau-driver nvidia-legacy-304xx-vdpau-driver
The following NEW packages will be installed:
  gstreamer1.0-libav handbrake i965-vd-driver intel-media-vd-driver libaacs0 libaom0 libass9
  libavcodec58 libavfilter7 libavformat58 libavutil56 libbdplus0 libbluray2 libbs2b0 libchromaprint1
  libcodec2-0.9 libdvdnav4 libdvdread7 libflite1 libgme0 libgsm1 libigdgmm1 liblilv-0-0 libmysofa1
  libnorm1 libopenmpt0 libpgm-5.2-0 libpostproc55 librubberband2 libserd-0-0 libshine3 libsnappy1v5
  libsord-0-0 libsratom-0-0 libssh-gcrypt-4 libswresample3 libswscale5 libva-drm2 libva-x11-2 libva2
  libvdpau1 libvidstab1.1 libx264-155 libx265-179 libxvidcore4 libzmq5 libzvi-common libzvi0
  mesa-vd-drivers mesa-vdpau-drivers ocl-icd-libopencl1 va-driver-all vdpau-driver-all
0 upgraded, 53 newly installed, 0 to remove and 0 not upgraded.
Need to get 46.4 MB of archives.
After this operation, 218 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
  
```

Figure 5.8 Screenshot of PLABUBUNTU: Displaying executing the required command in the Terminal window.

Step 5

Notice the **Handbrake** program has been installed.

Close the **Terminal** window.

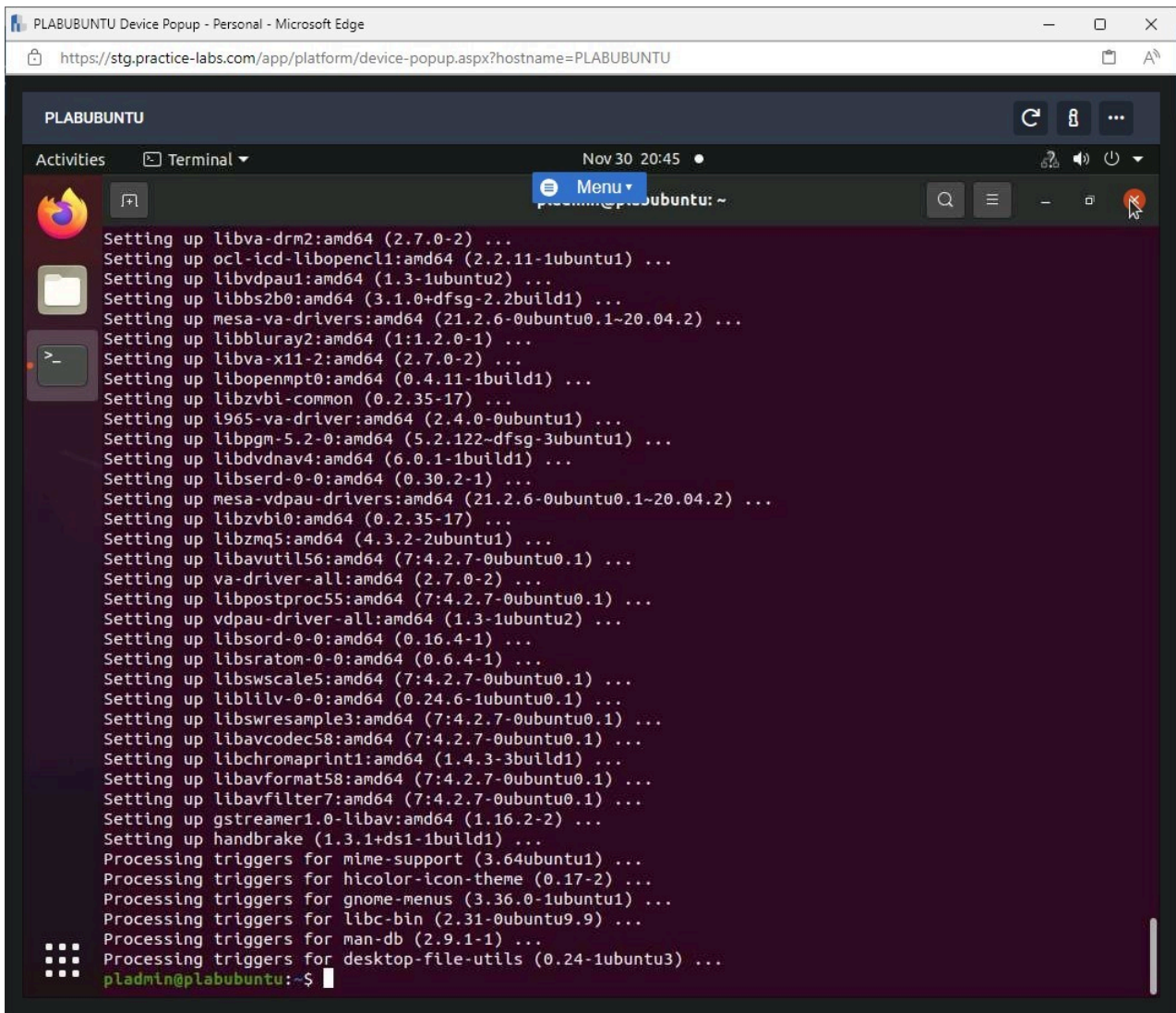


Figure 5.9 Screenshot of PLABUBUNTU: Displaying the command executed. Close icon on the top right corner of the window is selected.

Step 6

Select the **Show Applications** button in the bottom left corner.

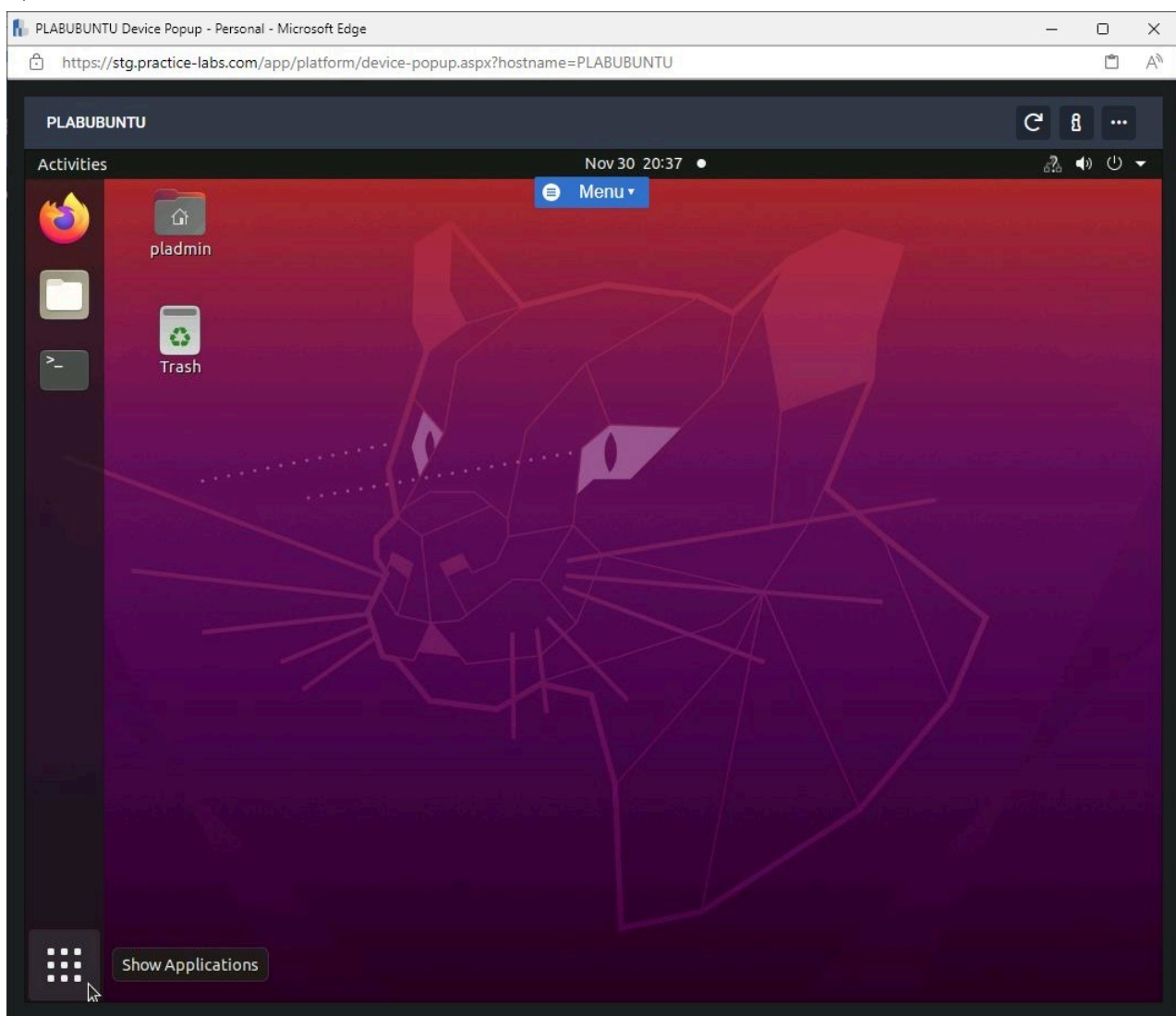


Figure 5.10 Screenshot of PLABUBUNTU: Displaying selecting the Show Applications button.

Step 7

Notice the **HandBrake** application now appears in the **Applications** menu.

Double-click on **HandBrake**.

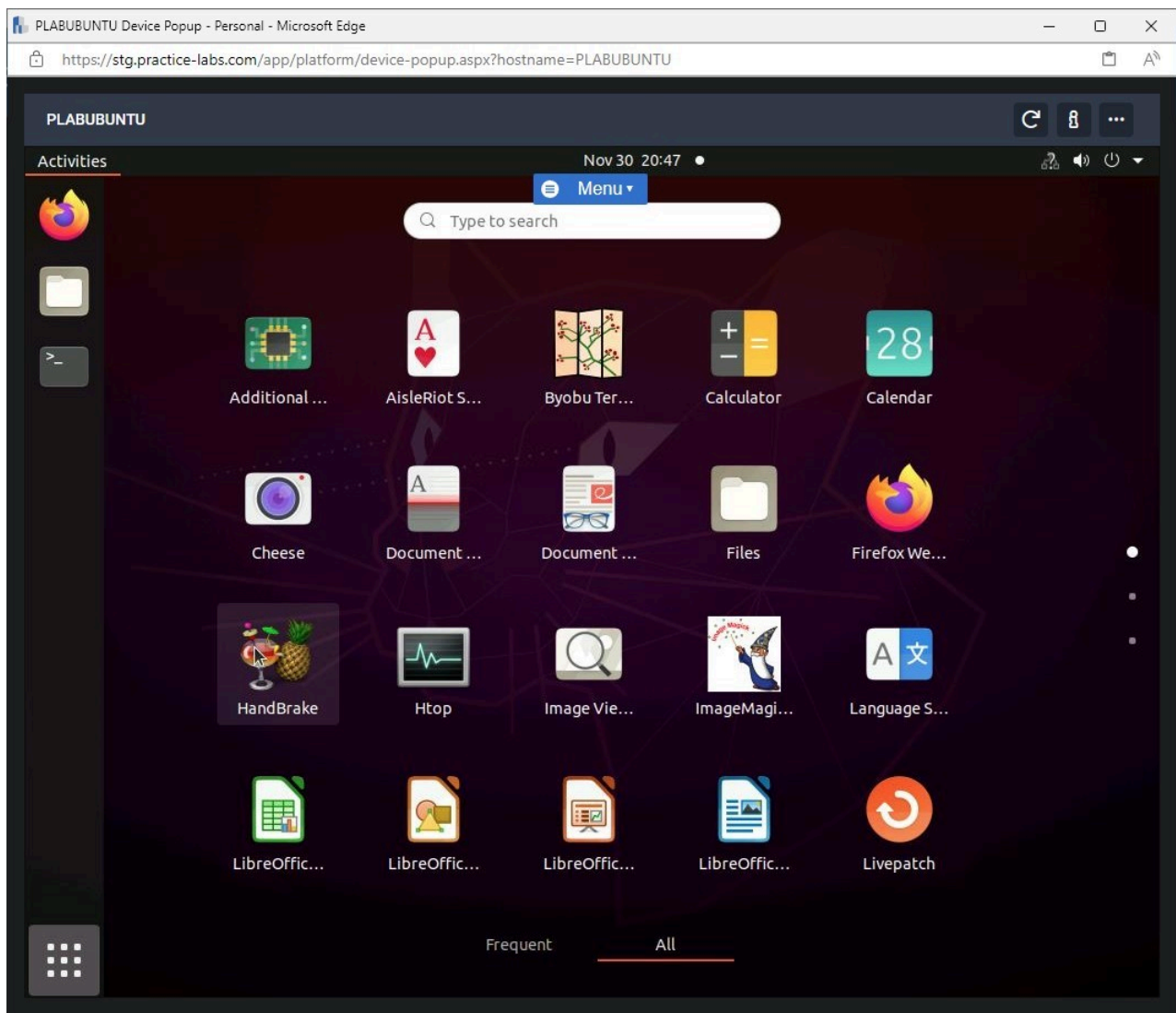


Figure 5.11 Screenshot of PLABUBUNTU: Displaying the Applications menu, showing the Handbrake application.

Step 8

The **Handbrake** application is functioning.

Close the **Handbrake** window.

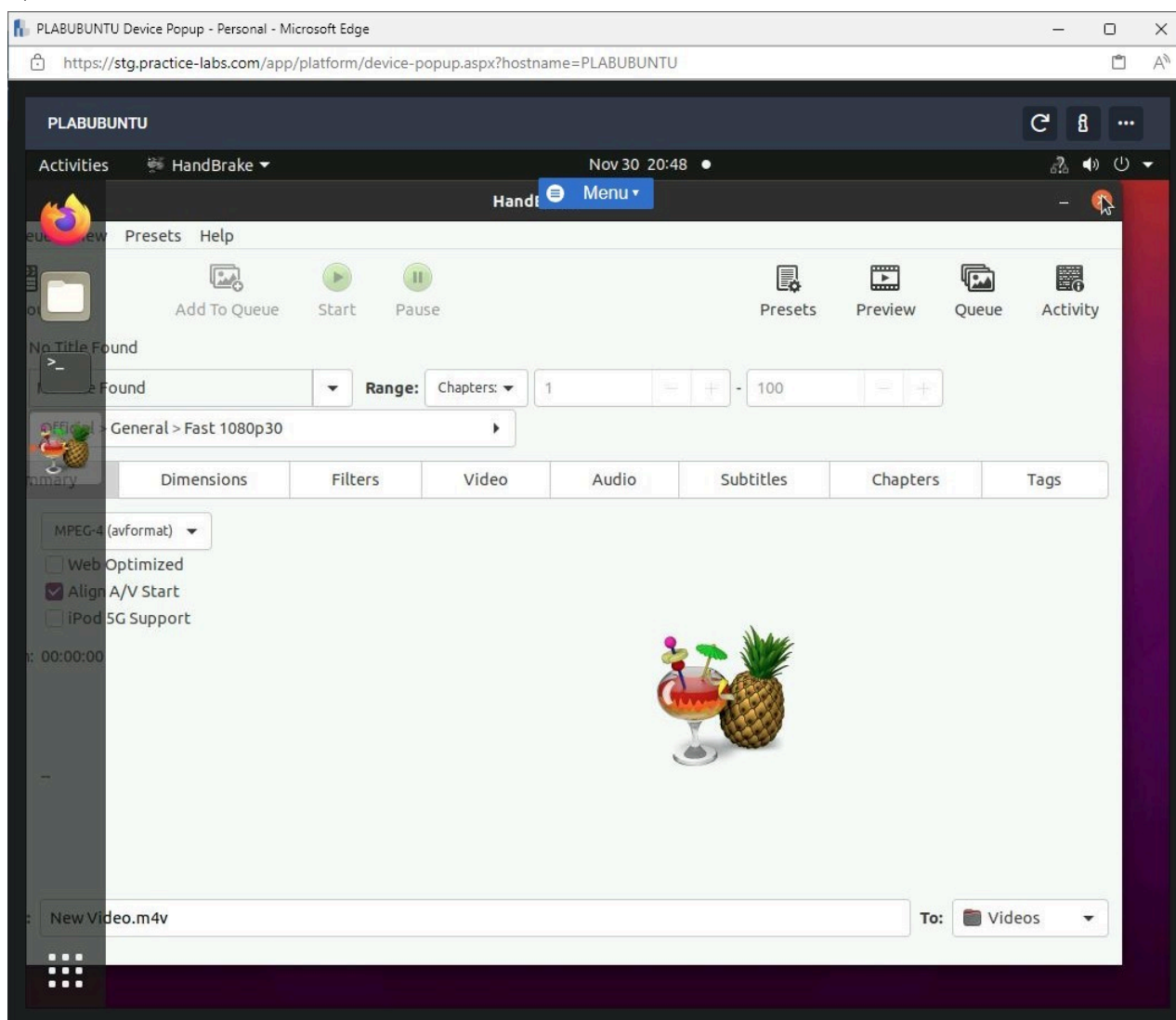


Figure 5.12 Screenshot of PLABUBUNTU: Displaying closing the Handbrake window.

Exercise 6 - Recovery

When the Windows operating system (OS) crashes, becomes damaged, or otherwise stops functioning as expected, the process of recovering and restoring it to its normal or last known good state is called Windows recovery. It's included in Windows by default and helps fix any problems that may have occurred while the OS was running. The Windows installation media or a separate system recovery backup of the system at an earlier point in time are the two primary means of doing a system recovery.

In this exercise, safe mode, snapshots and reloading the operating system will be discussed.

Learning Outcomes

After completing this exercise, you should be able to:

- Explore Safe Mode
- Work with Snapshots
- Reload OS

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABWIN10** - (Windows 10 - Domain Member Workstation)

Task 1 - Safe Mode

When Windows starts in safe mode, it uses just the minimum set of necessary files and drivers. If the issue does not occur in safe mode, then it is not related to the system's factory settings or the most fundamental device drivers. The safe mode debugging environment of Windows can be used to isolate the cause of a problem and fix the computer. Single User Mode is the name for safe mode on an Apple device.

In this task, you will learn about Safe mode in Windows.

Step 1

Connect to **PLABWIN10**.

Click the **Start** charm and type the following:

```
msconfig
```

Select **System Configuration** from the **Best match** pop-up menu.

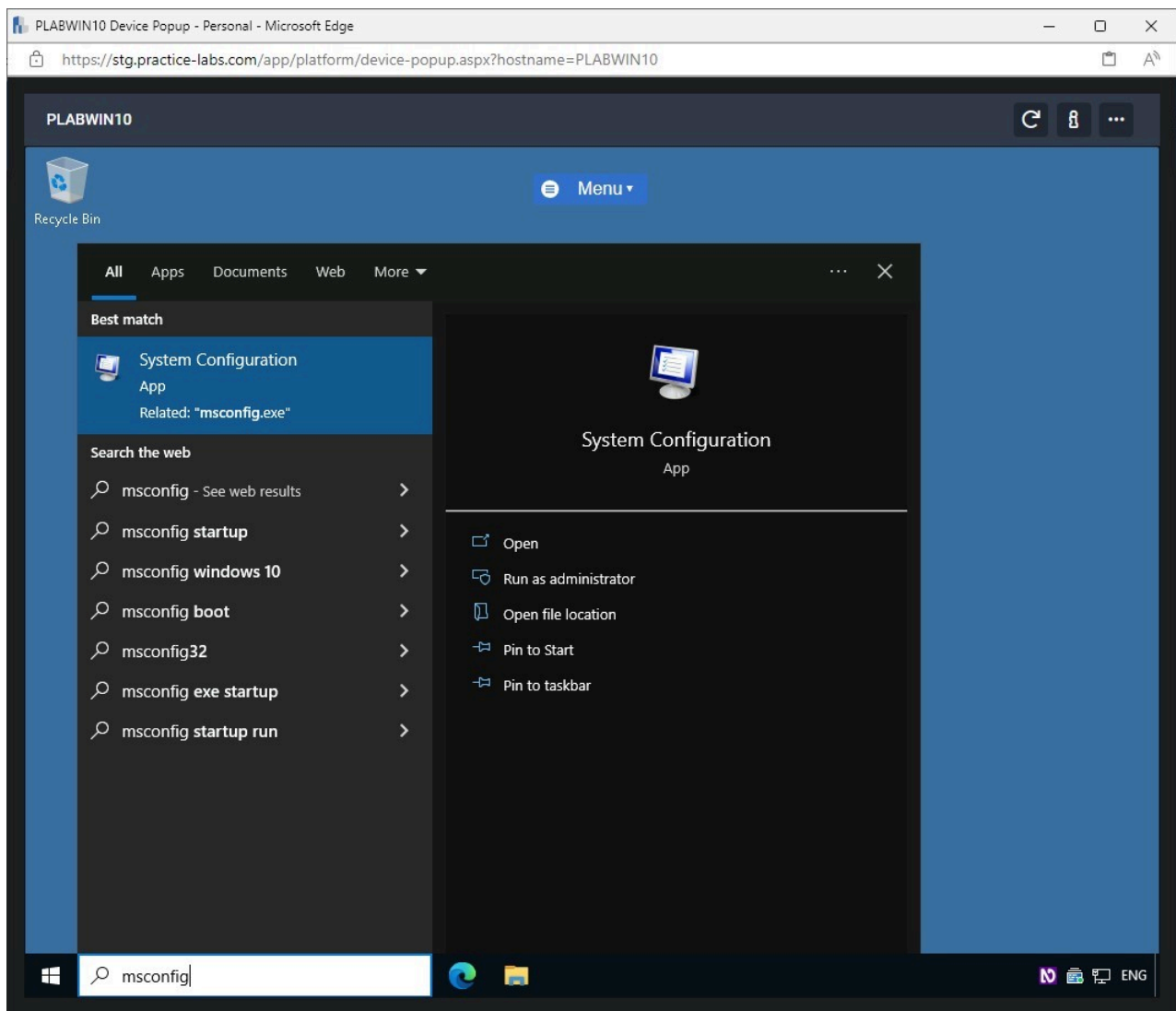


Figure 6.1 Screenshot of PLABWIN10: Displaying selecting System Configuration from the Best match pop-up menu.

Step 2

In the **System Configuration - General** tab, you can configure the following type of startups:

- Normal
- Diagnostic
- Selective

Select the **Boot** tab.

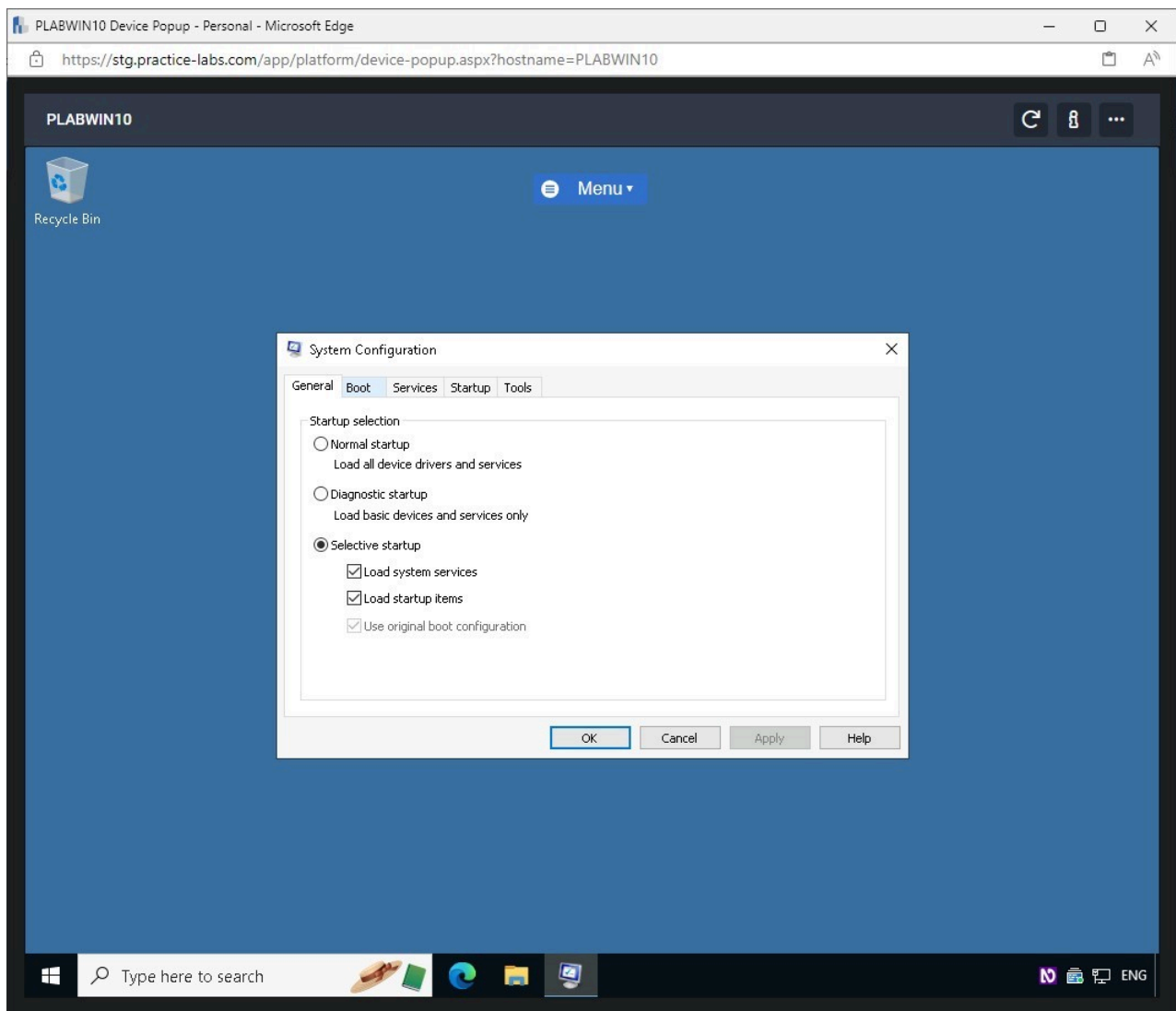


Figure 6.2 Screenshot of PLABWIN10: Displaying the System Configuration - General tab, showing the Boot tab selected.

Step 3

In the **System Configuration - Boot** tab, you can choose **safe boot** when the device restarts.

Note: A few solutions exist to aid troubleshooting, such as enabling networking to gain access to online diagnostic resources.

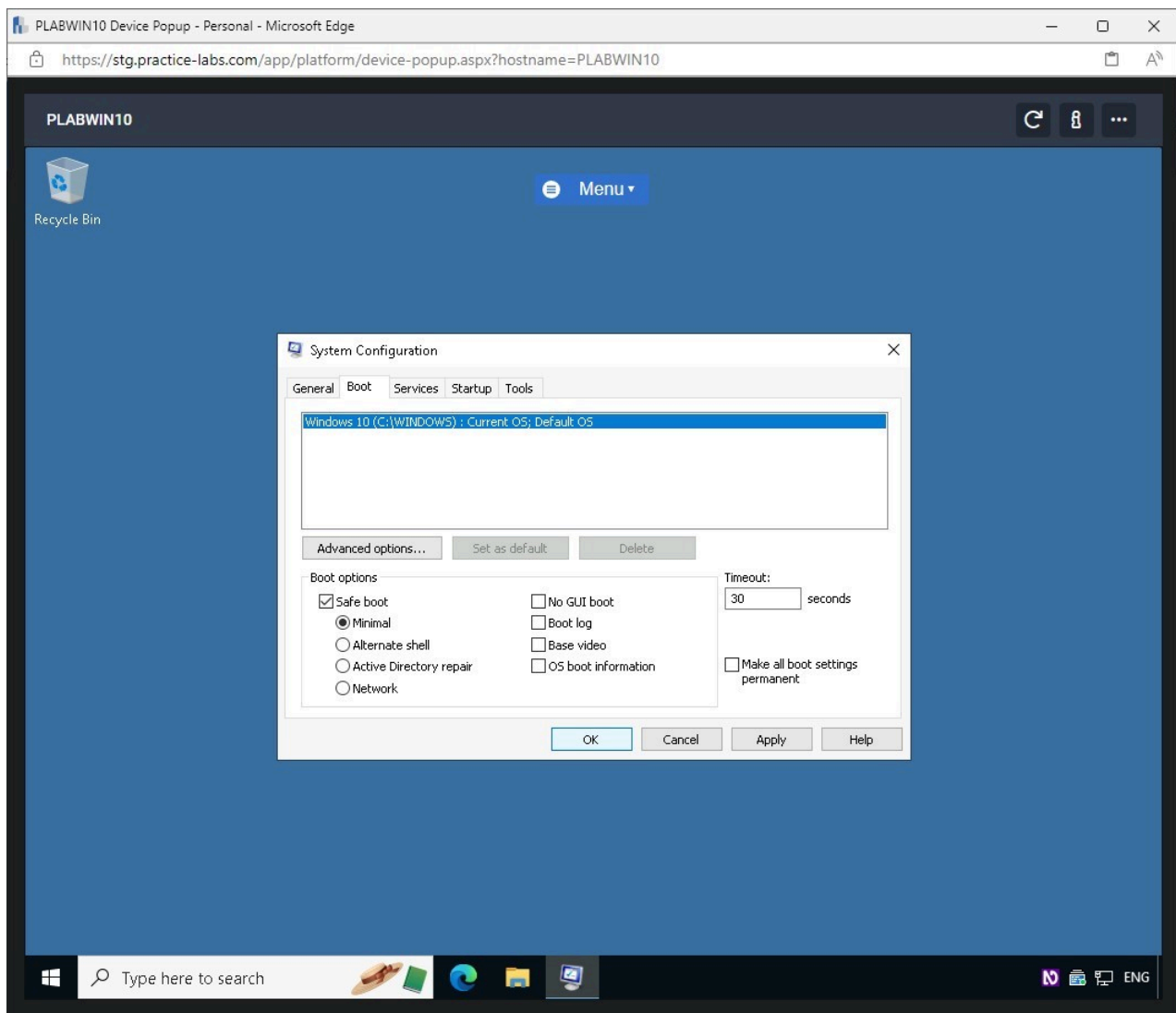


Figure 6.4 Screenshot of PLABWIN10: Displaying the System Configuration - Boot tab, showing the required settings performed and OK selected.

Step 4

In the **System Configuration** pop-up window, click **Restart**.

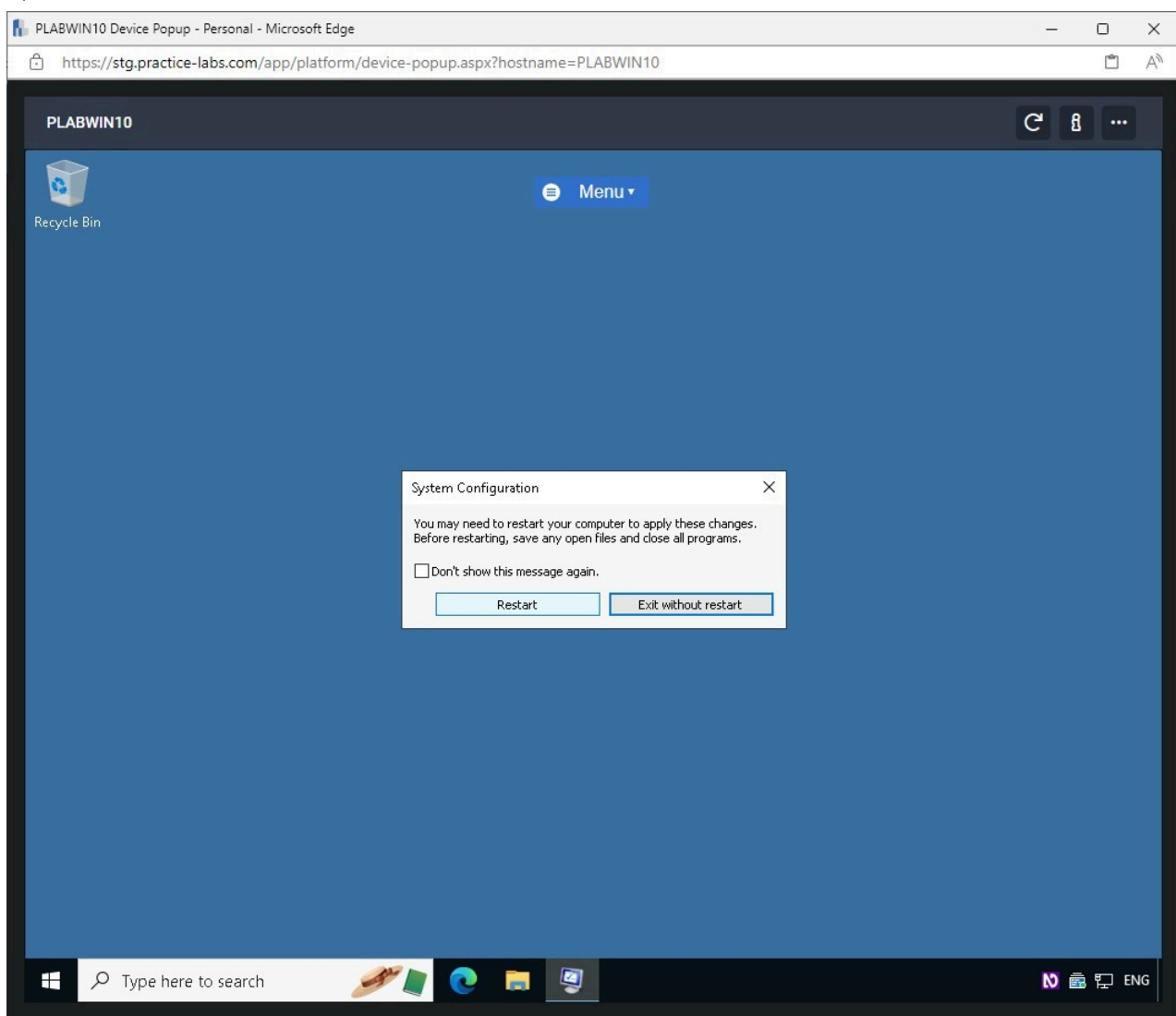


Figure 6.3 Screenshot of PLABWIN10: Displaying selecting Restart in the System Configuration pop-up window.

Step 5

To reconnect to the machine, access the secondary connection by selecting the ellipses (...) icon in the upper right corner and selecting **Console Mode**.

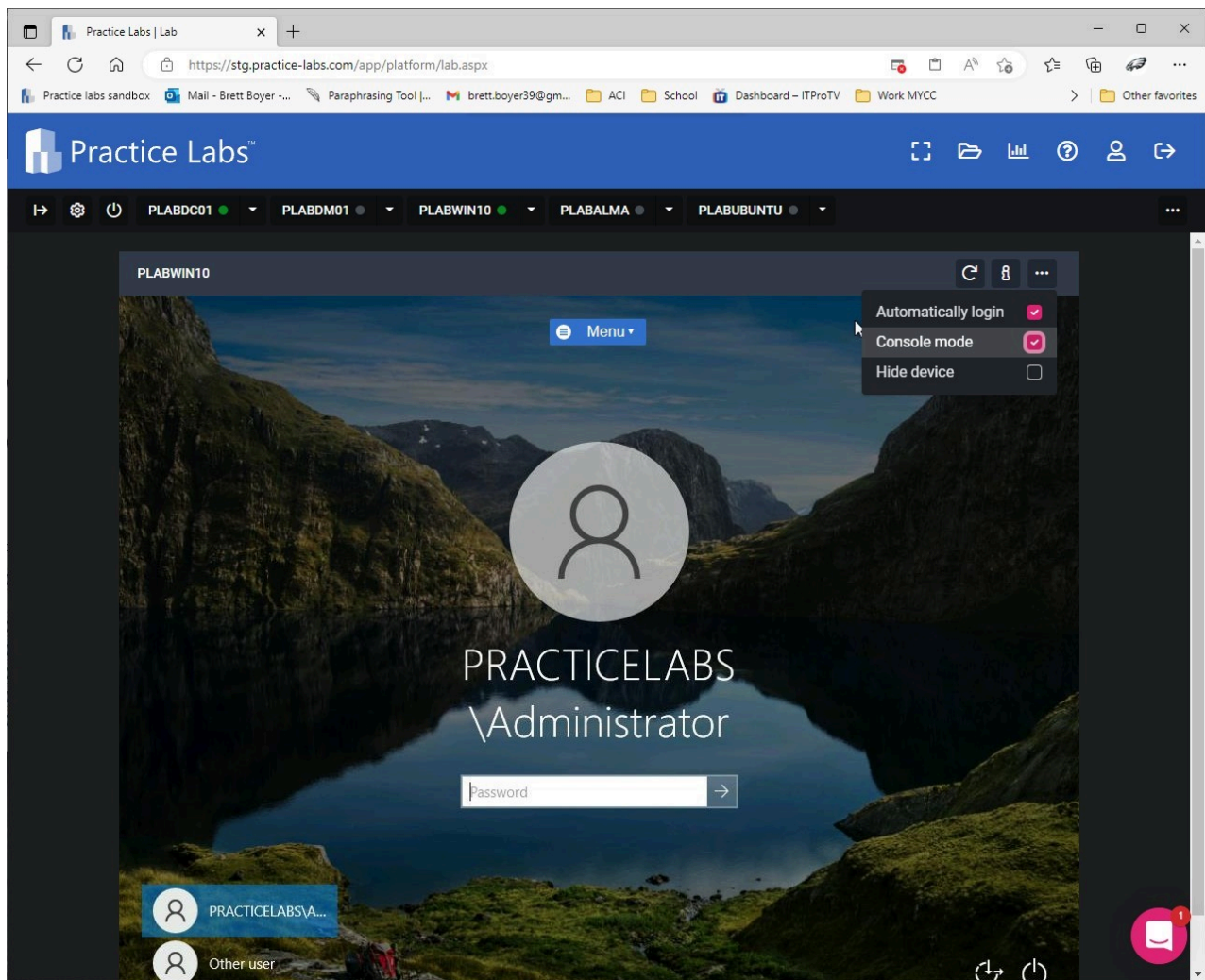


Figure 6.5 Screenshot of PLABWIN10: Displaying accessing the required drop-down menu and selecting Console mode.

Step 6

The user will be **PRACTICELABS\Administrator**.

Type the following password and press **Enter**:

Passw0rd

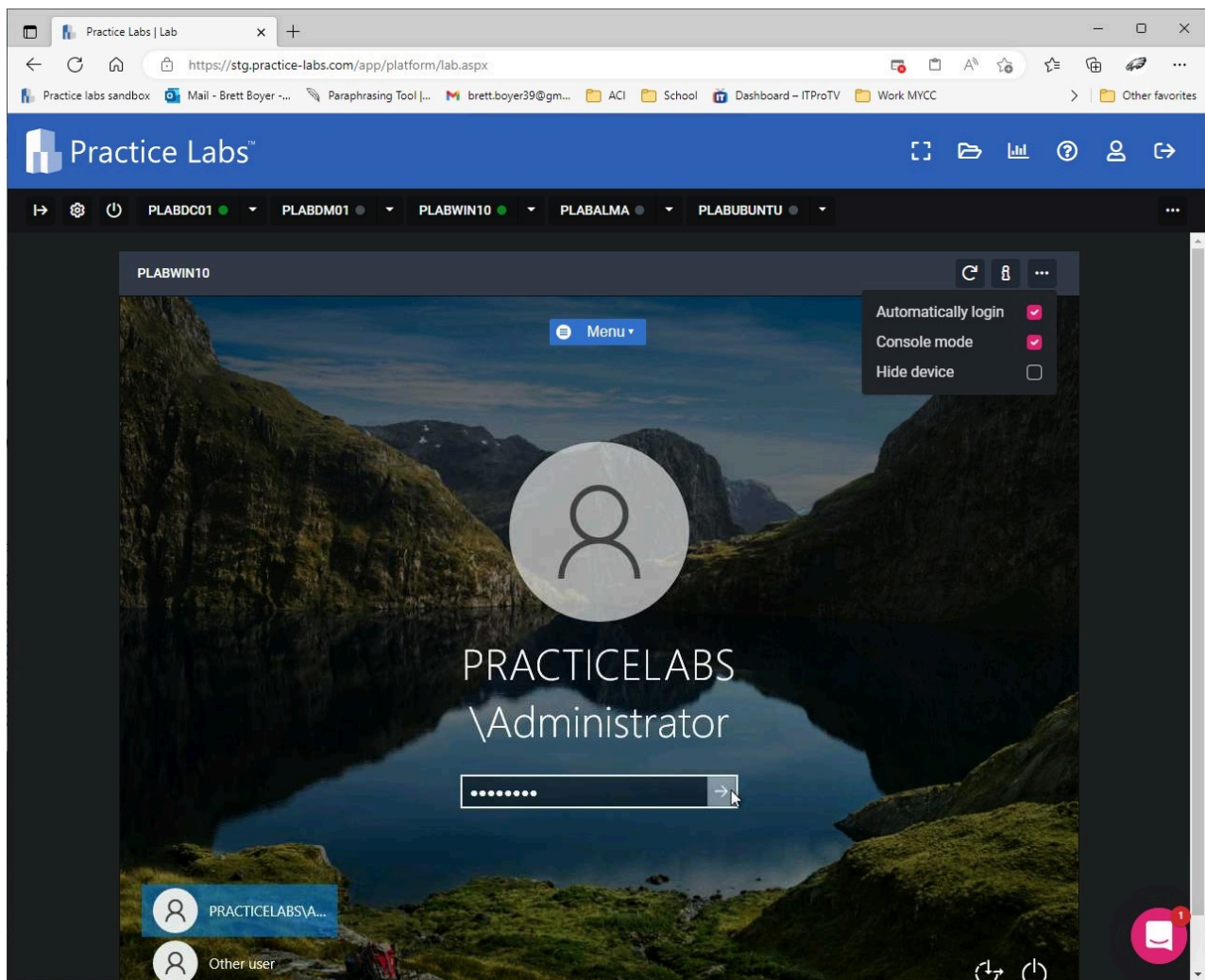


Figure 6.6 Screenshot of PLABWIN10: Displaying entering the required credentials in the login screen.

Step 7

The **Desktop** is displayed.

Notice there is no background, and it says **safe mode** in all 4 corners of the screen.

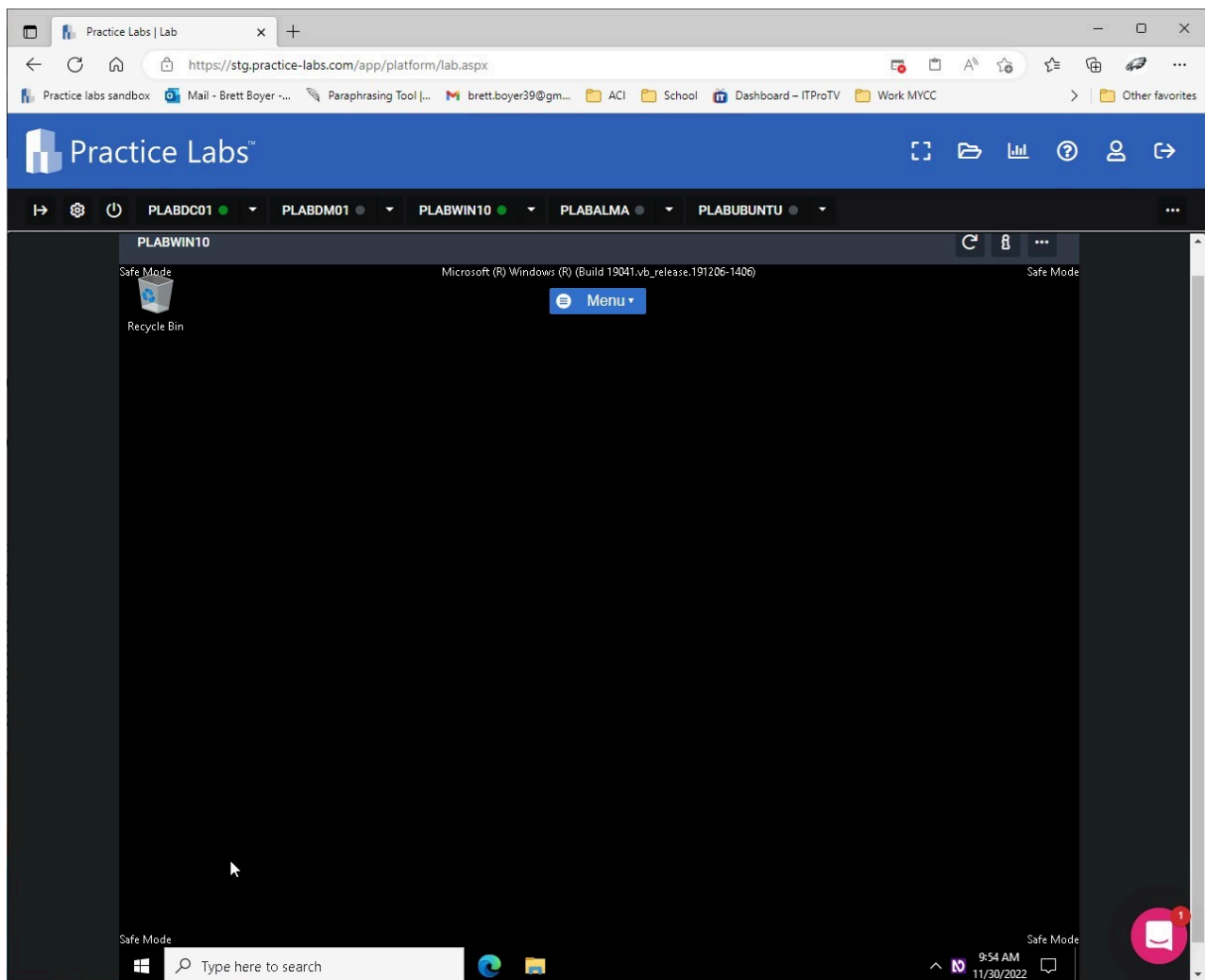


Figure 6.7 Screenshot of PLABWIN10: Displaying the Desktop in safe mode.

Step 8

Click the **Start** charm and type the following:

```
msconfig
```

Select **System Configuration** from the **Best match** pop-up menu.

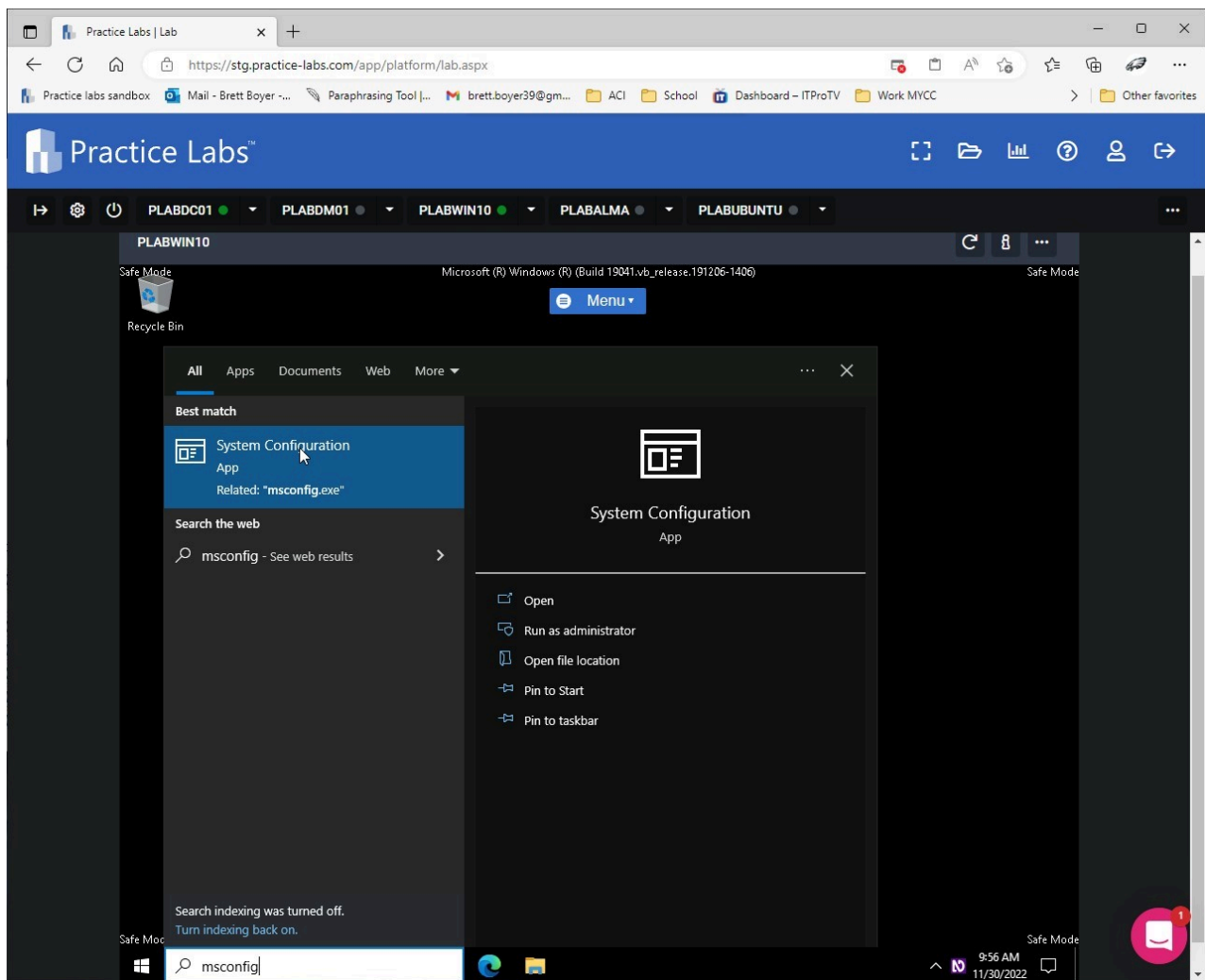


Figure 6.8 Screenshot of PLABWIN10: Displaying selecting System Configuration from the Best match pop-up menu.

Step 9

In the **System Configuration** window, select the **Boot** tab.

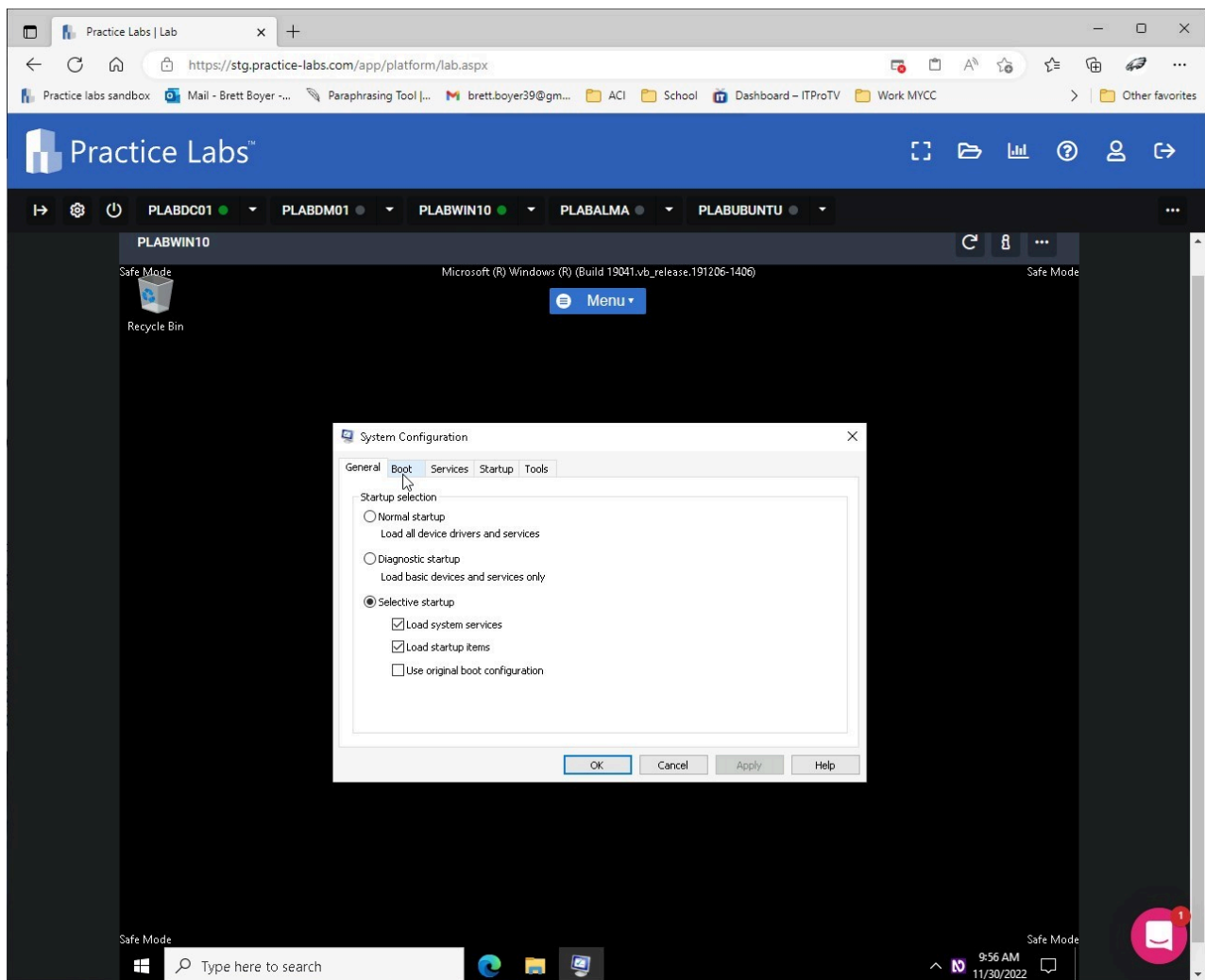


Figure 6.9 Screenshot of PLABWIN10: Displaying selecting the Boot tab in the System Configuration window.

Step 10

In the **System Configuration - Boot** tab, untick **Safe boot** and click **OK**.

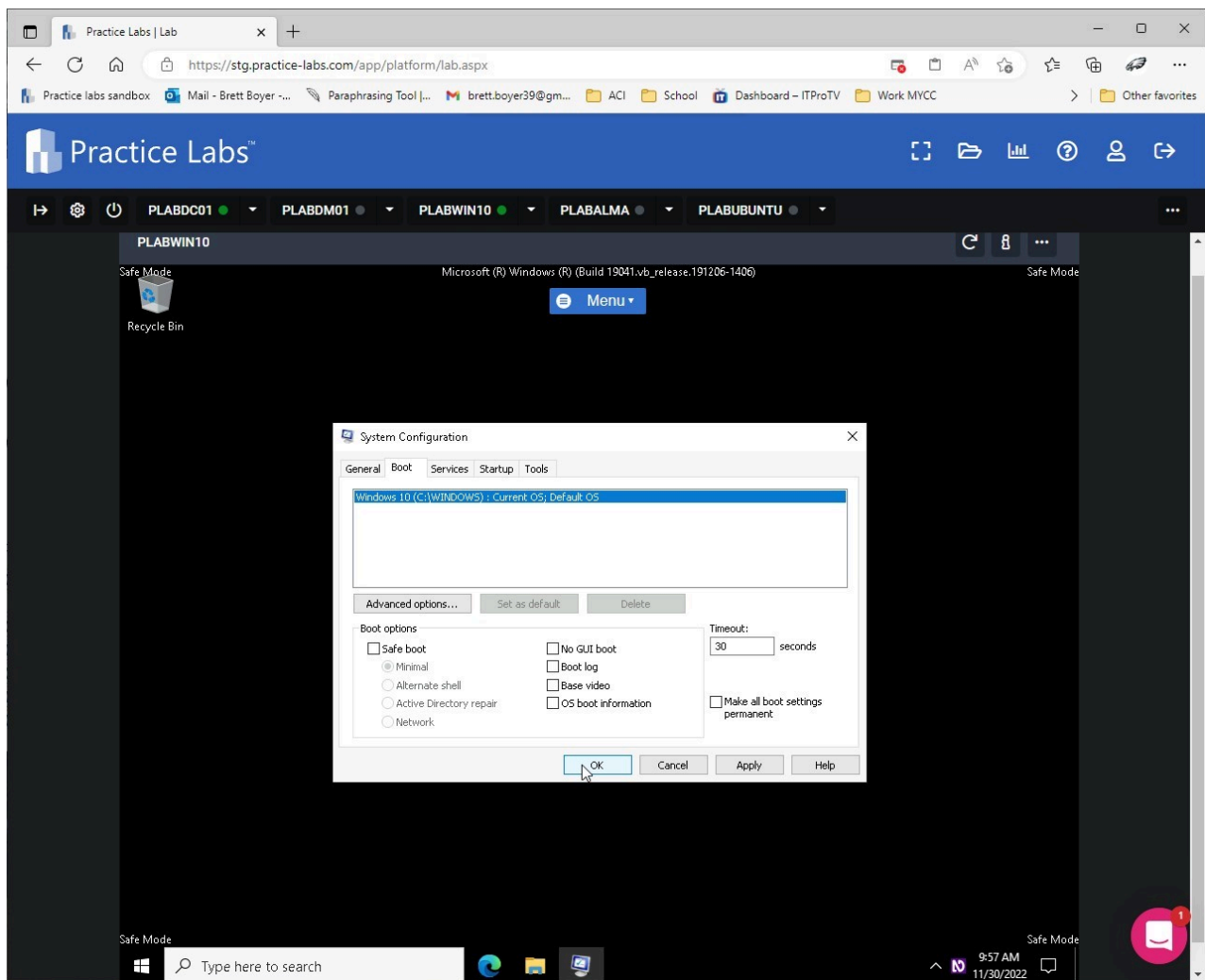


Figure 6.10 Screenshot of PLABWIN10: Displaying the System Configuration - Boot tab, showing the required settings performed and OK selected.

Step 11

From the **System Configuration** pop-up window, click **Restart**.

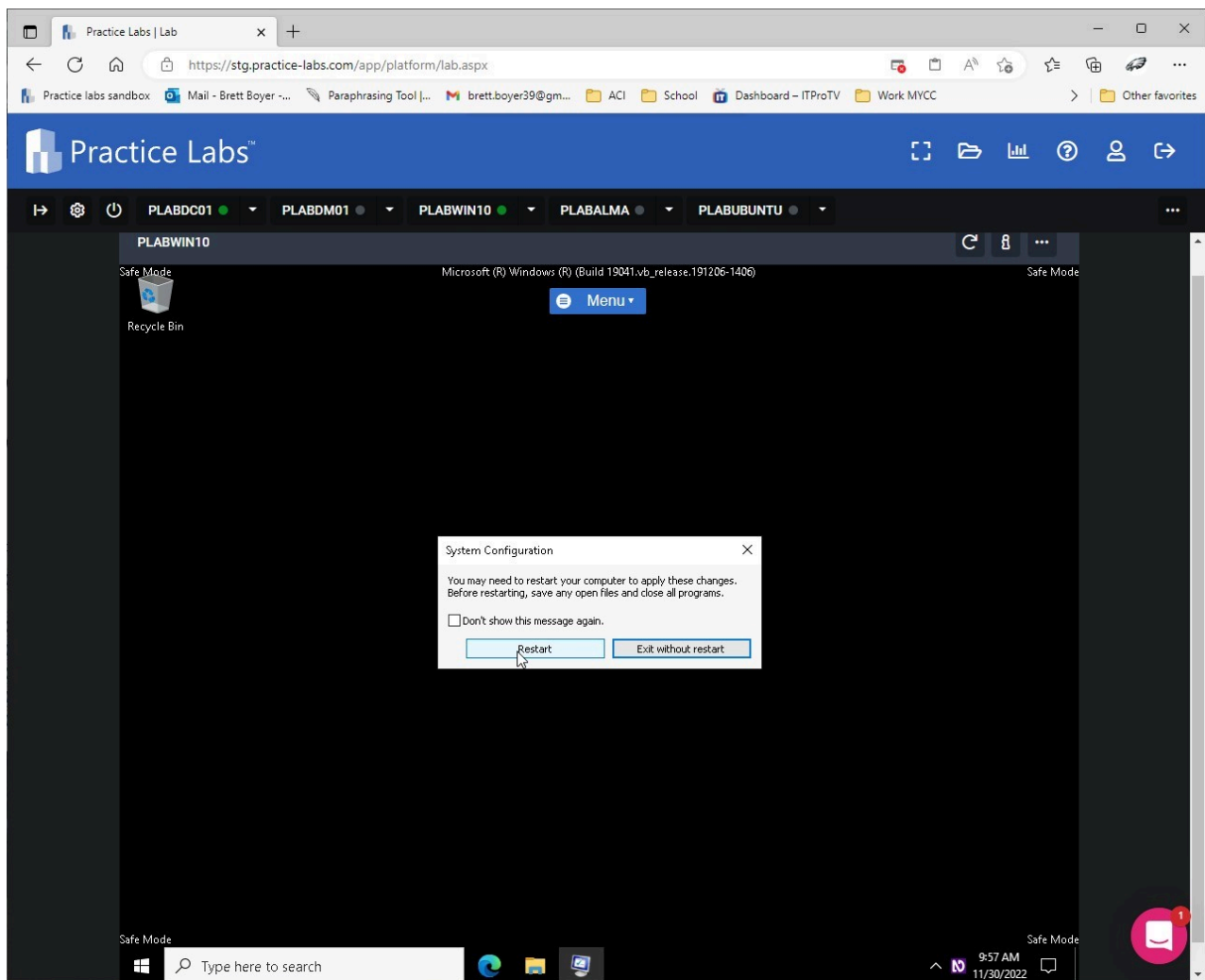


Figure 6.11 Screenshot of PLABWIN10: Displaying selecting Restart in the System Configuration pop-up window.

Step 12

After a minute, reconnect to the **PLABWIN10** device.

The **desktop** is no longer displayed in safe mode.

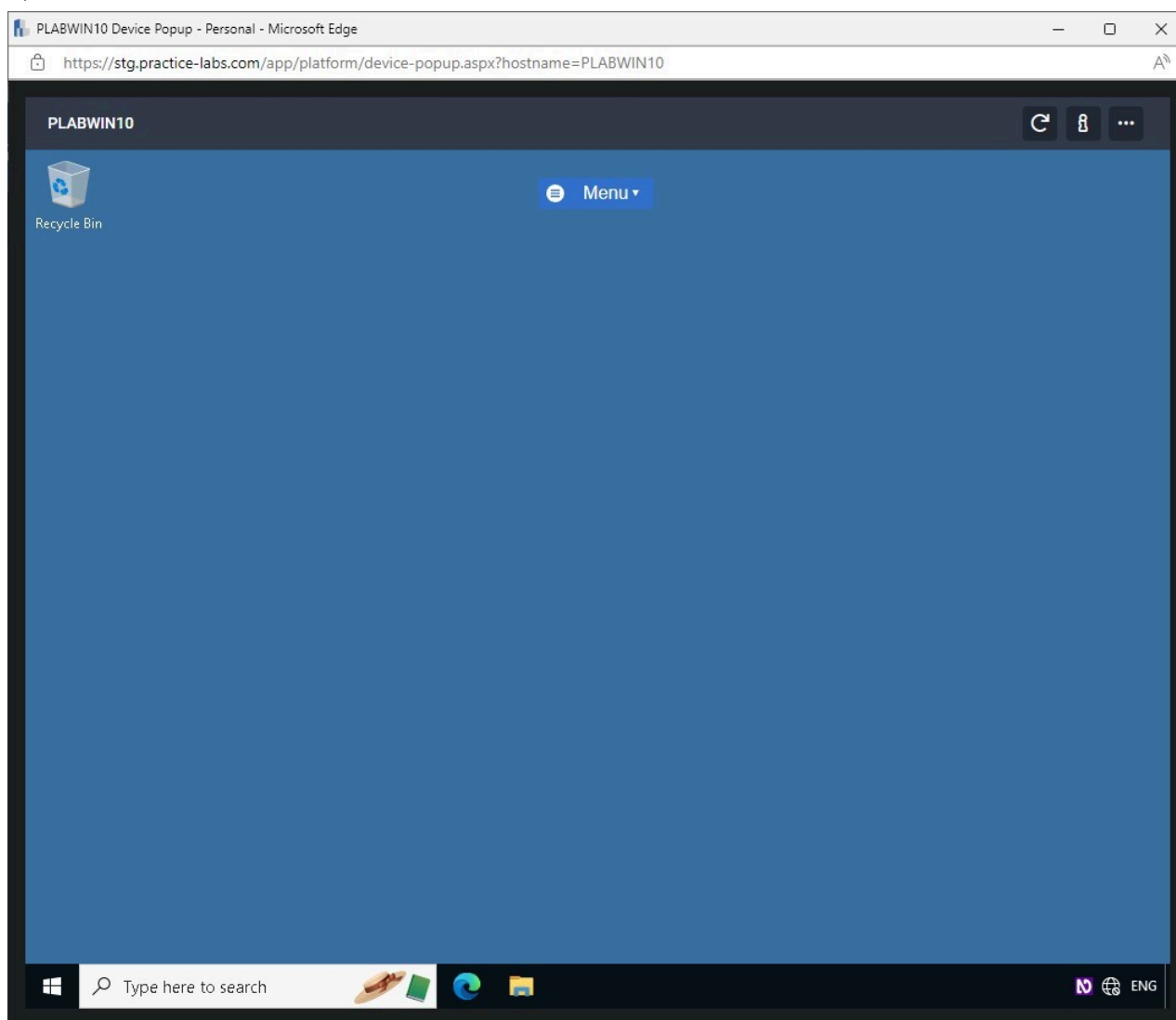


Figure 6.13 Screenshot of PLABWIN10: Displaying the PLABWIN10 desktop in normal mode.

Task 2 - Snapshots

A snapshot is like a picture that preserves the state of your server's data at a particular point in time. Snapshots are mostly metadata; therefore, they don't contain all of the hard drive's contents. Unlike full backups, which can take hours and usually need to be done overnight, snapshots can be made in seconds and as often as needed.

In this task, you will learn about Snapshots.

Step 1

Connect to **PLABWIN10**.

Click the **Start** charm and type the following:

recovery

Select **Recovery** from the **Best match** pop-up menu.

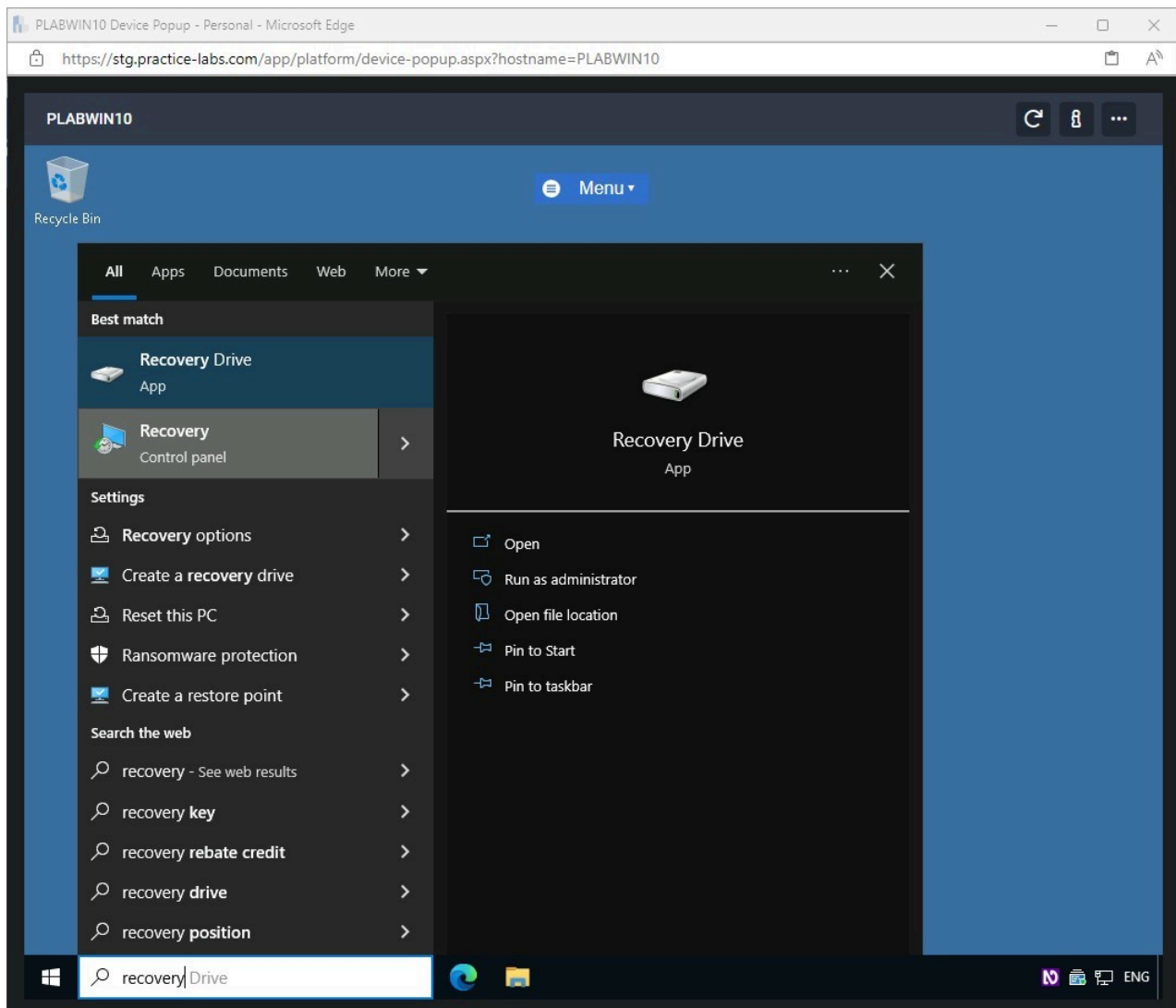


Figure 6.14 Screenshot of PLABWIN10: Displaying selecting Recovery from the Best match pop-up menu.

Step 2

In the **Recovery** window, select the **Configure System Restore** link.

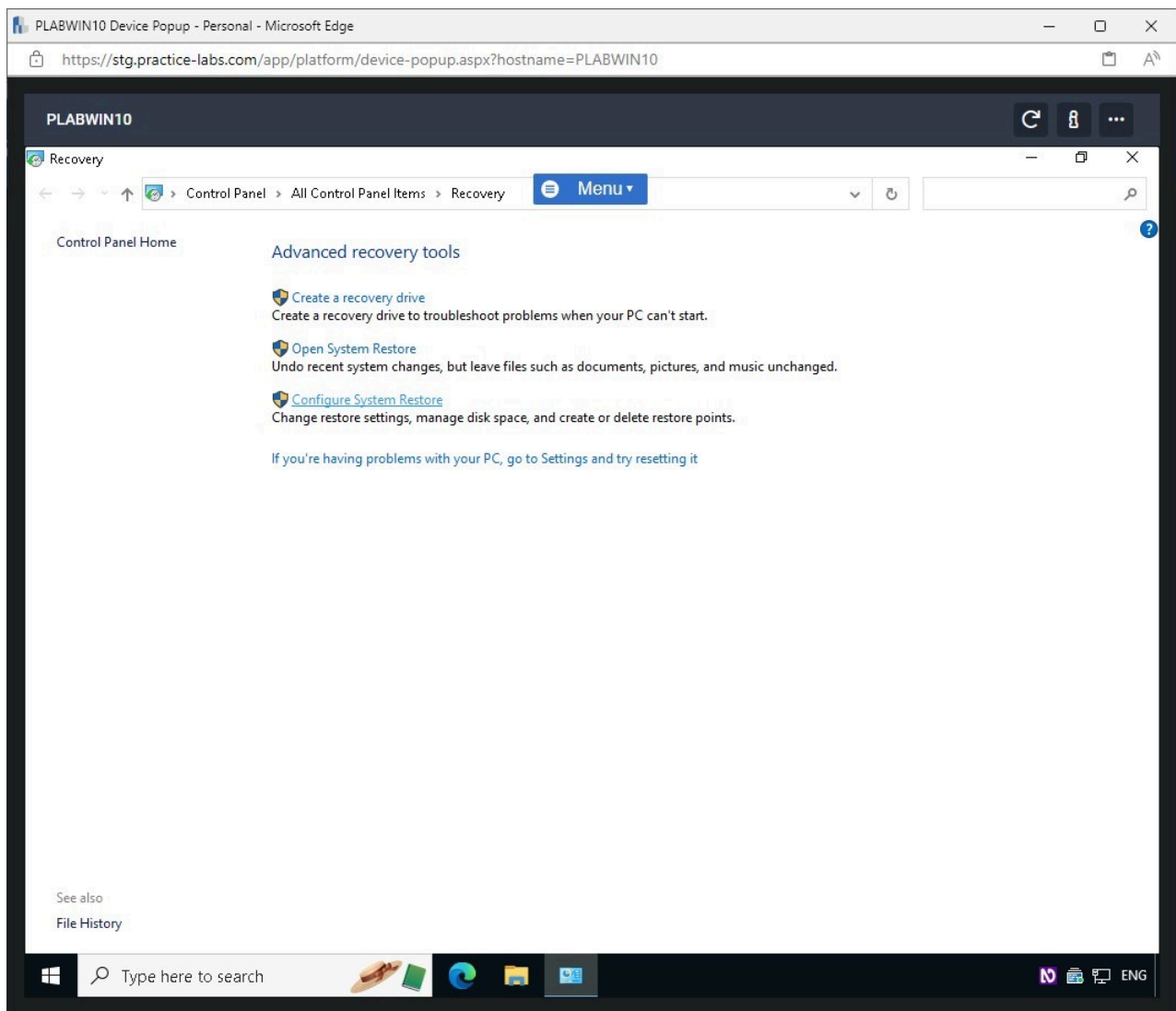


Figure 6.15 Screenshot of PLABWIN10: Displaying selecting the Configure System Restore link in the Recovery window.

Step 3

In the **System Properties** window, click **Configure**.

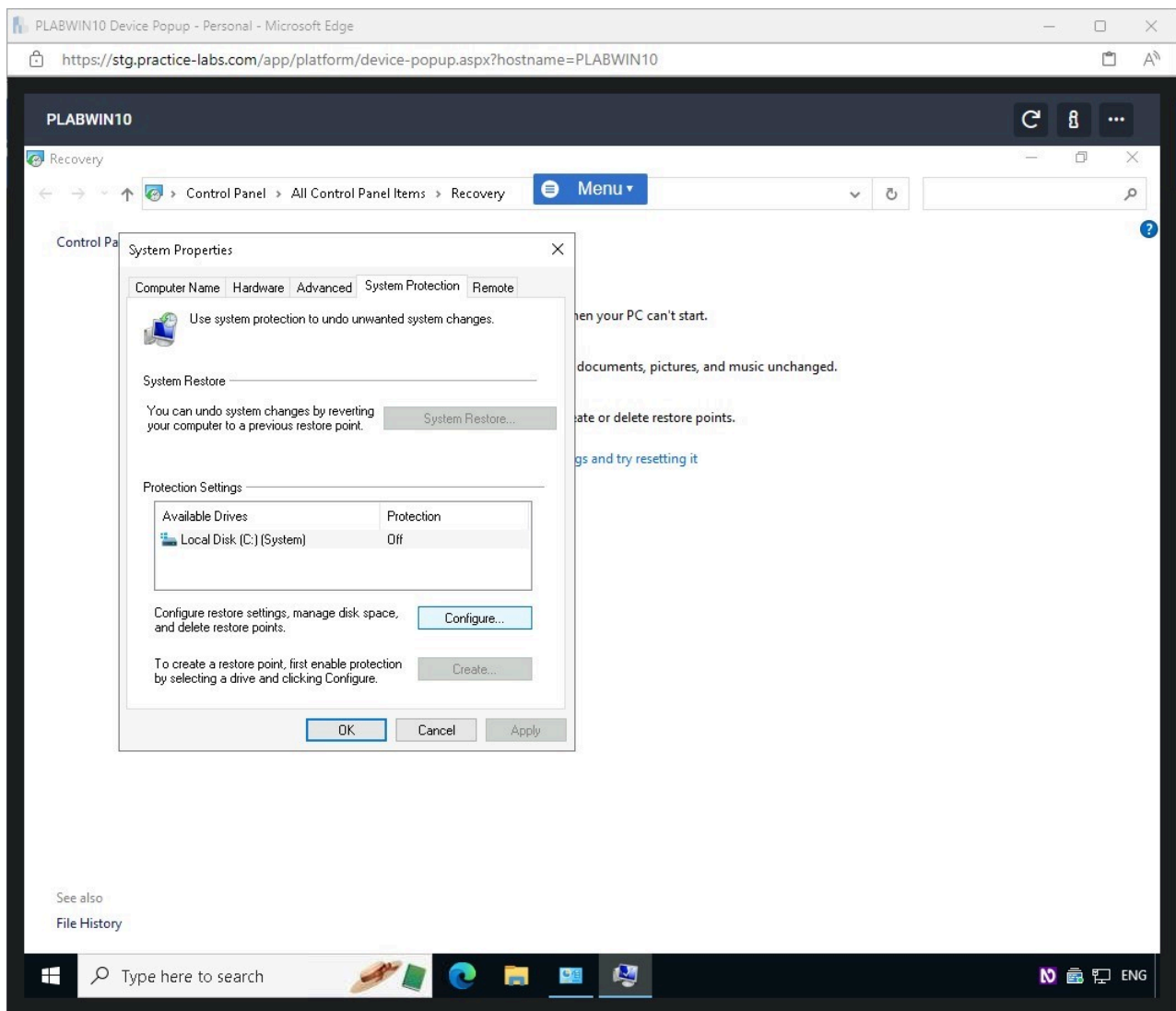


Figure 6.16 Screenshot of PLABWIN10: Displaying selecting Configure in the System Properties window.

Step 4

In the **System Protection for Local Disk (C:)** pop-up window, select the **Turn on system protection** option.

Change the **Max Usage** to **15%** to give us storage to create a restore point.

Click **OK**.

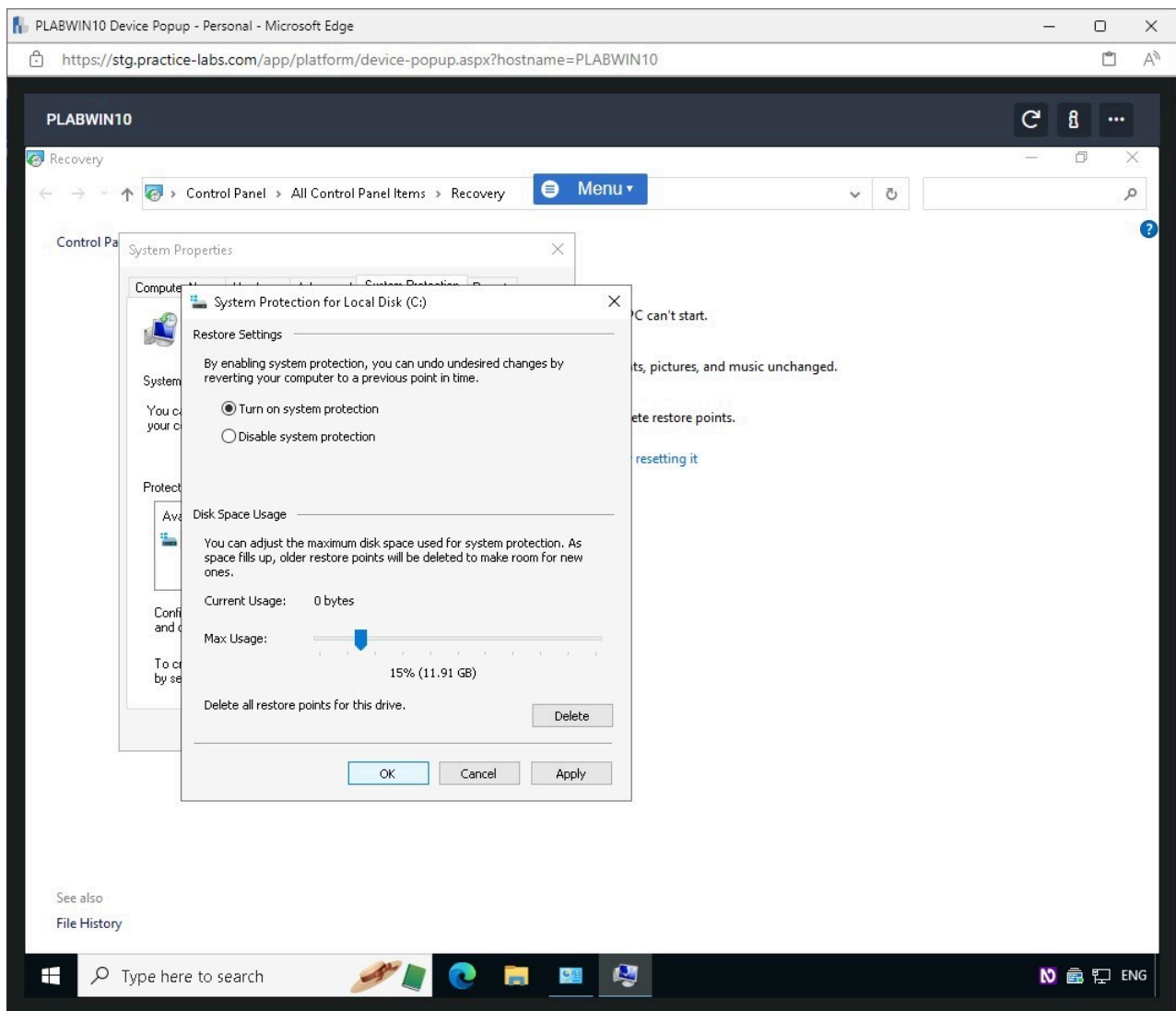


Figure 6.17 Screenshot of PLABWIN10: Displaying the System Protection for Local Disk (C:) pop-up window, showing the required settings performed and OK selected.

Step 5

Back on the **System Properties** window, click **Create**.

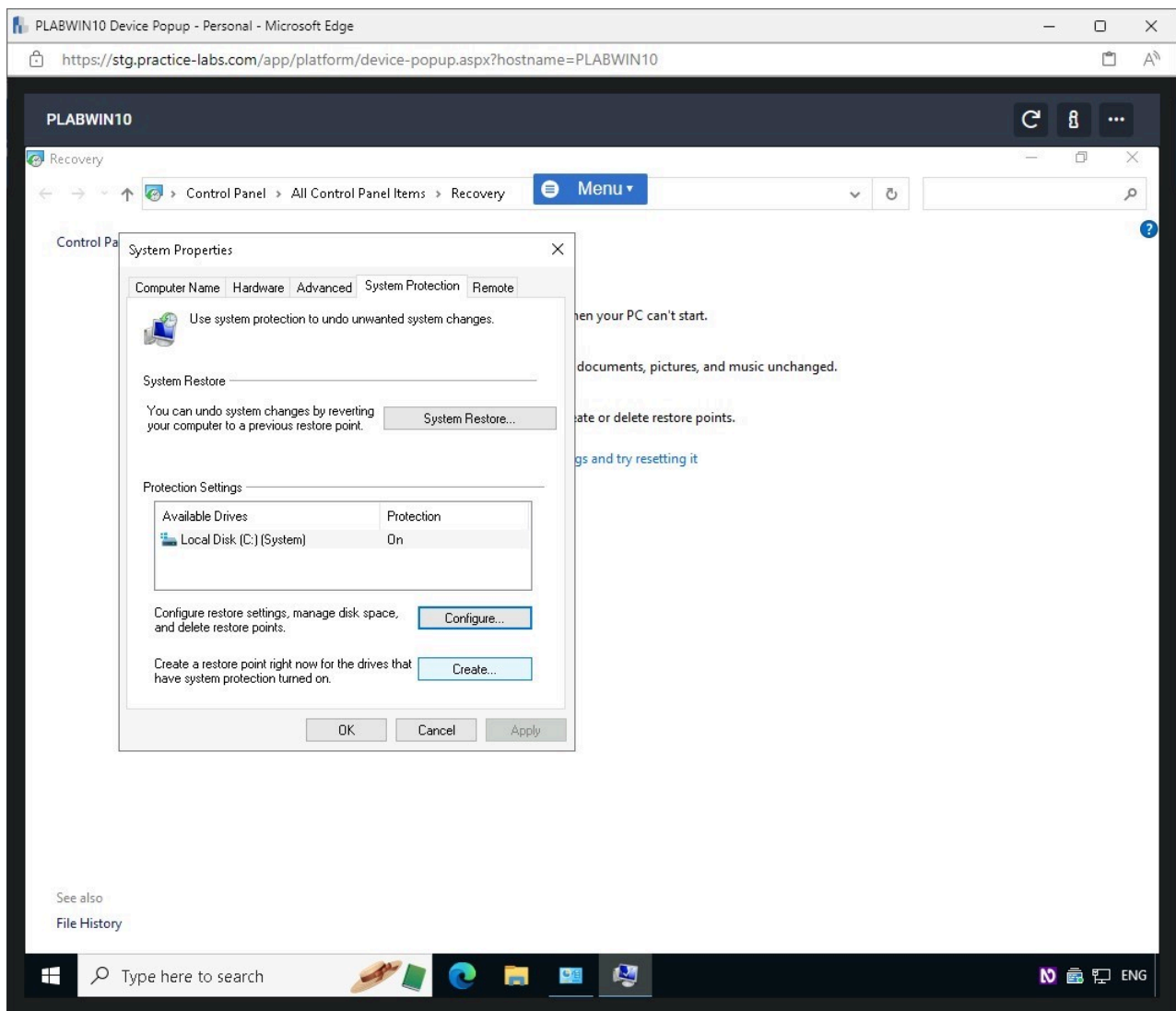


Figure 6.18 Screenshot of PLABWIN10: Displaying selecting Create in the System Properties window.

Step 6

In the **System Protection - Create a restore point** pop-up window, type the following name for the restore point:

Test Restore Point

Click **Create**.

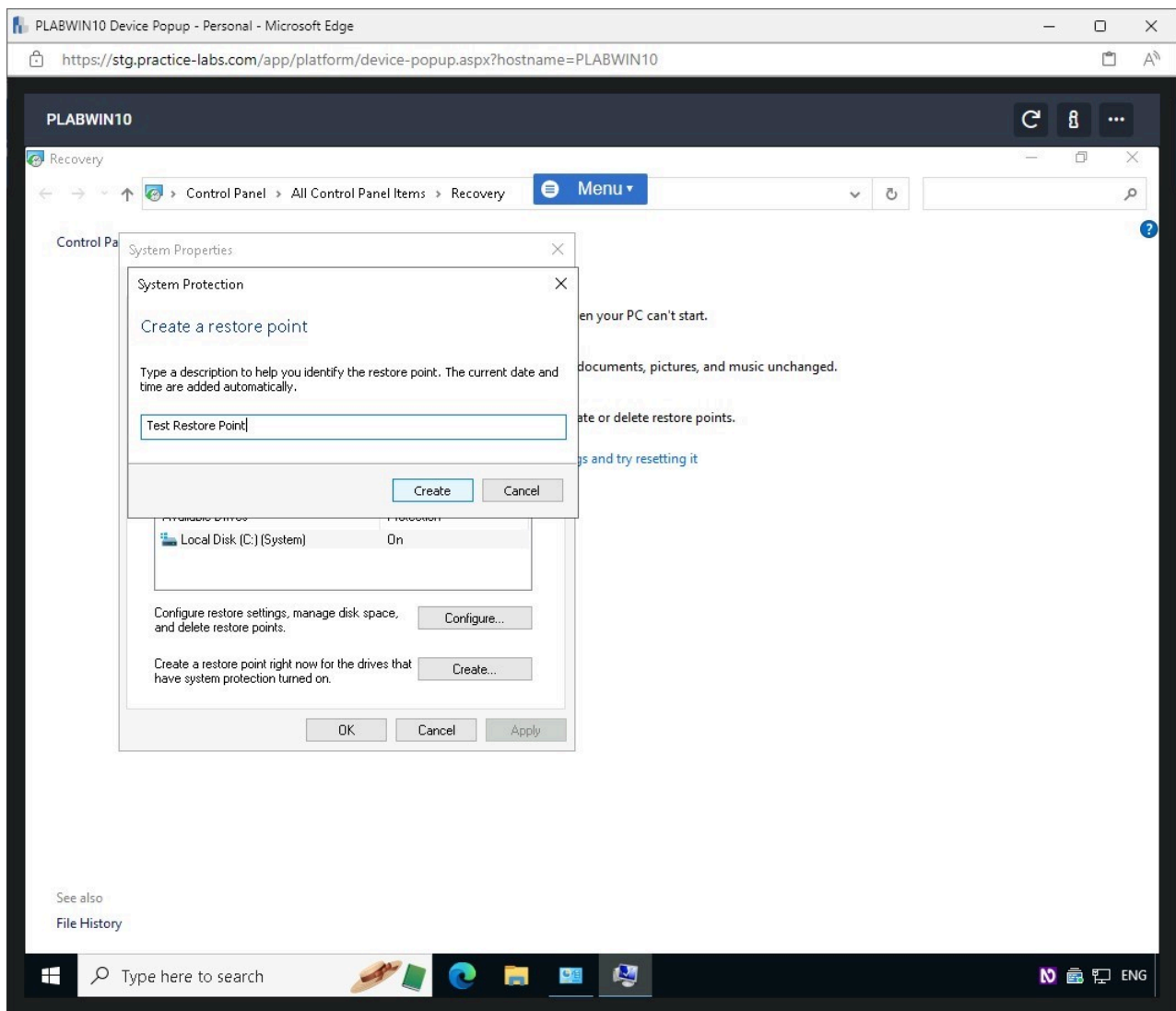


Figure 6.19 Screenshot of PLABWIN10: Displaying the System Protection - Create a restore point pop-up window, showing the required name typed in and Create selected.

Step 7

The **restore point** is now being created.

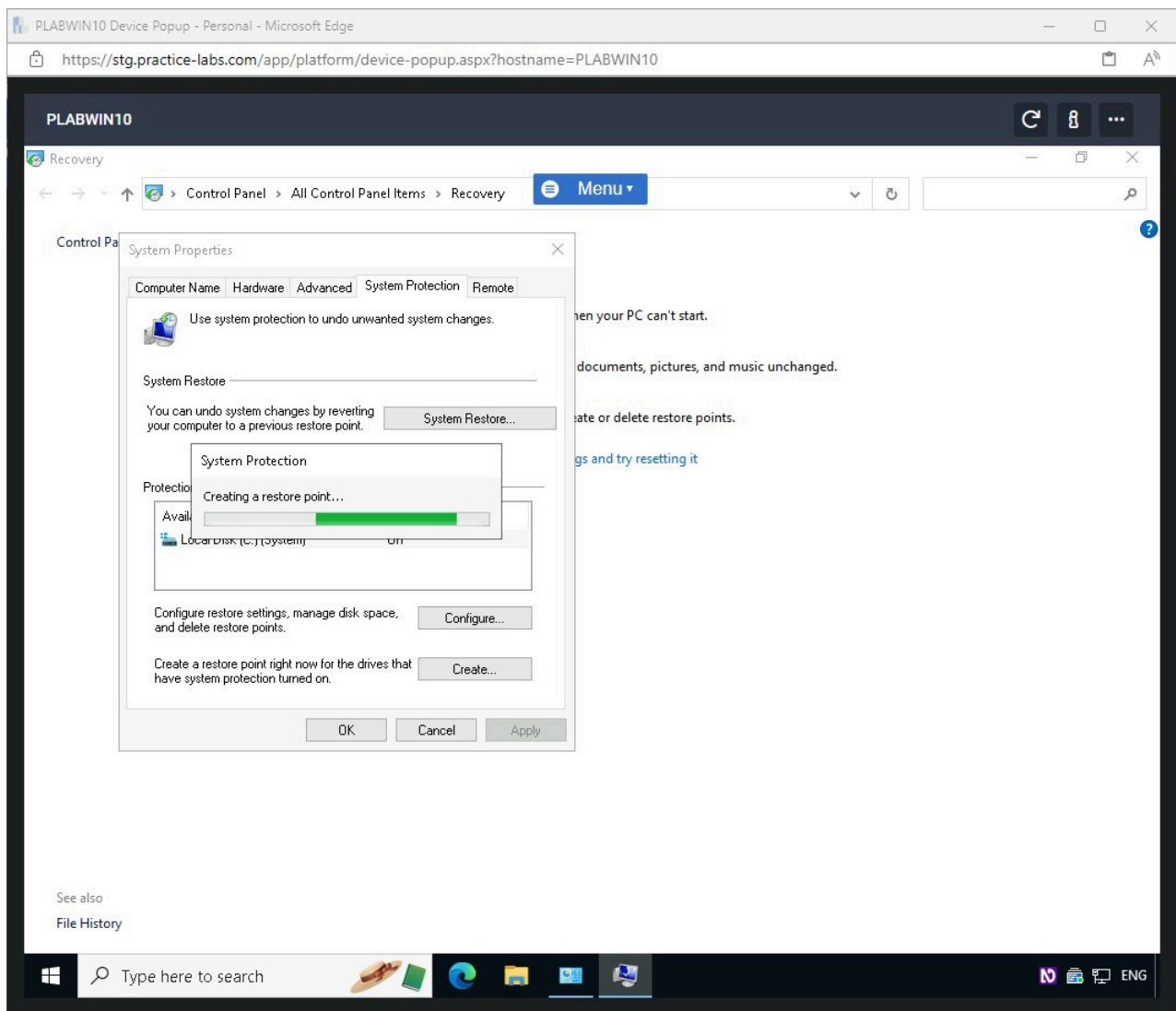


Figure 6.20 Screenshot of PLABWIN10: Displaying the System Protection pop-up window, showing the restore point creation in progress.

Step 8

Once the restore point is created successfully, click **Close** on the **System Protection** pop-up window.

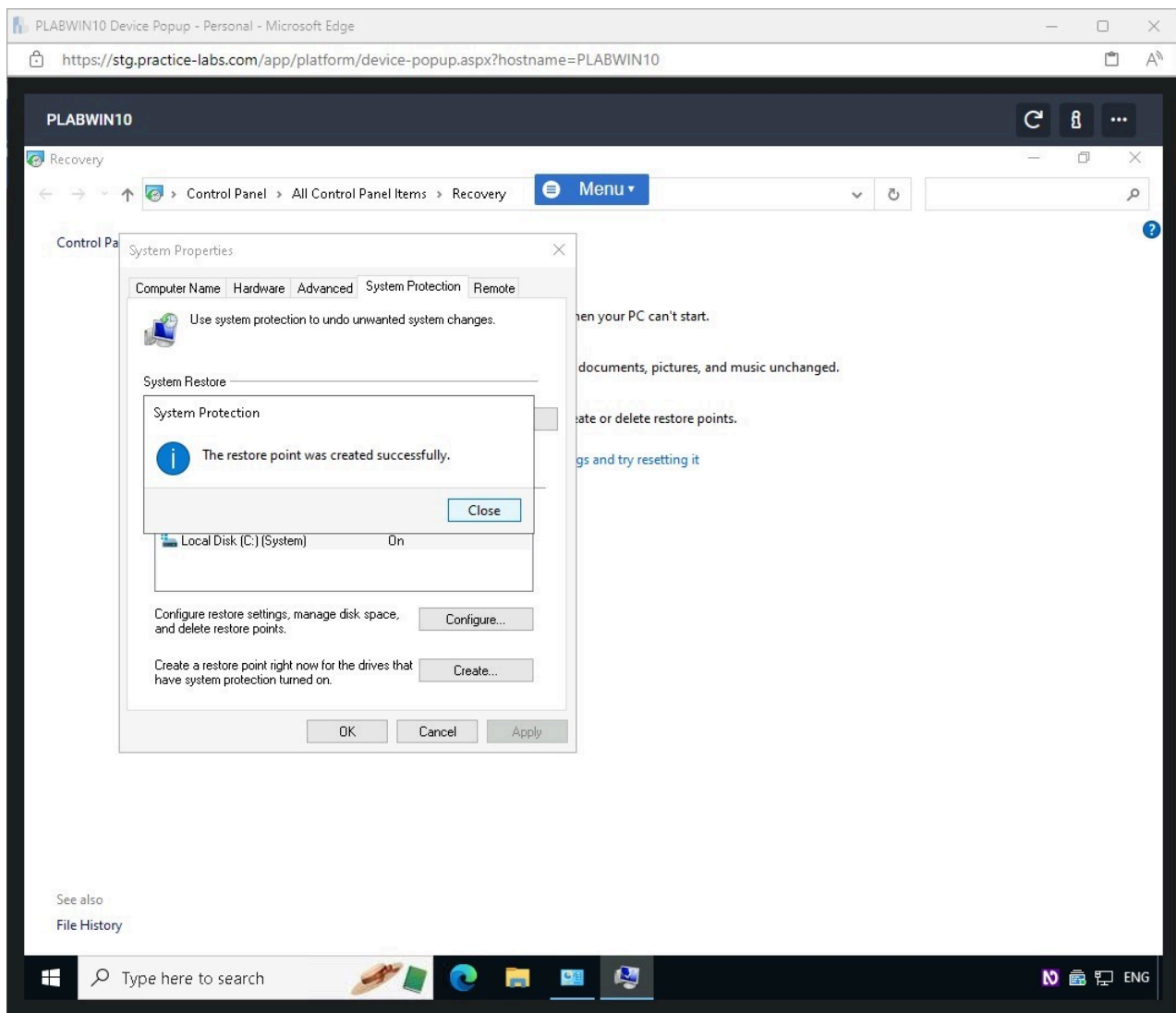


Figure 6.21 Screenshot of PLABWIN10: Displaying selecting Close in the System Protection pop-up window.

Step 9

Now that the restore point is created, you will implement it.

Back on the **System Properties** window, click **System Restore**.

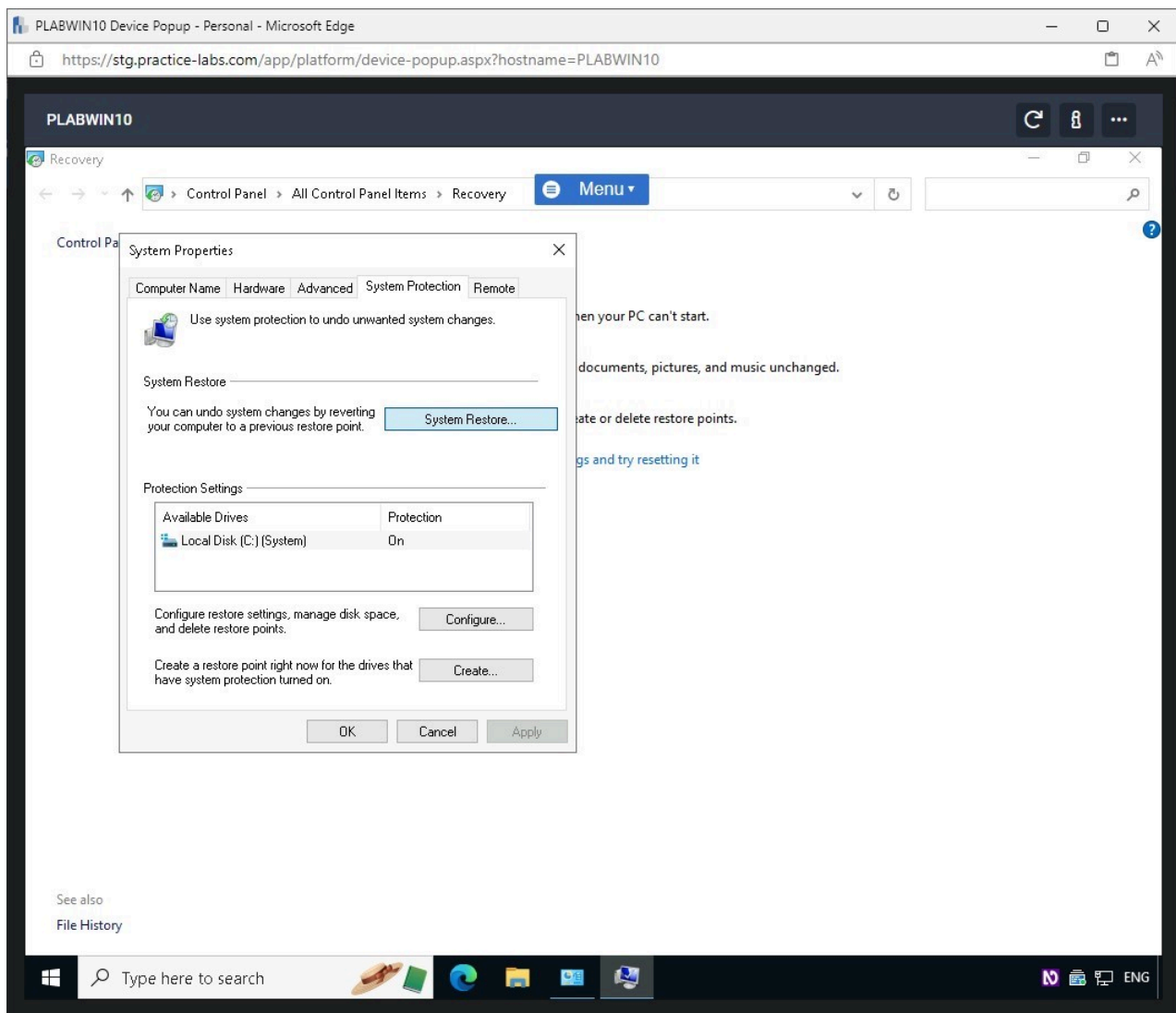


Figure 6.22 Screenshot of PLABWIN10: Displaying selecting System Restore in the System Properties window.

Step 10

In the **System Restore - Restore system files and settings** window, click **Next**.

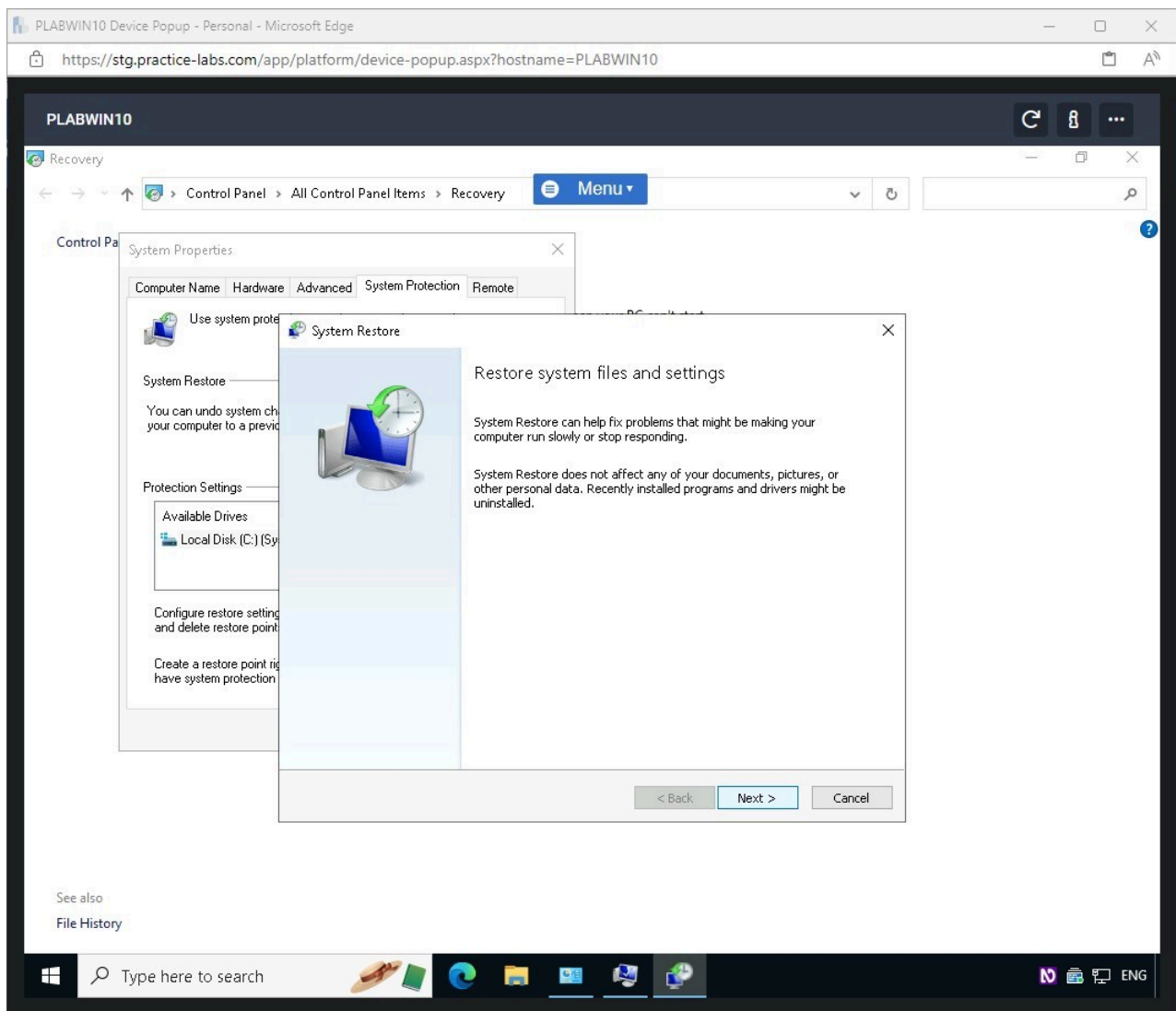


Figure 6.23 Screenshot of PLABWIN10: Displaying selecting Next in the System Restore - Restore system files and settings window.

Step 11

In the **Restore your computer to the state it was in before the selected event** page, select the **Test Restore Point** and click **Next**.

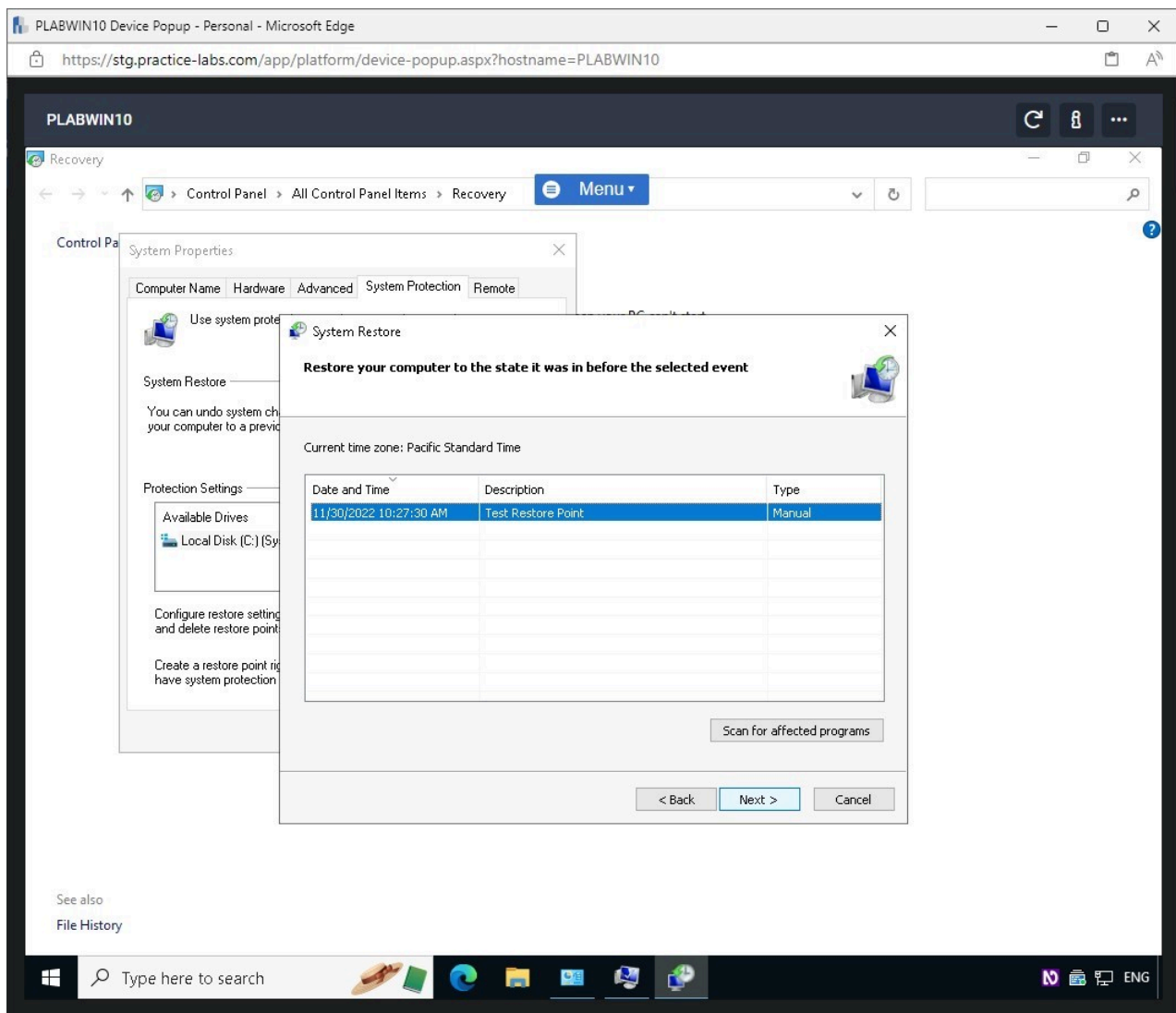


Figure 6.24 Screenshot of PLABWIN10: Displaying the Restore your computer to the state it was in before the selected event page, showing the Test Restore Point selected and Next highlighted.

Step 12

In the **Confirm your restore point** page, click **Finish**.

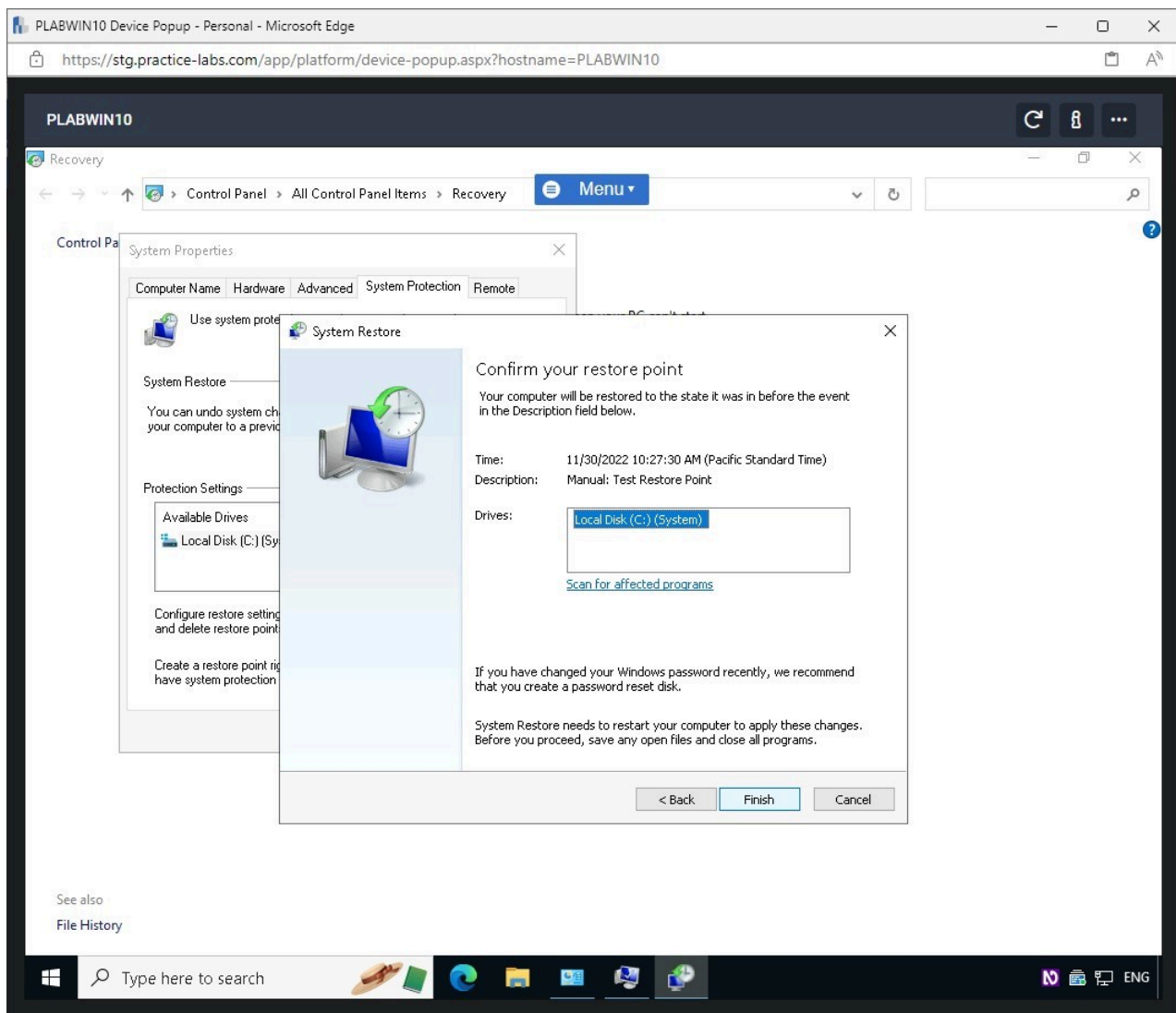


Figure 6.25 Screenshot of PLABWIN10: Displaying selecting Finish in the Confirm your restore point page.

Step 13

Click **Yes** on the warning pop-up message.

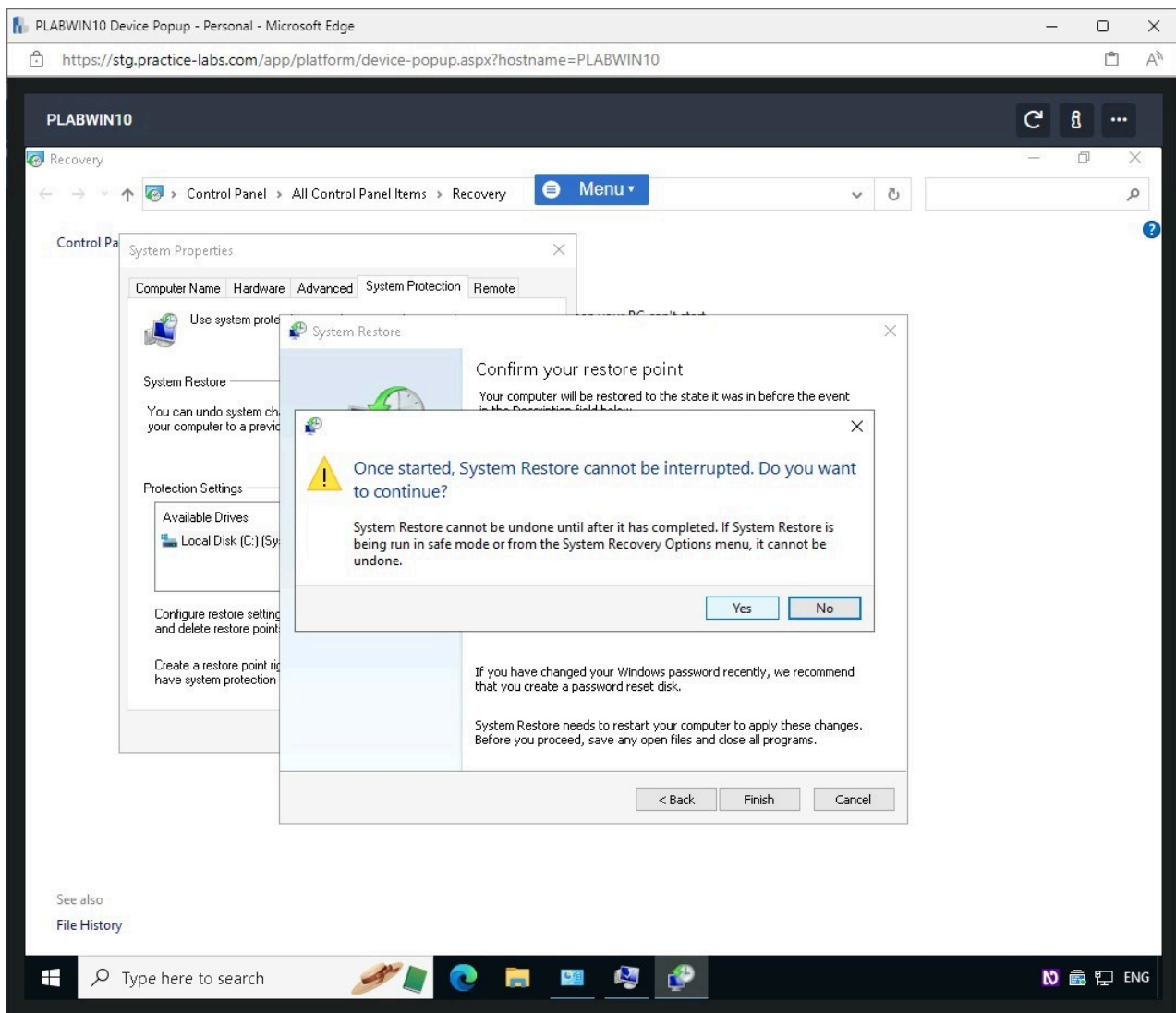


Figure 6.26 Screenshot of PLABWIN10: Displaying selecting Yes on the warning pop-up message.

Step 14

The **Restore** process initiates.

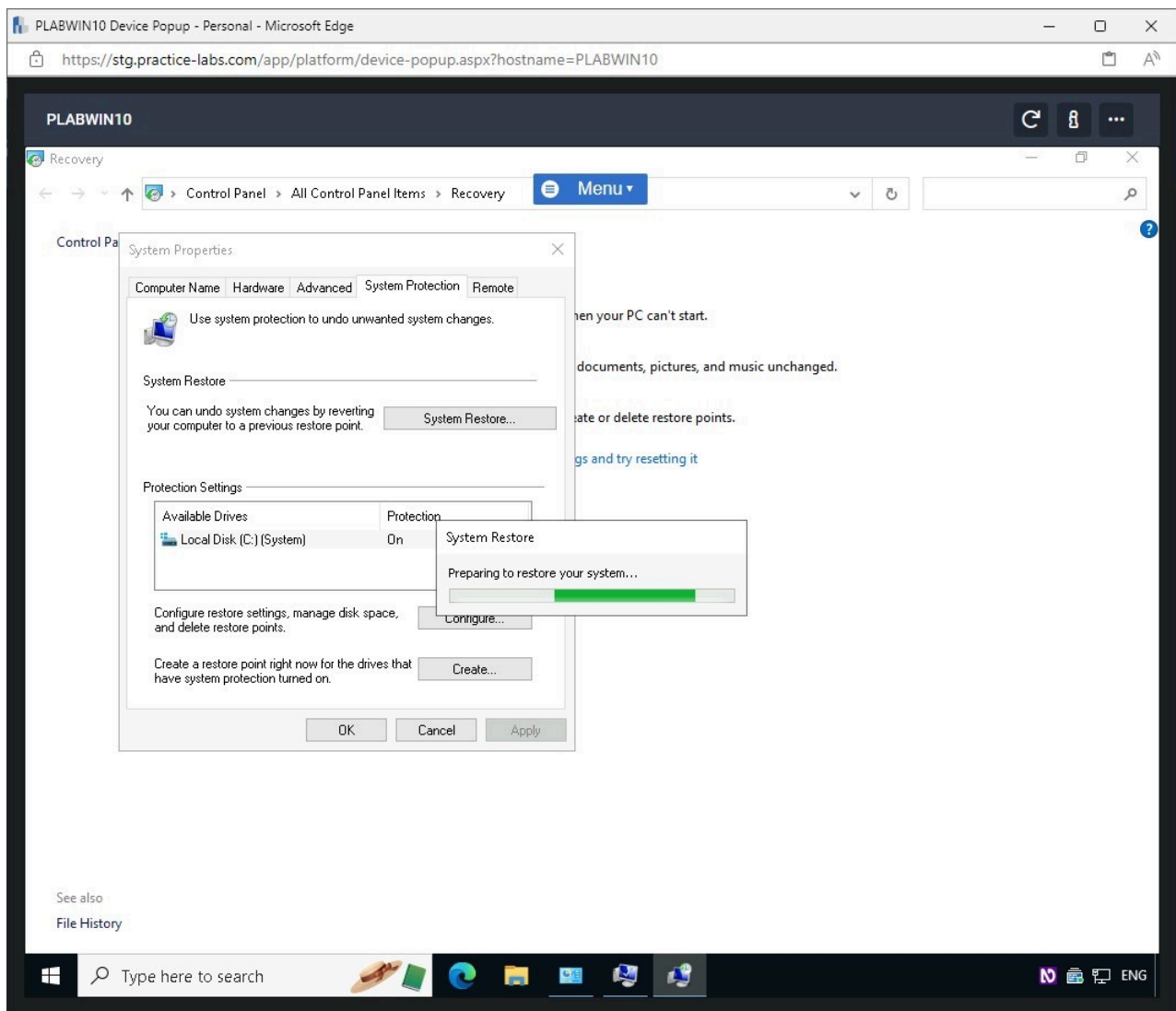


Figure 6.27 Screenshot of PLABWIN10: Displaying the System Restore pop-up window, showing the system restore in progress.

Step 15

The **PLABWIN10** device will disconnect.

This process will take **approximately 5 minutes** to complete.

Select the **refresh** option for the device to reconnect to the machine once the restore is complete.

If you want to check the progress, connect to the secondary connection using the console mode.

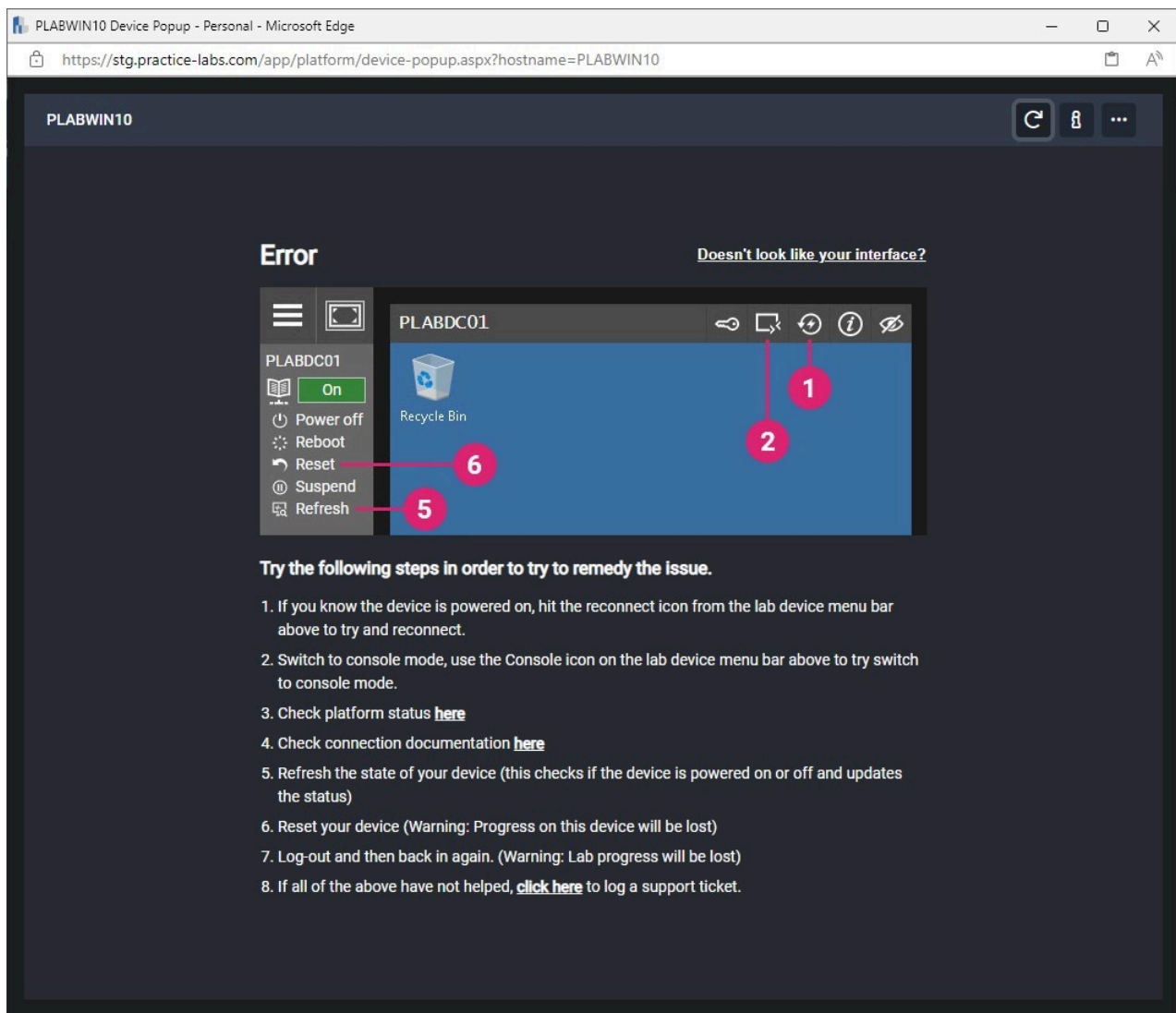


Figure 6.28 Screenshot of PLABWIN10: Displaying the machine disconnecting.

Step 16

Once connected to the **PLABWIN10** device, click **Close** on the **System Restore** pop-up window.

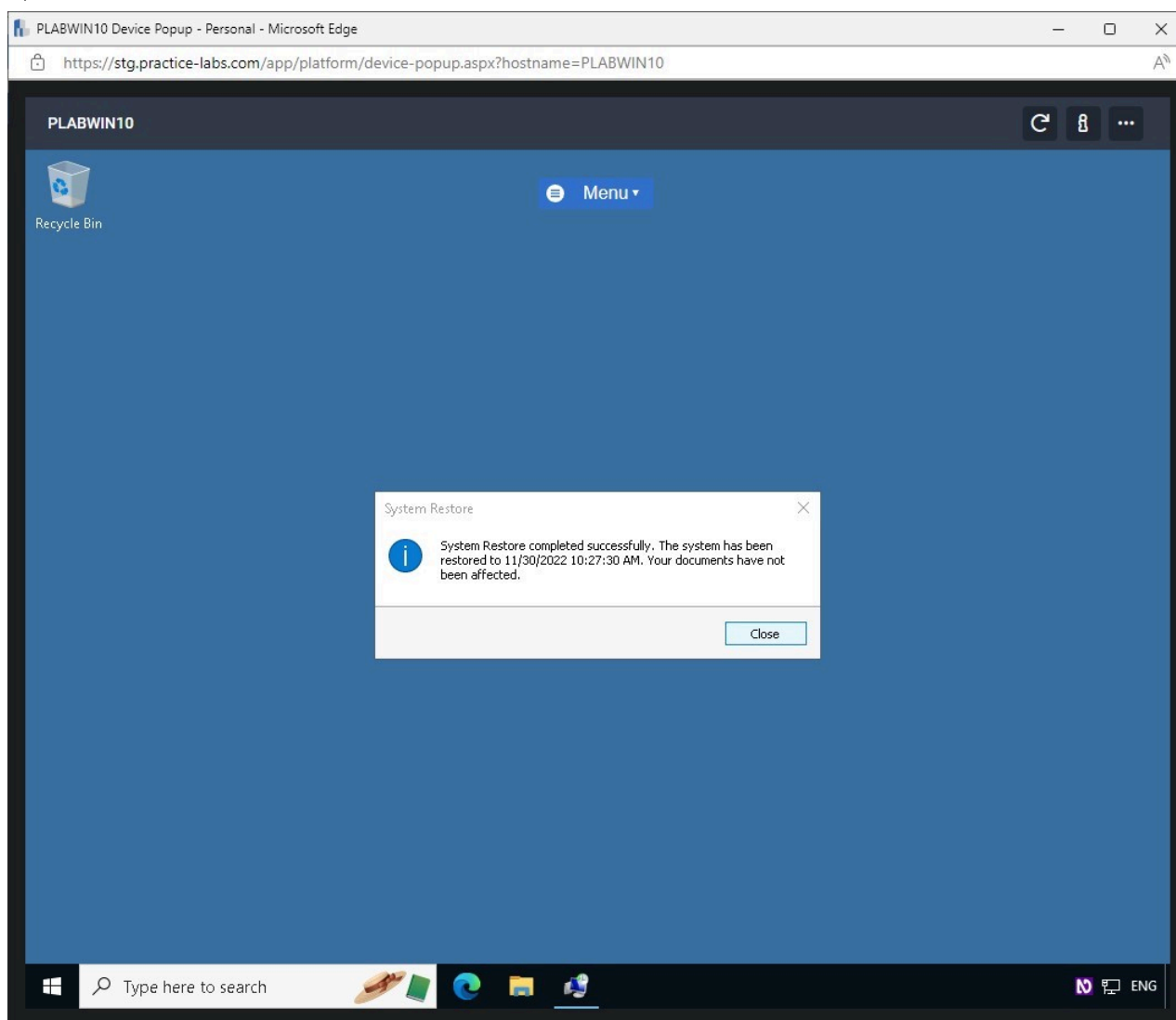


Figure 6.29 Screenshot of PLABWIN10: Displaying selecting Close on the System Restore pop-up window.

Task 3 - Reload OS

A device's OS may be reinstalled at any moment to return it to its factory settings and functionality. Users may also change a device's settings to use a different OS or application. A complete OS reload will wipe all data from the device and restore it to a "like new" state, as set up in the OS reloads configuration procedure. Any information stored on the device will be lost during an OS restart, so always ensure data is backed up before doing an OS reload.

In this task, you will reload the OS to its factory settings.

Step 1

Connect to **PLABWIN10**.

Right-click the **Start** charm and select **Settings**.

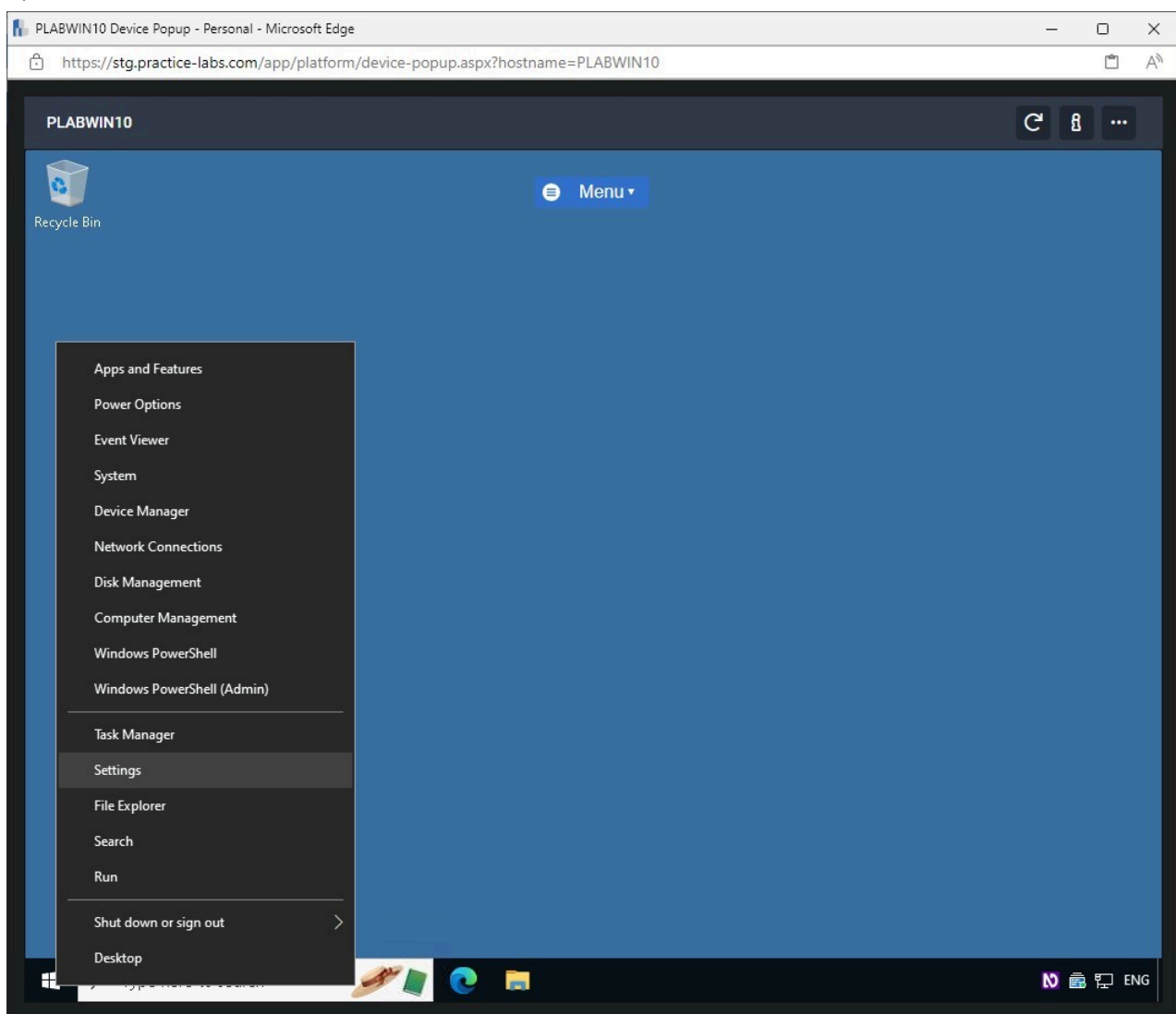


Figure 6.30 Screenshot of PLABWIN10: Displaying right-clicking the Start charm and selecting Settings.

Step 2

In the **Settings** window, select **Update & Security**.

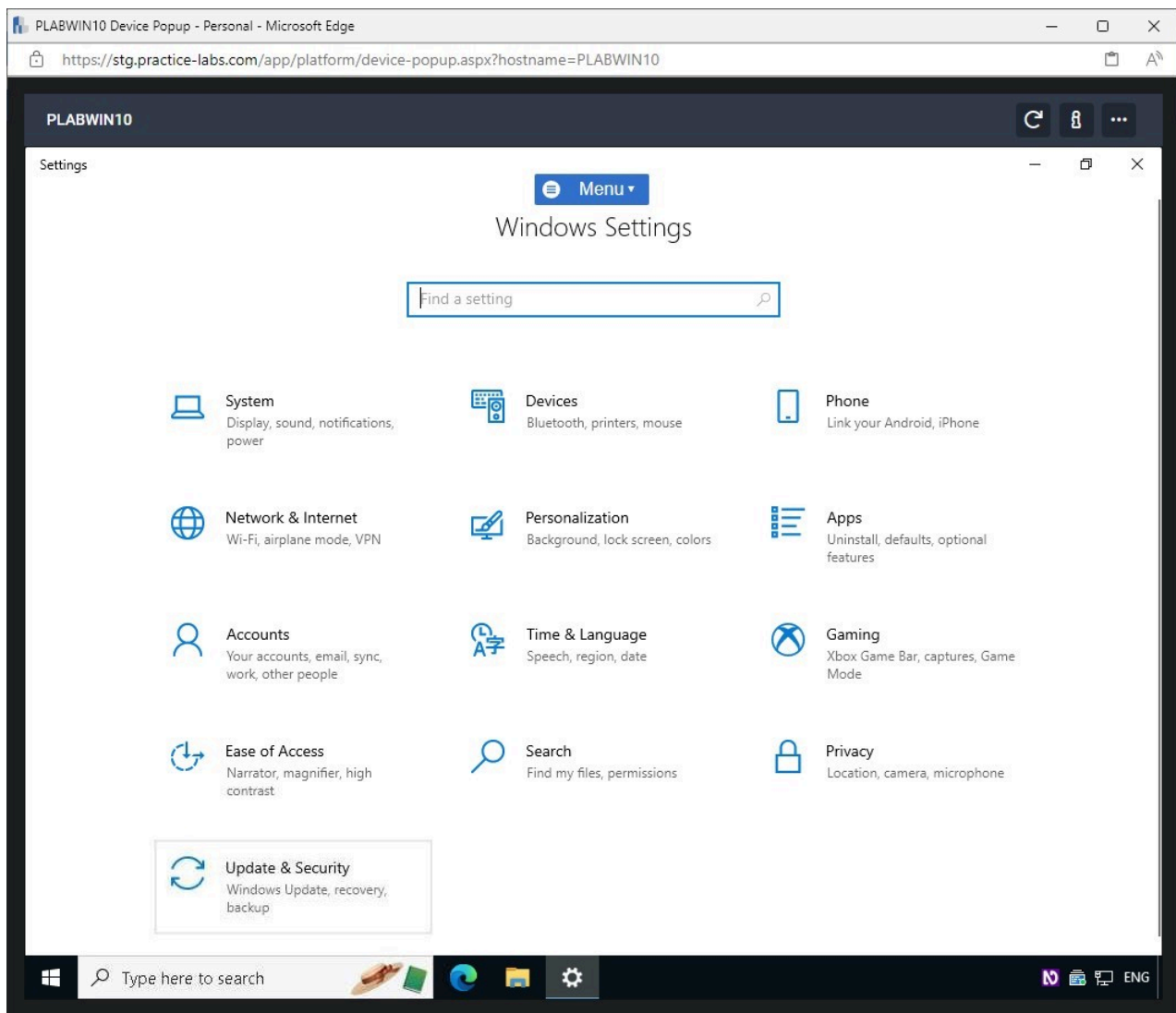


Figure 6.31 Screenshot of PLABWIN10: Displaying selecting Update & Security in the Settings window.

Step 3

In the **Settings** window, select **Recovery** on the left pane.

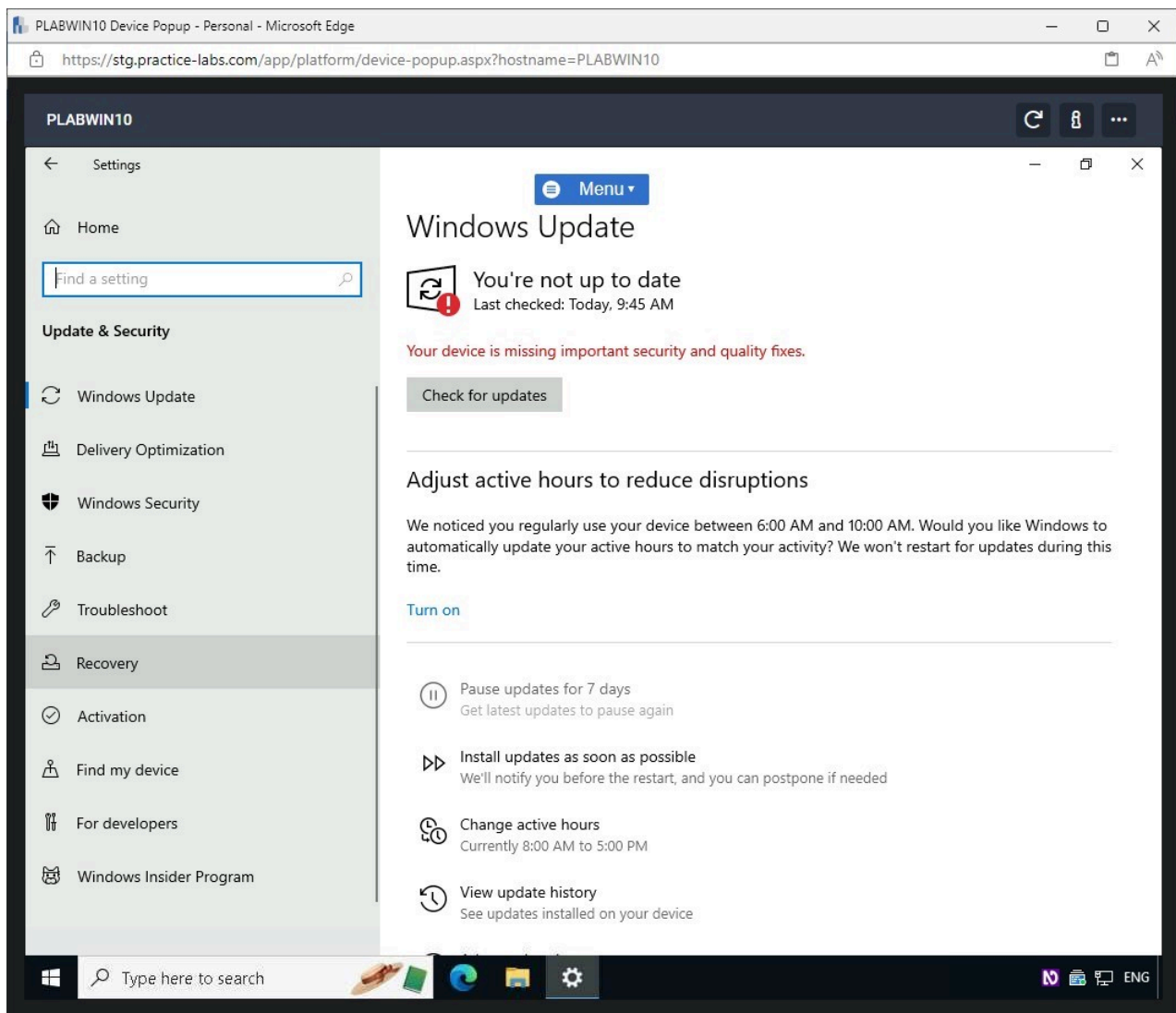


Figure 6.32 Screenshot of PLABWIN10: Displaying selecting Recovery in the Settings window.

Step 4

In the **Settings - Recovery** pane, click the **Get started** button under **Reset this PC**.

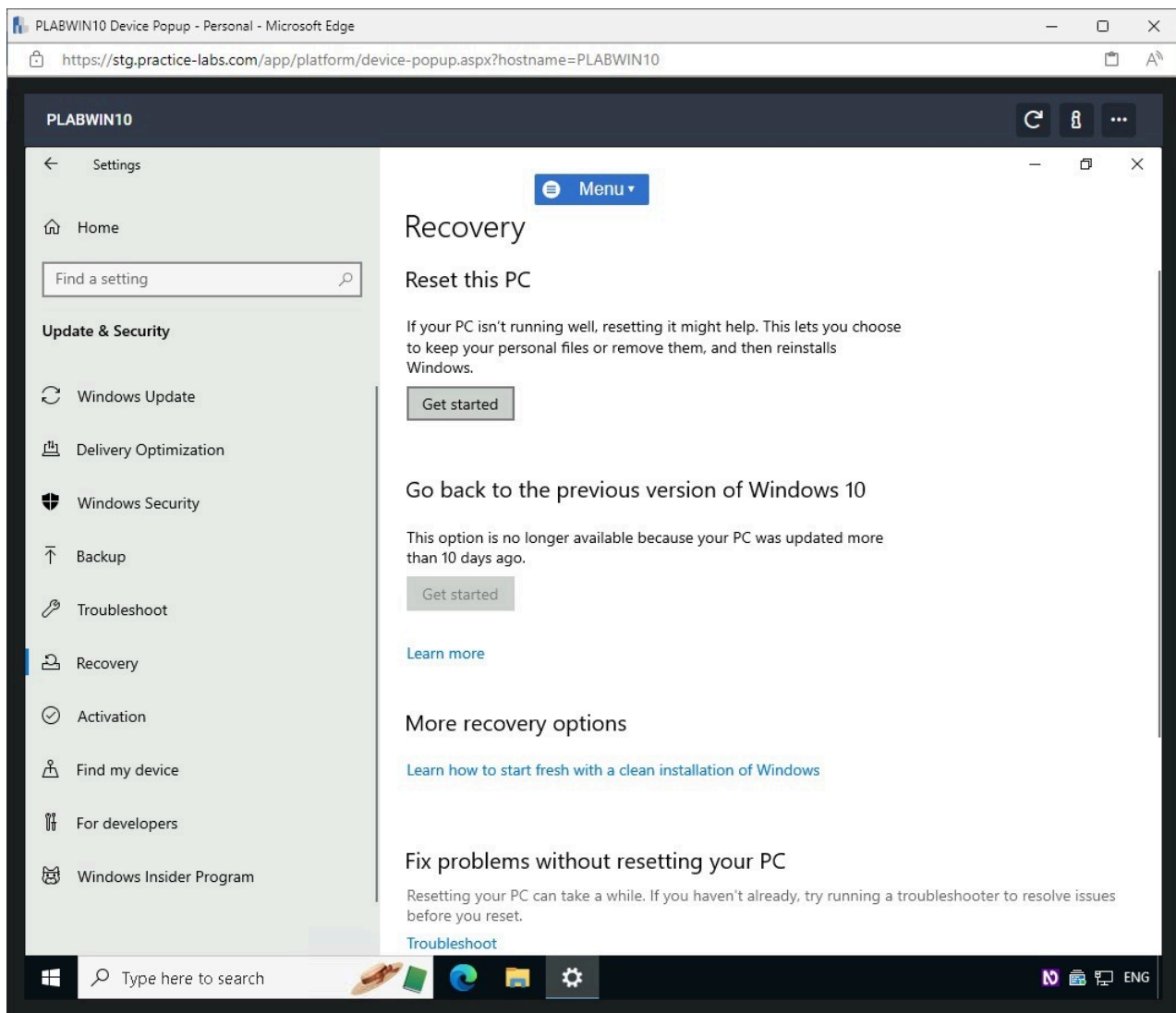


Figure 6.33 Screenshot of PLABWIN10: Displaying selecting the Get started button in the Settings - Recovery pane.

Step 5

In the **Reset this PC - Choose an option** window, you have two options

Keep my files is a partial reset where the user's files are saved.

Select **Remove everything** to begin a clean install.

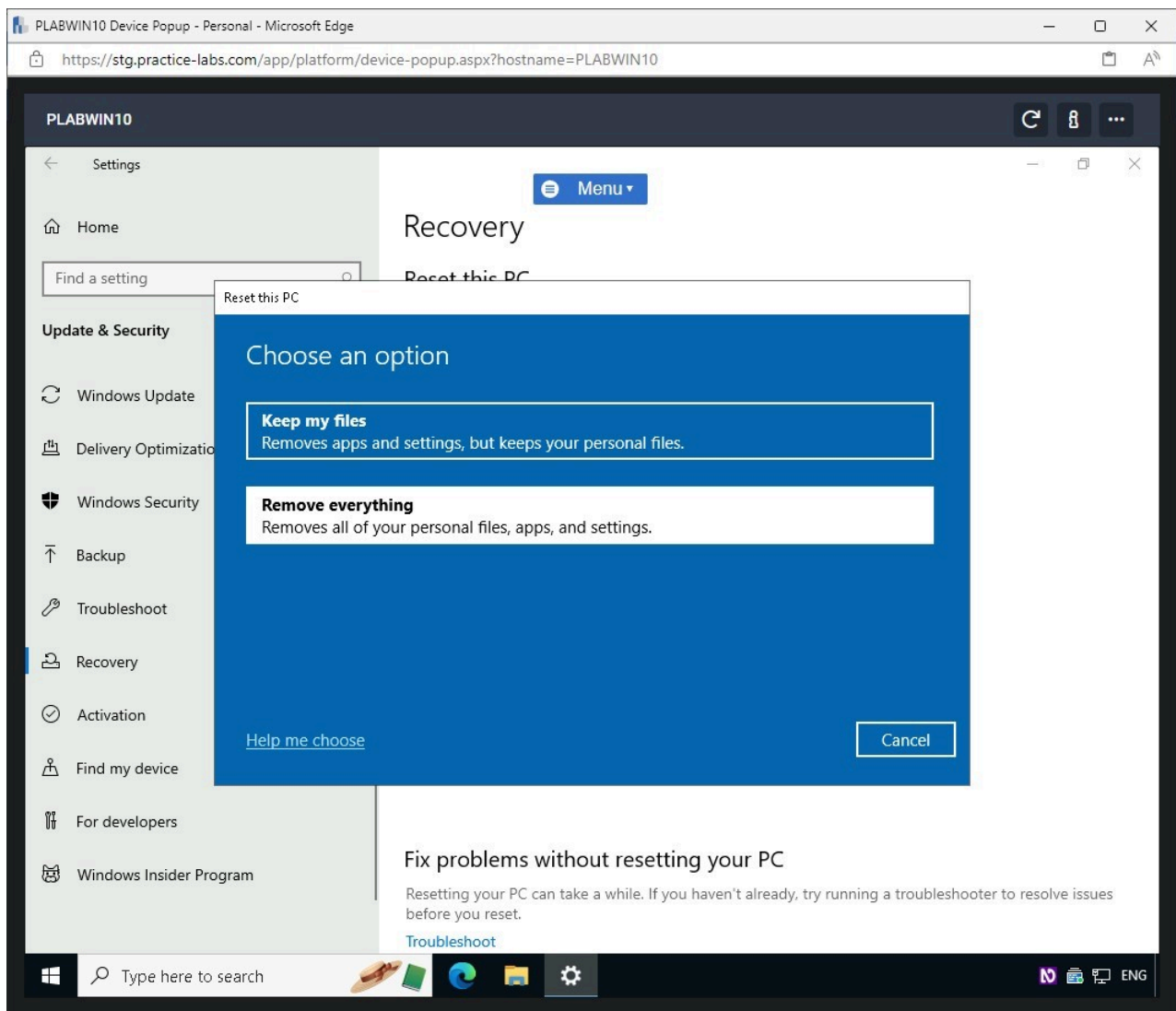


Figure 6.34 Screenshot of PLABWIN10: Displaying the Reset this PC - Choose an option window, showing Remove everything selected.

Step 6

In the **How would you like to reinstall Windows?** page, **Cloud download** will allow you to connect to a cloud service provider and get the Windows Operating System.

Select **Local reinstall**, which will access the recovery partition from the local hard drive.

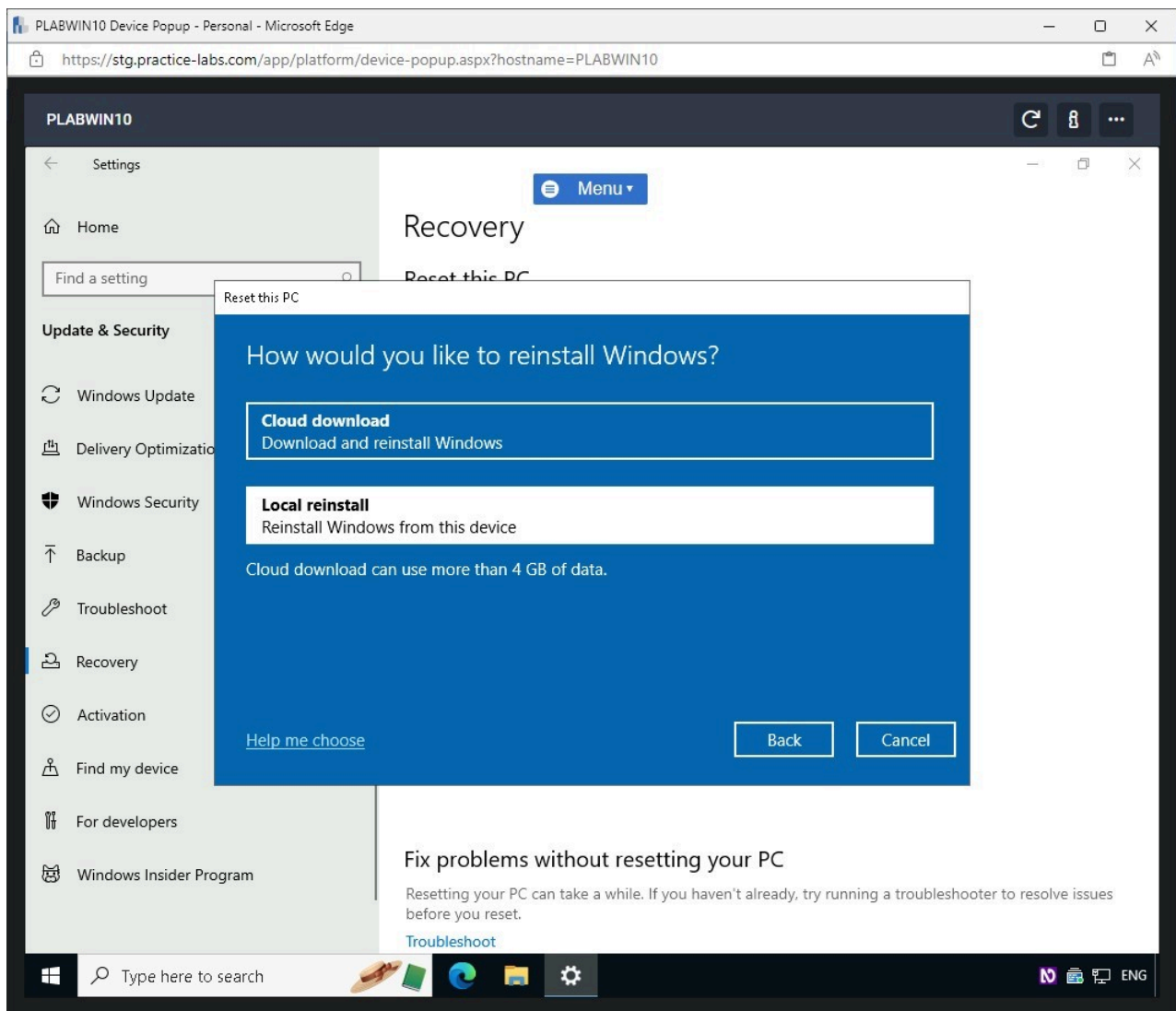


Figure 6.35 Screenshot of PLABWIN10: Displaying the How would you like to reinstall Windows? page, showing Local reinstall selected.

Step 7

On the **Additional settings** page, click **Next**.

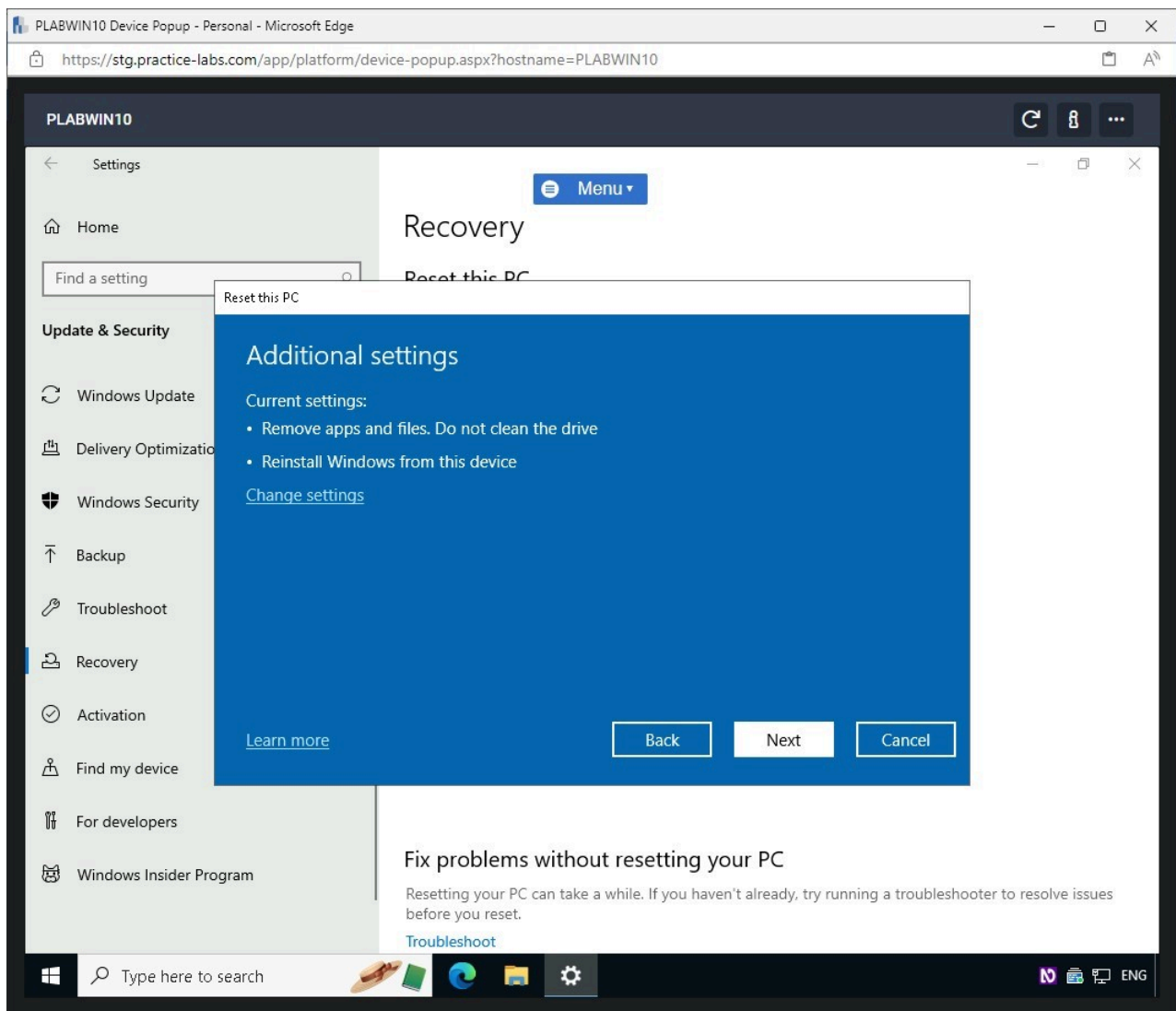


Figure 6.36 Screenshot of PLABWIN10: Displaying selecting Next on the Additional settings page.

Step 8

In the **Ready to reset this PC** page, click **Reset**.

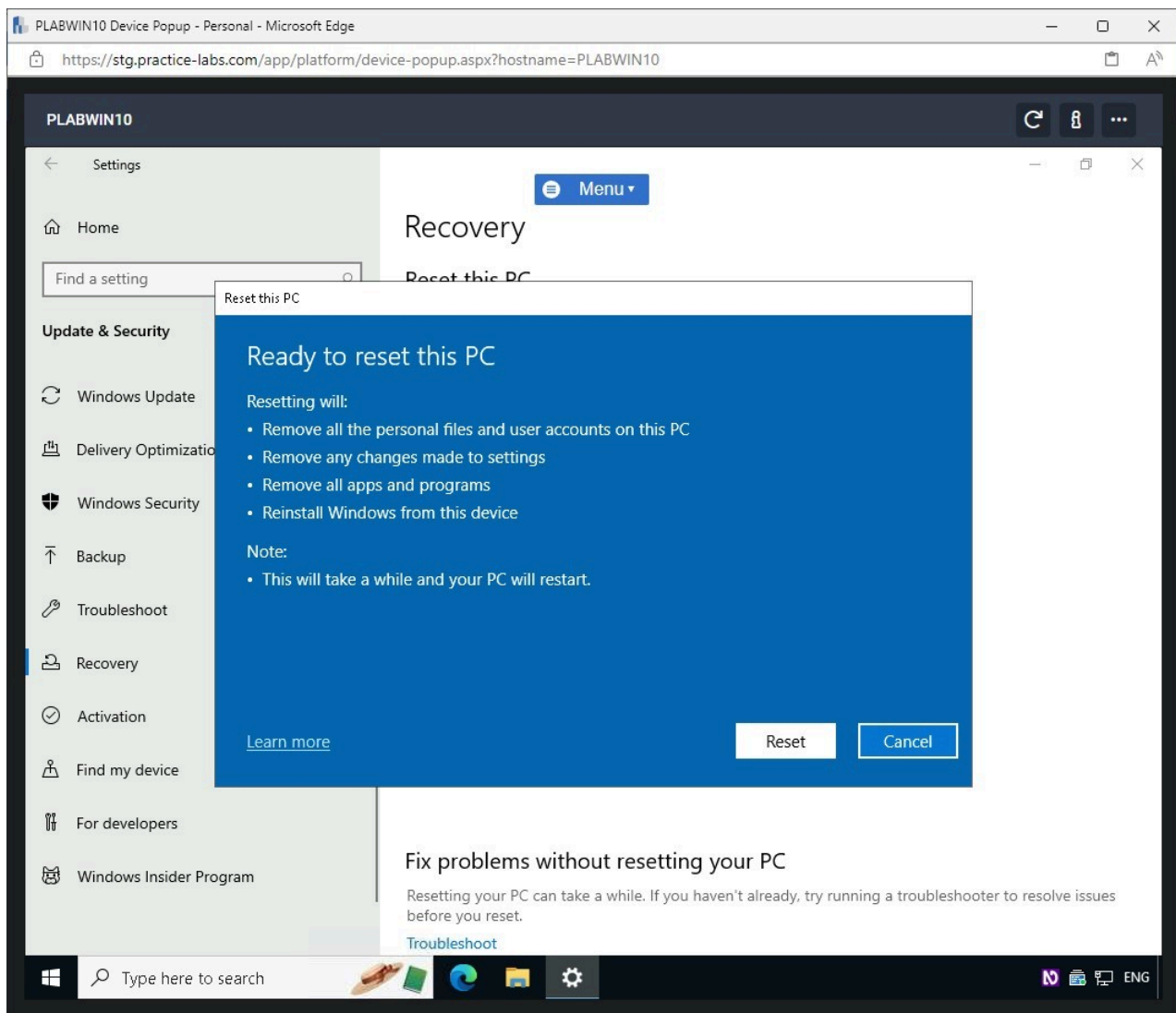


Figure 6.37 Screenshot of PLABWIN10: Displaying selecting Reset in the- Ready to reset this PC page.

Step 9

The reset process starts.

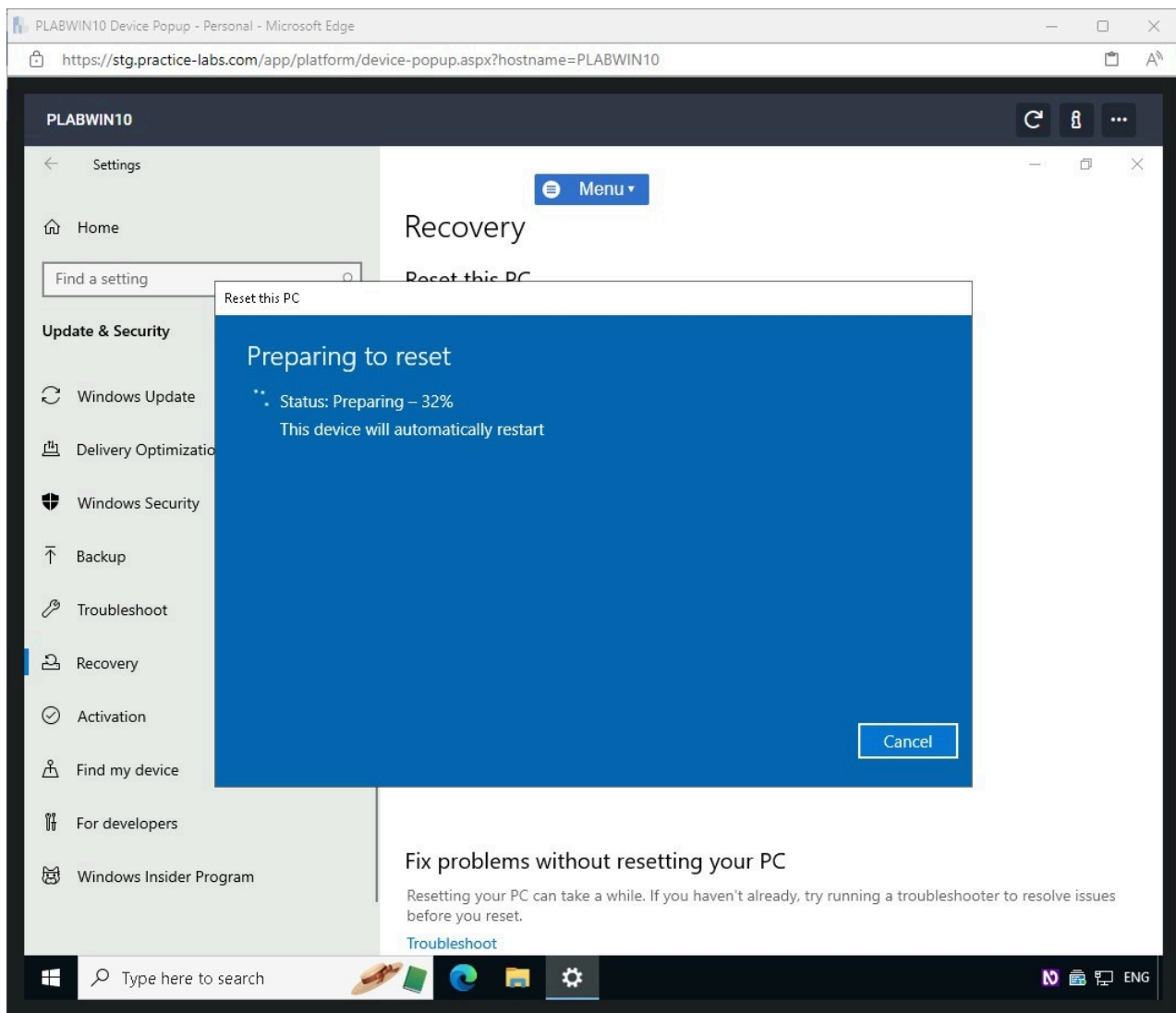


Figure 6.38 Screenshot of PLABWIN10: Displaying the reset progress in the Reset this PC window.

Step 10

To reconnect to the machine, access the secondary connection by selecting the ellipses (...) icon in the upper right corner and selecting **Console Mode**.

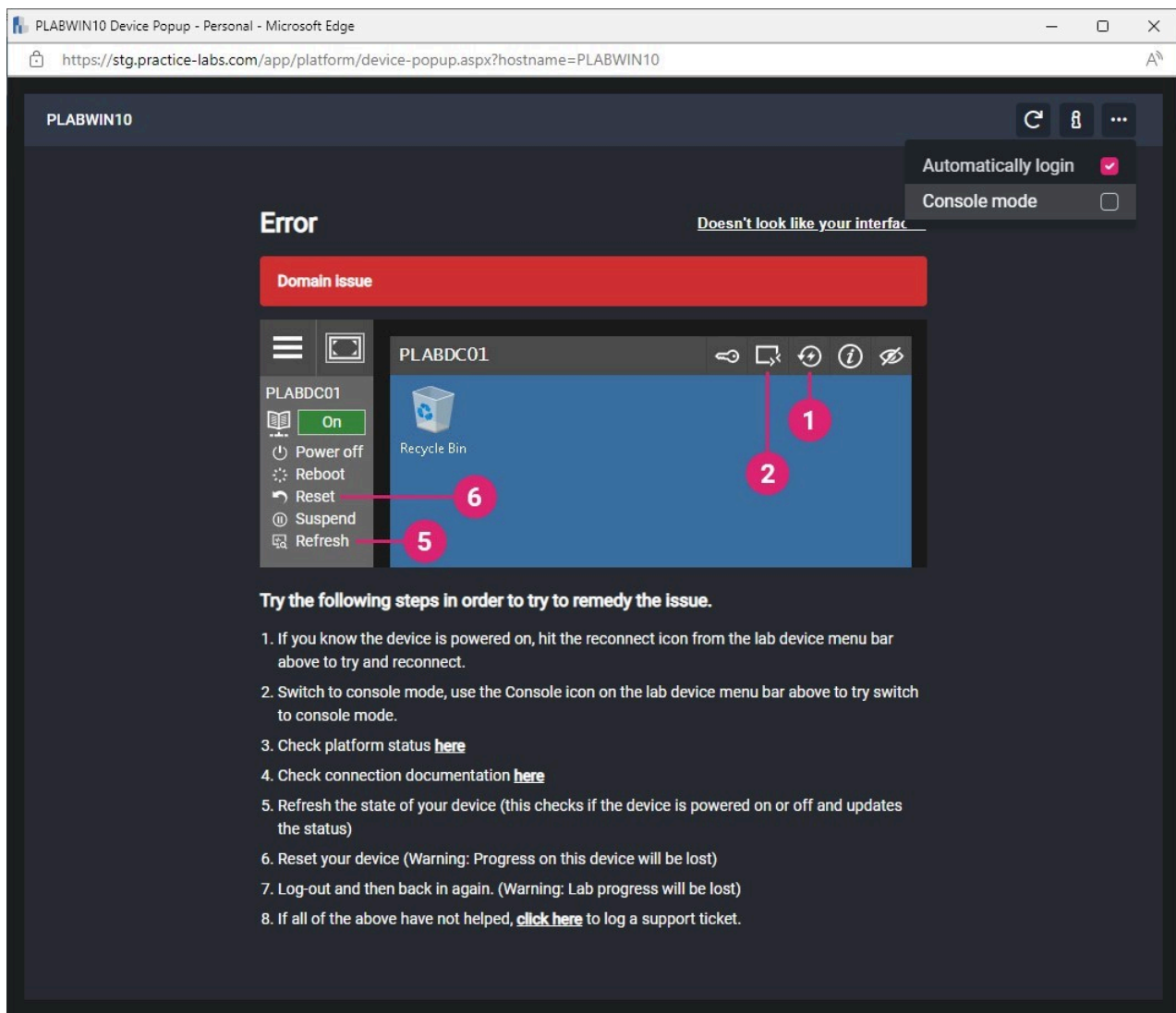


Figure 6.39 Screenshot of PLABWIN10: Displaying accessing the required drop-down menu and selecting Console mode.

Step 11

The reset process for the PC starts.

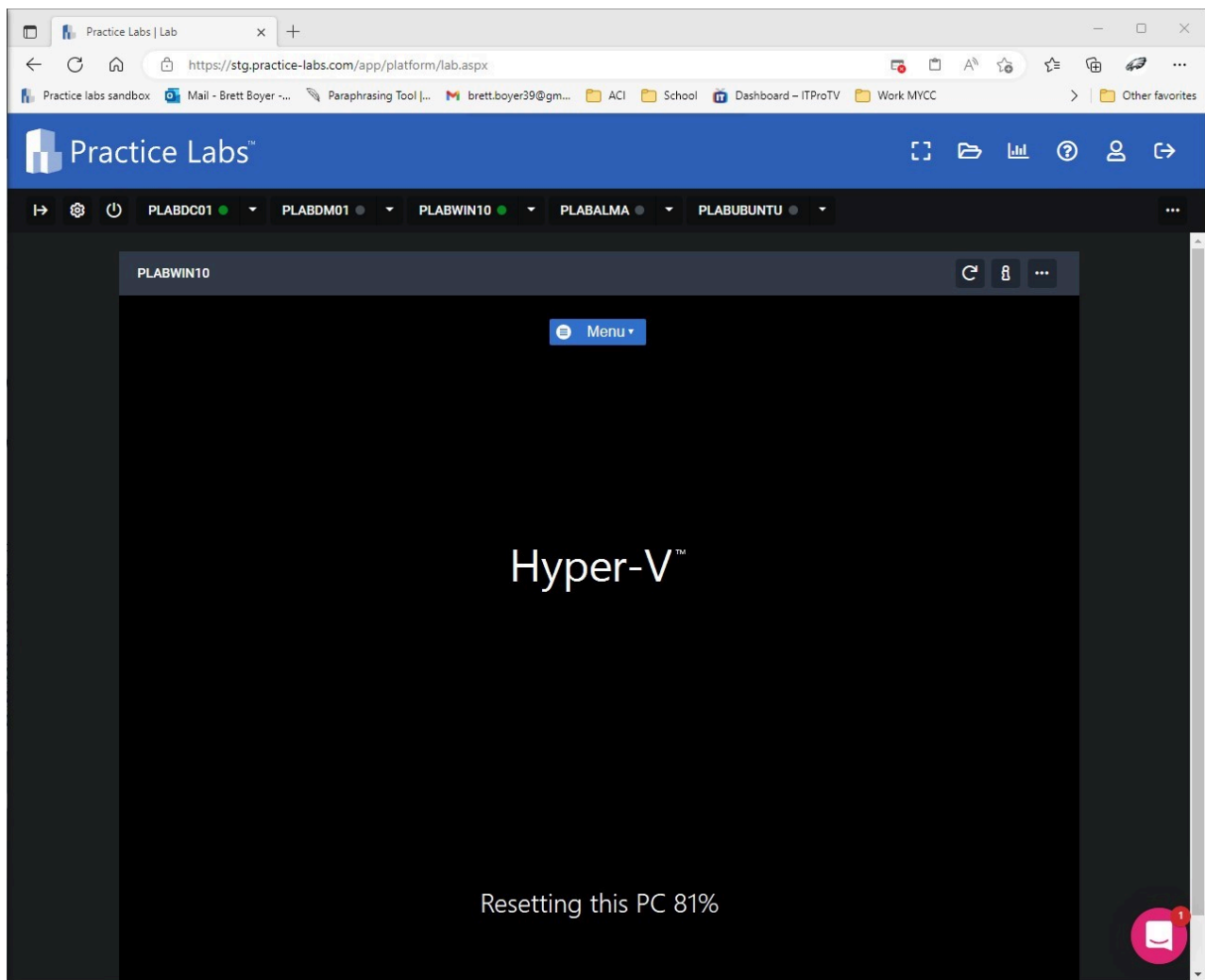


Figure 6.40 Screenshot of PLABWIN10: Displaying resetting this PC message.

Step 12

The device restarts and is now in the process of installing windows.

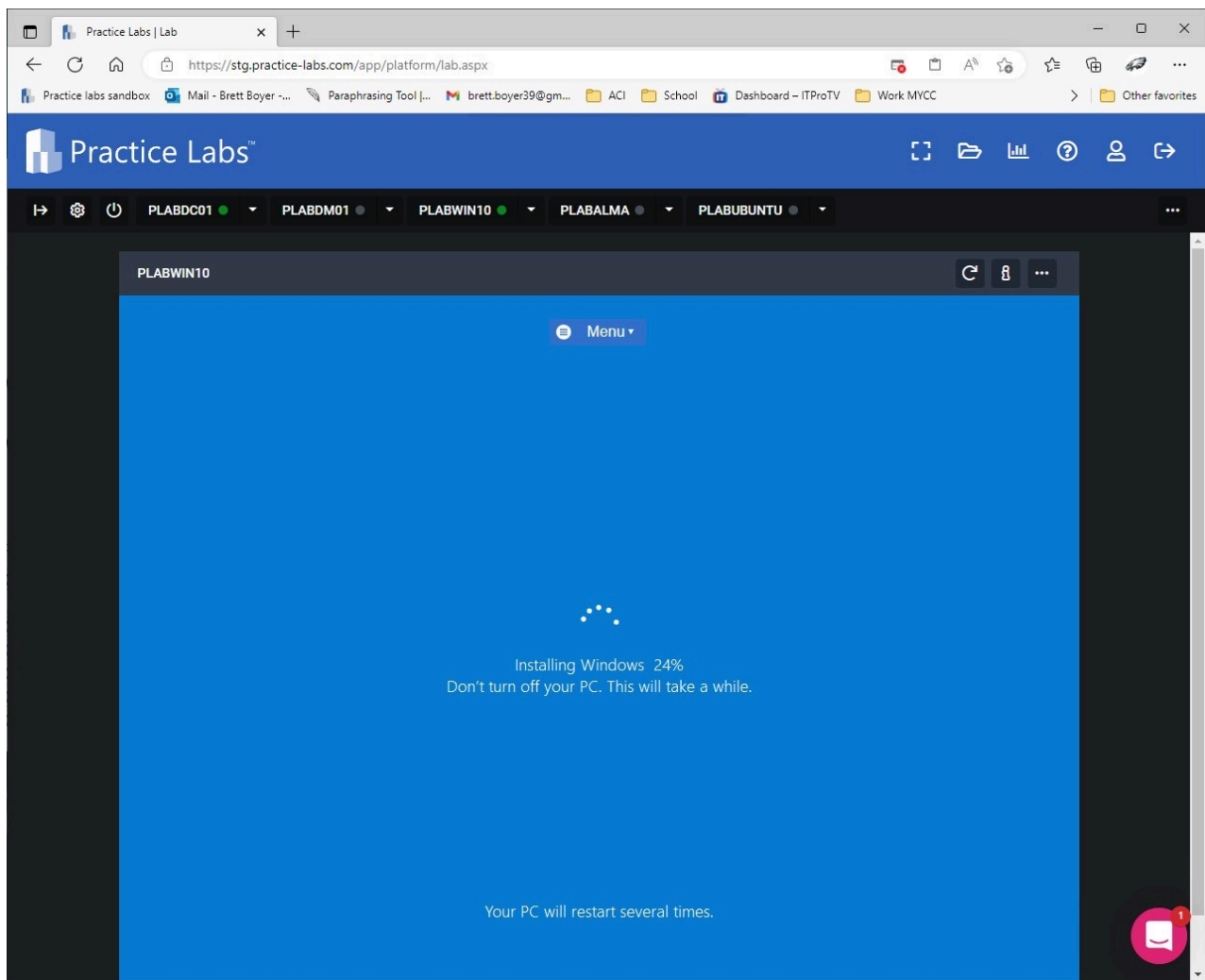


Figure 6.41 Screenshot of PLABWIN10: Displaying Windows installation in progress.

Step 13

In the **Region** window, select **United States** and click **Yes**.

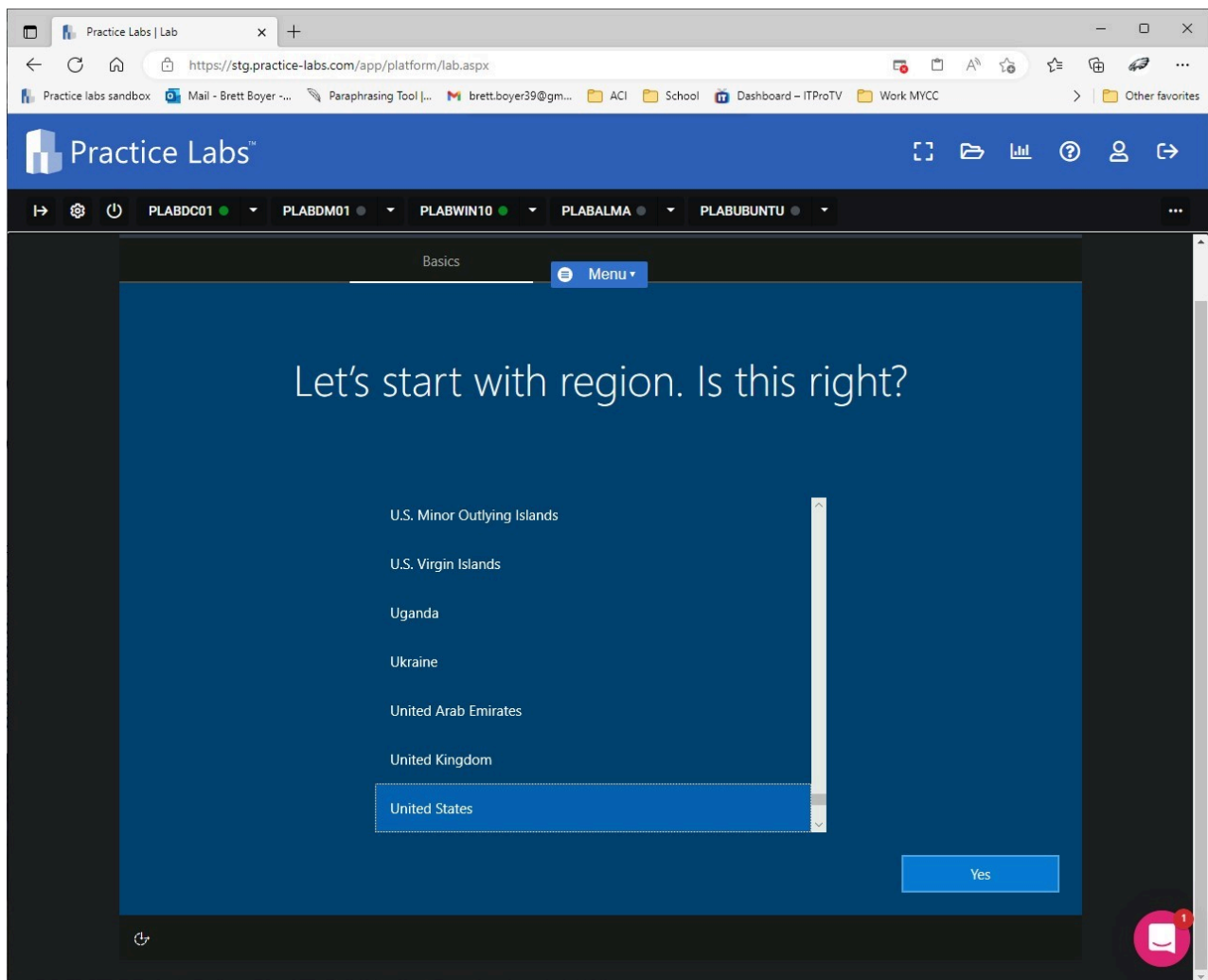


Figure 6.42 Screenshot of PLABWIN10: Displaying the region window, showing the required region selected and the Yes button highlighted.

Step 14

In the **keyboard layout** window, select **US** and click **Yes**.

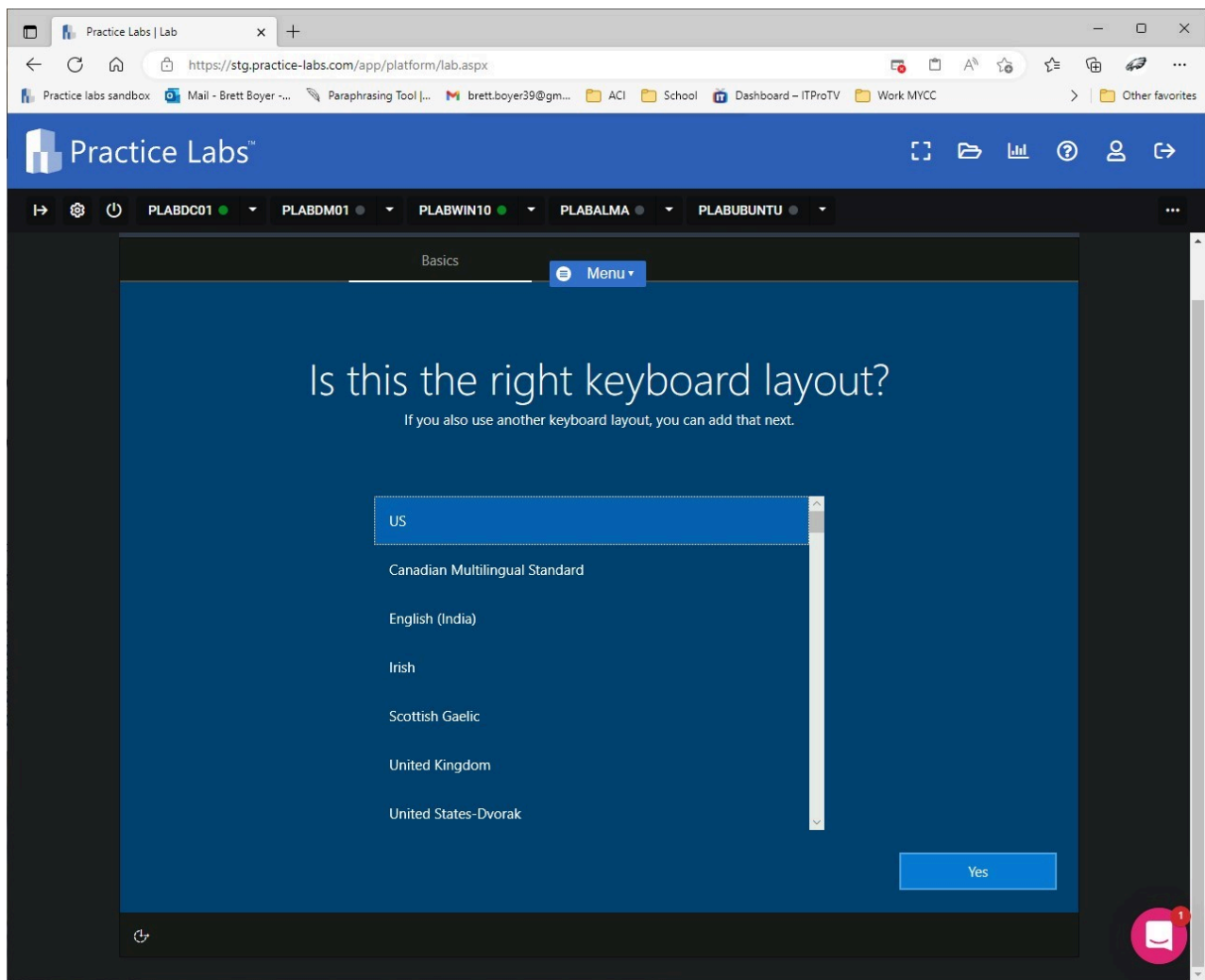


Figure 6.43 Screenshot of PLABWIN10: Displaying the keyboard layout window, showing the required layout selected and the Yes button highlighted.

Step 15

Click **Skip** on the **second keyboard layout** window.

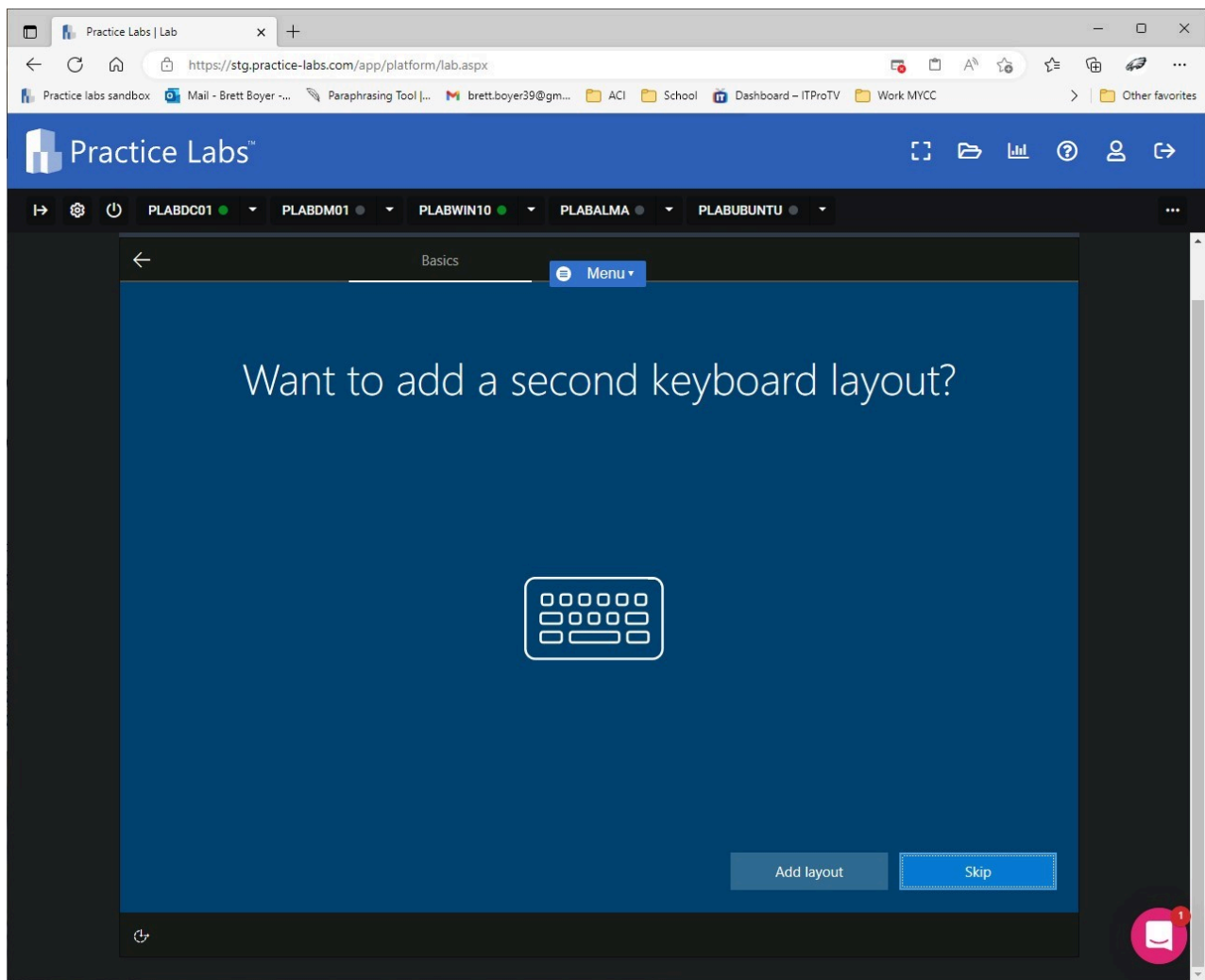


Figure 6.44 Screenshot of PLABWIN10: Displaying selecting Skip on the second keyboard layout window.

Step 16

In the **Let's connect to you a network** window, click **I don't have internet**.

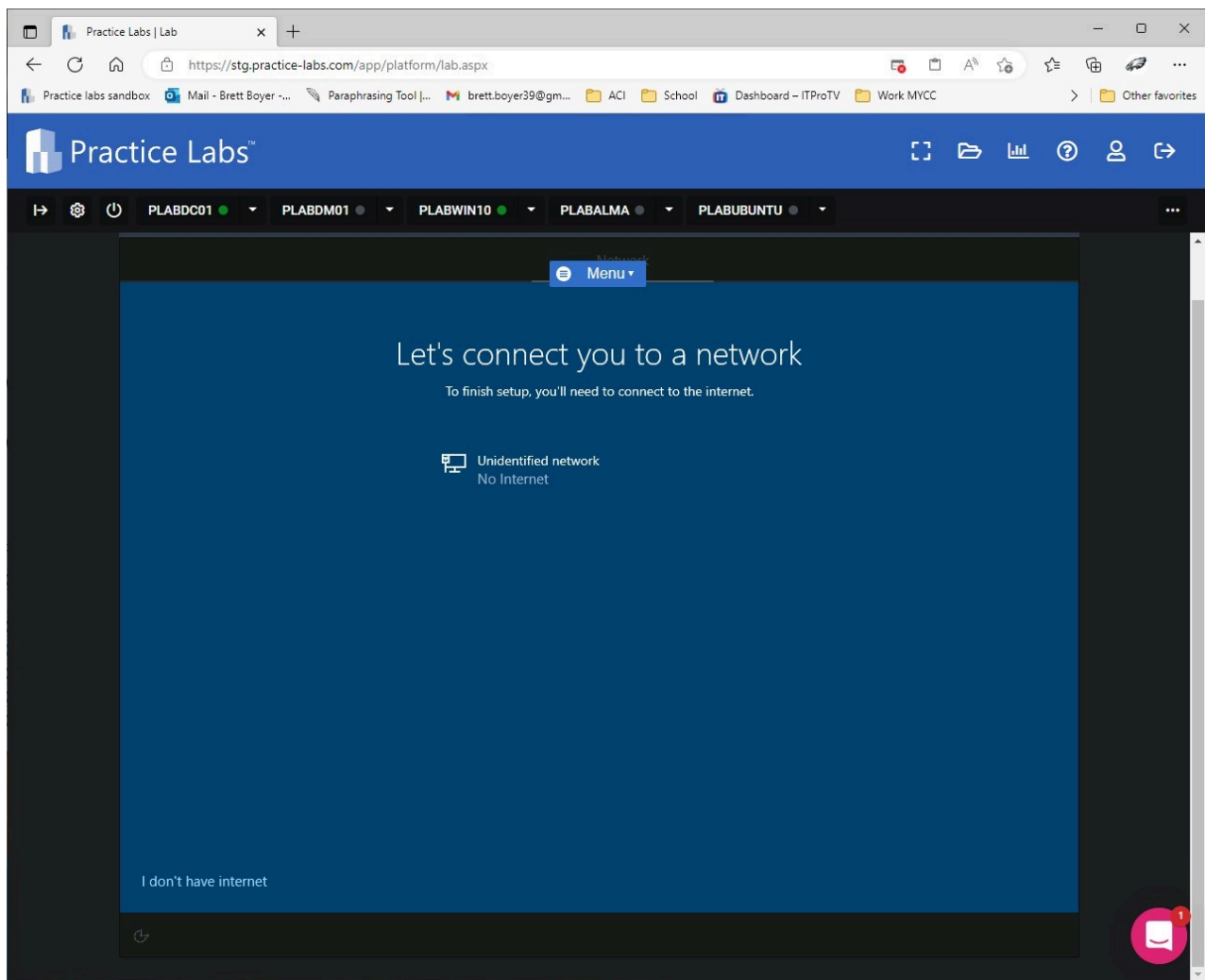


Figure 6.45 Screenshot of PLABWIN10: Displaying selecting I don't have internet In the Let's connect to you a network window.

Step 17

In the **There's more to discover...** window, select **Continue with limited setup**.

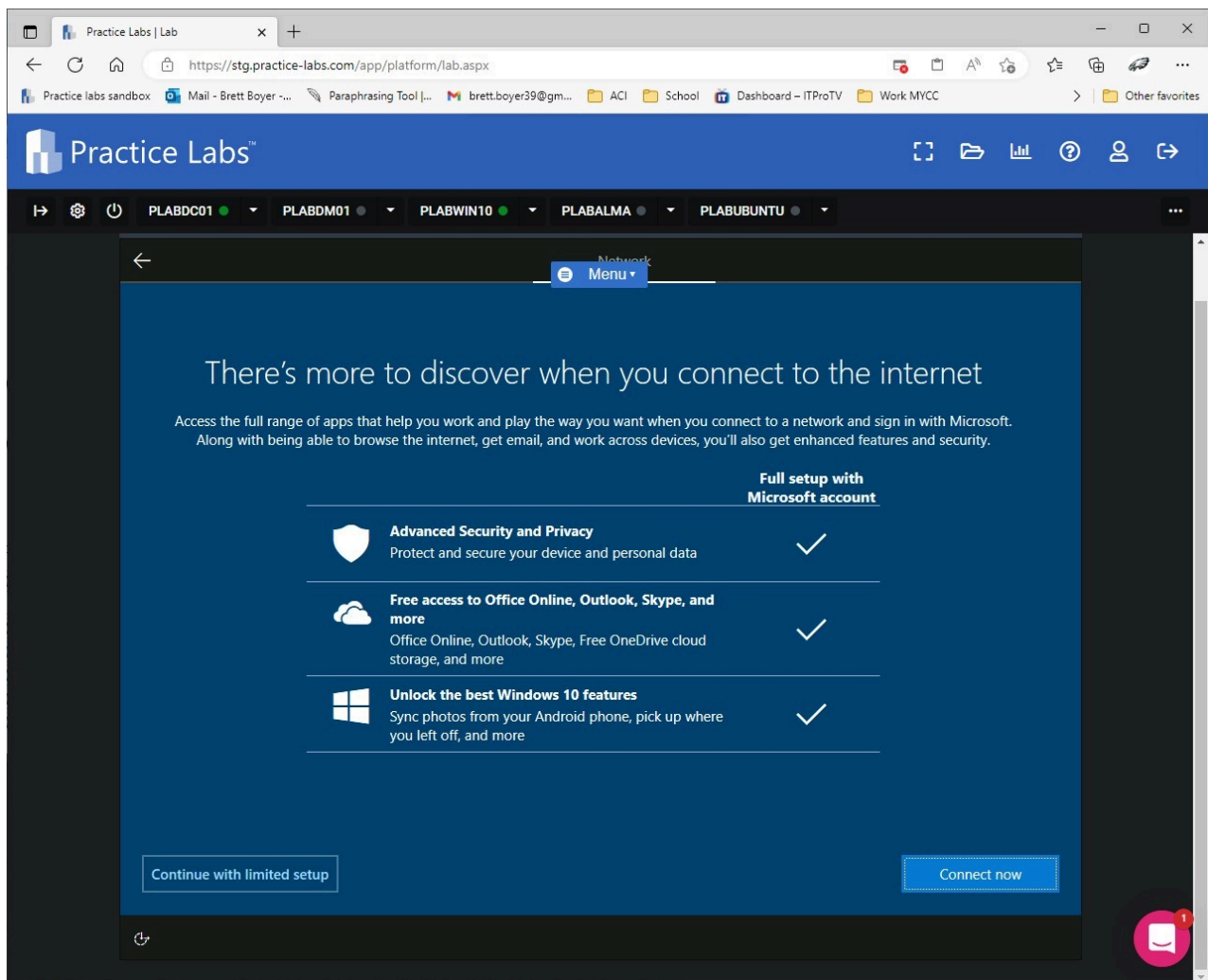


Figure 6.46 Screenshot of PLABWIN10: Displaying selecting Continue with limited setup in the There's more to discover... window.

Step 18

Click **Accept** on the **Windows License Agreement** window.

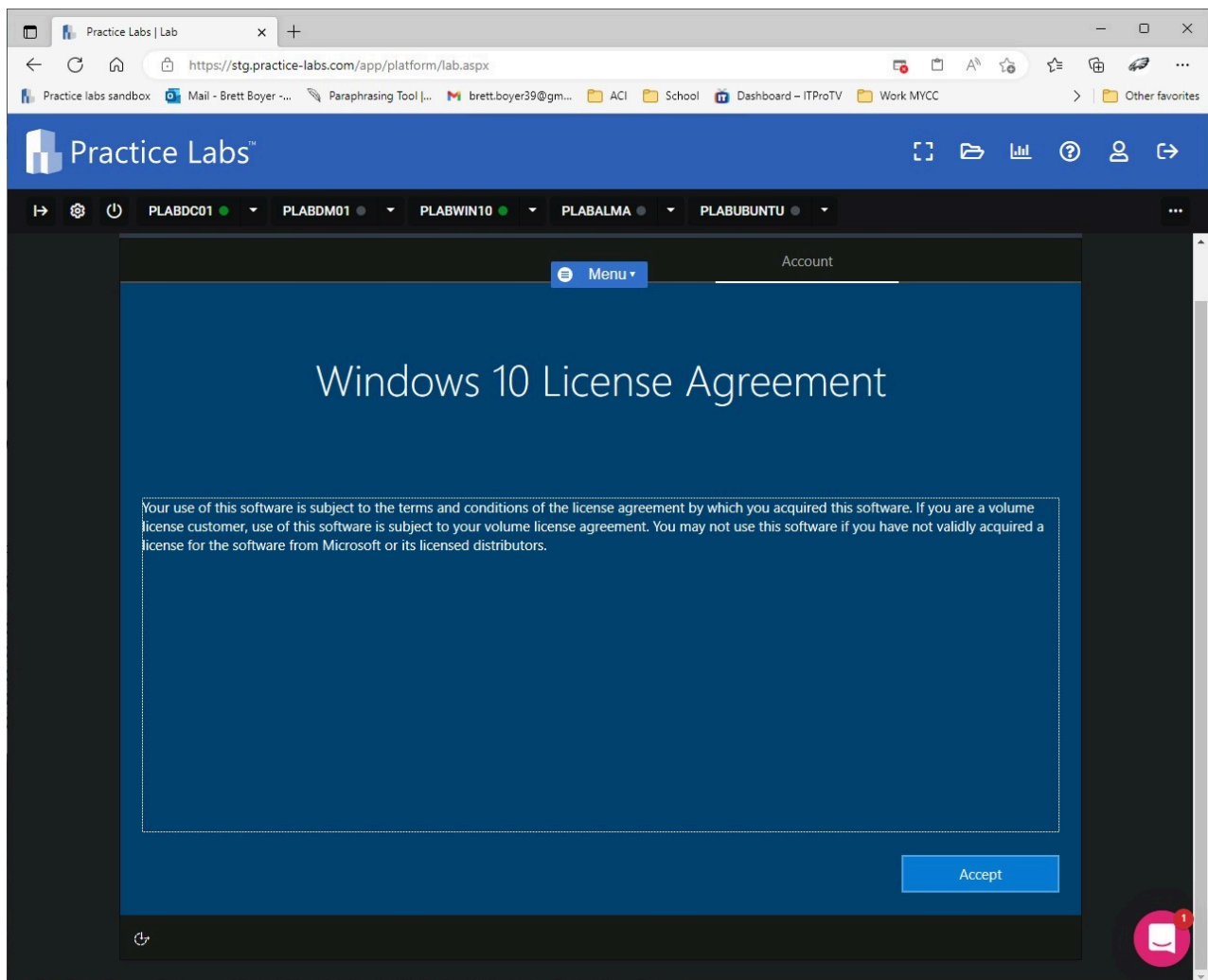


Figure 6.47 Screenshot of PLABWIN10: Displaying selecting Accept on the Windows License Agreement window.

Step 19

Type the following in the **Who's going to use this PC?** window:

Test User 1

Click **Next**.

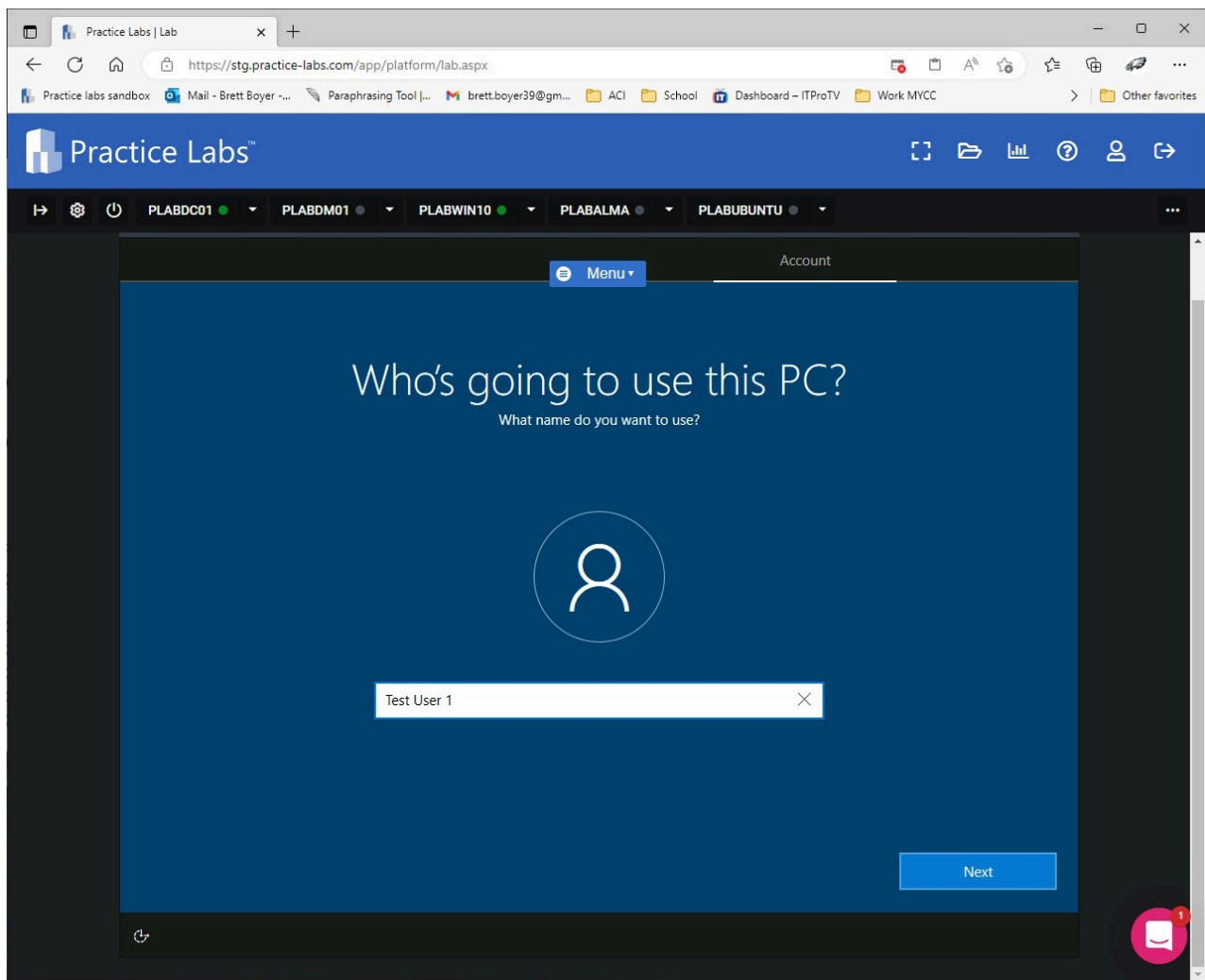


Figure 6.48 Screenshot of PLABWIN10: Displaying the Who's going to use this PC? window, showing the required username typed in and Next selected.

Step 20

Type the following in the **Create a super memorable password** text box:

Passw0rd

Click **Next**.

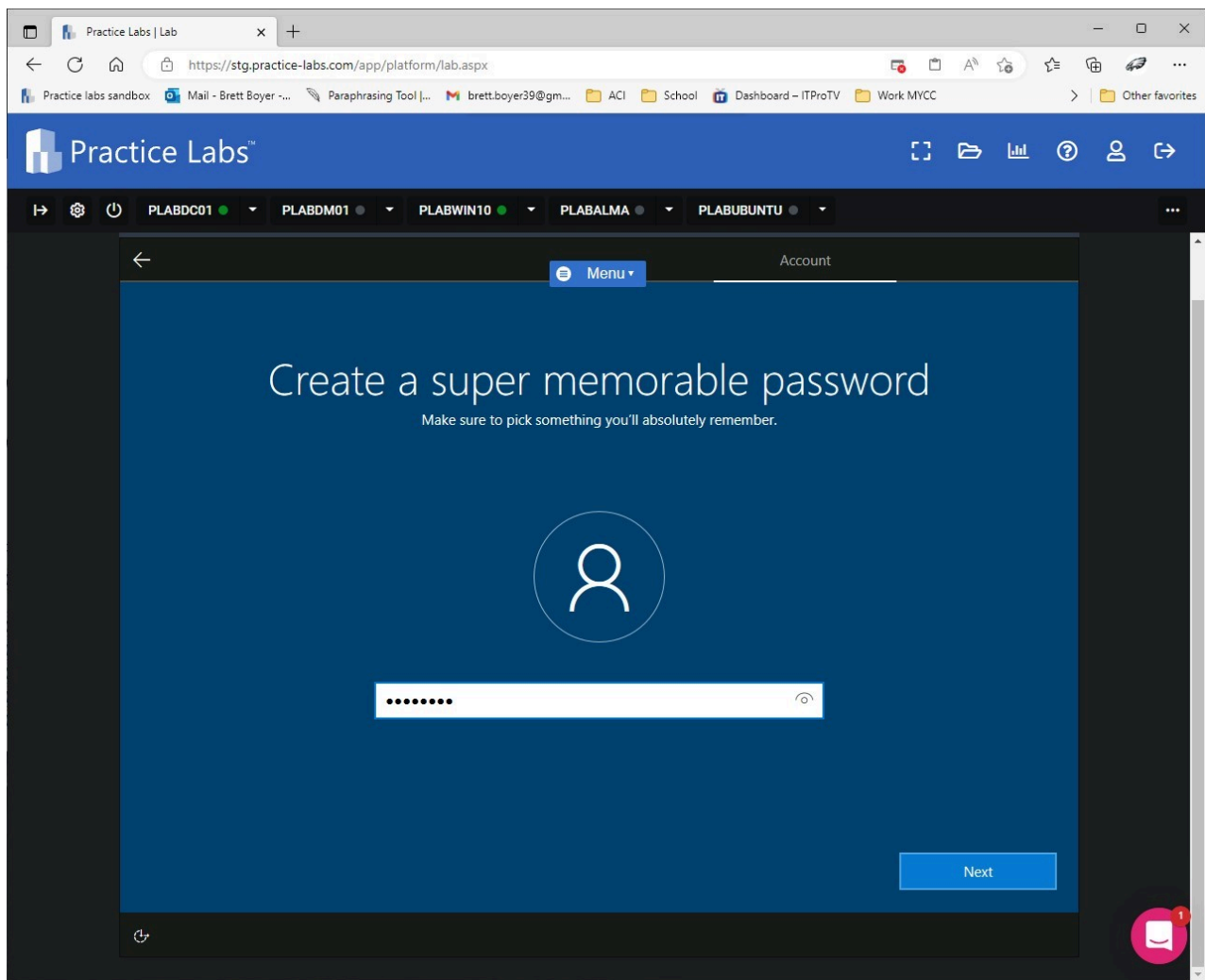


Figure 6.49 Screenshot of PLABWIN10: Displaying the Create a super memorable password window, showing the required password typed in and Next selected.

Step 21

Type the following in the **Confirm your password** text box:

Passw0rd

Click **Next**.

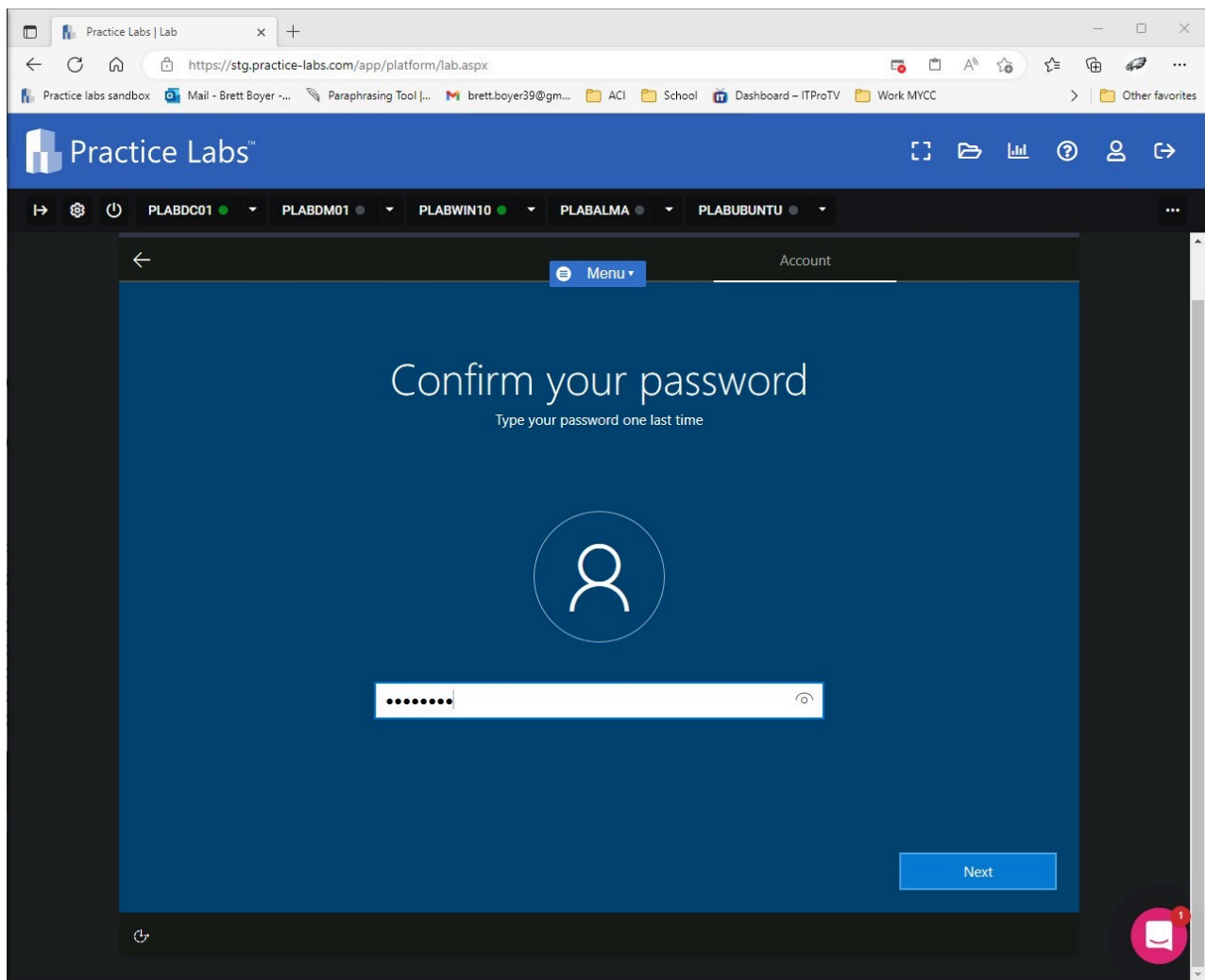


Figure 6.50 Screenshot of PLABWIN10: Displaying the Confirm your password window, showing the required password typed in and Next selected.

Step 22

In the **Create security questions for this account** window, click the drop-down, choose any question, and type anything in the text box.

Click **Next**.

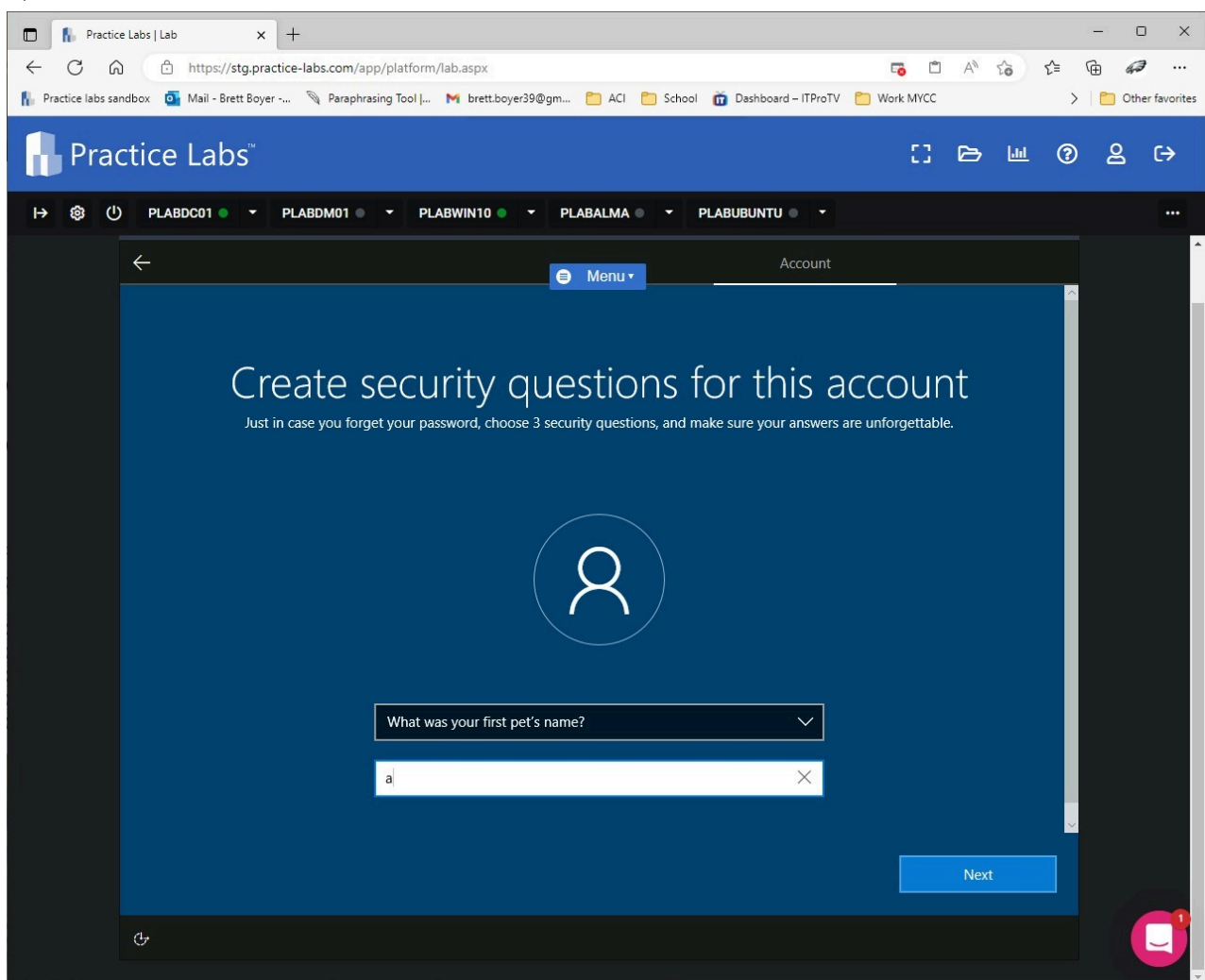


Figure 6.51 Screenshot of PLABWIN10: Displaying the Create security questions for this account window, showing the required settings performed and Next selected.

Step 23

In the **Create security questions for this account** window, click the drop-down, choose any question, and type anything in the text box.

Click **Next**.

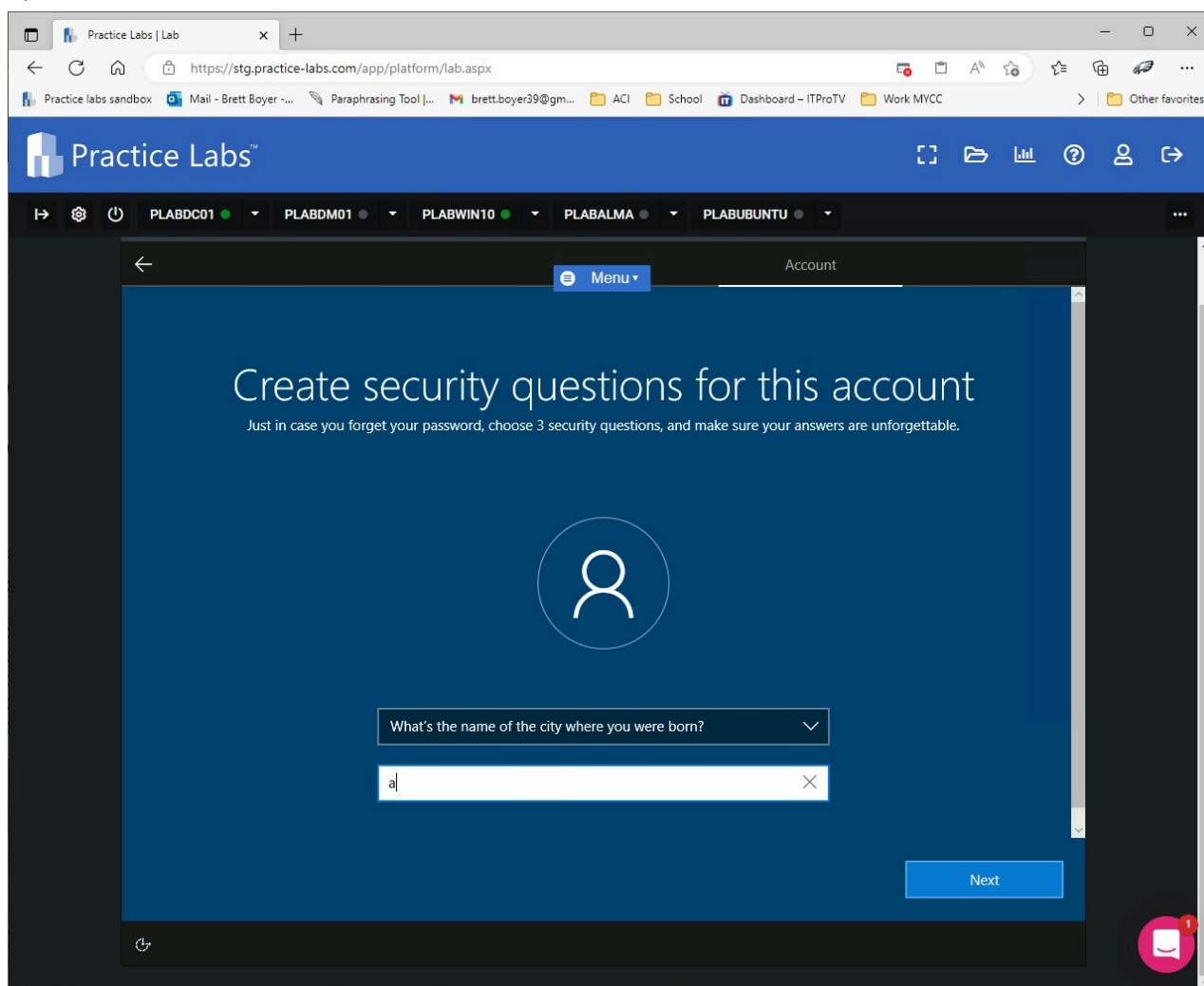


Figure 6.52 Screenshot of PLABWIN10: Displaying the Create security questions for this account window, showing the required settings performed and Next selected.

Step 24

In the **Create security questions for this account** window, click the drop-down, choose any question, and type anything in the text box.

Click **Next**.

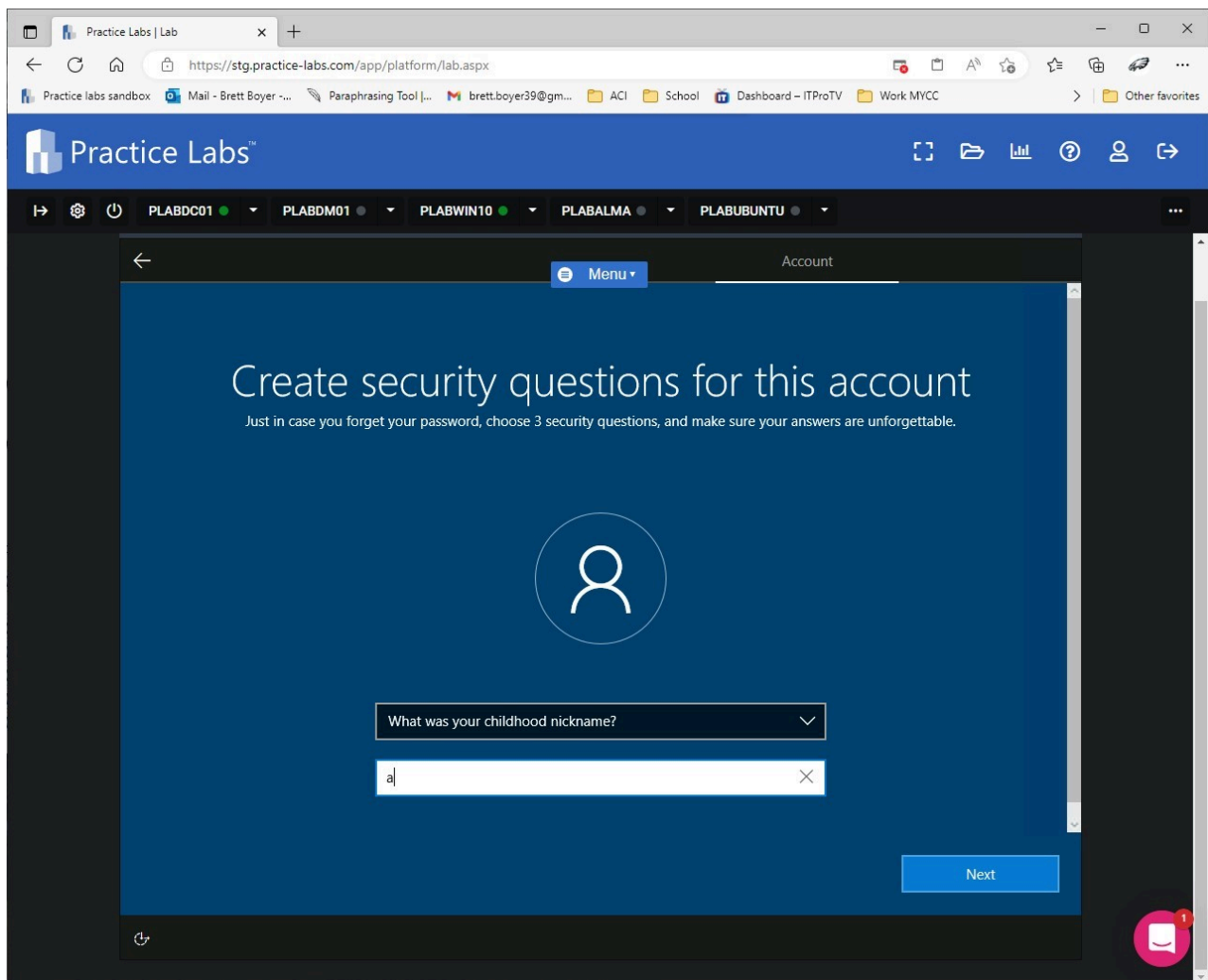


Figure 6.53 Screenshot of PLABWIN10: Displaying the Create security questions for this account window, showing the required settings performed and Next selected.

Step 25

Click **Accept** in the **privacy settings** window.

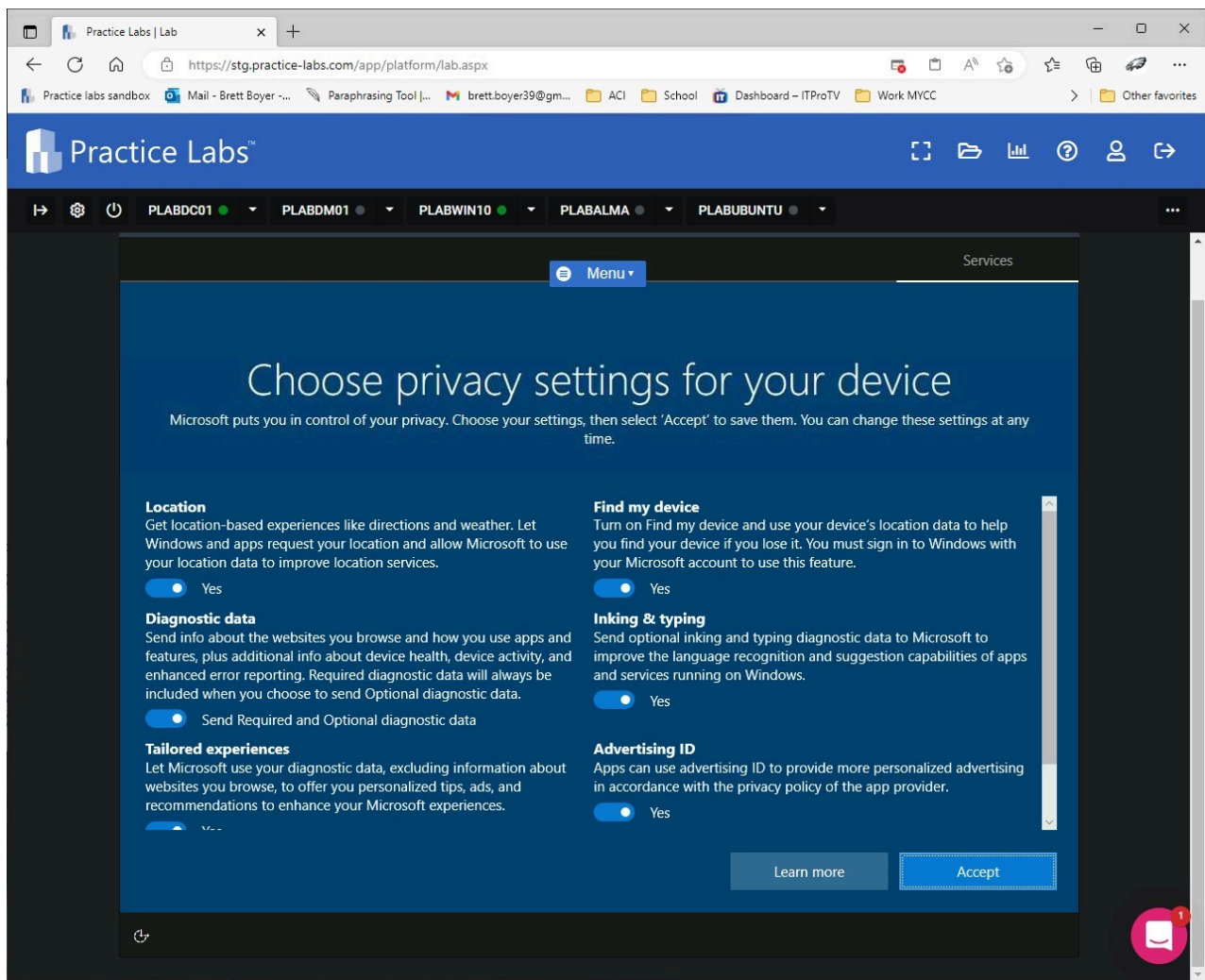


Figure 6.54 Screenshot of PLABWIN10: Displaying selecting Accept in the privacy settings window.

Step 26

Click **Not now** on the **Let Cortana help you get things done** window.

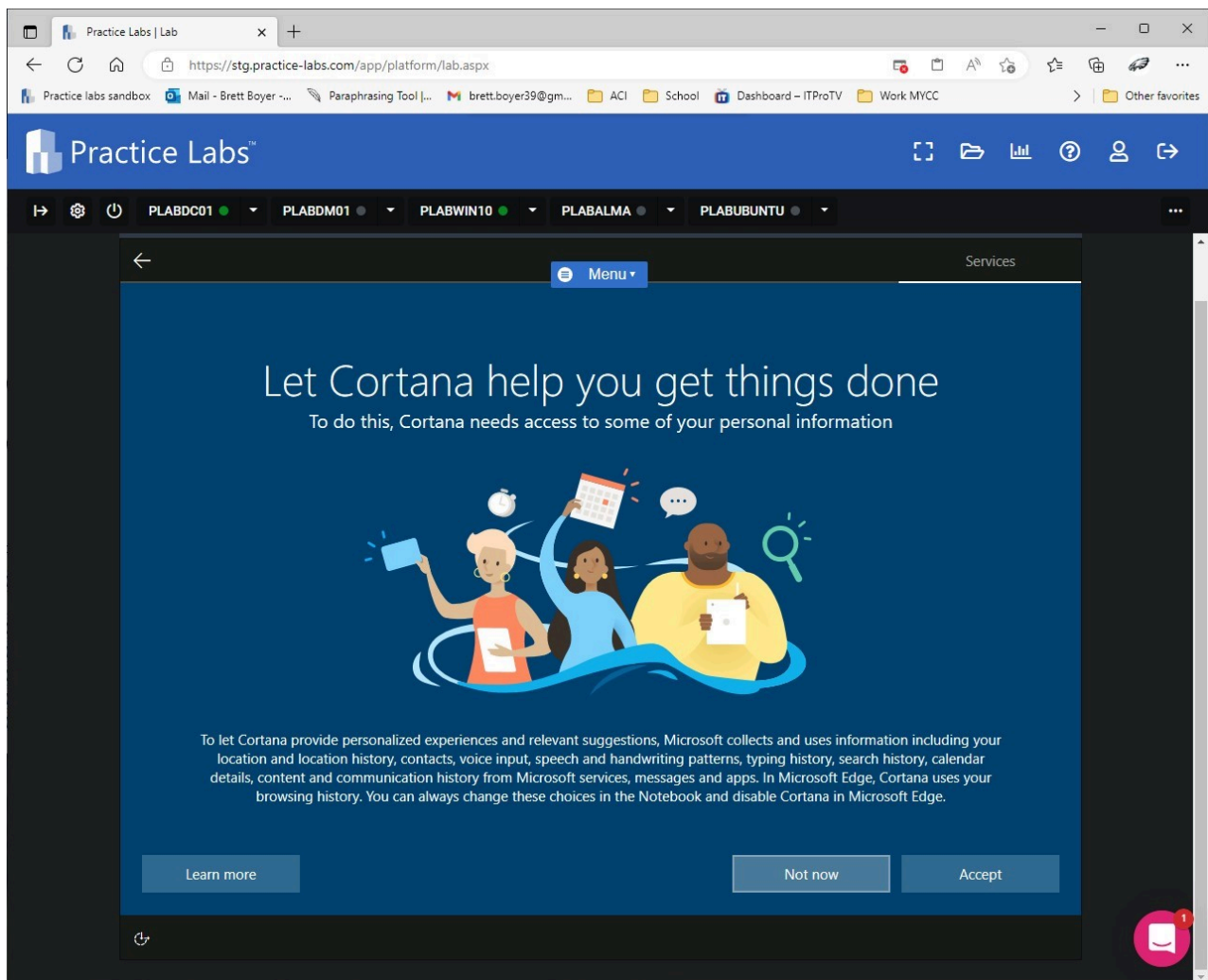


Figure 6.55 Screenshot of PLABWIN10: Displaying selecting Not now on the Let Cortana help you get things done window.

Step 27

The PC has now been **Reset** and is back to a **clean install**.

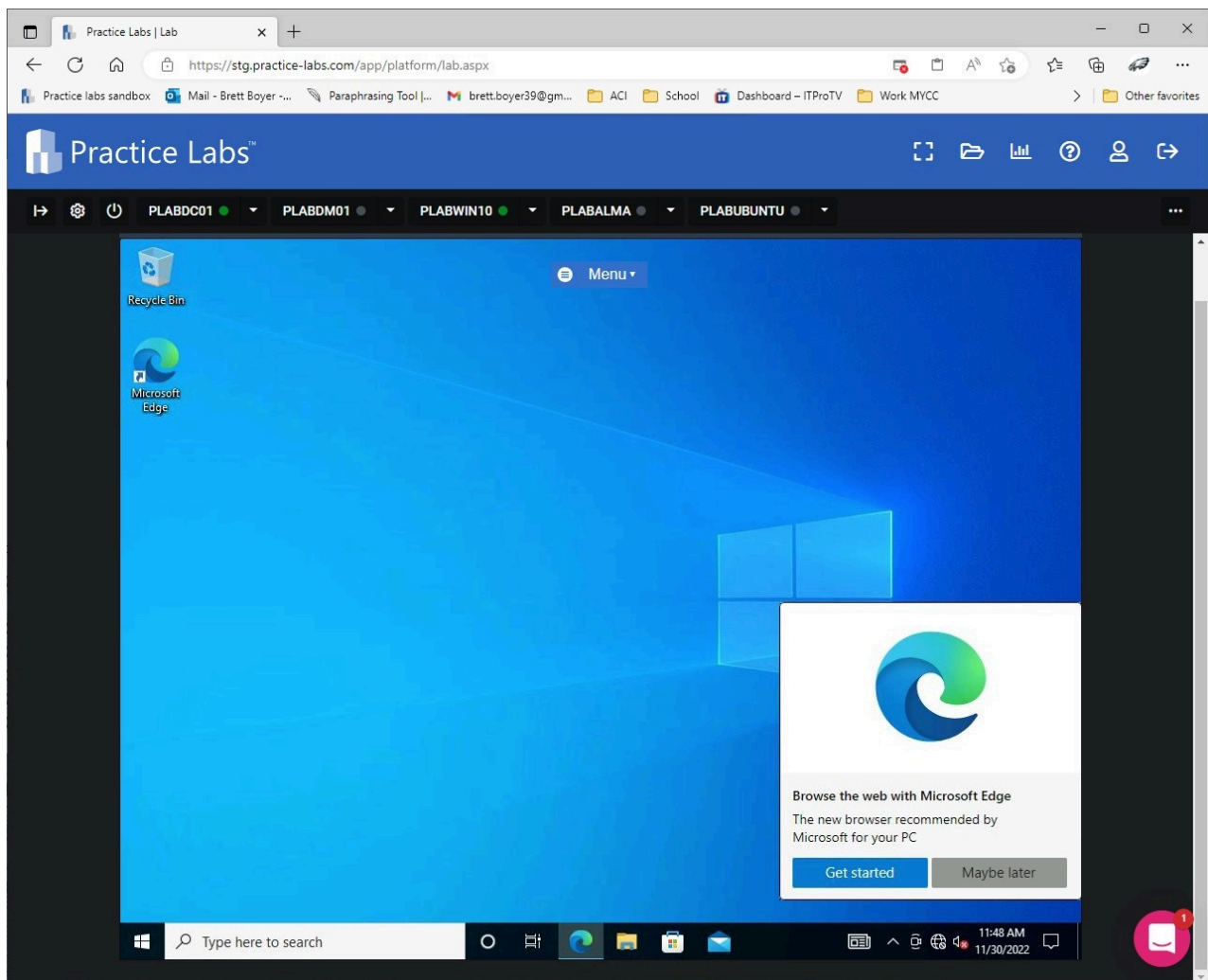


Figure 6.56 Screenshot of PLABWIN10: The computer has now been Reset and is back to a clean install.

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Server Operating Troubleshooting Techniques** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Common Problems & Causes
- Exercise 2 - Services and Processes

- Exercise 3 - Scheduled Reboots
- Exercise 4 - Services
- Exercise 5 - Privilege Escalation
- Exercise 6 - Recovery

You should now be able to:

- Create a Rule Blocking a Port
- Allow Secure Connection through the Windows Defender Firewall
- Allow a Program through the Windows Defender Firewall
- Block a Port with Group Policy Management
- Use Task Scheduler to Schedule Reboots
- Start and Stop a Service
- Use Run as Administrator
- Use the Sudo/Su Command
- Explore Safe Mode
- Work with Snapshots
- Reload OS

You should now have further knowledge of:

- Defining Common Problems
- Causes of Common Problems

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.